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A study of Unconventional Wings and Tails

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A. S.

*It is possible to fly without motors,
but not without knowdelge and skill.
Wilbur Wright*

*dedicato a tutti gli appassionati,
ingengeri aeronautici
e piloti
di oggi e di domani.*

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List of Symbols

Latin Symbols

$\mathcal{R}$	Aspect Ratio	
b	Wingspan	m
c	Chord	m
C_D	Wing Drag coefficient	
C_{Di}	Wing induced drag coefficient	
C_{Dv}	Wing viscous drag coefficient	
C_{dv}	Section viscous drag coefficient	
C_f	Friction coefficient	
C_L	Wing lift coefficient	
C_l	Section lift coefficient	
C_m	Wing Pitching moment coefficient	
C_n	Wing Yawing moment coefficient	
D	Drag	N
L	Lift	N
l	Tail arm	m
L/D	Lift-to-drag ratio	
m	Pitching Moment	N m
Ma	Mach Number	
n	Load Factor	
n	Yawing Moment	N m
q	Dynamic Pressure	Pa
R	Range	NM
Re	Reynold Number	
S_w	Wing surface area	m ²

SFC	Specific Fuel Consumption	lb/h/lbf
T	Thrust	N
V_∞	Freestream Velocity	m s^{-1}
V_a	Design maneuvering speed	m s^{-1}
V_c	Design cruise speed	m s^{-1}
V_d	Design dive speed	m s^{-1}
w	Downwash velocity	m s^{-1}
W/S	Wing loading	kg m^{-2}
W_e	Empty weight,	kg
W_f	Fuel weight,	kg
W_{py}	Payload weight,	kg
x	Coordinate along fuselage axis	m
Y	Side Force	N
y	Coordinate along right half-wing	m
z	Coordinate along aircraft symmetry plane	m

Greek Symbols

α	Angle of attack	°
α_0	Zero-lift angle of attack	°
α_g	Geometric twist	°
α_{2D}	Section angle of attack	°
β	Angle of sideslip	°
δ	Control surface deflection	°
η_q	Tail dynamic pressure efficiency	
Γ	Circulation	$\text{m}^2 \text{s}^{-1}$
Γ	Dihedral	°
γ	Viscous correction angle	°
Λ	Sweep	°
λ	Taper Ratio	
φ	Roll Angle	°
ρ	Air Density	kg m^{-3}
σ	Normal Stress	MPa

τ	Shear Stress	MPa
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Subscripts

α	Angle of attack derivative
β	Angle of sideslip derivative
δ	Control derivative
AC	Aerodynamic Center
CG	Center of Gravity
e	Elevator
f	Flaps
MAC	Mean aerodynamic chord
r	Rudder
t	Tail
TH	Horizontal tail
TV	Vertical tail
wb	Wing-body

Sommario

In questo elaborato si è andato a studiare le proprietà aerodinamiche della distribuzione di portanza "a campana" individuata da Prandtl nel 1933. Questa distribuzione è stata riscoperta più volte nel corso del XX secolo e recentemente utilizzata dalla NASA per un drone ad ala volante. La distribuzione a campana è ottenuta da un problema di ottimizzazione della resistenza indotta in cui non si fissa l'apertura alare ma il momento flettente alla radice dell'ala. Il risultato teorico è un aumento di apertura alare del 22% e una riduzione del 11% di resistenza indotta rispetto ad una ala ellittica di uguale portanza. Questo studio analizza quindi le proprietà di e fuori progetto della distribuzione a campana e una sua applicazione pratica su un business jet.

Il metodo sviluppato prevede di dimensionare l'ala aumentando l'apertura alare e introducendo una distribuzione di corde che approssima la forma a campana di Prandtl cosicché da mantenere la riduzione di resistenza aerodinamica e contenere l'aumento di momento flettente anche fuori dalla condizione di progetto. La nuova ala ottenuta presenta una riduzione del 5% di resistenza aerodinamica totale e un aumento del 7.4% della efficienza aerodinamica L/D nella condizione di crociera di riferimento. Nonostante un aumento di massa strutturale dell'ala del 18.17%, si è ottenuto complessivamente una riduzione del consumo di combustibile di circa del 4.8%.

Gli sviluppi della NASA su questa distribuzione di portanza suggeriscono la possibilità di eliminare o ridurre la dimensione della coda verticale, poiché si è dimostrato che la distribuzione a campana permette di correggere le virate autonomamente. Si è quindi deciso di esplorare la possibilità di sostituire la tradizionale coda a tre superfici con una coda a "V". Il metodo sviluppato differisce da quello tradizionale poiché cerca di trarre dalle normative dei criteri di progetto. Si è quindi ricorso a un codice di ottimizzazione "particle swarm" per individuare il punto di progetto con la minima resistenza aerodinamica e peso ma soddisfacente dei requisiti normativi di controllo e stabilità. Il risultato ha permesso complessivamente una riduzione del consumo di carburante del 10% nonostante l'aumento della massa a vuoto del 4.53%.

Parole chiave: Distribuzione di Portanza a Campana, Prandtl, Resistenza Indotta, Coda a V, Ottimizzazione, Progetto Aerodinamico, Riduzione Consumi, Riduzione Emissioni di Gas Serra, Riduzione Costi

Abstract

This study investigated the aerodynamic properties of Prandtl's 1933 bell lift distribution. It has been rediscovered several times in the 20th century by different authors and recently used by NASA to build a flying wing drone. This lift distribution is obtained from an induced drag optimization problem where root bending moment is adopted as a constraint rather than fixing wingspan. The theoretical result is an increment of 22% of wingspan and a reduction of 11% of induced drag compared to an equivalent elliptical wing. This study analyzes the design and out of design point characteristics of the bell lift distribution and its application on a business jet.

The developed method sizes the wing by increasing the wingspan and introducing a chord distribution that approximates the bell shape in order to maintain the induced drag reduction and constrain the root bending moment over a wide range of operating points. The new wing obtained yielded a 5% reduction of drag and an increase of 7.4% of lift-to-drag ratio L/D in the reference cruise condition. Despite the 18.17% wing mass increment, an overall a reduction of 4.8% of burned fuel was obtained.

NASA's developments on this lift distribution suggest the opportunity of reducing or removing the vertical tail since they proved Prandtl's bell distribution does not have adverse yaw. Hence it was investigated the idea of replacing the traditional tail with a vee-tail. The developed design method is different from the classical tail volume method because it synthesizes design constraints from airworthiness regulations. A Particle Swarm Optimization code was used to find the optimum point with the least drag and weight yet still compliant with the regulatory requirements of stability and control. The overall result yielded a reduction of fuel burn of 10% with a empty weight increment of 4.53%.

Keywords: Bell Lift distribution, Prandtl, Induced drag, Vee-Tail, Optimization, Aerodynamic Design, Fuel Saving, Reduction of Greenhouse Emissions, Cost Reduction

Chapter 1

Introduction

Market forecasts predict that in the next 20 years the commercial aviation industry will grow by an average of 4.3 % every year [1], with an expected demand of 39'210 new airplanes, both for passenger and freight. Most of the demand will be for single-aisle airplanes, 76 % of the total deliveries, as the main drivers of traffic growth are continental flights in China and the United States.

However this increase in demand would intensify the commercial aviation effects on climate change, whose current contribution, according to the IPCC [2], is estimated between 2 % and 3 % of total greenhouse emissions CO₂. Aviation also contributes to the emission of Ozone-depleting gases NO_x, and cancerogenous substances like soot and carbon monoxide.

Due to these concerns, the European Union has been taking action in the last decade by introducing a carbon-emission trading scheme in order to incentivize the reduction of greenhouse emission. A similar approach has been developed by the ICAO for international air travel, including measures to promote alternative fuels [3].

By far the most straightforward method to reduce aircraft emissions is reducing fuel consumption. For a transport aircraft, most of the mission time is spent in cruising, thus for a qualitative analysis it's possible to use the Breguet range equation to estimate the burned fuel weight for a given range (Eq. 1.1).

$$W_f = (W_{py} + W_e) \left[\exp \left(\frac{R \cdot SFC}{V(L/D)} \right) - 1 \right] \quad (1.1)$$

While it assumes a specific fuel consumption independent from flight speed and a cruise-climb mission, the range equations indicates several possible ways to reduce fuel burn:

- Reduce the aircraft empty mass W_e , which means increase structural efficiency;
- Reduce the specific fuel consumption SFC of the engines;
- Increase the airplane aerodynamic efficiency L/D ;
- Optimize the cruise trajectory and climb program in order to maximize $V(L/D)$.

All the past and current research efforts to improve aircraft efficiency can be divided into these four main categories. This is evident in Boeing's most recent clean-sheet design the 787 Dreamliner in which most of the new technologies aimed at reducing weight by adopting large integrated fuselage barrels, made with composites, and reducing SFC by replacing an engine-driven pneumatic system with a fully electric one. On top of these major advancements, a new wing with raked-wingtips and new laminar-flow engine nacelles contributed to higher aerodynamic efficiency [4][5]. In universities and institutions around the world several new aircraft configurations and technologies are being tested and evaluated for increasing aircraft efficiency.

From the point of view of aerodynamics Hybrid Laminar Flow Control (HLFC) has been in development at NASA since the 1980s, which promises a reduction up to 15% in fuel burn [6] by means of applying suction only at the forward portion of the wing, and exploiting NLF design for the remaining wing portion. Boeing has been testing the technology on a 757 testbed [7], which is being introduced in the newest development of the 787 family [8]. The European-sponsored Clean Skies project is also researching and flight testing laminar-flow technologies [9].

New propulsion developments include the introduction of geared turbofans, with bypass ratios of 10 and more, and the investigation of hybrid-electric (HE) propulsion. Current projects include Airbus E-Fan X demonstrator [10] for a future HE regional transport and the European-sponsored MAHEPA Project [11] for applications of HE technology in general aviation. Electric propulsion is also being taken into consideration, but it's still a technology at its infancy, with loiter times of less than one hour.

Another interesting concept is Boundary-Layer Ingestion (BLI) propulsion, currently being developed by MIT with the D8 Aircraft [12] and the TU Delft-led CENTERLINE project [13]. Drag is reduced by "filling" the fuselage wake with the jet exhaust of a propulsive device. An estimated 9% fuel burn reduction is targeted by this new aircraft layout.

Regarding structures and weight, active flutter suppression is being experimented with Lockheed Martin X-56 unmanned aircraft [14]. Currently stiffening, hence adding weight, is the approach used to suppress flutter at high speeds: active suppression would open the possibility to use lighter structures for the range of flying speeds. Another possibility is the re-introduction of strut-braced wings, concept that has been investigated at Virginia Tech [15]. Strut-bracing would enable higher wing aspect ratios than a conventional cantilevered wing, thus less induced drag. According to the study, for an hypothetical long range widebody aircraft, up to 9% MTOW reduction can be expected with fuel-burn savings between 11 and 16%. On a similar approach of distributing load-paths, boxed wings and joined-wing aircrafts have also been studied for possible weight reductions [16].

All these projects and investigations seek to improve the environmental-friendliness of the transport aircraft by means of the strategies outlined in the Breguet equation. This thesis work is not an exception as it is set to explore a new configuration which would promise drag and weight reductions. The idea is to apply Prandtl's 1933 Bell Spanload [17] on a modern business jet and replace its tail surfaces from a conventional tail to a V-tail.

In his 1933 paper, Prandtl sought to find the lift distribution with the lowest

induced drag given the total lift and structural weight (assumed proportional to the wing bending moment). This optimization problem is more practical than an arbitrary wingspan constraint, which yields the well-known elliptic lift distribution. Prandtl's result predicts a 11% reduction of induced drag compared to a elliptical spanload wing with the same lift and root bending moment. Similar results were found independently by Jones [18], while Klein and Viswanathan extended the approach for a given root bending moment [19]. Wind-tunnel testing by Wroblewski and Ansell [20] confirmed experimentally the induced-drag reduction while maintaining the same lift and bending moment.

According to the recent findings of Albion Bowers at NASA [21], the bell spanload presents an upwash in the outboard region which would provide a "induced thrust", similar to the mechanism of drag reduction of winglets. This induced thrust can be exploited to reduce the adverse yaw, or completely eliminate it. By adopting the bell spanload, Bowers et al. were able to build several flying-wing RC models, named PRANDTL-D, and measure a positive yawing moment due to aileron, confirming their hypothesis [22]. It is suggested that by reducing the requirements for the tail to correct adverse-yaw, it's possible to reduce both drag and weight of an airplane, besides adopting an optimized wing.

Carrizalez published a conference paper in which they sought to analyze this possibility [23]. Their theoretical analysis focused on a small single engine GA aircraft, already fitted with a V-tail (Cirrus SF-50). The combination of weight reduction and L/D improvement of 14%, yielded a 17% range improvement, without dramatically changing the aircraft flying dynamics. While promising, the study does not completely eliminate adverse yaw. Another limitation is the analysis carried out on a single-engined airplane whose thrust is aligned with its centerline: this configuration for jet aircraft is not common and thus not very representative.

Therefore the goal of this thesis work is to analyze the general performance improvements of a practical airplane, that is a multiple-engined aircraft whose engines do not lie in the plane of symmetry. Indeed vertical tail surfaces usually are sized for a critical engine failure scenario or lateral gusts, not just the adverse yaw requirement. If the bell spanload lowers the requirements for a vertical tail, then it's possible to adopt a V-tail with a lower dihedral angle, sized to counteract an engine failure scenario. A design procedure for replacing a conventional tail with a V-Tail is presented in [24] and [25].

Chapter 2 introduces to the theoretical properties of the bell spanload and the wing design procedure. Then the aerodynamic computational algorithm is described in Ch. 3. It combines a Vortex Lattice code with 2D Panel analysis for estimating lift and drag of a wing or wing and tail configuration. Agreement with experimental data is also presented.

This aerodynamic code is used to evaluate the characteristics and design properties of the bell lift distribution. Chapter 4 investigates the effects on drag and structural mass of design variables such as design lift coefficient $C_{L,des}$, aspect ratio $\mathcal{R}$, taper ratio λ and chord distribution $c(y)$. Comparisons are made between wings with an elliptical spanload and their respective wing with the new bell spanload.

These design guidelines found from this investigation are then used and applied in Ch. 5 on conventional wings. First application is on two wings tested by NACA in 1947, from report TN 1270: performance and mass results are compared to their

respective baseline. Differences and similarities to the ideal results with elliptical wings are also discussed. Finally the procedure is applied on the wing of the Pilatus PC-24: three different wing designs are proposed and analyzed from the point of view of fuel consumption and CO₂ emissions.

Chapter 6 deals with the design method and application of a Vee-tail on the Pilatus PC-24. Aircraft regulations are used as a guideline to synthesize constraints from which the tail is sized with. Two possible control surface arrangements are presented and the optimum is found by employing a Particle Swarm Optimization algorithm. Finally the performance of the new wing and tail combination is analyzed.

In conclusion Ch. 7 recaps all the learned results from the investigations and provides some points for further development.

Chapter 2

Wings of non-elliptical lift distribution

2.1 General Introduction

As discussed in Ch. 1, the objective of this study is to analyze a new aircraft configuration in order to reduce pollution and fuel burn. This configuration combines two proposed drag-reducing technologies: a non-elliptic lift distribution wing, by applying Prandtl's 1933 lift distribution, and a two-surface Vee tail.

The bell lift distribution provides a reduction of induced drag by extending the wingspan, without an excessive bending moment increase. It also presents the characteristic of self-correcting turns, therefore reducing the necessity of rudder control in turning flight. Hence it is proposed to replace the conventional three-surface tail with a two-surface tail with dihedral, what is usually called a Vee-tail. However the design of such a tail must comply with the controllability and stability of the airplane as prescribed by airworthiness code and regulations. These topics are discussed in Ch. 6, while the theoretical characteristics and state-of-the art regarding the bell lift distribution is discussed in this chapter.

New wings and tails also have an effect on the total weight budget of the airplane. Because these new surfaces are designed with non-conventional characteristics, it seems more appropriate to calculate weight, or a representative quantity, directly from a estimation of structural panel thickness rather than using statistical equations, as it is usually done in a conceptual stage of aircraft design. This method is discussed in Appendix A.

The main goal is to have a reduction of wetted area from a smaller tail and lower induced drag from the new wing design. A smaller tail may also produce a reduction in total structural weight, and therefore another point of saving. As pointed out in Ch. 1, both drag reduction and weight saving lead to lower fuel burn and greater range. These changes are compared to the baseline aircraft performance in order to evaluate benefits and possible disadvantages.

2.2 Theoretical Background

Prandtl's Lifting Line Theory (LLT) is the standard aerodynamic tool for analyzing the characteristics of high-aspect ratio wings with moderate sweep. The classical solution for minimum induced drag is the elliptic distribution of lift [26] (Eq.2.1), which has since been taught in standard aerodynamic texts. The theoretical downwash obtained from this distribution is constant with value 2.2.

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{y}{b/2}\right)^2} \quad (2.1)$$

$$w(y) = -\frac{\Gamma_0}{2b} \quad (2.2)$$

However this lift distribution is derived from the simple hypothesis of a fixed span, which is an arbitrary constraint for a practical optimization problem. Unless the airplane has practical requirements that constrain the wingspan (such as the size airport facilities), then a more useful constraint would be the weight of the wing itself. Indeed the main trade-off in wing design is the wingspan: the greater, the lower the induced drag, however it would require an heavier structure in order to bear the increased bending moments.

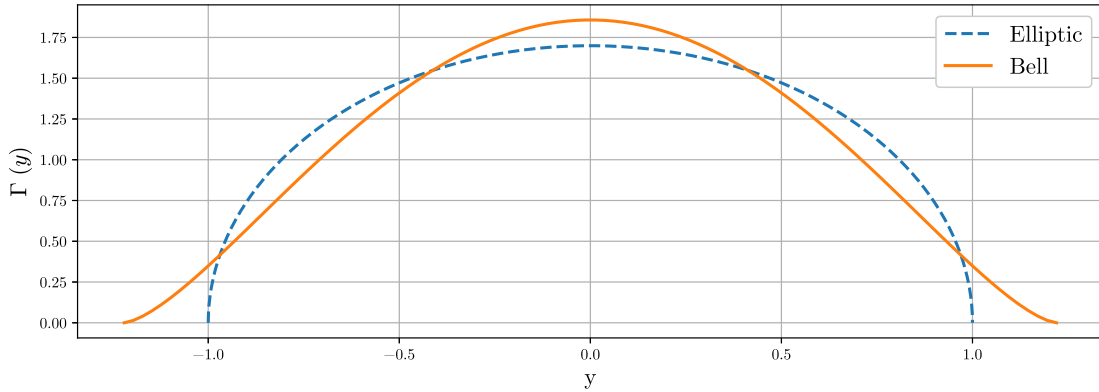


Figure 2.1: Spanloads.

Prandtl thought about this issue and published in 1933 [17] a new minimum induced drag solution in which the span constraint was replaced with a root bending moment constraint.

The new lift distribution is bell-shaped and sized with a weight constraint, assumed proportional the root bending moment. Prandtl's new solution has a wingspan extension of 22% and a drag reduction of 11% compared to an equivalent elliptical wing with the same lift. The circulation distribution and its downwash are shown in equation 2.3 and 2.4. The two lift distributions are shown together in Fig. 2.1, while the downwash distributions are shown in Fig. 2.2. A translation of Prandtl's 1933 paper was published in 2020 by Hunsaker and Philips[27].

$$\Gamma(y) = \Gamma_0 \left(1 - \left(\frac{y}{b/2}\right)^2\right)^{\frac{3}{2}} \quad (2.3)$$

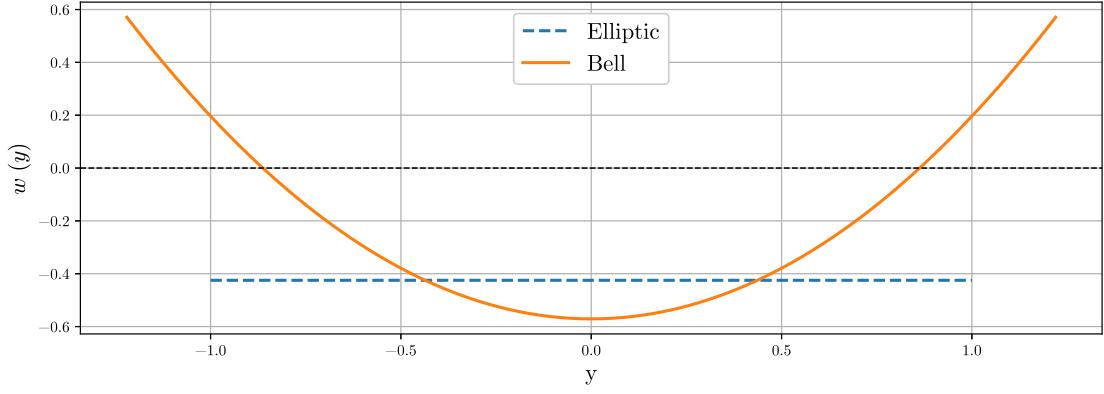


Figure 2.2: Downwash distributions.

$$w(y) = \Gamma_0 \frac{3}{2b} \left(\left(\frac{y}{b/2} \right)^2 - \frac{1}{2} \right) \quad (2.4)$$

The downwash distribution of this solution is not constant but parabolic and, interestingly, it has an upwash region at the wingtips. The vortex core is therefore shifted inboard, positioned at $y/b = 0.704$. In this upwash region the section lift vector is rotated forward, and not backwards, generating an local induced thrust and reducing the net induced drag, besides the drag-reduction effect of the span extension. This drag-reduction mechanism is similar to the ones found in winglets.

From the point of view of structural loading, the bell-shaped distribution shifts the bulk of the lift near the root and unloads the tips, reducing the root bending moment increase from the span extension. Bending moment distributions resulting from the bell and equivalent elliptic lift distributions are shown in Fig. 2.3, which are nearly identical.

In turning flight, adverse yaw can be mitigated or removed by properly designing the ailerons to exploit the upwash region in order to achieve a coordinated turn. This has been proved both by the Horten Brothers gliders and NASA's Prandtl-D experiment[21][22].

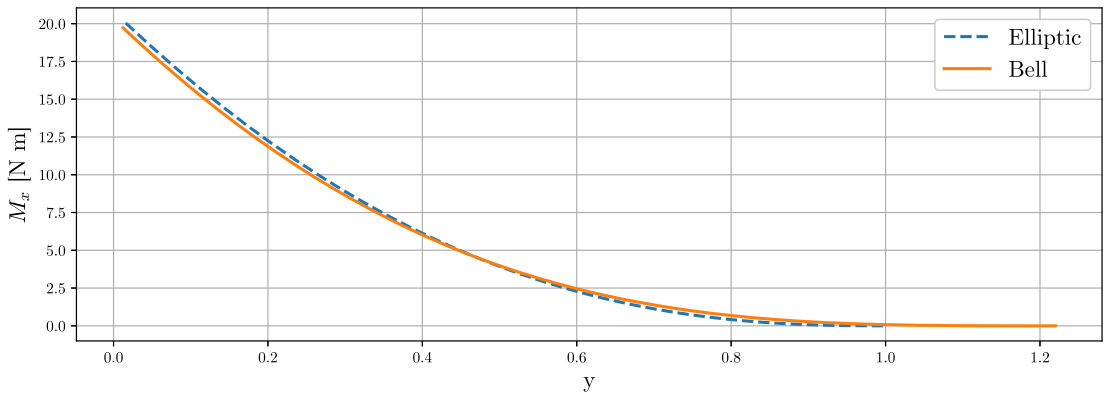


Figure 2.3: Wing bending moments

The design problem is to find the chord distribution or the geometric twist

distribution for a given wing shape that would yield the bell lift distribution at the design lift coefficient.

An analytical solution for geometric twist can be derived for a straight wing of high aspect ratio by substituting 2.3 into the general LLT equation 2.5, where α_g is the geometric twist of the wing, while α_0 is the zero lift angle of attack of section y .

$$\alpha_g(y) = \frac{2\Gamma(y)}{C_{L\alpha} V_\infty c(y)} + \alpha_0(y) + \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{\frac{\Gamma(\eta)}{d\eta}}{y - \eta} d\eta \quad (2.5)$$

By inserting 2.3 in the previous equation, Carrizalez et al. [23] found Eq. 2.6 as an analytical equation of the twist distribution α_g which would produce a bell-shaped lift distribution for a given lift L and flight condition (velocity V and freestream air density ρ).

$$\alpha_g(y) = \alpha_0(y) - \frac{8L}{\pi b^2 \rho V^2} \left(\left(\frac{y}{b/2} \right)^2 - \frac{1}{2} \right) + \frac{32L}{3\pi b \rho V^2 C_{L\alpha}(y) c(y)} \left(1 - \left(\frac{y}{b/2} \right)^2 \right)^{3/2} \quad (2.6)$$

2.3 Iterative wing design method

Equation 2.6 is useful for straight and tapered wings, but is inaccurate for swept wings and more complex geometries. For a more general application a numerical solution is required, by means of iterating with a Vortex Lattice code. In this study a procedure was developed by using the robust MIT AVL code developed by Drela and Youngren[28].

With a given wing geometry, the bell lift distribution has to be sized according to the wingspan and the design lift. The root circulation Γ_0 is calculated for a given lift and wingspan b with Eq. 2.7. If a lift coefficient C_L is specified, Γ_0 is then calculated with Eq. 2.8

$$\Gamma_0 = \frac{L}{\rho V \int_{b/2}^{-b/2} \left(1 - \left(\frac{y}{b/2} \right)^2 \right)^{3/2} dy} = \frac{16}{3\pi} \frac{L}{b \rho V} \quad (2.7)$$

$$\Gamma_0 = \frac{\frac{1}{2} \rho V^2 S_w C_L}{\rho V \int_{b/2}^{-b/2} \left(1 - \left(\frac{y}{b/2} \right)^2 \right)^{3/2} dy} = \frac{8V S_w C_L}{3\pi b} \quad (2.8)$$

Then the target lift distribution is calculated for each span station of the vortex lattice mesh. By applying the Kutta-Joukowski theorem for each section, the local lift coefficient and the vorticity are correlated with Eq. 2.9. For each iteration, the lift distribution calculated with the VLM code and the target one are compared: the difference is scaled and added to the twist distribution. This process is iterated until convergence is reached. A diagram of this algorithm is shown in Fig. 2.4.

$$\Gamma(y) = \frac{C_l(y) c(y)}{2} V \quad (2.9)$$

A good approximation of the twist distribution is easily produced with 20 or more points. The geometry of the wing does affect the effectiveness of the bell lift distribution. Indeed best results are achieved with high aspect ratio wings. If the designer is starting from a clean-sheet design, a procedure would be to design a standard wing and then extend the wingspan by 22% while maintaining the same wing area.

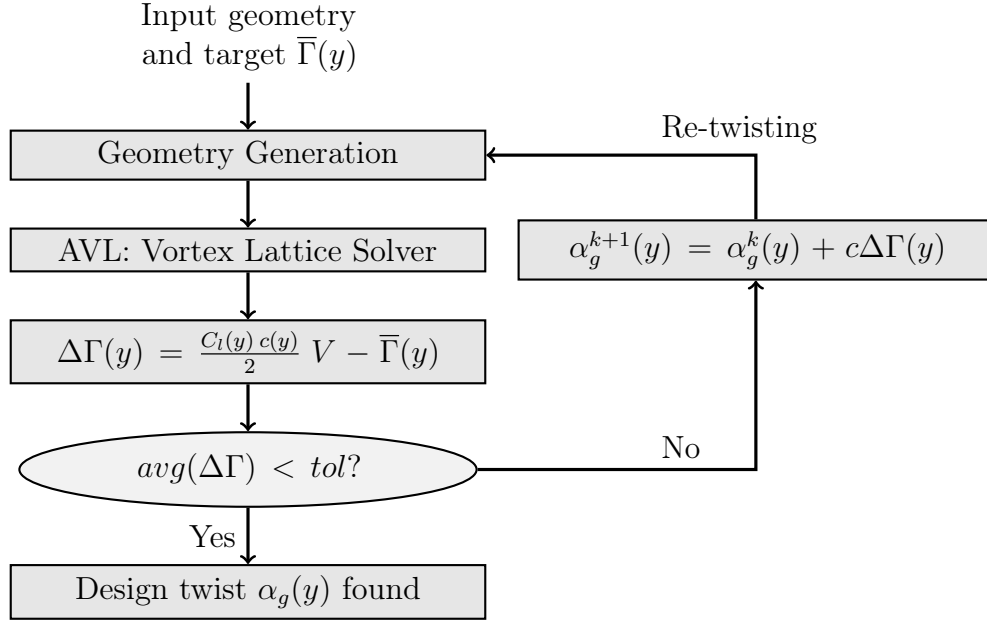
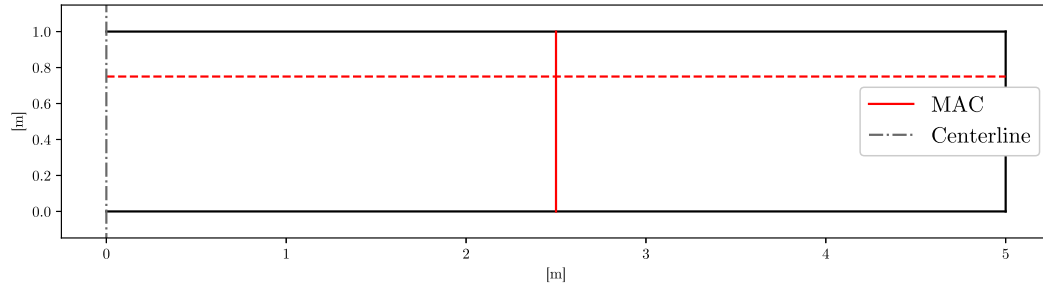


Figure 2.4: Iterative twisting procedure

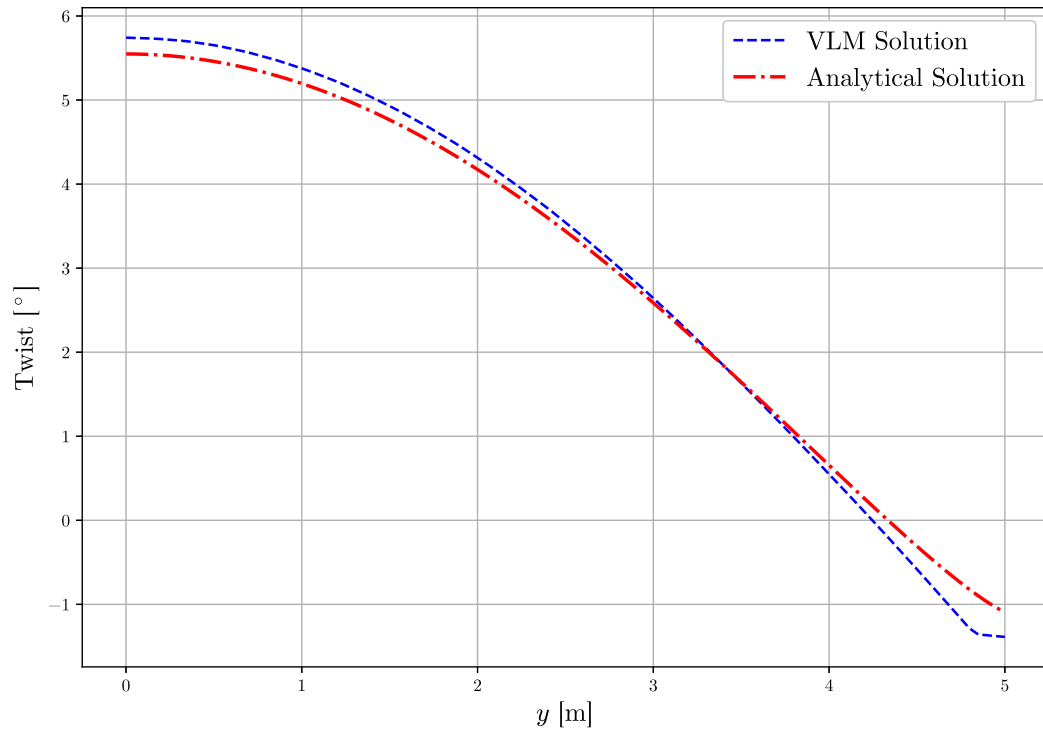
The results of this procedure are shown in Figures 2.5 - 2.6, where the numerical lift distribution is compared with the analytical solution of Eq. 2.6. The wing geometry is shown in the first image of each page. All are wings have $\mathcal{R} = 10$. Variations on taper, sweep and dihedral are introduced over the basic unswept rectangular wing. Overall there's an agreement between the two solutions, with a discrepancy of 1 degree at the root of the wing for both the rectangular and the tapered wing.

In Fig. 2.7b is presented the twist distribution for the same wing with 30° sweep. The large difference from the analytic solution at the root is due to the lower lift generated as an effect of sweep. Therefore more positive geometric twist is required in this region to obtain the prescribed bell distribution. Indeed as discussed above, Eq. 2.6 was derived from LLT, which is inaccurate for wings with large sweep. Finally in Fig. 2.8 the effect of dihedral is presented. Not much difference is visible from the straight wing case, except for a discrepancy at the very root. This is probably caused by a numerical artifacting.

The iterative procedure is inviscid and thus does not account for the untwisting caused by the boundary layer displacement thickness. This can be approximated with a strip-theory approach and a 2D airfoil solver. As it will be discussed in the

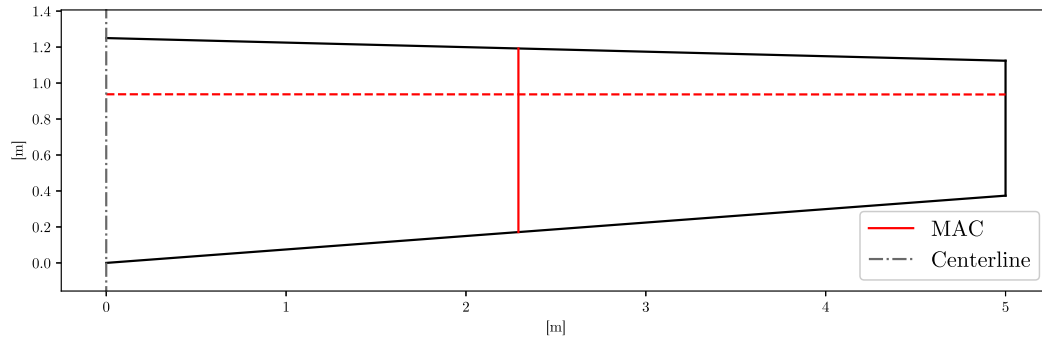


(a) Half wing geometry plot

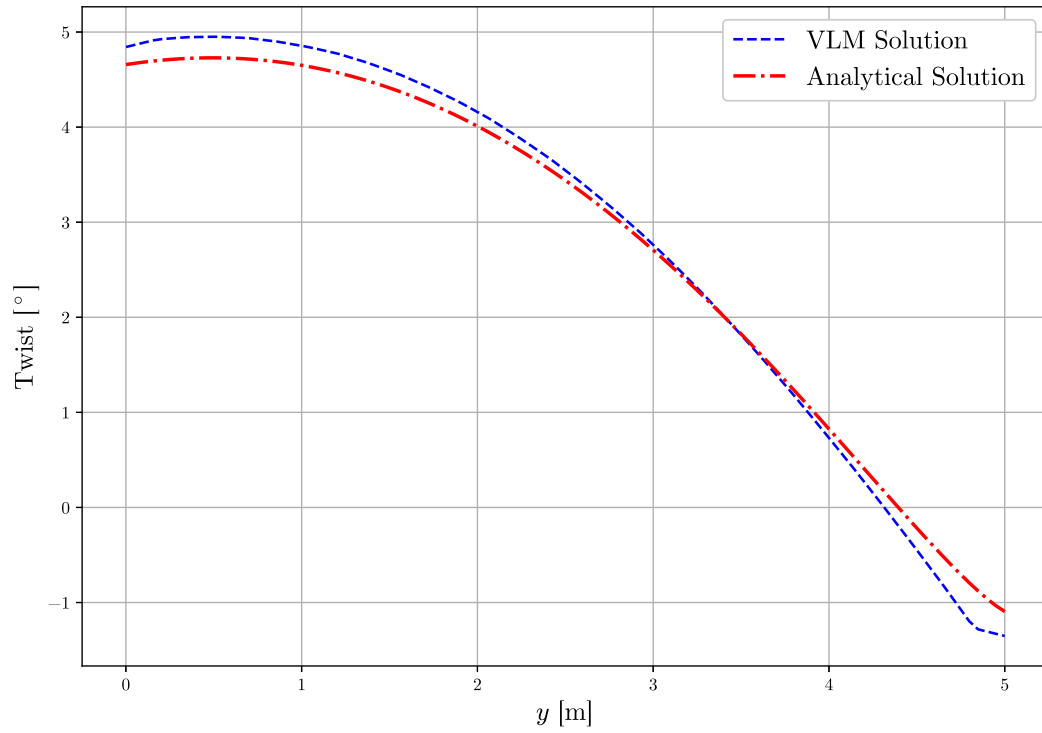


(b) Twist Distribution

Figure 2.5: $\mathcal{R} = 10$ Straight Wing

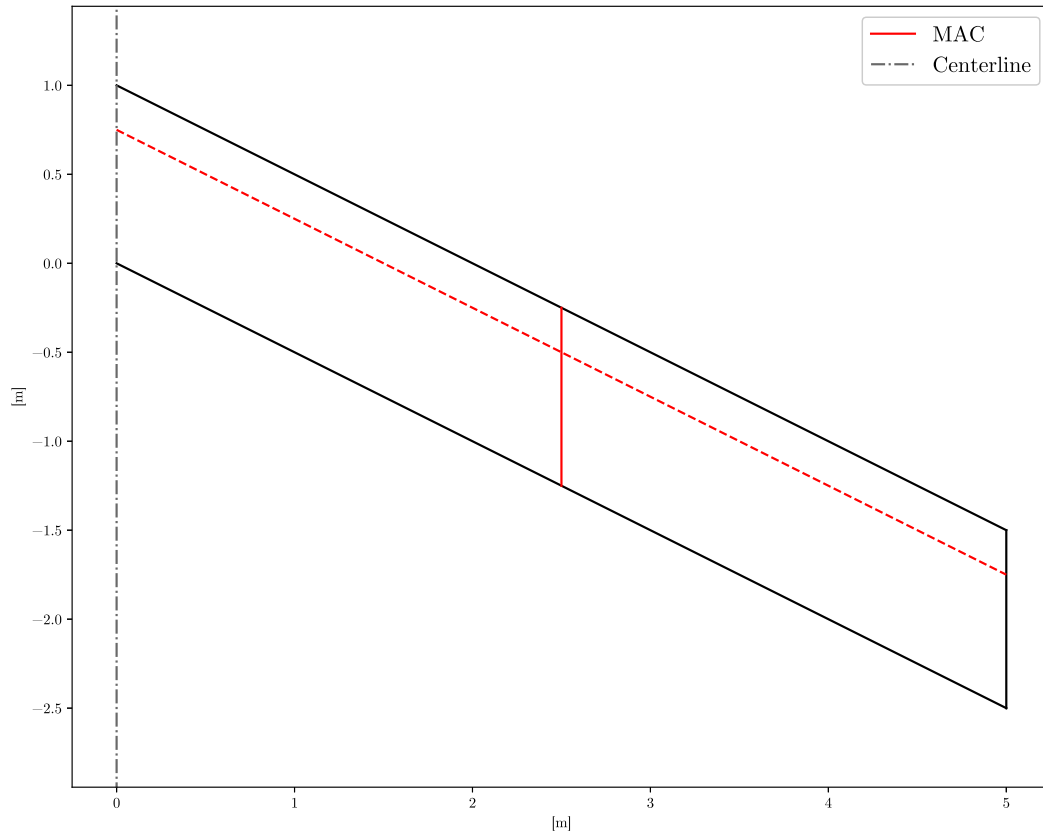


(a) Half wing geometry plot

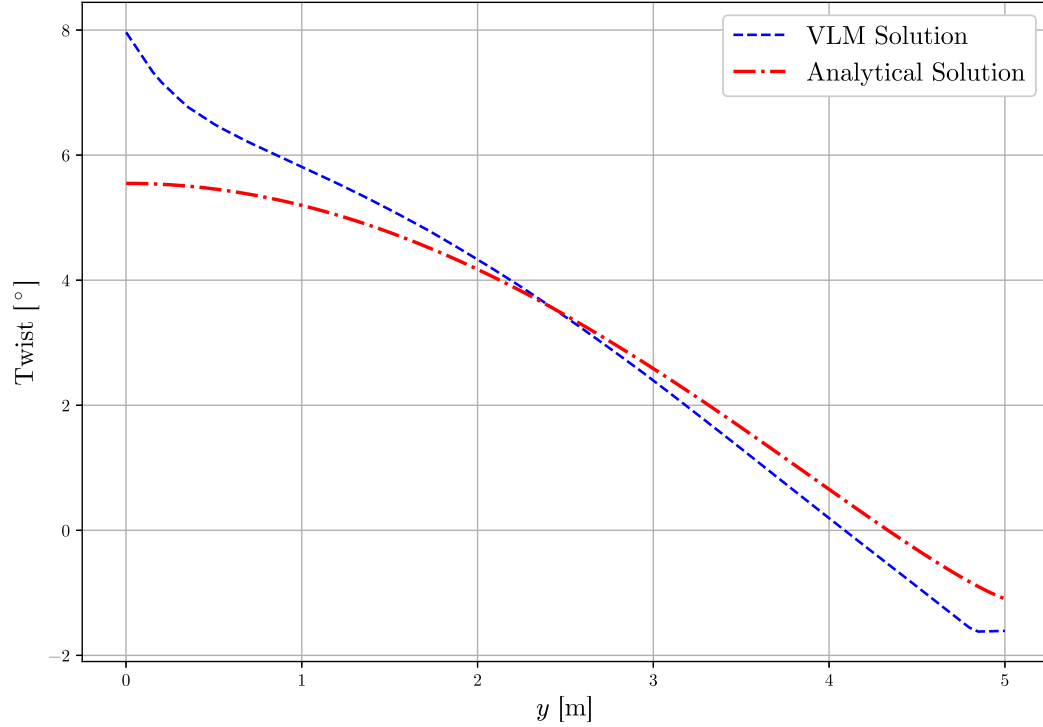


(b) Twist Distribution

Figure 2.6: $\mathcal{R} = 10$ wing with $\lambda = 0.6$

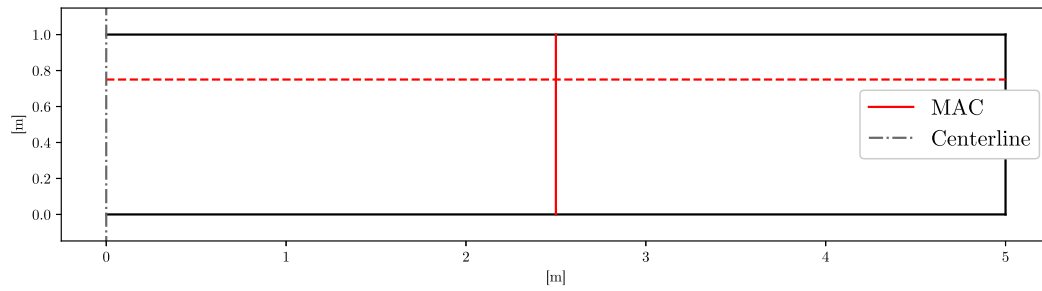


(a) Half wing geometry plot

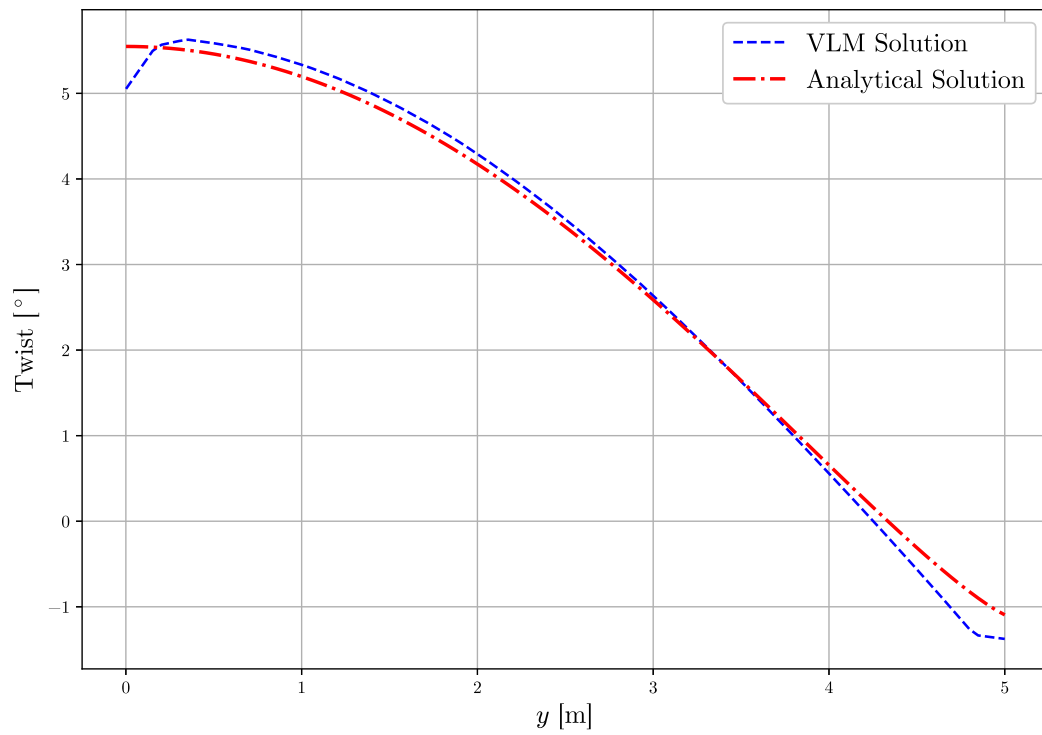


(b) Twist Distribution

Figure 2.7: $\mathcal{R} = 10$ wing with 30° sweep



(a) Half wing geometry plot



(b) Twist Distribution

Figure 2.8: $\mathcal{R} = 10$ wing with 10° dihedral

following chapter (Ch. 3), this has been implemented for viscous drag estimation. Wing flexibility is another unaccounted effect in this approach: under load the wing would flex and twist, changing its shape. Therefore the aerodynamic performance estimated with the rigid jig shape would be different from the actual performance with the deformed shape.

Estimation of this would require a structural model of the wing like a Finite Element approach and an iterative procedure to obtain the correct deformed shape, before calculating the viscous effects. Chaparro and Fujiwara developed a computational framework to evaluate the aerodynamic performance of a wing including structural flexibility, boundary layer thickness effects and transonic effects [29].

Chapter 3

Aerodynamic Computational Algorithm

In order to calculate and assess the aerodynamic properties of the wings and tails of this project, a proper aerodynamic code was built. It is then tested against experimental data to evaluate its accuracy.

3.1 Methodology

The aerodynamic computational algorithm in this study relies on the combination of two open-source codes: Athena Vortex Lattice (AVL)[28] and XFOIL[30]. The first is a highly flexible vortex lattice code capable to model multiple surfaces in a non-planar disposition. It also has the capability to calculate aerodynamic derivatives and dynamic modes, if provided with inertial properties of the aircraft. The second is the de-facto standard code in 2D subsonic airfoil aerodynamic analysis. It provides a strongly-coupled boundary layer solver for calculating the viscous drag of the airfoil. A Python script was developed to generate AVL geometry, run an analysis and calculate viscous drag of a full configuration by employing strip-theory and XFOIL.

In order to obtain good coupling between the two solvers and account for viscous effects on lift, the script iterates by correcting the geometric twist in order to include the boundary layer displacement effect responsible for producing a lower C_L for the same angle of attack, compared to the inviscid solution. This procedure is based on the work of Chaparro and Fujiwara [29], and shown in the block diagram of Fig. 3.1.

AVL provides symmetry when defining lifting surfaces. Because the viscous correction is applied on one half of the wing and then mirrored, it was opted to limit viscous calculations to cases with vertical symmetry. This limitation is not binding however, as unsymmetric cases (sideslip and rudder control) do not require viscous analysis in this study.

Geometric twist for each section is calculated as a sum of the prescribed geometric twist α_g plus the "viscous twist" γ resulting from boundary layer displacement (Eq. 3.1).

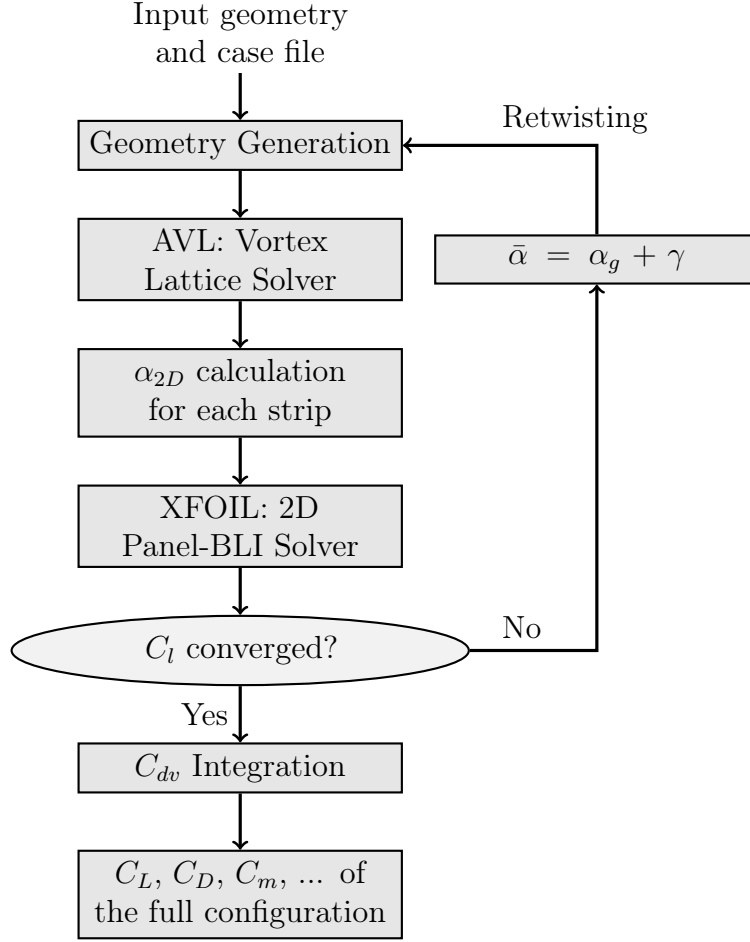


Figure 3.1: Aerodynamic code block diagram

$$\bar{\alpha}(y) = \alpha_g(y) + \gamma(y) \quad (3.1)$$

The value of γ has to be calculated iteratively and updated until convergence on lift is achieved. After the VLM calculation, local Reynolds number and angle of attack are calculated for each section. The former is calculated with Eq. 3.2, in which $C_{l\alpha}$ and α_0 are calculated with XFOIL for the local interpolated airfoil, including Mach number effects (via Karman-Tsien compressibility correction). The VLM code accounts for compressibility with the classical Prandtl-Glauert correction.

$$\alpha_{2D}(y) = \alpha_0(y) + \frac{C_{l,vlm}(y)}{C_{l\alpha}(y, Ma)} - \gamma(y) \quad (3.2)$$

Then viscous analysis for the local airfoil is executed and a new value of γ calculated with Eq. 3.3. The viscous effect is updated each iteration until the difference between the 2D lift $C_{l,2D}$ and the VLM lift $C_{l,vlm}$ is less than a user-specified tolerance. The geometry is generated again with the new viscous twist added and the process repeated until convergence of lift.

$$\gamma_{i+1}(y) = \gamma_i(y) + \frac{C_{l,2D}^i(y) - C_{l,vlm}^i(y)}{C_{l\alpha}(y, Ma)} \quad (3.3)$$

Viscous drag is then integrated along each lifting surface with Eq. 3.4, in which S_{ref} is the reference area (usually the wing), while S is the area of the surface (wing or tail) being integrated. This integral is in practice calculated as a weighted sum of the viscous drag on each strip CC_{dv}^k and the strip area A^k .

$$C_{D,v} = \frac{S}{S_{ref}} \left(\frac{1}{S} \int_0^b C_{d,v}(y)c(y)dy \right) = \frac{S}{S_{ref}} \left(\frac{1}{S} \sum_{k=0}^{N_{strips}} C_{dv}^k A^k \right) \quad (3.4)$$

Drag from non-lifting surfaces, such as fuselage or engine pods, is estimated with the wetted-area approach, using Schlichting's friction coefficient formula (Eq. 3.5). It is assumed that all the boundary layer is turbulent for a conservative drag estimation. Wetted areas are corrected with a proper form factor FF coefficient, different for each type of geometry (fuselage, pods, etc). Form factor formulas are available from several standard textbooks: for this study Raymer's formulas are used[31].

$$C_f = \frac{0.455}{(\log_{10} Re)^{2.58} (1 + 0.144 M^2)^{0.65}} \quad (3.5)$$

The total drag for a whole aircraft is finally calculated by summing all contributions (Eq. 3.6): induced drag C_{Di} , viscous drag from N_l lifting surfaces and viscous drag from N_{nl} non-lifting components.

$$C_D = C_{Di} + \sum_{k=0}^{N_l} C_{Dv}^k + \sum_{j=0}^{N_{nl}} \frac{S_{wet}^j C_f FF^j}{S_{ref}} \quad (3.6)$$

Besides calculating drag and lift, this algorithm is also used with finite differences for calculating the aerodynamic derivatives necessary for the methods in Sec. 6.2. Indeed it is possible to specify control surfaces deflections besides aerodynamic angles.

3.2 Experimental validation

To validate the accuracy of this algorithm, wind tunnel data from NACA TN 1270 [32] is used. Two wing geometries from this report are used, whose geometric properties are shown in Tab. 3.1. For each of them, two different discretizations schemes were applied: equal discretization and half-sine discretization, which concentrates points near the wingtips. Tab. 3.2 and 3.2 provide the average error on lift and drag coefficients for the different discretizations analyzed. On average the discrepancy is about of 11% for drag and 5% for lift. It was opted for the analysis in this work to use a 30 point half-sin discretization because it resolves better the wingtip induced drag, especially for the bell lift distribution.

Lift curves and Polars for both wings are shown in Fig. 3.2 and 3.3. Most of the discrepancy in calculated drag is present at high lift coefficients, above $C_L = 0.75$. More points help reducing the error but marginally. Experimental lift data agrees mostly with the numerical data, except for little discrepancy at the lowest values of C_L . Adding more points do not reduce this error.

	Wing 1	Wing 2
S_w	2.6 m^2	2.08 m^2
b	4.572 m	4.572 m
c_r	0.8126 m	0.65 m
c_t	0.325 m	0.26 m
Root Airfoil	NACA 4416	NACA 4420
Tip Airfoil	NACA 4412	NACA 4412
Washout angle	-4.5°	-3.5°
$\mathcal{R}$	8.02	10
λ	0.4	0.4
Re	$4.32 \cdot 10^6$	$3.49 \cdot 10^6$

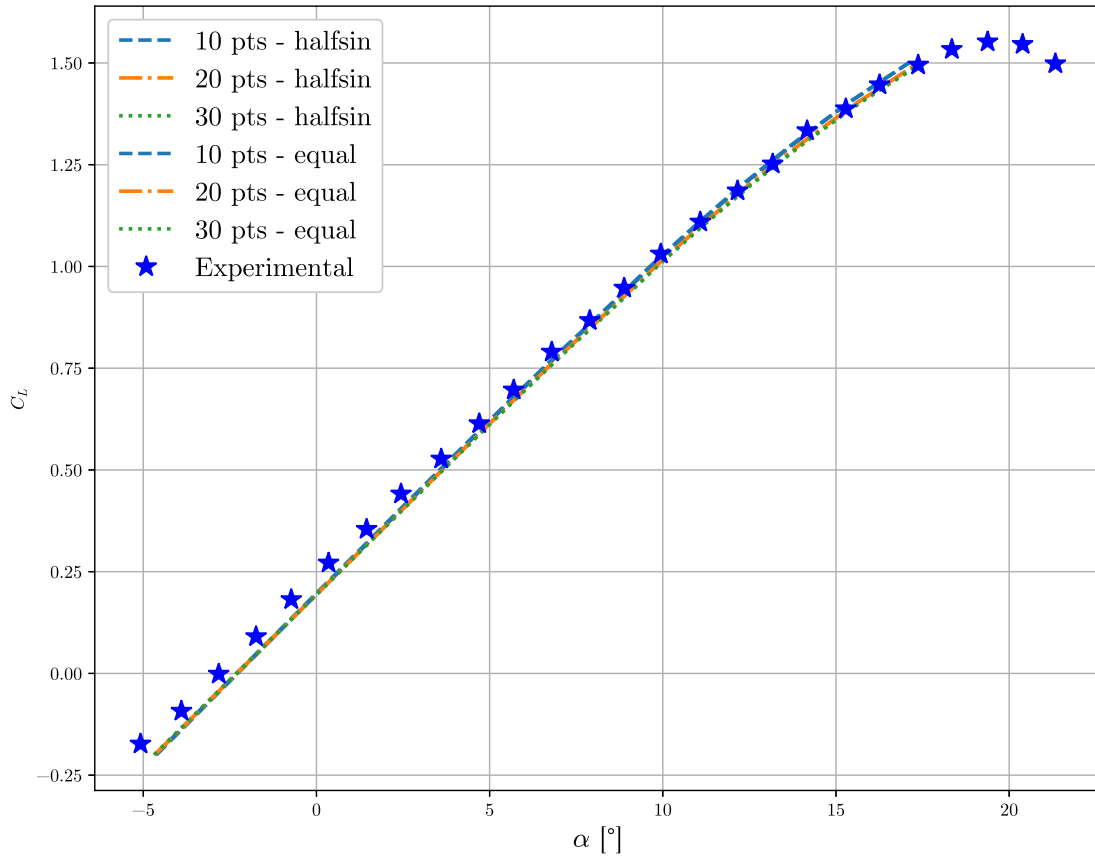
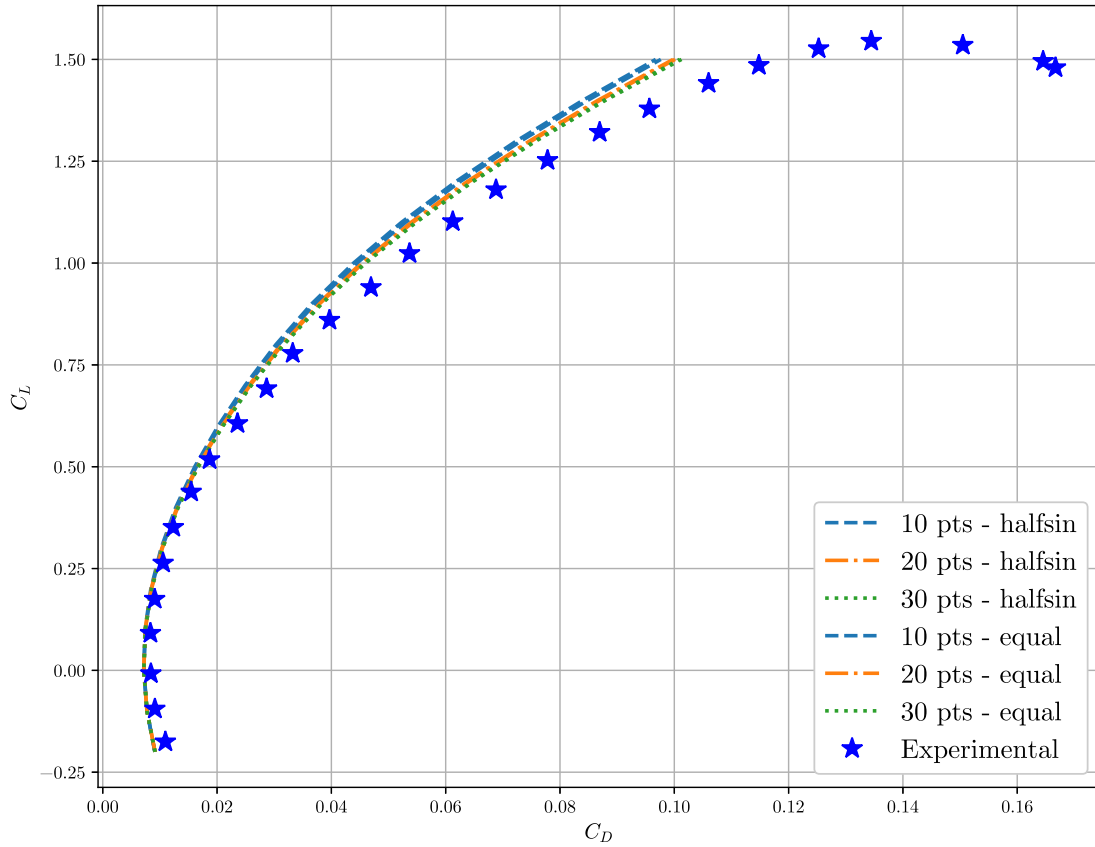
Table 3.1: Geometric properties of tested NACA wings.

	Half-sin Spacing		Equal Spacing	
N° Points	% e_{C_L}	% e_{C_D}	% e_{C_L}	% e_{C_D}
10	4.23	13.85	3.918	12.81
20	5.07	12.10	4.94	11.56
30	5.42	10.95	5.33	11.16

Table 3.2: Average discrepancy from experimental data for Wing 1

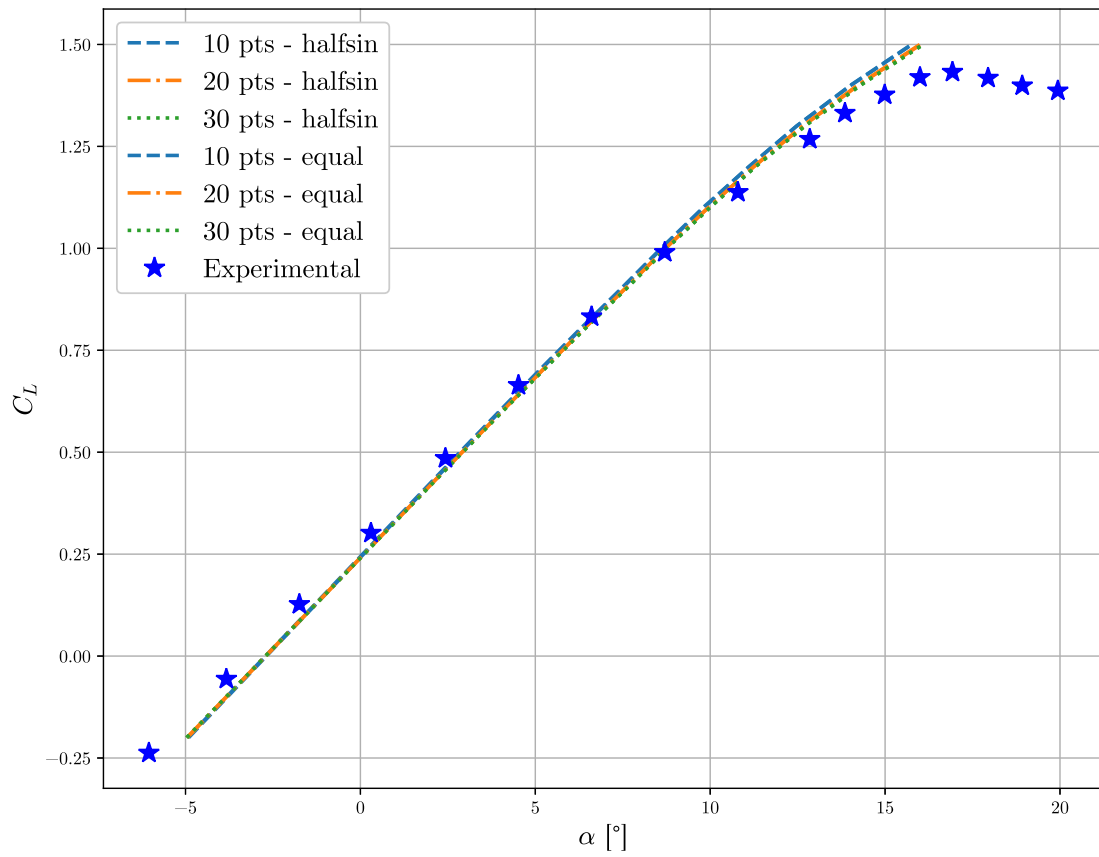
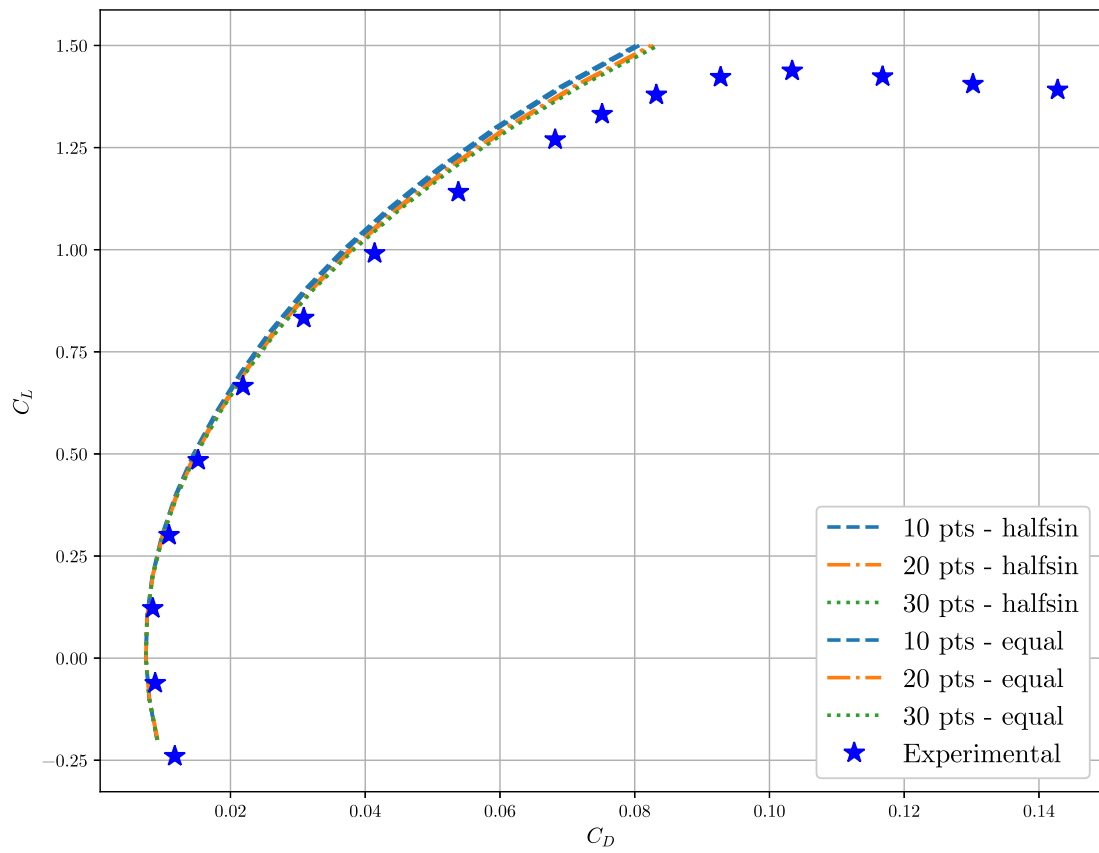
	Half-sin Spacing		Equal Spacing	
N° Points	% e_{C_L}	% e_{C_D}	% e_{C_L}	% e_{C_D}
10	4.35	12.91	4.01	12.11
20	4.68	11.45	4.55	10.91
30	4.81	10.71	4.75	10.51

Table 3.3: Average discrepancy from experimental data for Wing 2

(a) $C_L - \alpha$ 

(b) Polar

Figure 3.2: Comparison of aerodynamic data for Wing 1

(a) $C_L - \alpha$ 

(b) Polar

Figure 3.3: Comparison of aerodynamic data for Wing 2

Chapter 4

Parametric study

Prandtl's original paper was two pages long and did not fully explore the properties of this lift distribution. Furthermore authors discovering Prandtl's work and applying on their projects take for granted the hypothesis that weight is unchanged with this design. This has to be checked however, as the structural weight is determined by the critical load and not by a cruise load. Another point of investigation is the performance of the bell lift distribution outside of its design point, as Prandtl's result is essentially a fixed-point optimization. This will be analyzed in the first section of this chapter.

Several parameters characterize the wing and it is be important to analyze and understand their effect on this new lift distribution. Therefore the rest of the chapter explores the variation of drag and weight with the new lift distribution by varying the design lift coefficient $C_{L,des}$, the aspect ratio of the original wing $\mathcal{A}$ and the taper ratio λ . Each case compares a wing with elliptical distribution with their respective bell-designed wing.

In all these analyses, the redesign procedure follows two steps:

1. a new planform is generated by extending the original span by 22% and maintaining the same surface S_{wing} ;
2. the wing is twisted until the bell-lift distribution is achieved at the prescribed lift coefficient.

4.1 Performance outside of the design point

A simple straight tapered wing was defined for analyzing the characteristics of the bell lift distribution on all the practical lift coefficients. The geometric characteristics are defined in Tab. 4.1, whose original planform is shown in Fig. 4.2. The wing is re-designed by keeping the taper ratio and the surface constant (Fig. 4.3).

The code was setup with a chord-normalized Reynolds number Re_L of $2.8 \cdot 10^6$. The twist distribution was calculated by setting a design airstream velocity $V_\infty = 60 \text{ m/s}$ and a design lift coefficient of $C_{L,des} = 0.4$, the result is shown in Fig. 4.1.

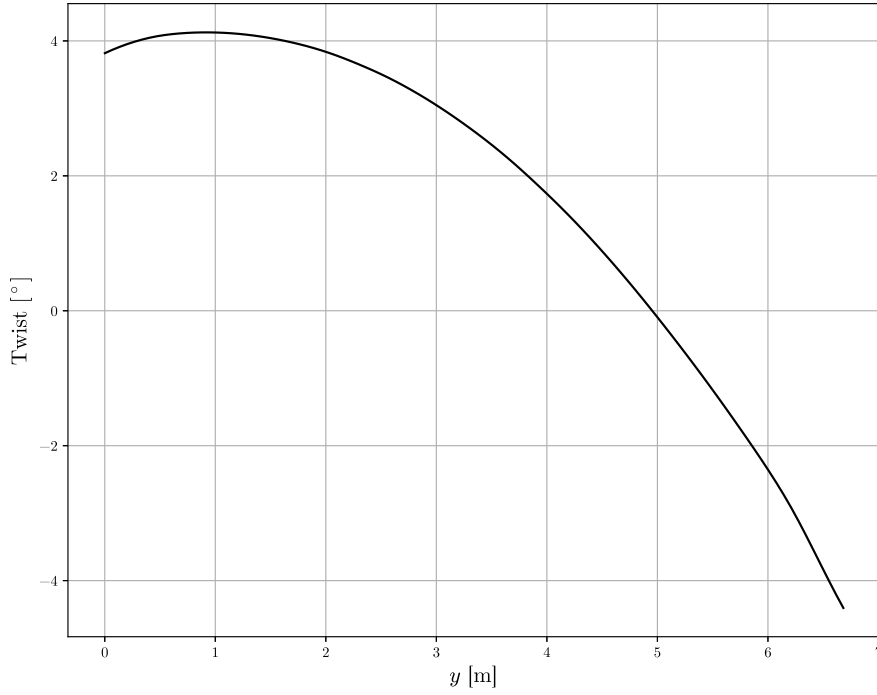


Figure 4.1: Twist distribution of Sec. 4.1 wing.

$\mathcal{R}$	8
S_w	15 m^2
λ	0.5
Γ	0°
Λ	0°
Root Airfoil	NACA 2415
Tip Airfoil	NACA 2412

Table 4.1: Geometric properties of the baseline wing

The calculated lift curve and drag polars are respectively shown in Fig. 4.4 and Fig. 4.5. Because the new wing has an higher aspect ratio $\mathcal{R} = 11.9$, the lift curve is steeper than the baseline wing. The drag polar of the new bell spanload wing is more flattened and steeper than the elliptical one, crossing each-one at $C_L = 0.3625$. Above of such value the new wing has less drag, while under the original wing is still better. Therefore this lift distribution benefits flight conditions where high lift is present, such as climbing or maneuvering. Hence the cruise lift coefficient should be used as design value for the spanload. The effect of $C_{L,des}$ on the performance of the wing is investigated further in Sec. 4.2.

Fig. 4.6 shows the relative trend of drag reduction of the new wing relative to the baseline, showing more clearly the results already visible in the polar plot. The relative variation is calculated with Eq. 4.1, where a positive value represents drag reduction compared to the reference polar.

$$R_{\%,C_D}(C_L) = 100 \frac{C_D(C_L) - C_{D,ref}(C_L)}{C_{D,ref}(C_L)} \quad (4.1)$$

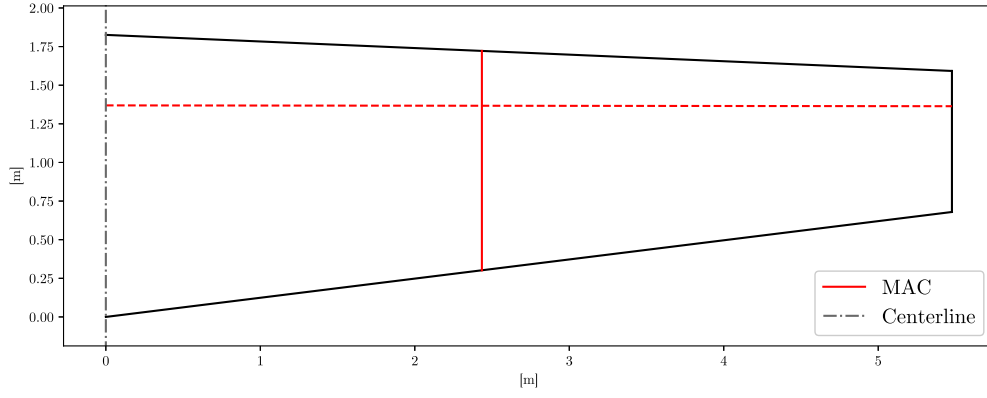


Figure 4.2: Elliptic spanload wing geometry of Sec. 4.1.

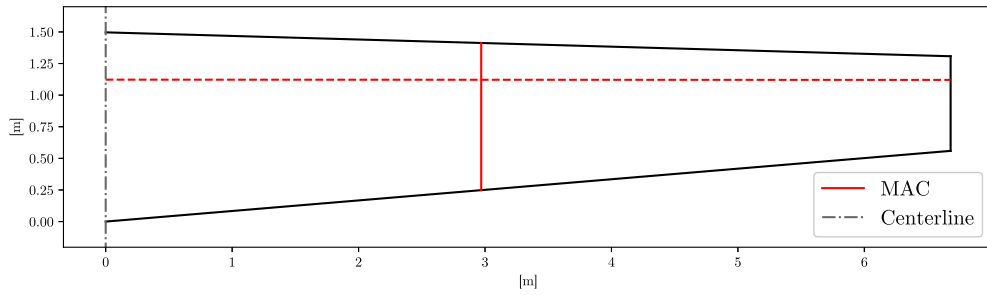


Figure 4.3: Bell-spanload wing geometry of Sec. 4.1.

Instead in Fig. 4.7 the value of L/D as a function of lift is shown. The new lift distribution gives an increment of aerodynamic efficiency above the design lift coefficient, while being slightly lower under it. This interesting result is similar to the behavior of winglets or other wingtip devices for the reduction of induced drag.

	C_D	C_{Di}	C_{Dv}	L/D	e	$\mathcal{R}$
Elliptic	0.0117169	0.0062669	0.0054500	34.13	1.0161	8
Bell	0.0112353	0.0056312	0.0056041	35.60	0.7595	11.9

Table 4.2: Aerodynamic data at design lift coefficient

Tab. 4.2 provides the aerodynamic data at the design lift coefficient $C_L = 0.4$, with a breakdown of the drag components. As predicted by Prandtl's theory, the induced drag is lower, with a calculated reduction of 10.14 %, close to the theoretical 11% if numerical error is considered. This drag reduction yields an increased L/D , which is fundamental in reducing burned fuel as discussed in Ch. 1.

Interestingly the Oswald efficiency factor e for the new wing is less than the optimal value of 1. This does not contradict the results because the efficiency factor accounts for aerodynamic efficiency of a non-elliptical spanload to a wing with an elliptical spanload of the same aspect ratio. Let's recall the classical formula for induced drag:

$$C_{Di} = \frac{C_L^2}{\pi \mathcal{R} e} \quad (4.2)$$

While the bell wing has $e < 1$, its aspect ratio $\mathcal{R}$ is greater than the original elliptical wing, yielding a lower a global factor $k = \frac{1}{\pi \mathcal{R} e}$. Prandtl's result can be considered an *aerostructural optimization* rather than a simple aerodynamic optimization because it sought to reduce drag while constraining the bending moment of the lift distribution.

The change of e with the lift coefficient is shown in Fig. 4.8: as C_L increases, the efficiency factor becomes closer to the elliptical wing value. This is caused by how the shape of the lift distribution changes with C_L as the geometric twist distribution provides the bell shape only for a specific flight condition. When lift is increased, the extremities of the wing generate more lift disrupting the bell shape and becoming more elliptical. While this is beneficial aerodynamically, this means the bending moment is no longer constrained to the elliptical wing value.

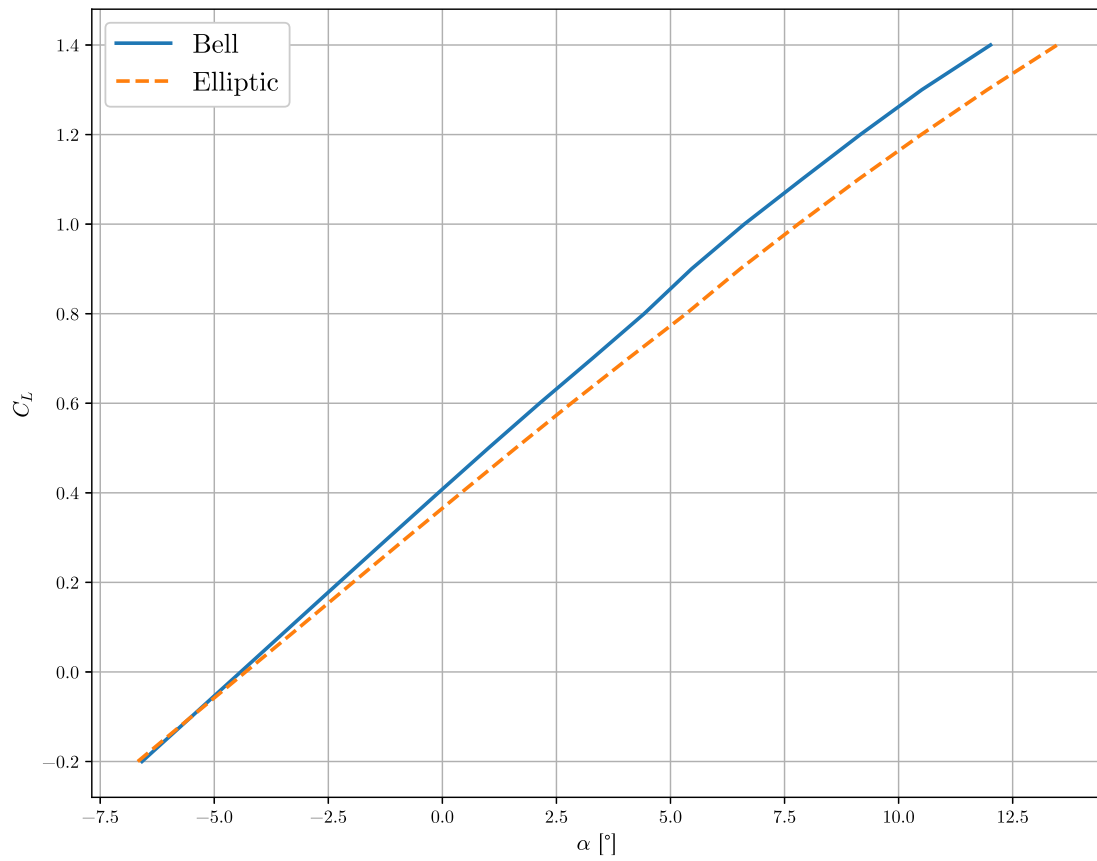
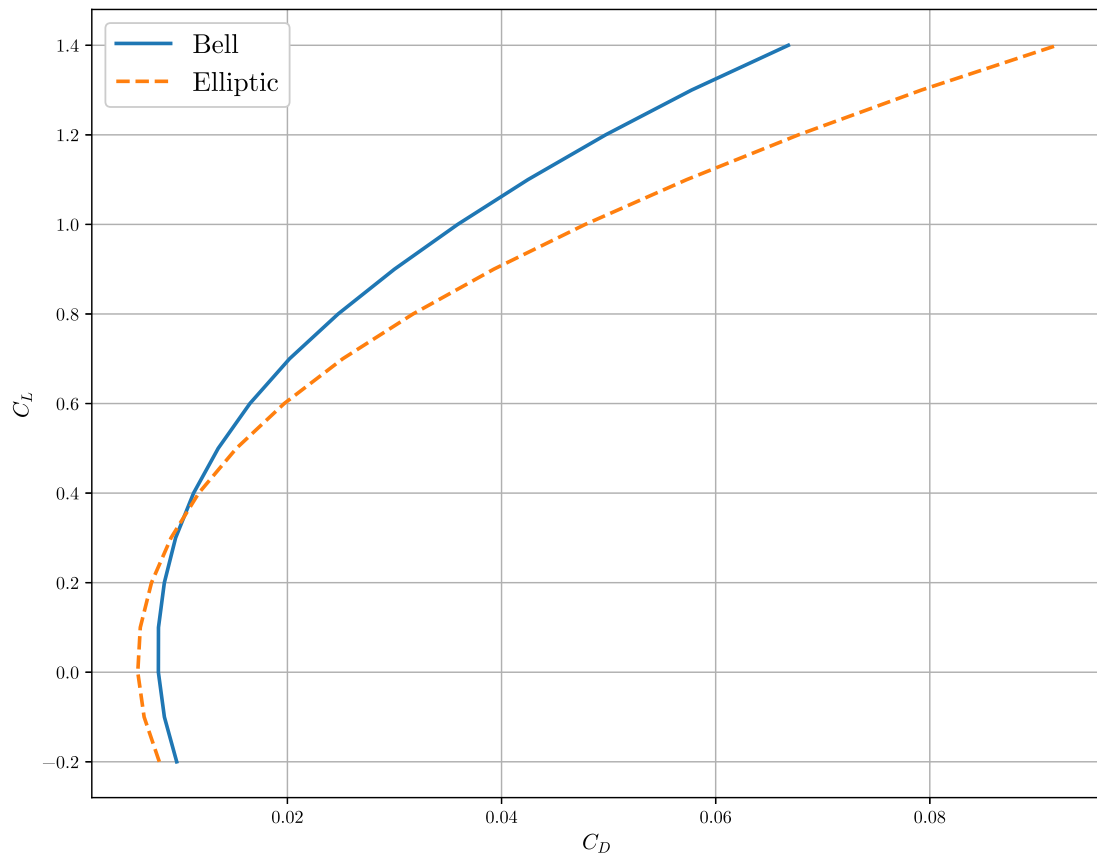
Finally Fig. 4.9 shows the combined effect of e and $\mathcal{R}$ as a function of lift, confirming what was explained above. At low lift coefficients, e is not compensated by the aspect ratio. Indeed at $C_L = 0$, even if the global wing has no lift, the twist distribution generates local lift near the extremities which produce local induced drag. This effect is discussed further in the next section.

In order to evaluate the wing mass, a direct mass estimation method was developed by assuming the wing structure composed of skin panels and a single shear web. The wing is divided into sections, in which panel and web thicknesses are calculated: the skin panels resist bending loads while the web resists only shear loads. This simplified procedure is called "skin-panel method", explained in detail in Appendix A.

While it is not bound to the elliptic lift distribution, like statistical methods, the drawback of this approach is the underestimation of mass as many structural details and non-structural components are ignored. Therefore these calculated results are not realistic per se, but useful for comparing relative structural mass variations. They will be defined as *representative mass* to be not confused with the actual structural mass calculated with statistical equations (like in Section 5.2). For realistic mass estimations a more detailed approach is required, one that would involve a full structural design.

Another issue is defining the critical loads for properly estimating wing masses. Ideally it would require to design a whole aircraft around the analyzed wing configuration for providing such data. Instead for the sake of comparison, values of wing loading W/S and limit load factor n_{max} were chosen to calculate the design velocities of a $V - n$ diagram by using Roskam's methods[33]. Tab. 4.3 provides such data and the calculated equivalent airspeeds. These loads are then multiplied by a factor of safety of 1.5 to obtain the respective ultimate loads.

By using the defined wing loading and load factor, the lift coefficients and thus wing loads are calculated. The resulting representative masses are shown in Tab. 4.4 with the respective percentile variations. It is clear that the flight condition at maximum load factor and cruise design speed V_c is the most critical load in

**Figure 4.4:** Lift curve**Figure 4.5:** Polar

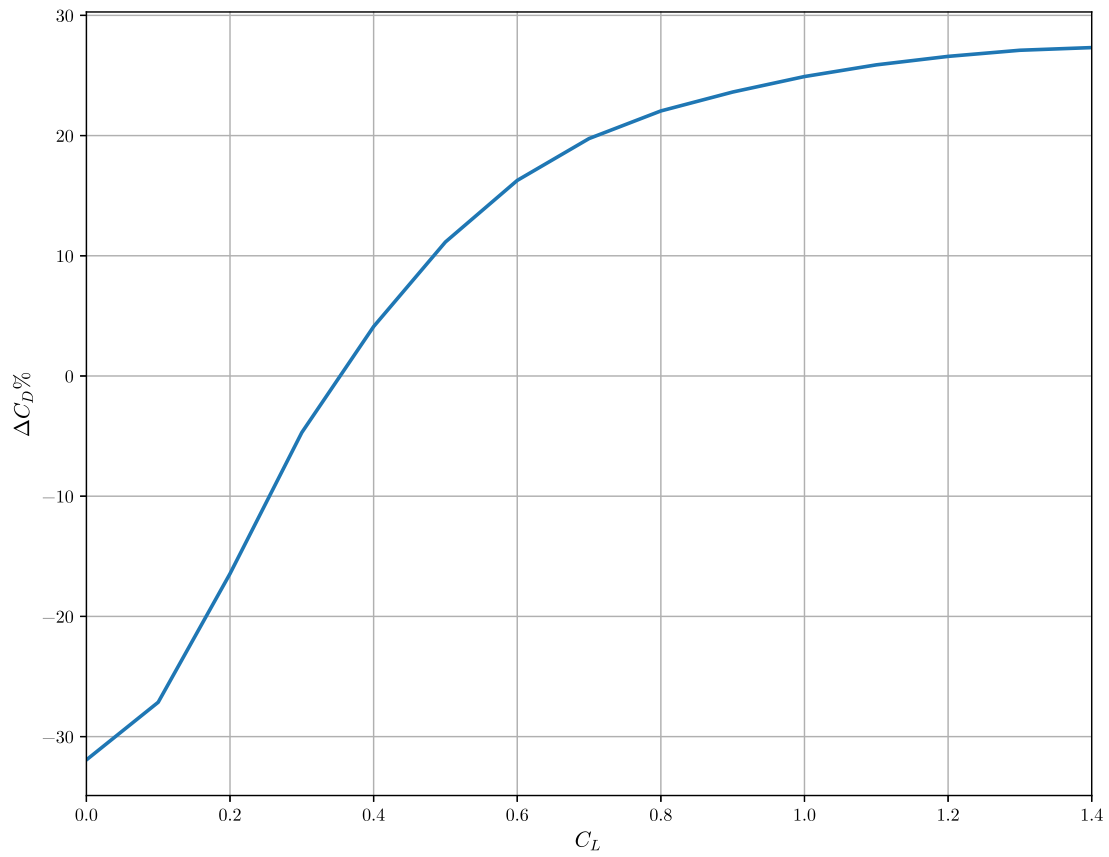


Figure 4.6: Relative Drag Reduction

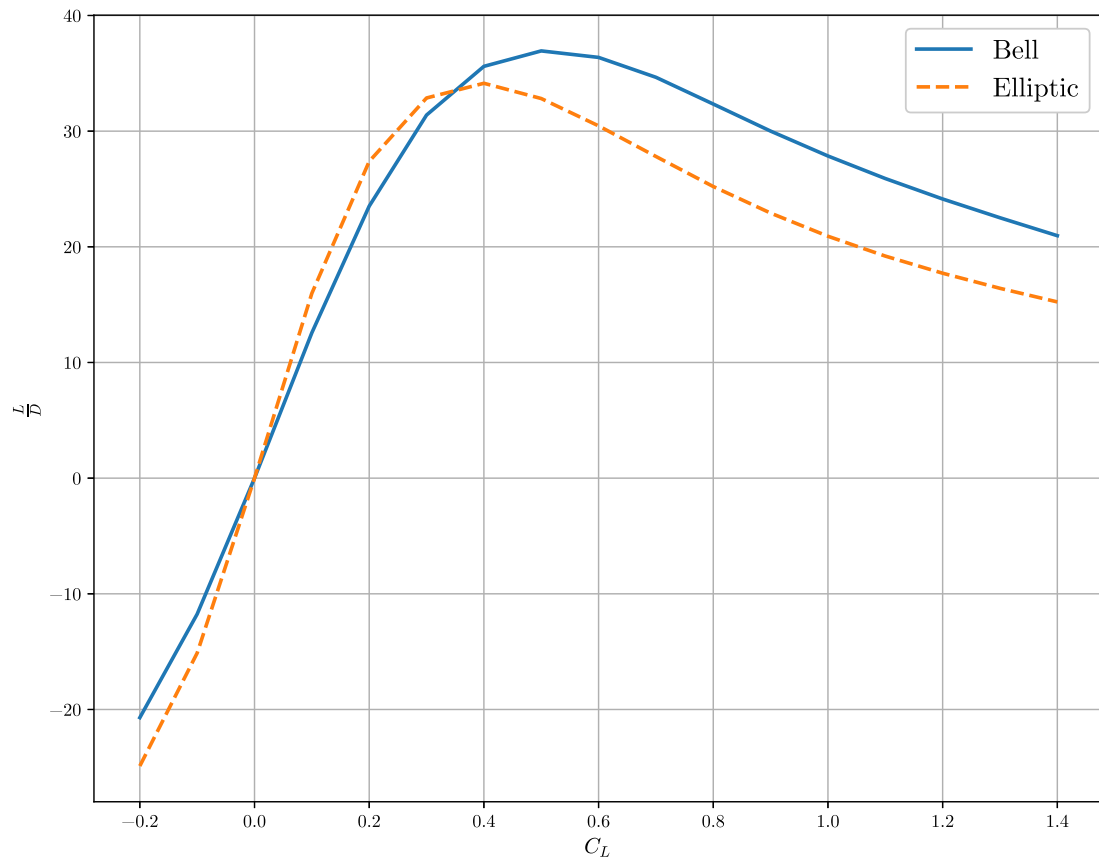
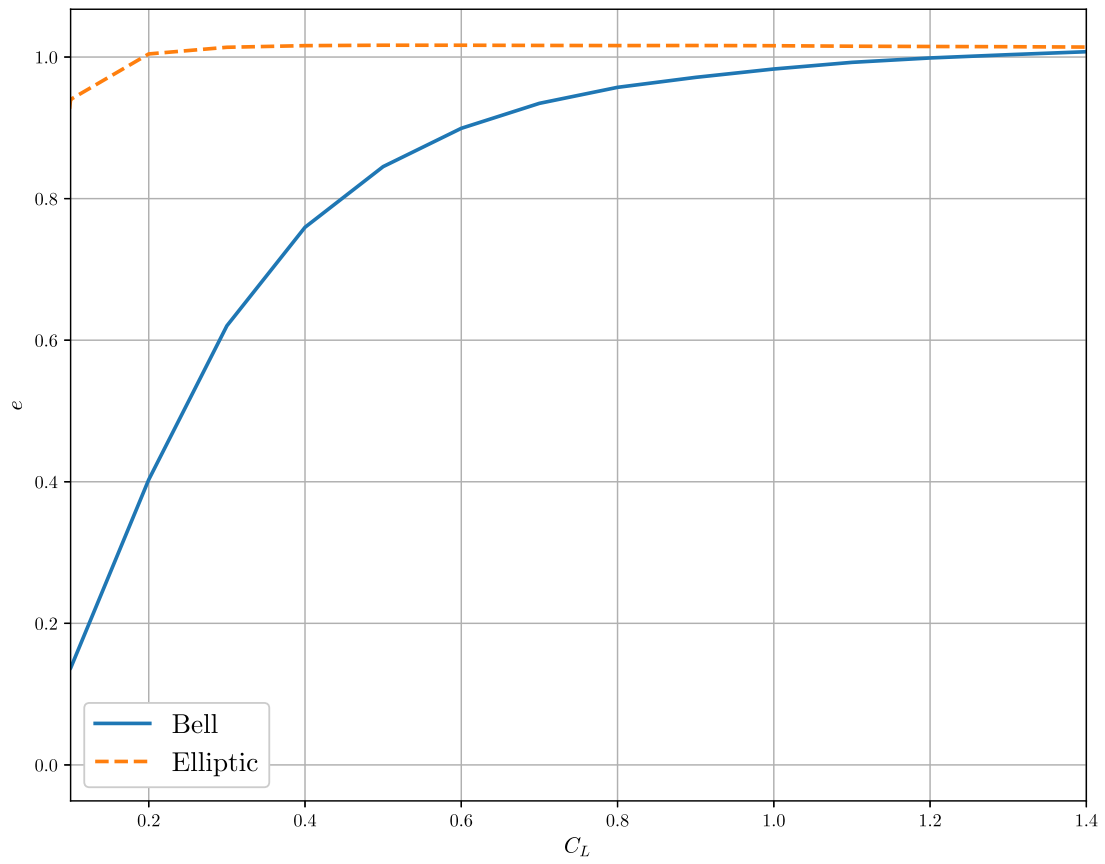
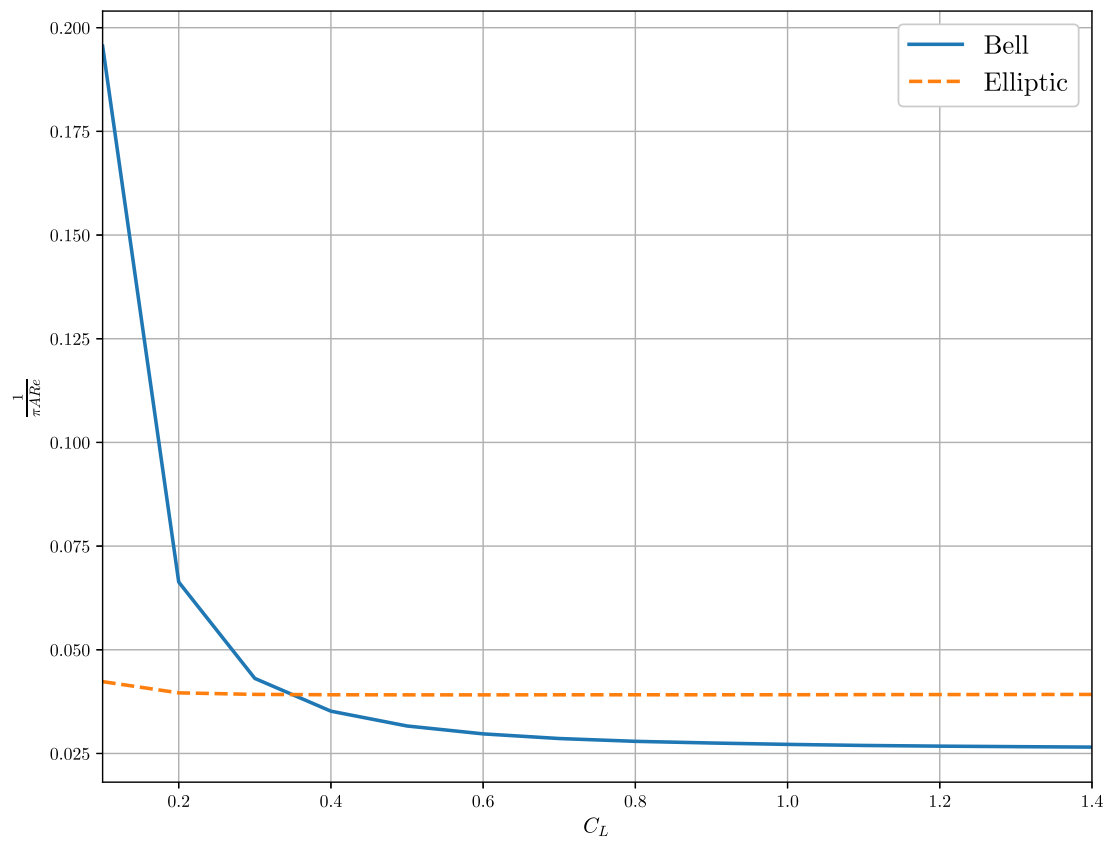


Figure 4.7: Lift to Drag ratio

**Figure 4.8:** Oswald's Efficiency Factor**Figure 4.9:** $1/(\pi AR e)$

W/S	n_{Max}	V_a	V_c	V_d
200 kg/m^2	3.5	85.32 m/s	104.41 m/s	130.51 m/s

Table 4.3: Data for Mass Estimation

this case. Therefore it shall be used in the next investigations as the mass sizing condition.

	Elliptic	Bell	% Variation	C_L
V_a	79.59 kg	133.52 kg	67.76	1.4
V_c	87.51 kg	139.67 kg	59.59	1.028
V_d	86.84 kg	126.08 kg	45.18	0.658

Table 4.4: Calculated representative masses

The wing bending moments for each loading case are shown in Figures 4.10-4.13. At the design point, the bending moments are almost coincident. However, the cruise condition is not a limit load, therefore it cannot be used to estimate structural mass. Because the root bending moment of the redesigned wing for these load cases is greater than the baseline one, structural masses have increased. It is interesting however to notice how the increment changes depending on which load case is used. The load at V_a yields the highest mass increase (67.76%) because the lift coefficient is the furthest from the design lift coefficient. Indeed if the lift distribution is inspected, the shape will be no longer bell-like but more similar to an elliptical lift distribution. Thus the wingtips would become lifty adding to the overall bending moment. The structural mass increment is not affected by the dynamic pressure q , as it is the same for both wings, but by how much the lift coefficient at the examined flight condition differs from the design lift coefficient. Hence the lowest percentile increment is at V_d . It is interesting to investigate how the mass of the wing changes with the variation of the wing design parameters. This is done in the following sections of this chapter.

Let's finally analyze the spanwise forces at the design lift coefficient (Fig. 4.14). The downwash distribution of the bell-spanload wing changes sign at $y = 4.6809 \text{ m}$, that corresponds to $y/b = 0.701$. From this point to the wingtip, the induced drag is negative, producing therefore *induced thrust* as shown in Fig. 4.14c. This effect concurs with the increased wingspan to reduce induced drag. As tested by NASA's Prandtl-D experiment, by introducing an aileron in this upwash region, it is possible to exploit the induced thrust to eliminate adverse yaw.

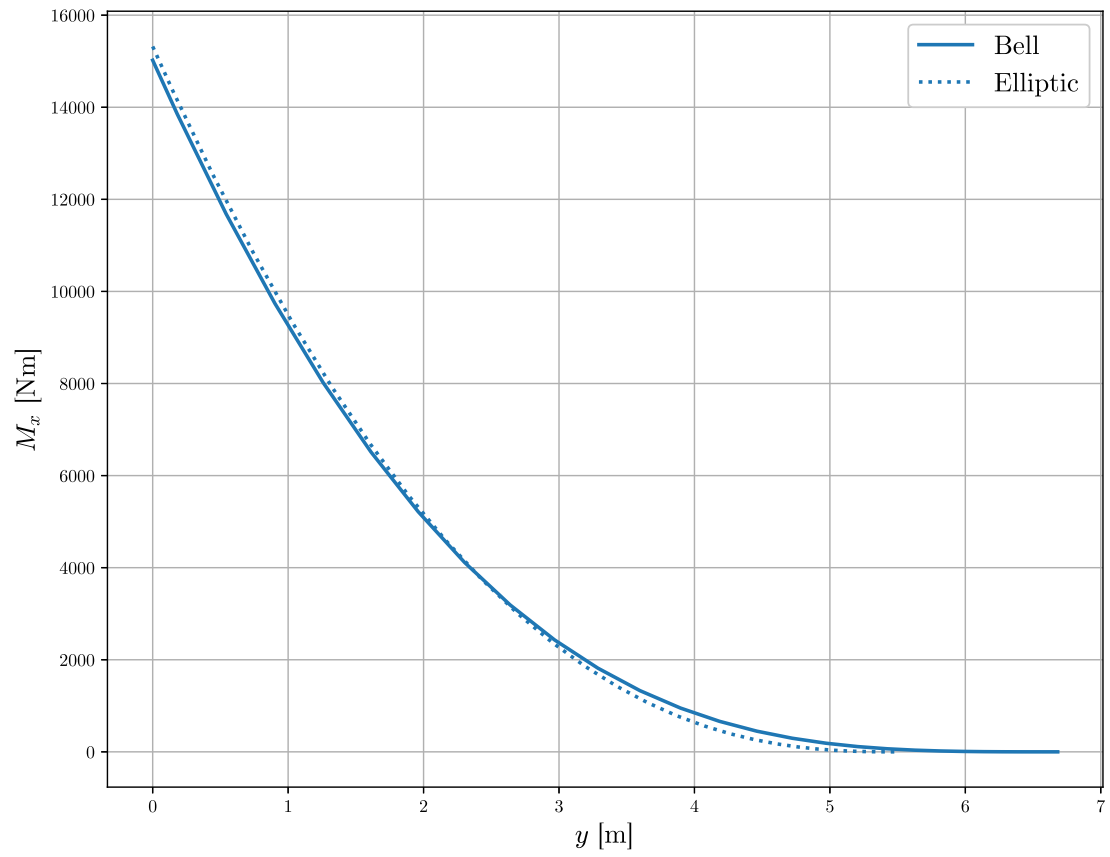


Figure 4.10: Bending moment at design point

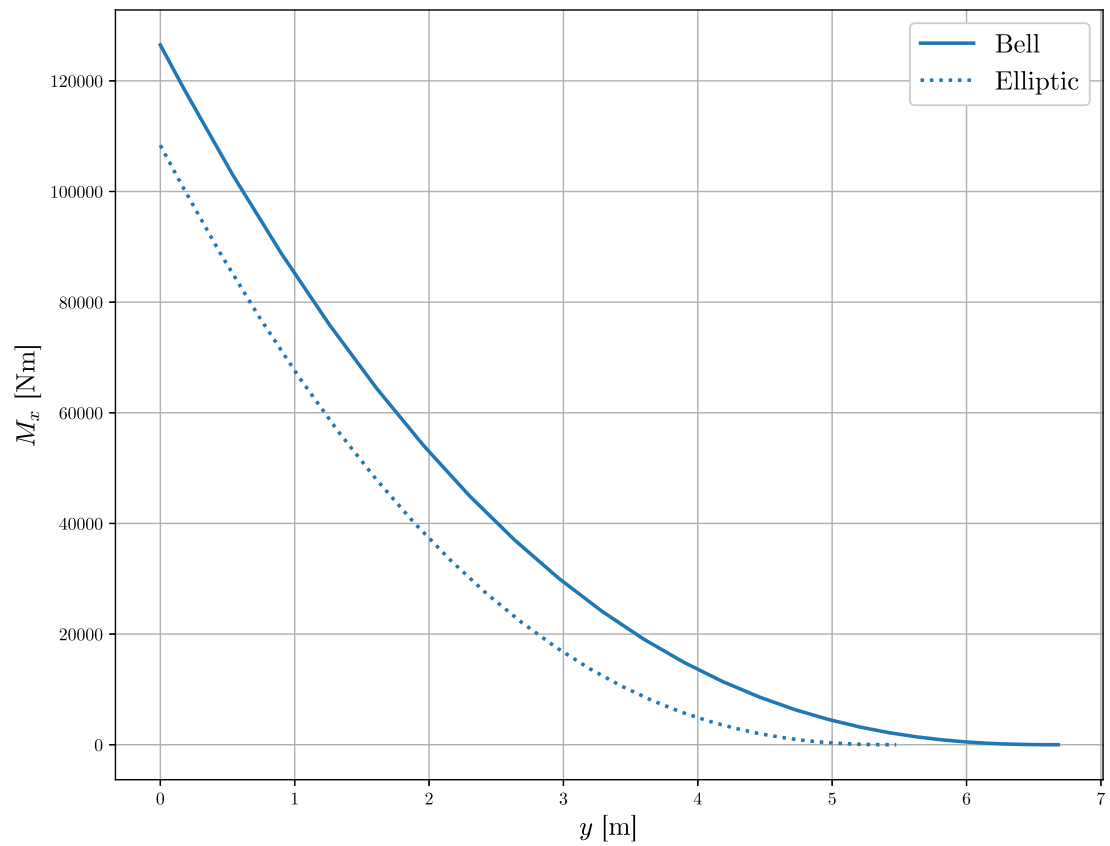


Figure 4.11: Bending moment at V_a

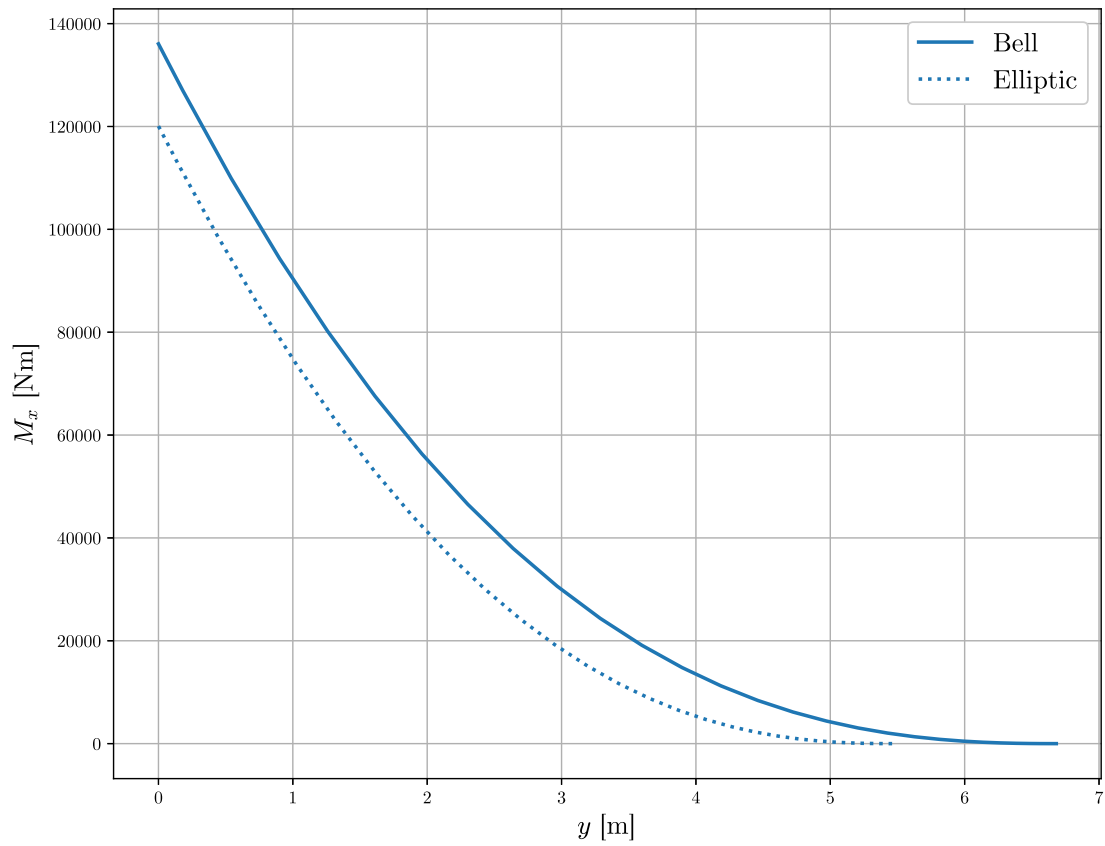


Figure 4.12: Bending moment at V_c

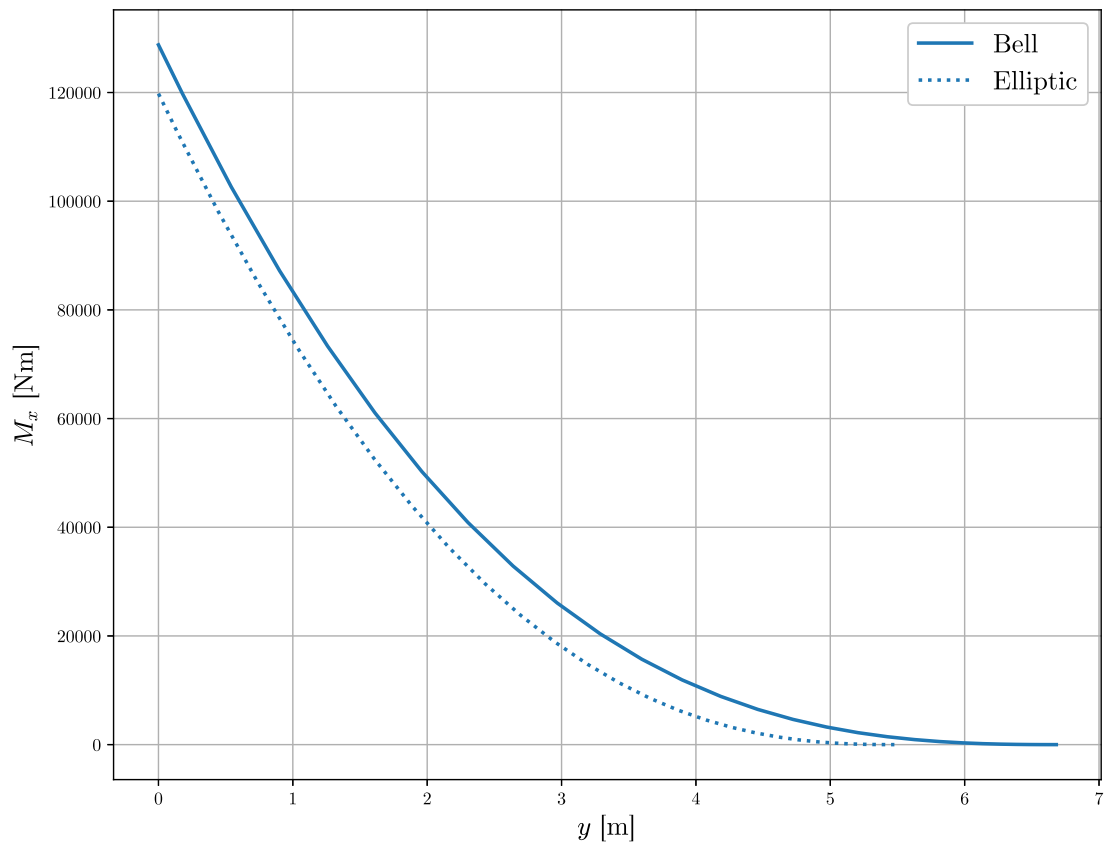
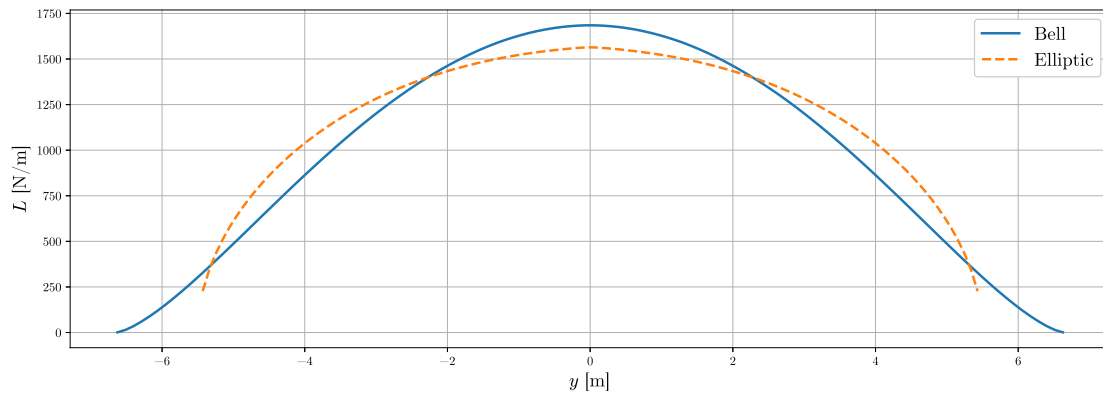
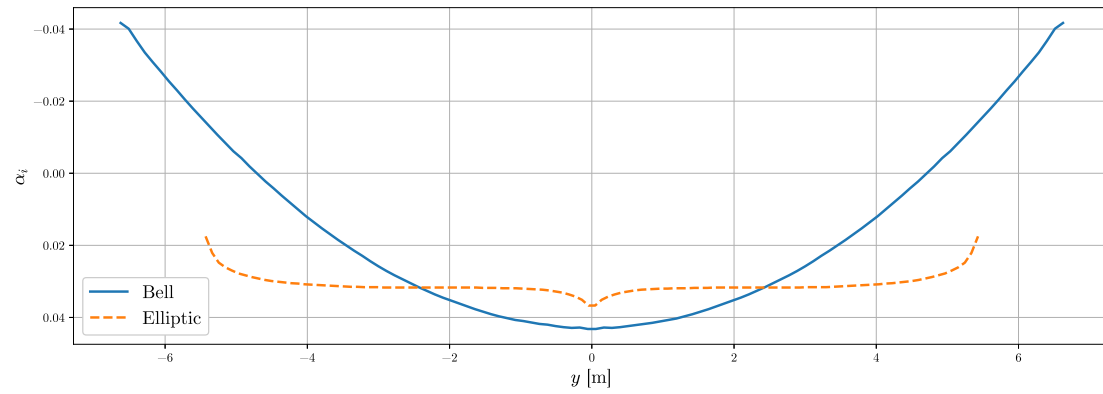


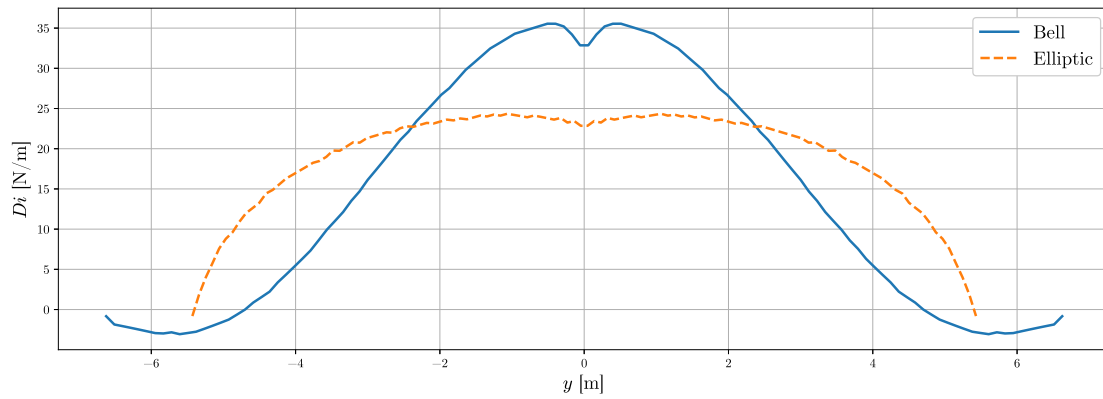
Figure 4.13: Bending moment at V_d



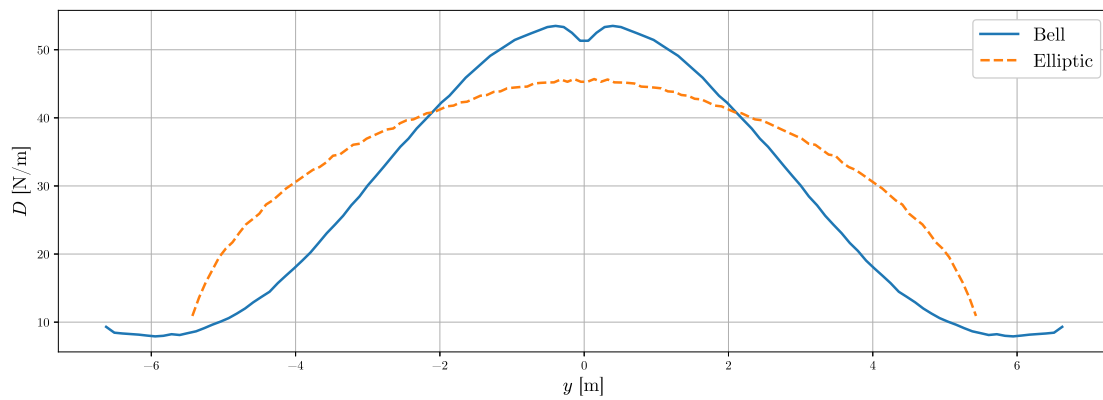
(a) Lift distribution



(b) Induced angle distribution



(c) Induced Drag distribution



(d) Drag distribution

Figure 4.14: Spanwise distribution of forces at design lift coefficient

4.2 Effects of $C_{L,des}$

The effect of the $C_{L,des}$ was studied on a fixed wing geometry, whose parameters are shown in Tab. 4.1. The design lift coefficient was set with a design airstream velocity of 60 m/s. Values 0.2, 0.4 and 0.6 were tested.

Aerodynamic results are shown in Figures 4.15-4.17. The twist distributions were designed to produce the desired $C_{L,des}$ at $\alpha = 0$, in order to eliminate any effect caused by the shift of the zero lift angle of the wing. All three designs show an increase of the $C_{L,\alpha}$ derivative which is to be expected from the increase of aspect ratio. Drag of the new designs (Fig. 4.15b - 4.17b) is higher at lift coefficients below the design one. The break-even point is slightly below the design lift coefficient. The calculated L/D is shown in Fig. 4.24, alongside the calculated relative change with respect to the baseline wing (Fig. 4.25). The maximum L/D is greater if the design lift coefficient is low, however it is shifted at a higher lift coefficient. The relative increment of L/D at the design lift coefficient grows with $C_{L,des}$. This would suggest to set the design lift coefficient below than the cruise one to benefit the highest L/D possible.

These polars show total drag, that is the sum of induced and viscous drag. In order to evaluate the reduction of induced drag alone and then of total drag, the graph shown in Fig. 4.22 was developed. The induced drag reduction is at 10% for all the three tested cases, close to Prandtl's theoretical value of 11%. The discrepancy is the effect of using a numerical Vortex-lattice scheme which is not continuous, unlike the lifting line equation: this is visible in the Oswald efficiency factor of the elliptical wing which is slightly above the theoretical value of 1 (Fig. 4.23).

However if viscous drag is incorporated in the comparison, drag reduction becomes smaller. It is no longer independent from $C_{L,des}$, but increases with the value of design lift coefficient. The cause is in the relative contribution of each drag component.

	e	b	e	b	e	b
C_L	0.2	0.2	0.4	0.4	0.6	0.6
$C_{D,v}$	0.005719	0.005830	0.005450	0.005604	0.005960	0.005591
$C_{D,i}$	0.001567	0.001409	0.006267	0.005631	0.014106	0.012588
C_D	0.007285	0.007238	0.0117170	0.0112354	0.0196967	0.018548
% viscous	78.49	80.53	46.51	49.87	28.38	32.13
% induced	27.45	19.46	53.48	50.12	71.61	67.86

Table 4.5: Drag breakdown for different $C_{L,des}$.

Table 4.5 provides a drag breakdown of all the tested cases at their respective design lift coefficient, where e indicates the elliptical spanload wing case and b the re-designed bell spanload wing case.

As expected from classical wing theory, the induced component of total drag is small relative to the viscous component for small lift coefficients. Therefore the possible benefit from reducing induced drag is practically non-existent if the design lift coefficient is too small. Interestingly, even if the wetted area is kept constant,

the re-designed wing introduces a small increase of few drag counts of the viscous drag. This effect may be the result of a local increment of angle of attack near the root of the wing, compared to the elliptical lift distribution, which would increase locally the profile viscous drag.

If $C_{L,des}$ is high, like in Fig. 4.17, drag at small lift coefficients is almost doubled compared to the respective elliptical wing, even when there is no lift. The issue is in the twist distribution, as root and tip sections are heavily twisted. Therefore even if globally the wing is not producing lift, locally a portion of the wing is generating positive lift, while another portion is generating negative lift. Since the excursion of the twist distribution is ample, the sectional lift coefficient has peaks of $C_l = -0.32$ and 0.28 (Fig. 4.30). Because the shape lift distribution is not bell-shaped, lift does not taper at the wingtip, instead it abruptly falls to zero after the negative peak. And since the induced angle is proportional to the derivative of the vorticity distribution, and induced drag is proportional to the induced angle, near the wingtip the induced drag is the highest. Therefore it is better to not exceed with $C_{L,des}$ to avoid such drag penalty.

As predicted from Prandtl's theoretical results, the bending moments of the elliptical and bell distributions match (Fig. 4.26). Instead at the limit load cases, the same flight conditions used in Sec. 4.1, the bending moment of the new wing is higher. By comparing the bending moment distribution at V_a (Fig. 4.27), at V_c (Fig. 4.28) and V_d (Fig. 4.29), it is clear that the second case introduces the highest loads on the wing.

$C_{L,des}$ affects the bending moments depending on which load case is considered: the spread of root bending moment between each design is greater at V_d and the lowest at V_a . As it was discussed before, the lift distribution morphs into an elliptical-like distribution when C_L is greater than the design lift coefficient. Therefore for high C_L , like the V_a condition, the shape of the lift distribution has become more similar for all the three cases. Another observation is how the root bending moment for $C_{L,des} = 0.6$ is always below the 0.2 design. This is again an effect of the shape of the lift distribution: the furthest is the design lift coefficient from the flight condition one, the higher is going to be its off-design load. The conclusion is for constraining the structural weight increase, $C_{L,des}$ should be high and equivalent to the flight condition of the ultimate load. However, this lift coefficient is usually much higher than the cruise one of which we seek to minimize drag and maximize L/D . Therefore a trade-off has to be performed to choose a proper value of $C_{L,des}$: between increased weight and reduced drag.

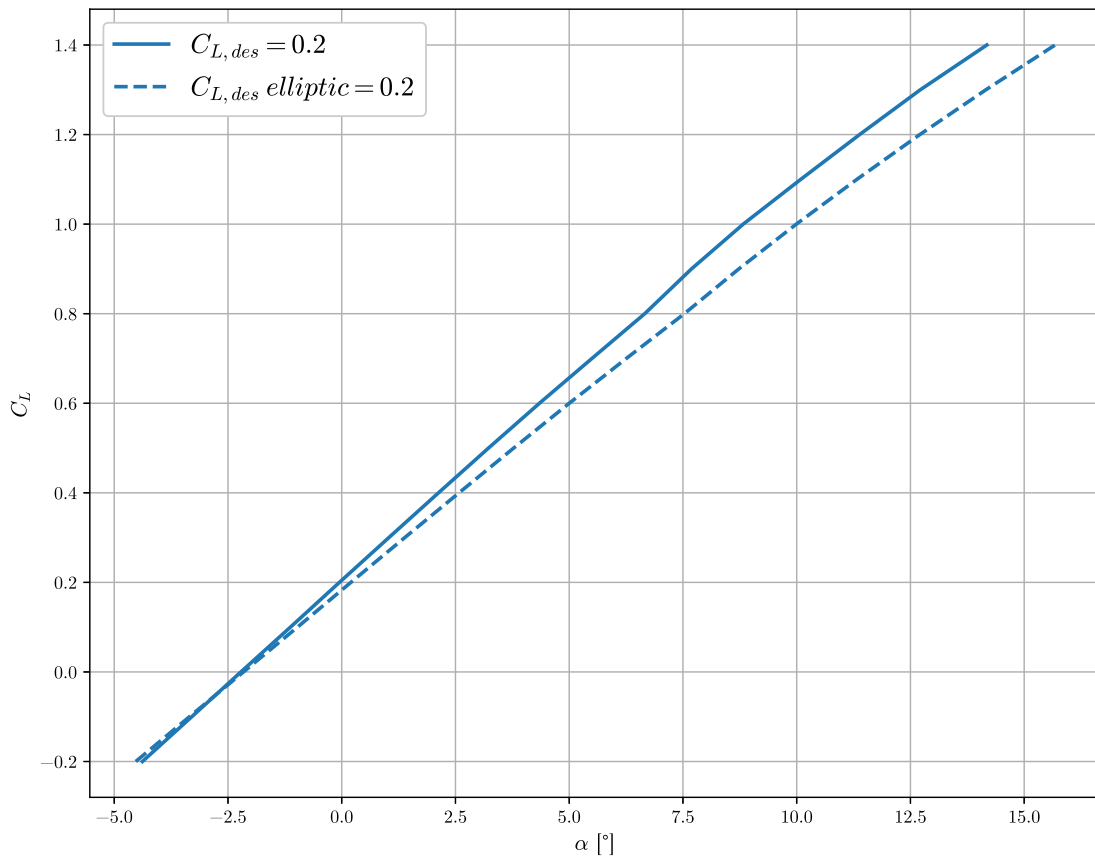
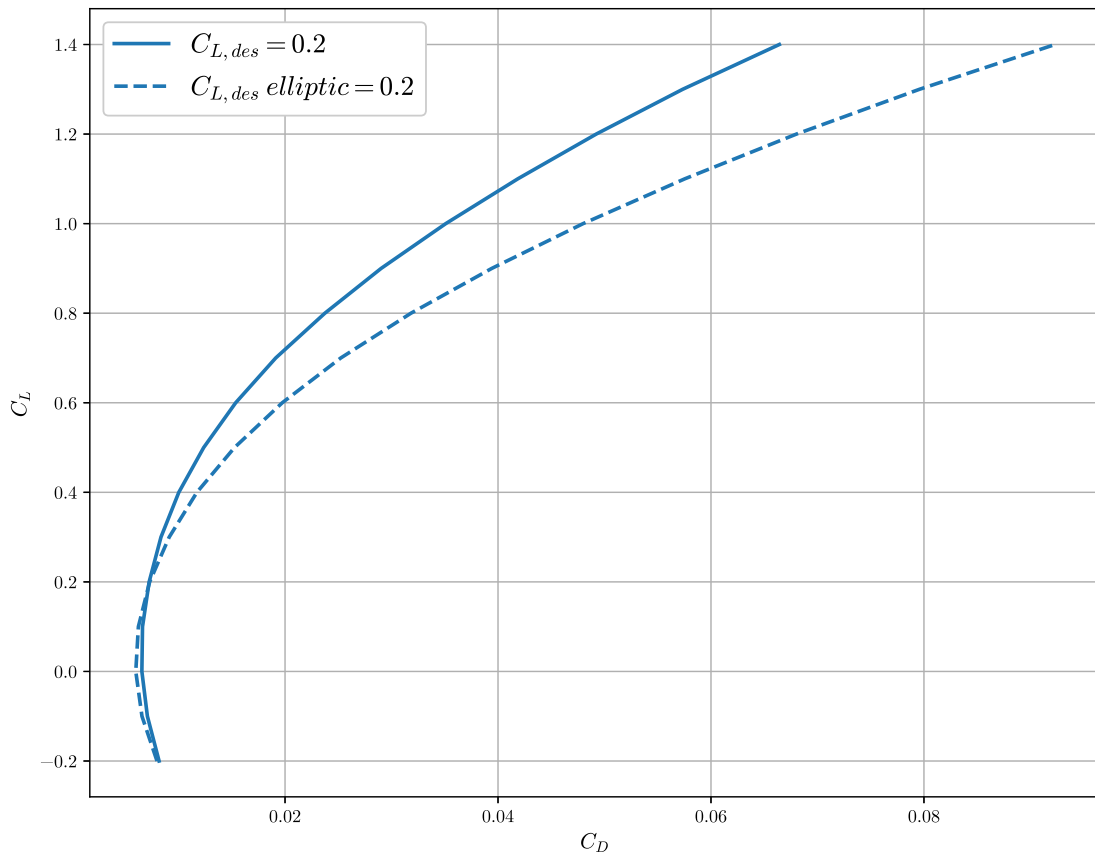
$C_{L,des}$	0.2	0.4	0.6
Elliptical	81.46 kg	87.51 kg	93.22 kg
Bell	137.20 kg	139.67 kg	139.56 kg
% Variation	68.42	59.59	49.71

Table 4.6: Representative wing mass for different $C_{L,des}$

Representative wing mass was calculated with the assumptions of Sec. 4.1, by using data in Tab. 4.3. The structural design flight condition was found to be at V_c .

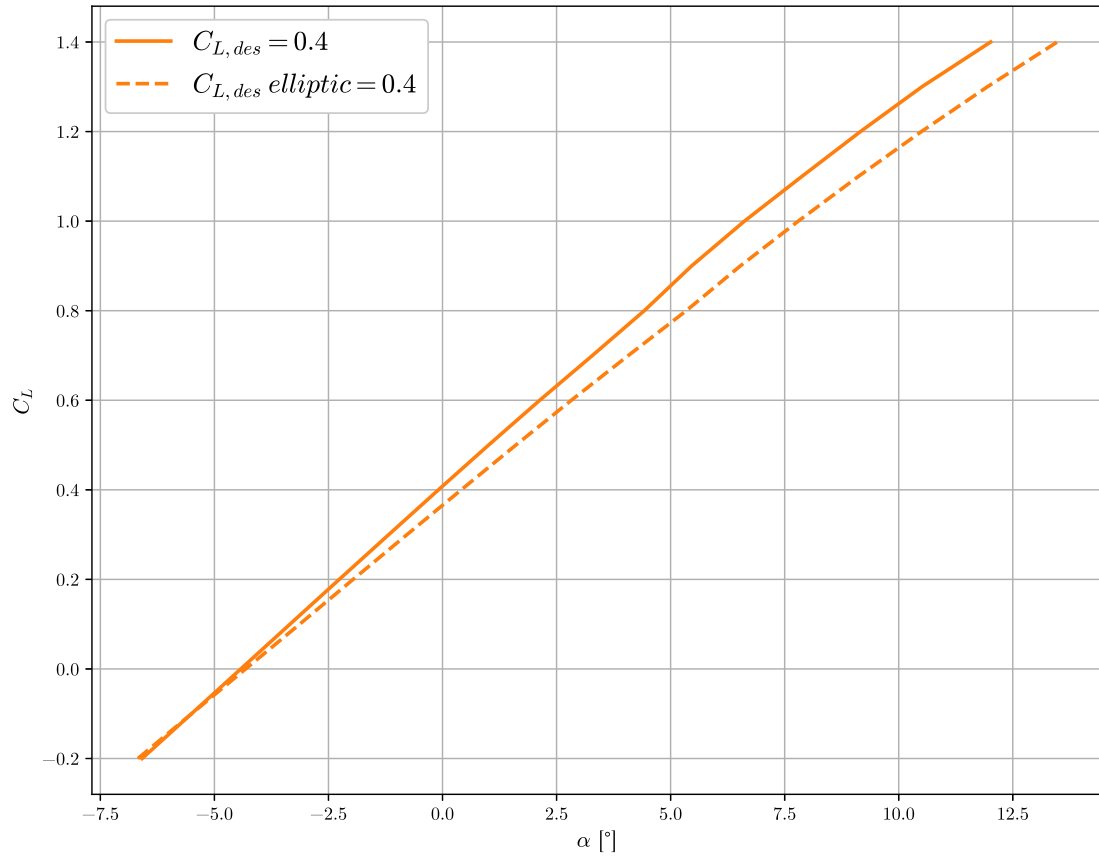
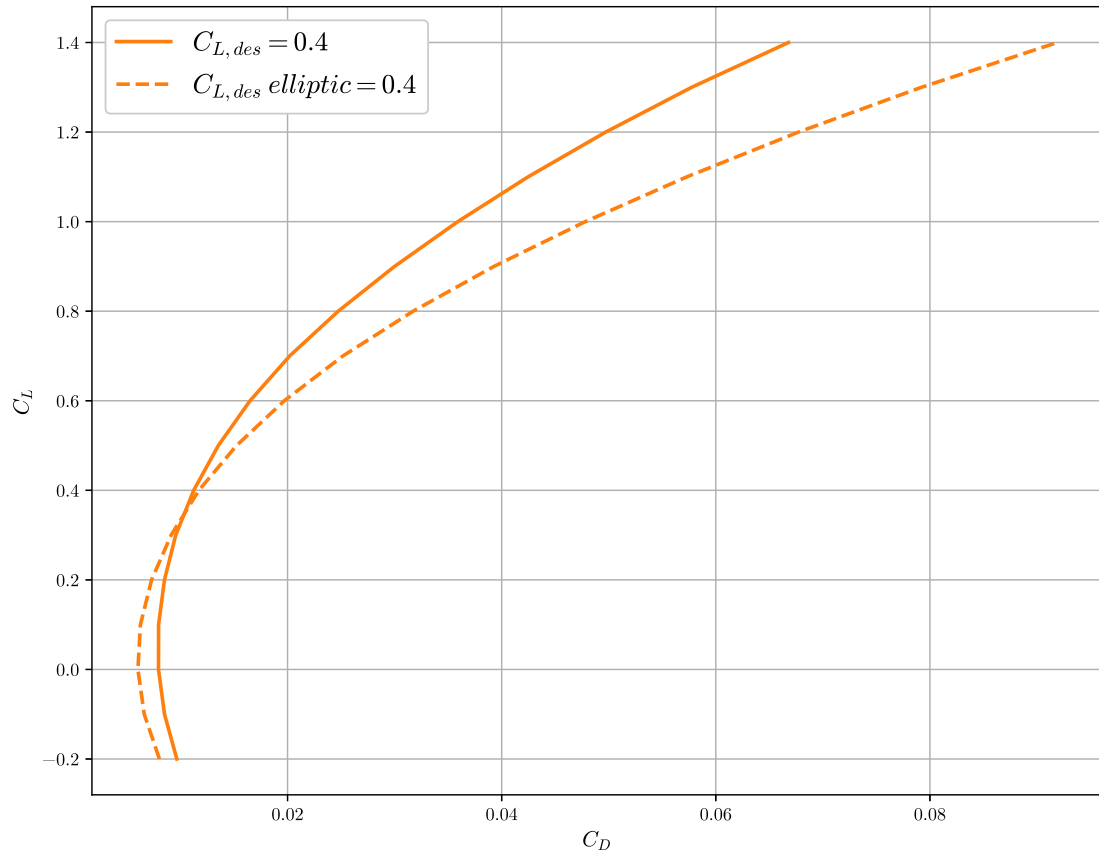
Table 4.6 shows the results and their respective variations. The mass increments are coherent with the observed behavior in the bending moments: the highest mass corresponds to the lowest design lift coefficient, for which the root bending moment is the furthest from the respective elliptical wing. The mass variation is 68.42%, that is the weight increases more than one half the original wing.

The C_l distributions, normalized by the local chord, are shown in Fig. 4.18. Bell and Elliptical lift distributions are obtained at their design point. The respective downwash distributions are shown in Fig. 4.19. The shape of the wake is parabolic, as Prandtl found in his paper, with the upwash region present at the outer portion of the wing. However the upwash region reduces as $C_L > C_{L,des}$, the vortex core shifts closer to the wingtips. Indeed at $C_{L,max}$, the upwash region is completely absent (Fig. 4.21) with an irregular shape. The C_l distribution itself, at the maximum lift coefficient, becomes more similar to an elliptical one than a bell lift distribution (Fig. 4.20). This is responsible for the increase of root bending moment, as the outer portion of the wing is more loaded. The distribution becomes more elliptical the further $C_{L,des}$ is from the maximum lift coefficient.

(a) $CL - \alpha$ 

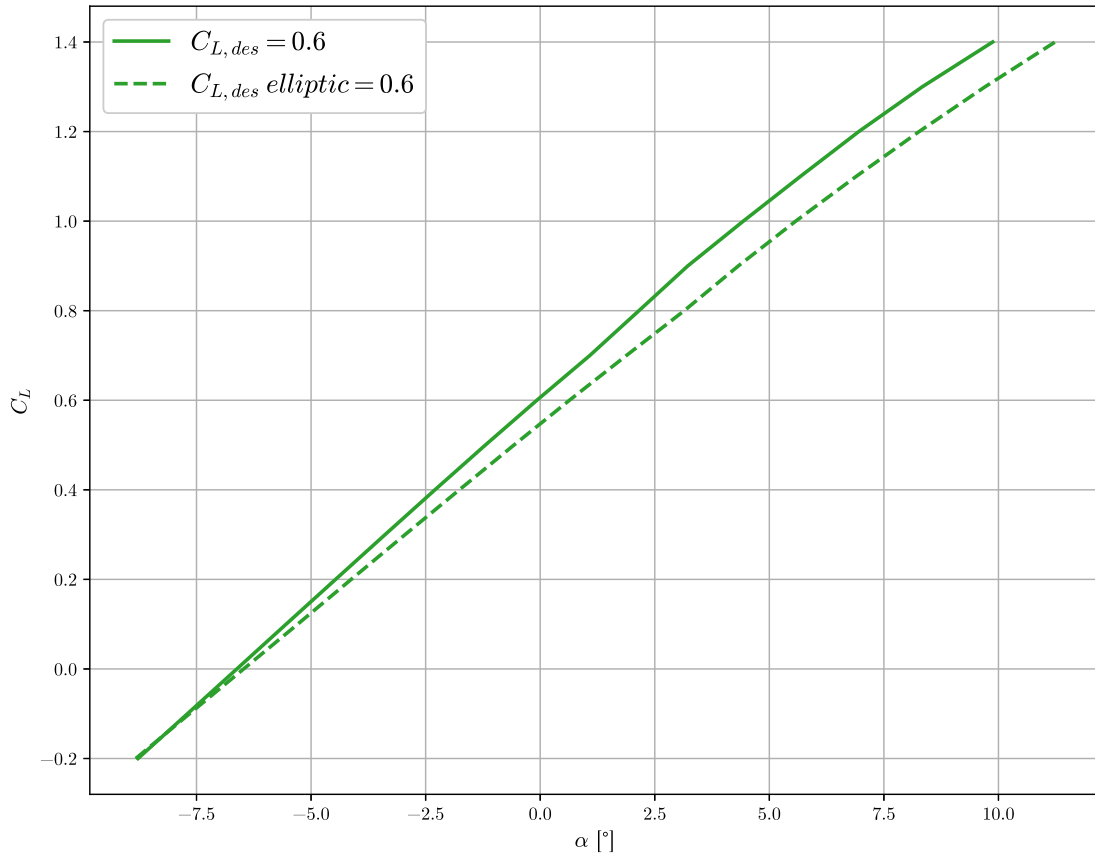
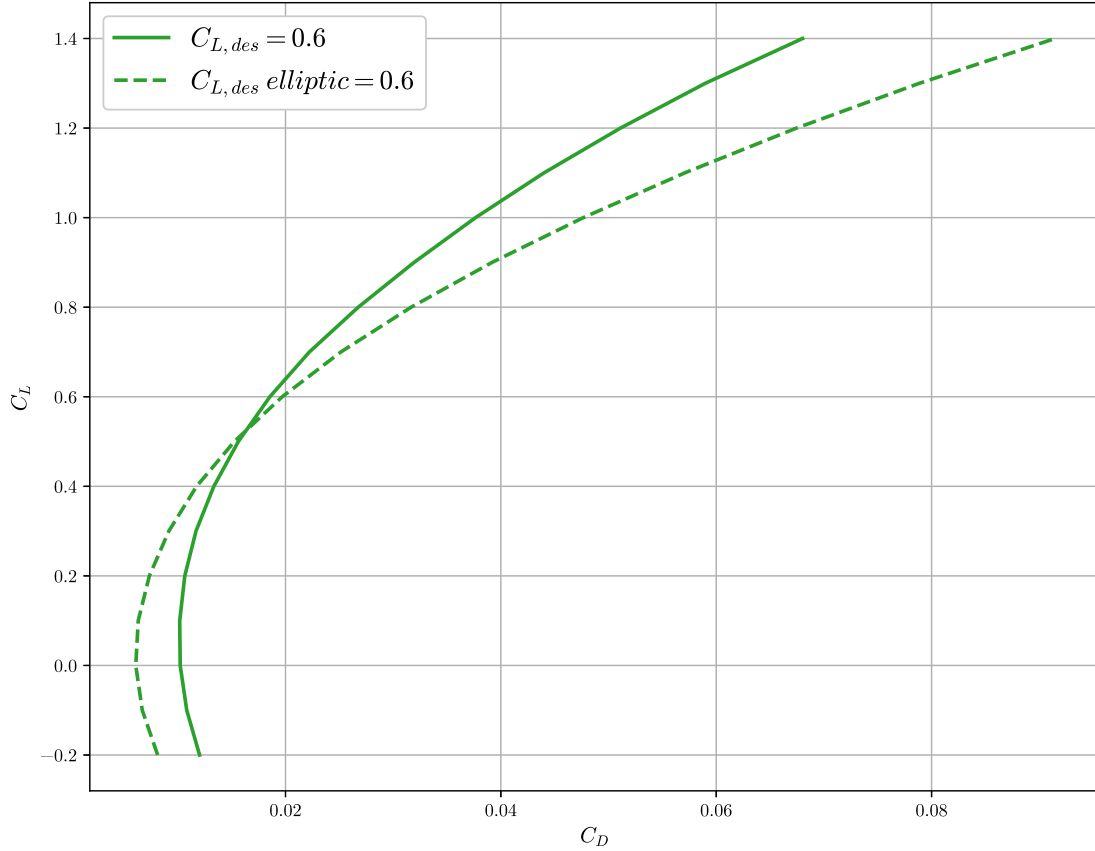
(b) Polar

Figure 4.15: Aerodynamic data at $C_{L,des} = 0.2$

(a) $CL - \alpha$ 

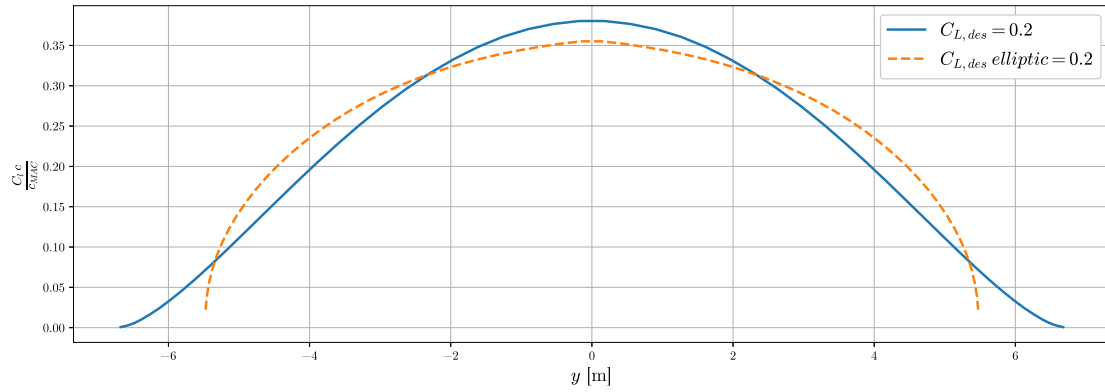
(b) Polar

Figure 4.16: Aerodynamic data at $C_{L,des} = 0.4$

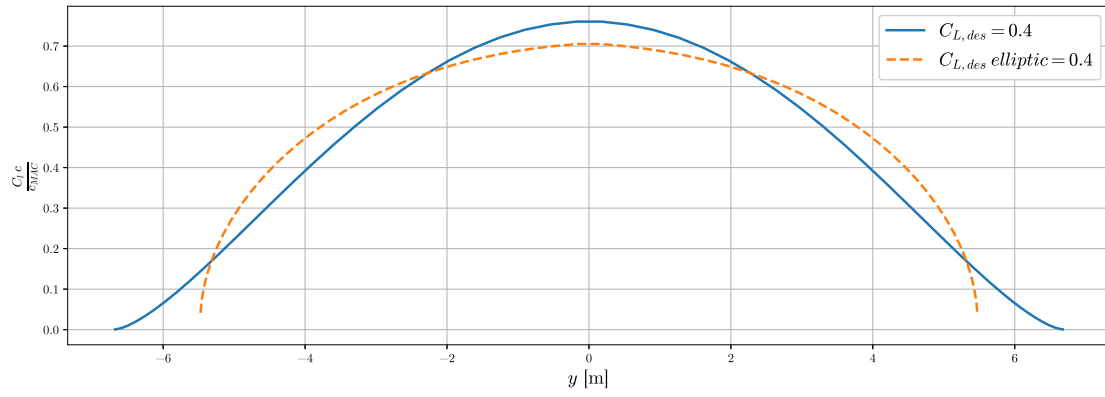
(a) $CL - \alpha$ 

(b) Polar

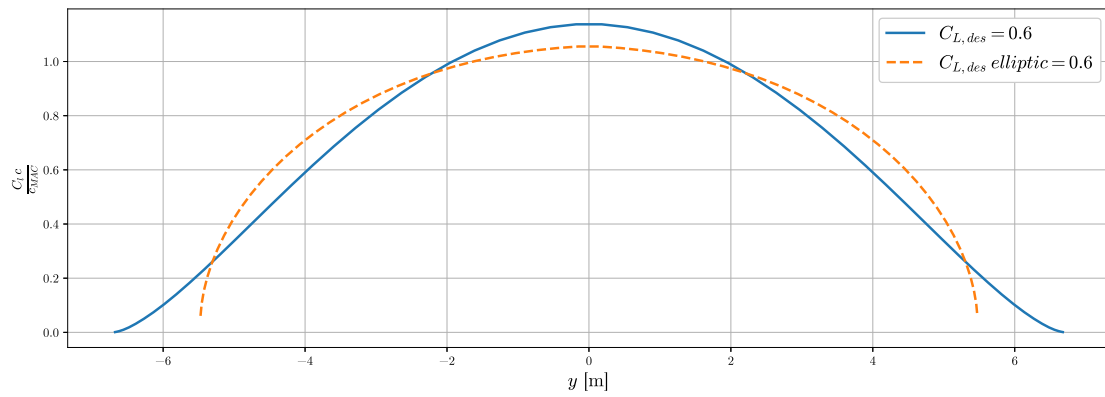
Figure 4.17: Aerodynamic data at $C_{L,des} = 0.6$



(a)

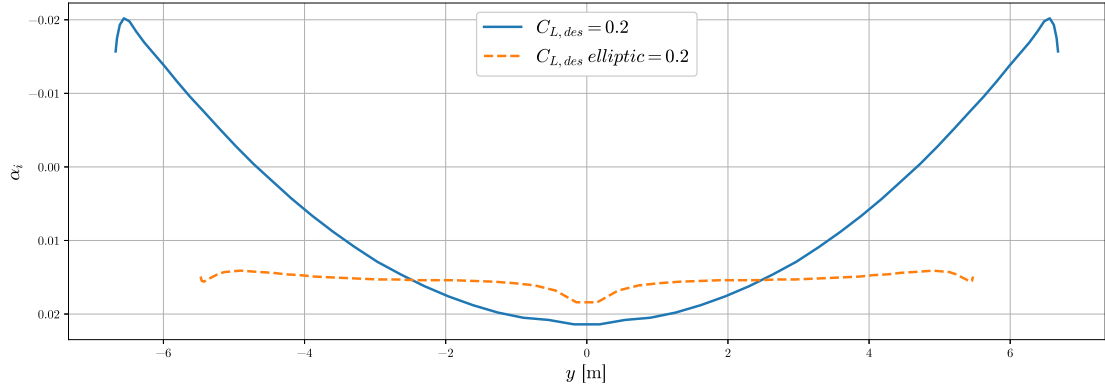


(b)

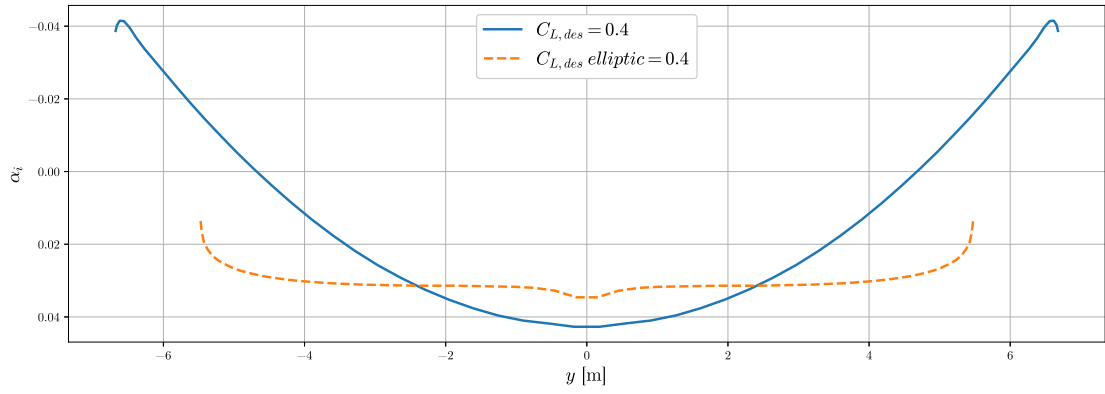


(c)

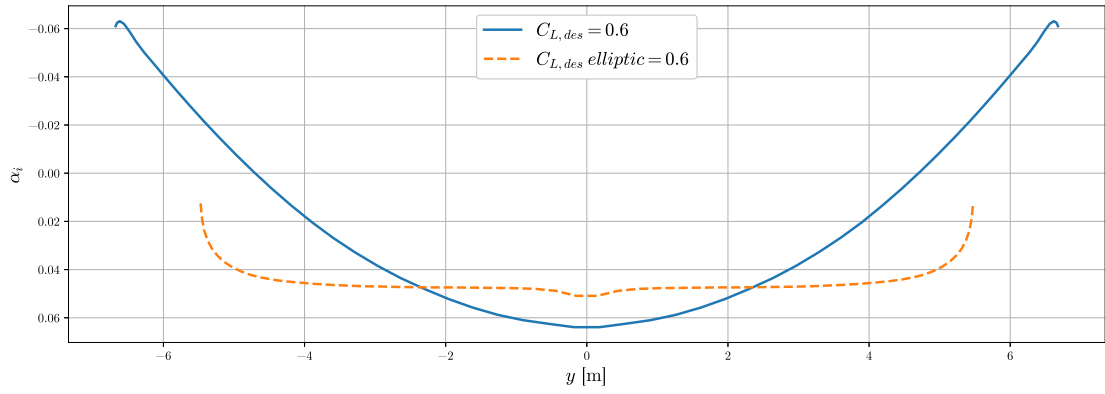
Figure 4.18: C_L distribution at the design point for different $C_{L,des}$



(a)

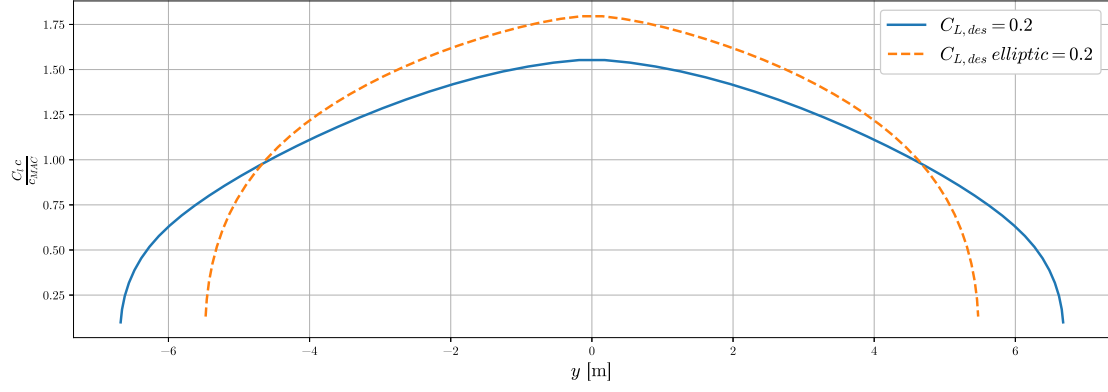


(b)

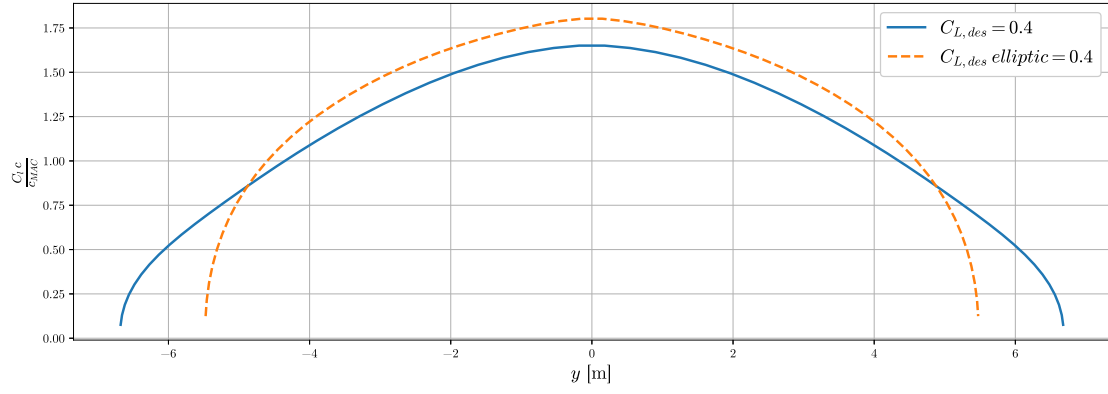


(c)

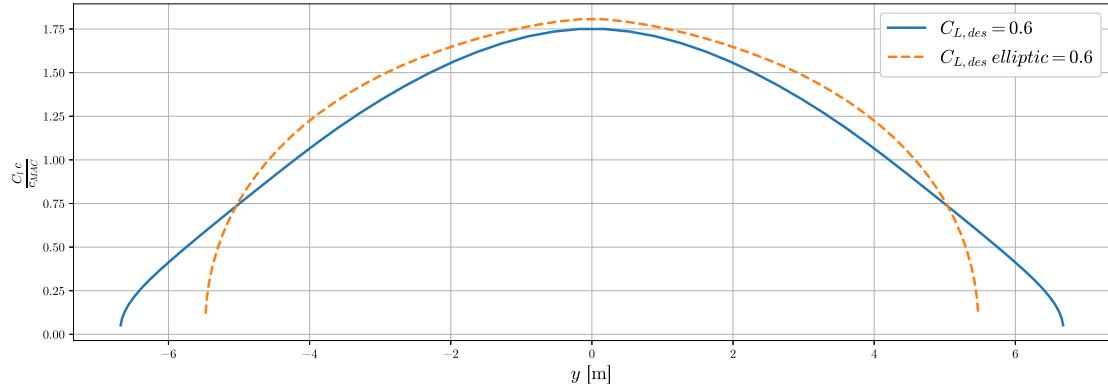
Figure 4.19: Downwash distribution at the design point for different $C_{L,des}$



(a)

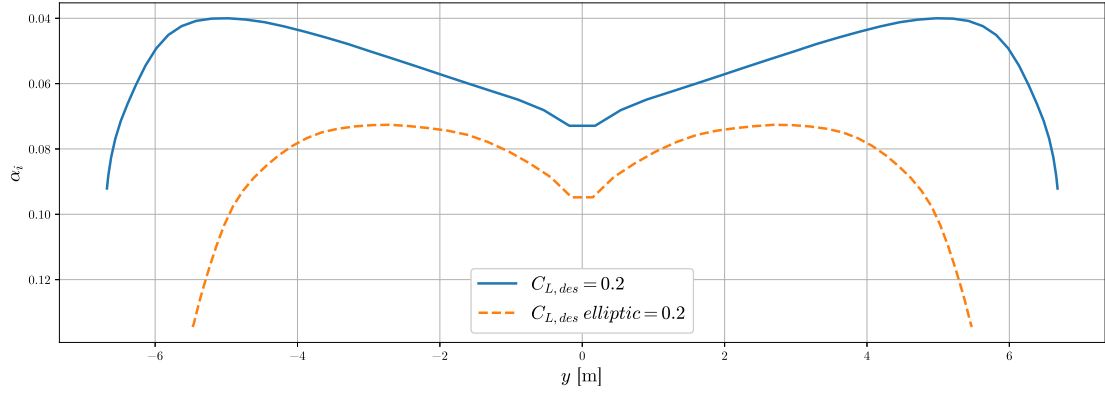


(b)

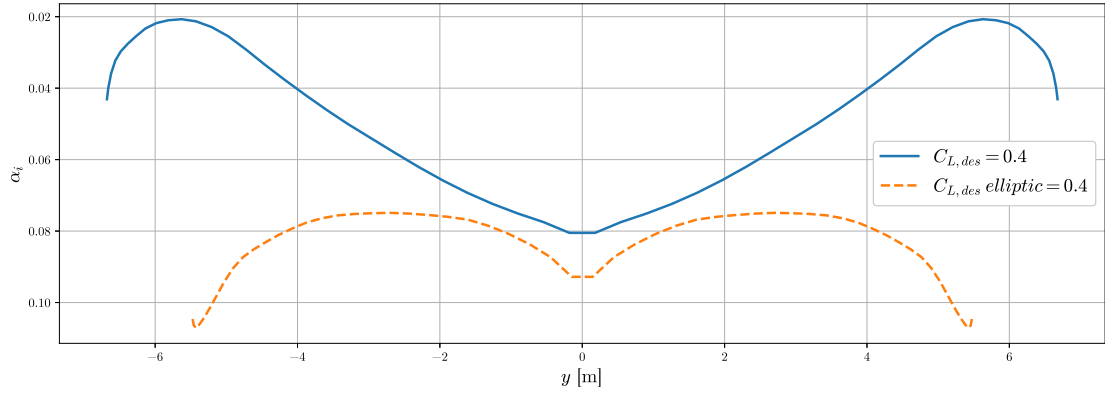


(c)

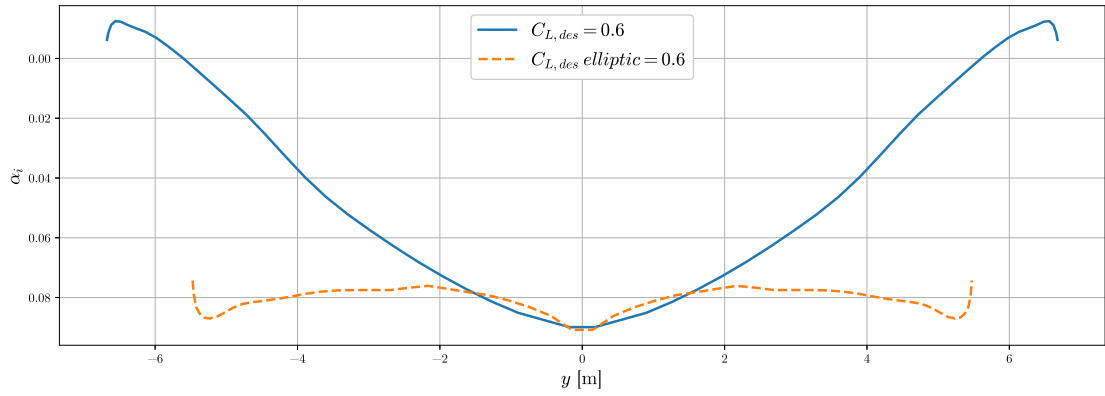
Figure 4.20: C_L distribution at V_c for different $C_{L,des}$



(a)



(b)



(c)

Figure 4.21: Downwash distribution at V_c for different $C_{L,des}$

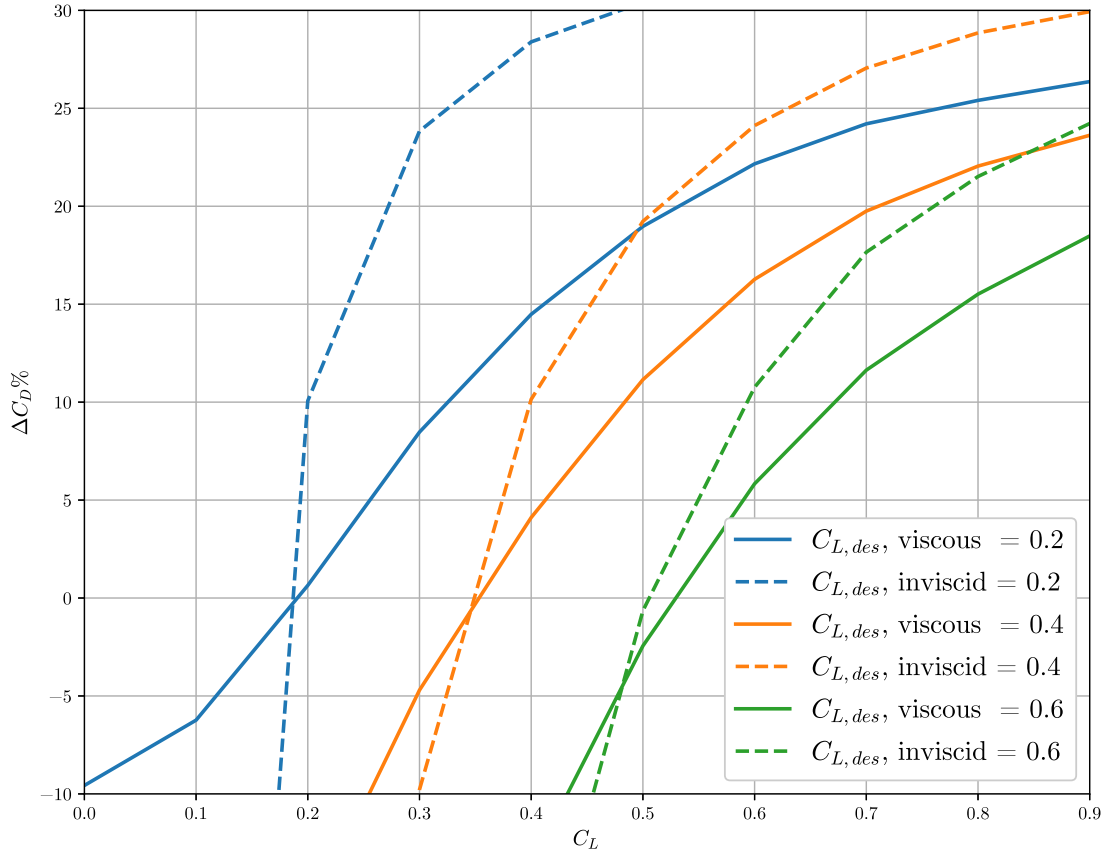


Figure 4.22: Percentile drag reduction for different $C_{L,des}$.

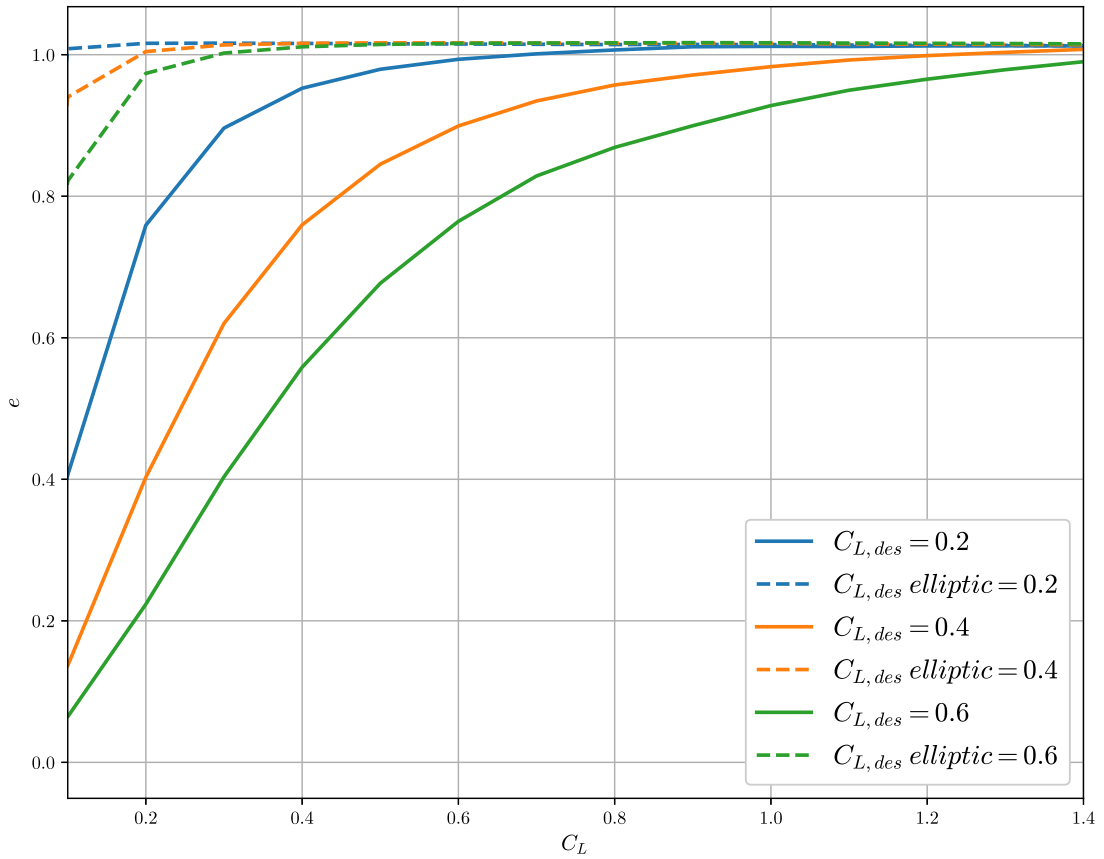


Figure 4.23: Oswald efficiency factor e for different $C_{L,des}$.

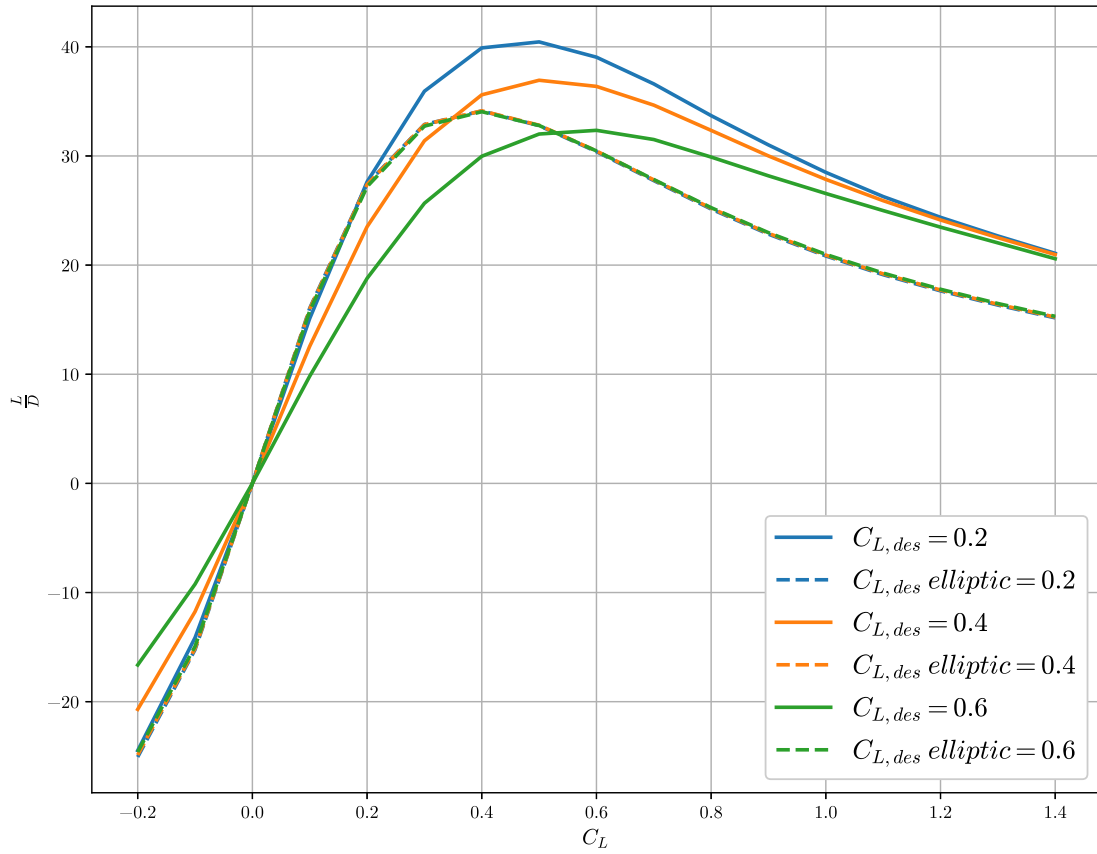


Figure 4.24: L/D for different $C_{L,des}$.

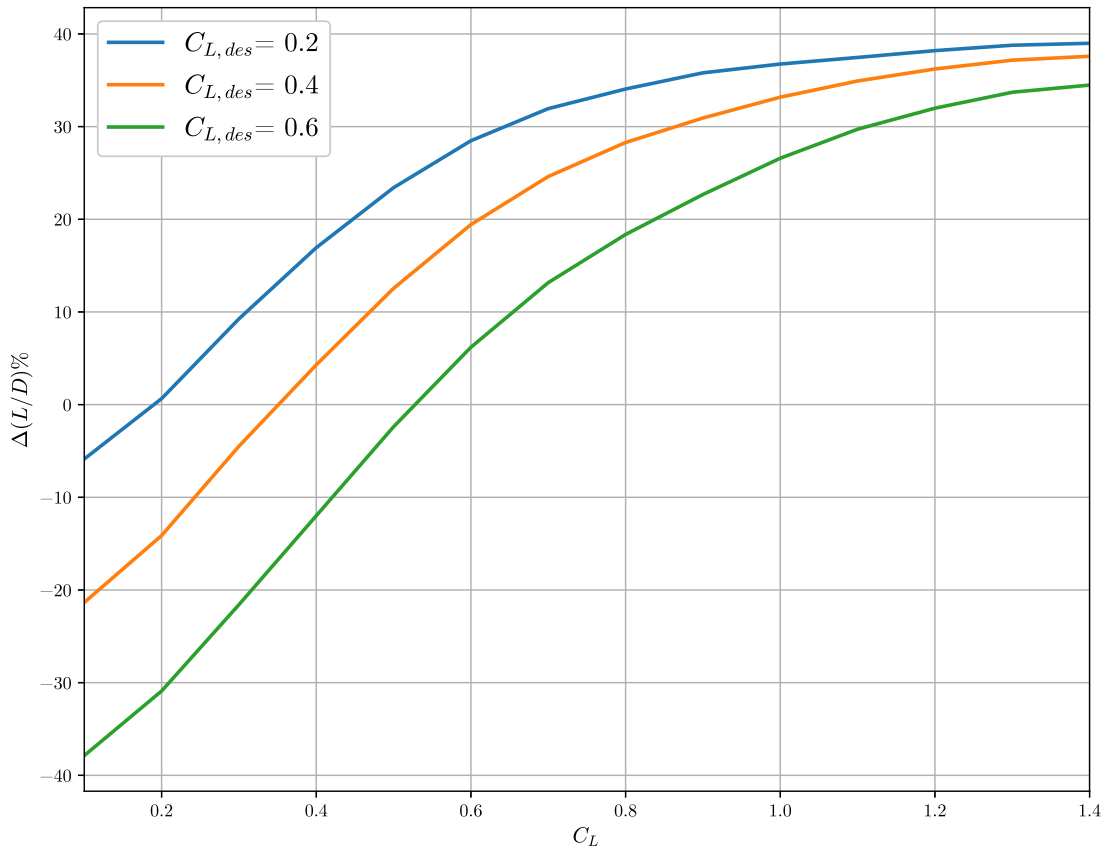


Figure 4.25: Relative change of L/D for different $C_{L,des}$.

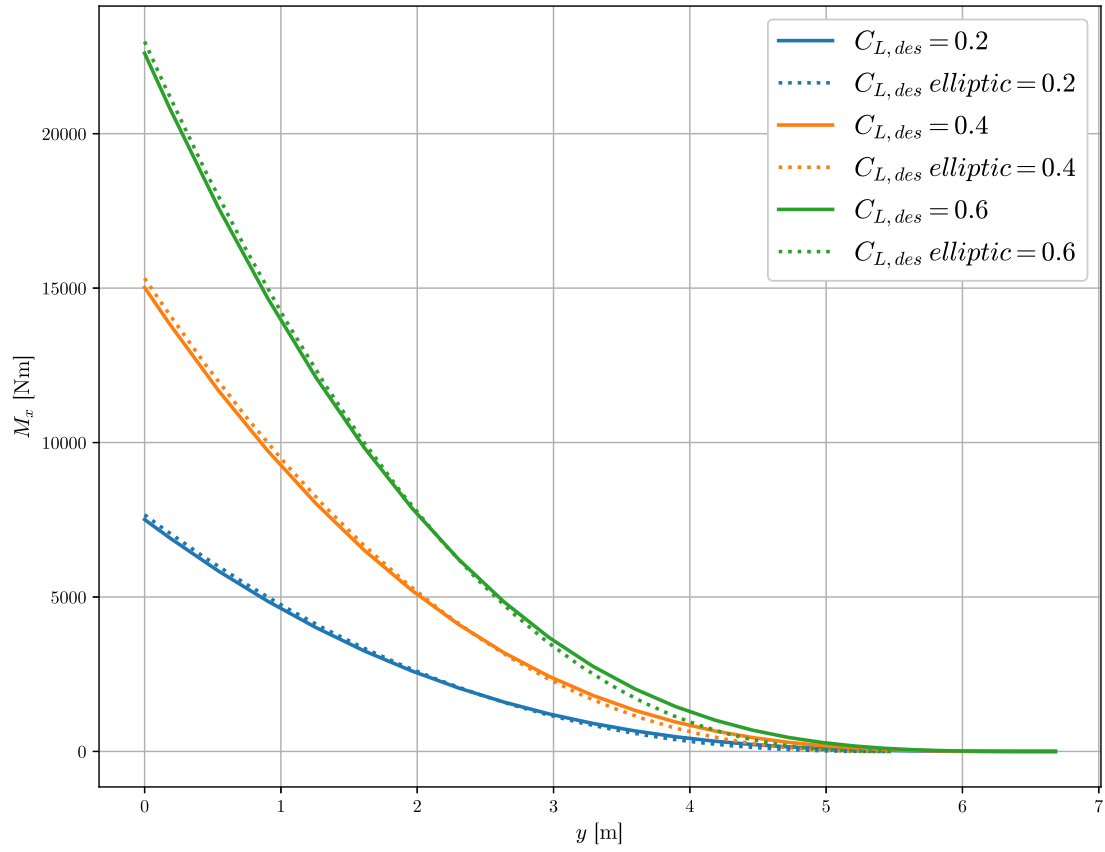


Figure 4.26: Bending moment at design point for different $C_{L,des}$.

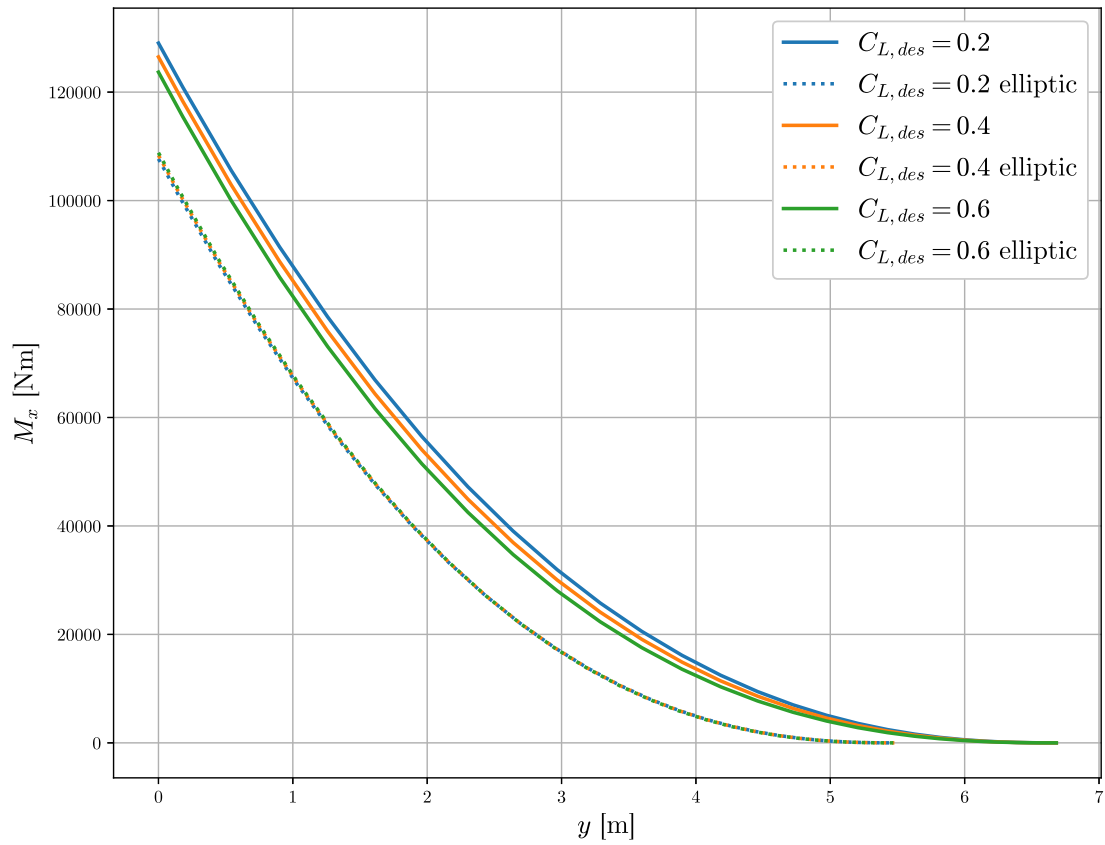


Figure 4.27: Bending moment at V_a for different $C_{L,des}$.

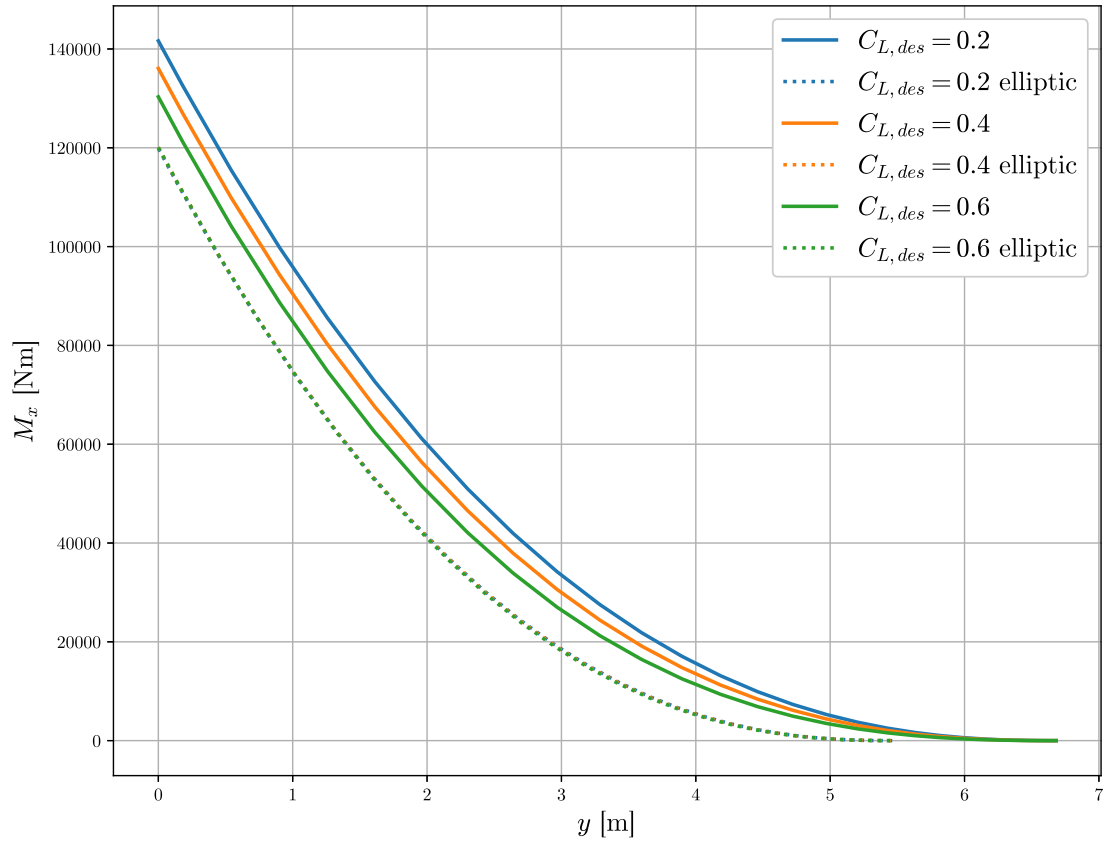


Figure 4.28: Bending moment at V_c for different $C_{L,des}$.

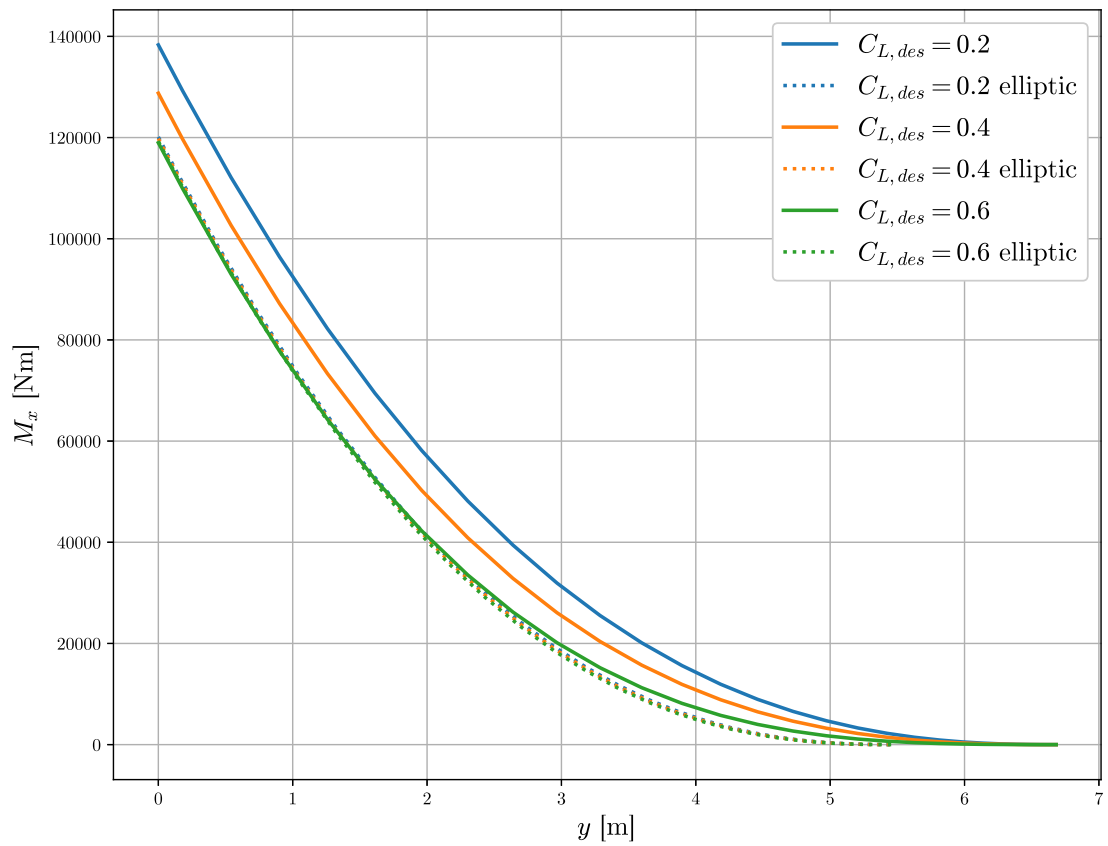


Figure 4.29: Bending moment at V_d for different $C_{L,des}$.

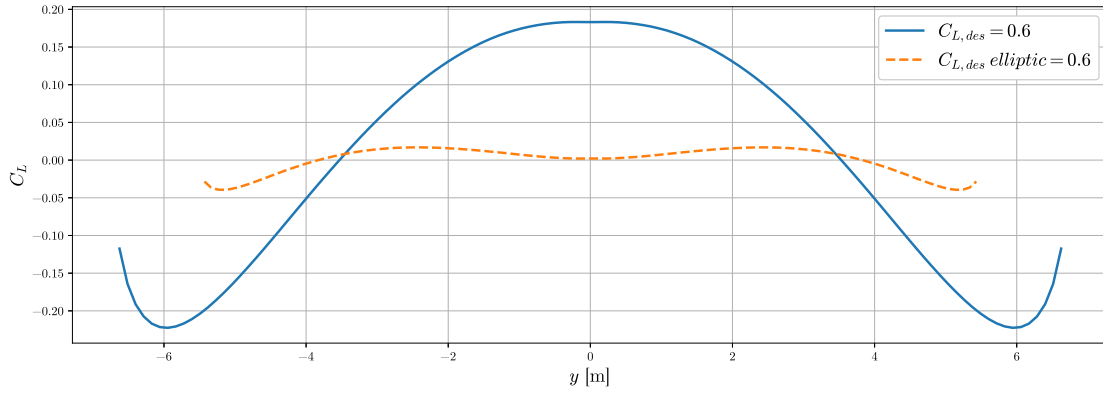


Figure 4.30: C_L distribution at $C_L = 0$, for wing with $C_{L,des} = 0.6$

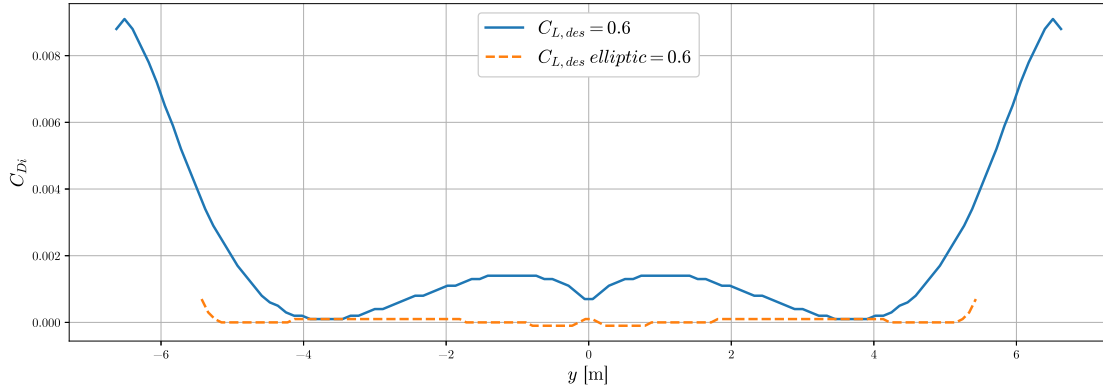


Figure 4.31: C_{di} distribution at $C_L = 0$, for wing with $C_{L,des} = 0.6$

4.3 Effects of $\mathcal{R}$

To analyze the effect of aspect ratio, five different wings were tested and redesigned. The freestream setup was identical to the one used in Sec. 4.1 but $\mathcal{R}$ was changed with the values of 4, 6, 8, 10 and 12.

Drag polars from this batch of analysis are shown in Fig. 4.32-4.36. The $C_L - \alpha$ curves are similar to the ones shown in 4.15a - 4.17a, with the derivative $C_{L,\alpha}$ increasing as the aspect ratio is increased.

The variation of induced drag between elliptical lift distribution and the respective bell distribution has a small oscillation around the 10% improvement, as shown in Fig. 4.38. However when viscosity is introduced, the beneficial effect diminishes with the increase of the aspect ratio of the starting wing. If $\mathcal{R}$ is less than 8, drag reduction is about 6%. A similar figure was calculated by Ansell and Wroblewski in their testing of Prandtl's bell spanload: indeed their baseline elliptical wing had an aspect ratio of 6.5 and $C_{L,des}$ of 0.46[20].

Fig. 4.39 shows the value of L/D for each tested case, as a function of lift coefficient. For small aspect ratios, the new redesigned wing does not improve upon the maximum L/D . However according to Fig. 4.40, these wings have the most relative improvement on L/D relative to their baseline. Depending on the $\mathcal{R}$ of the baseline wing, the improvement goes from 7.17 % to 2.54 %.

	$C_{D,v}$	$C_{D,i}$	C_D	% viscous	% induced
$\mathcal{R} = 4$, e	0.005397	0.012547	0.017945	30.08	69.92
$\mathcal{R} = 4$, b	0.005512	0.011232	0.016744	32.92	67.08
$\mathcal{R} = 6$, e	0.005425	0.008359	0.013784	39.36	60.64
$\mathcal{R} = 6$, b	0.005560	0.007495	0.013055	42.59	57.41
$\mathcal{R} = 8$, e	0.005451	0.006267	0.011718	46.51	53.49
$\mathcal{R} = 8$, b	0.005604	0.005636	0.011240	49.86	50.14
$\mathcal{R} = 10$, e	0.005480	0.005013	0.010494	52.23	47.77
$\mathcal{R} = 10$, b	0.005640	0.004516	0.010156	55.54	44.46
$\mathcal{R} = 12$, e	0.005505	0.004177	0.009683	56.86	43.14
$\mathcal{R} = 12$, b	0.005676	0.003766	0.009442	60.11	39.89

Table 4.7: Drag breakdown for different $\mathcal{R}$ at $C_L = 0.4$

The diminishing of drag reduction with aspect ratio is an effect of the drag breakdown. Table 4.7 provides the components of viscous and induced drag for each tested case. When $\mathcal{R}$ is less than 8, induced drag contributes more than 50% of the total drag; therefore reducing it with Prandtl's bell lift distribution yields significant results. However when $\mathcal{R}$ is already high, that is induced drag plays a marginal role in the total drag, reducing induced drag has a minor impact on the total drag reduction.

Bending moments at the design point are again close (Fig. 4.41), while at the structural design loads it no longer matches (Fig. 4.42-4.44). Small decrement of the root bending moment with the increment of $\mathcal{R}$ is observed for V_c and V_d , while very little change is found for V_a (Tab. 4.8).

AR	V_a	V_c	V_d
4	17.88	15.07	10.39
6	17.87	14.97	10.09
8	17.89	14.82	9.80
10	17.89	14.64	9.61
12	17.87	14.51	9.40

Table 4.8: % M_x increase as function of $\mathcal{R}$

$\mathcal{R}$	4	6	8	10	12
Elliptical	35.04 kg	55.63 kg	80.83 kg	109.80 kg	142.01 kg
Bell	53.08 kg	88.24 kg	130.35 kg	178.41 kg	231.65 kg
% Variation	51.49	58.61	61.27	62.49	63.12

Table 4.9: Representative wing masses for different $\mathcal{R}$

Calculated representative masses with the loading case V_c is shown in Tab. 4.9. As expected the mass increment is greater the larger is the aspect ratio of the original wing, because the lifting sections at the wingtip have a greater bending arm even with an identical lift distribution shape for all the tested wings.

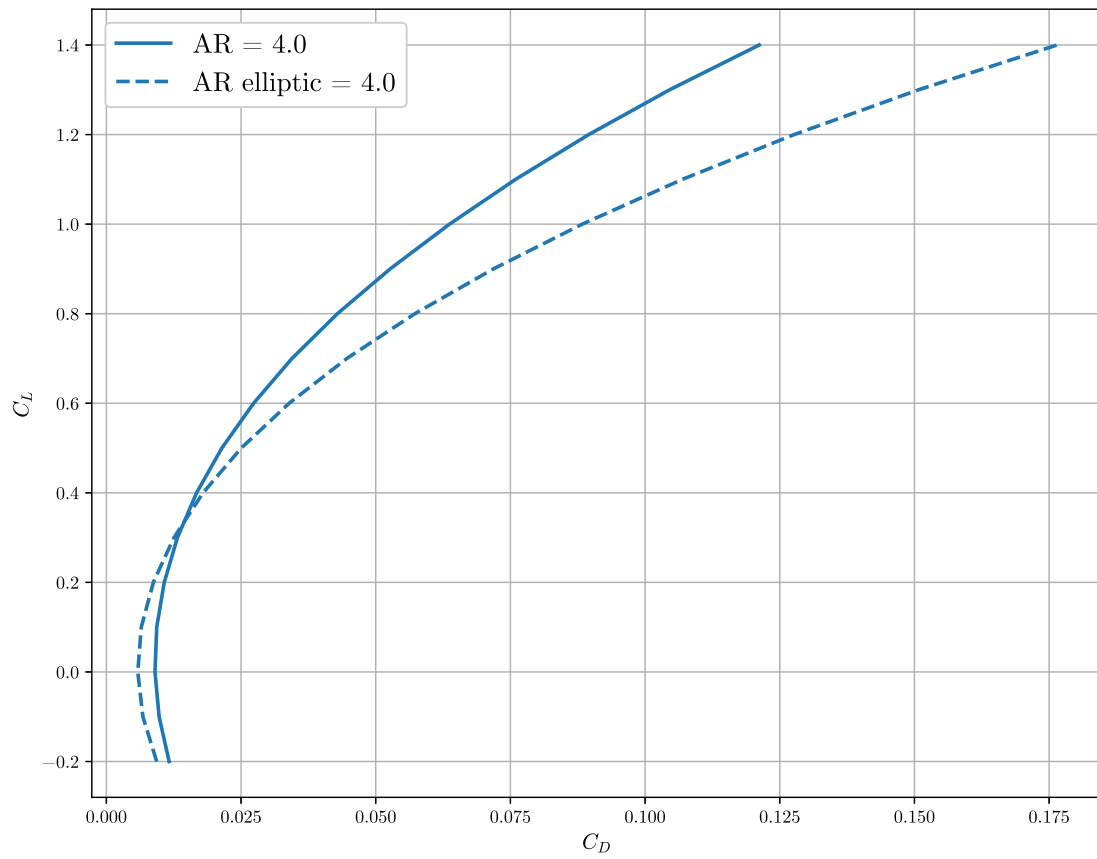


Figure 4.32: Drag polar for $\mathcal{R} = 4$.

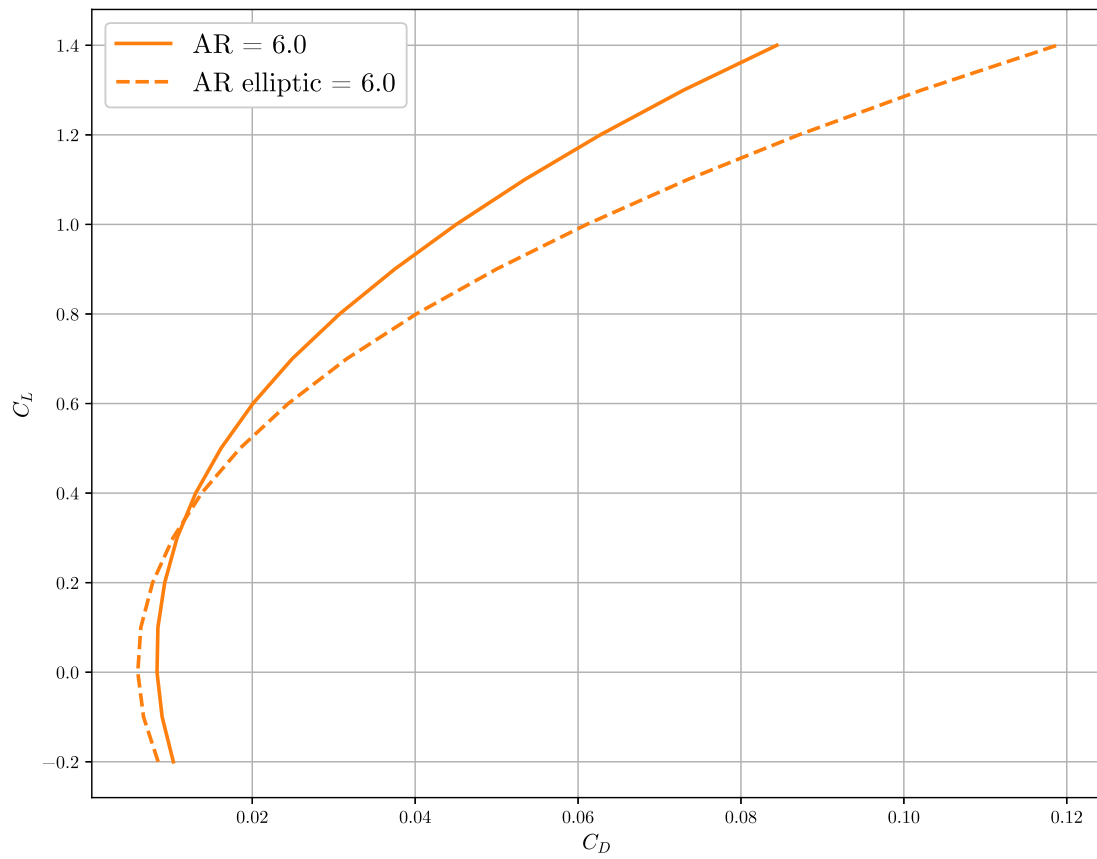


Figure 4.33: Drag polar for $\mathcal{R} = 6$.

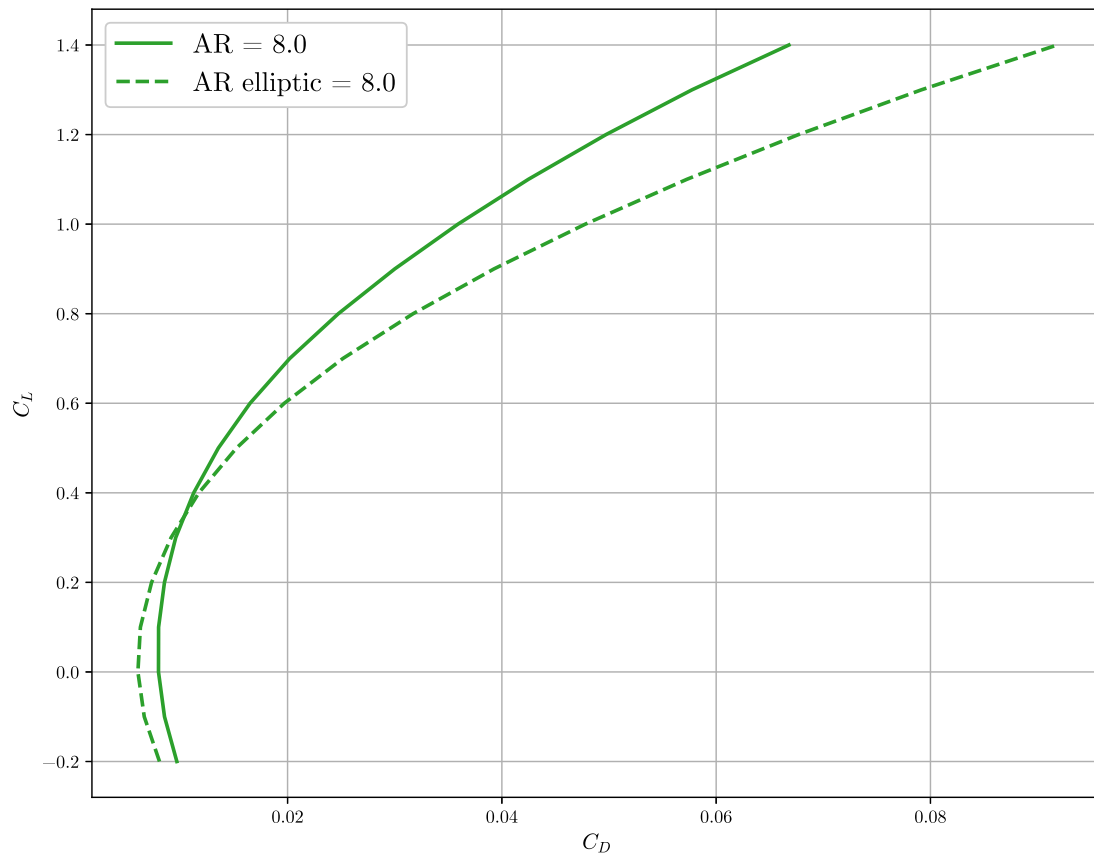


Figure 4.34: Drag polar for $AR = 8$.

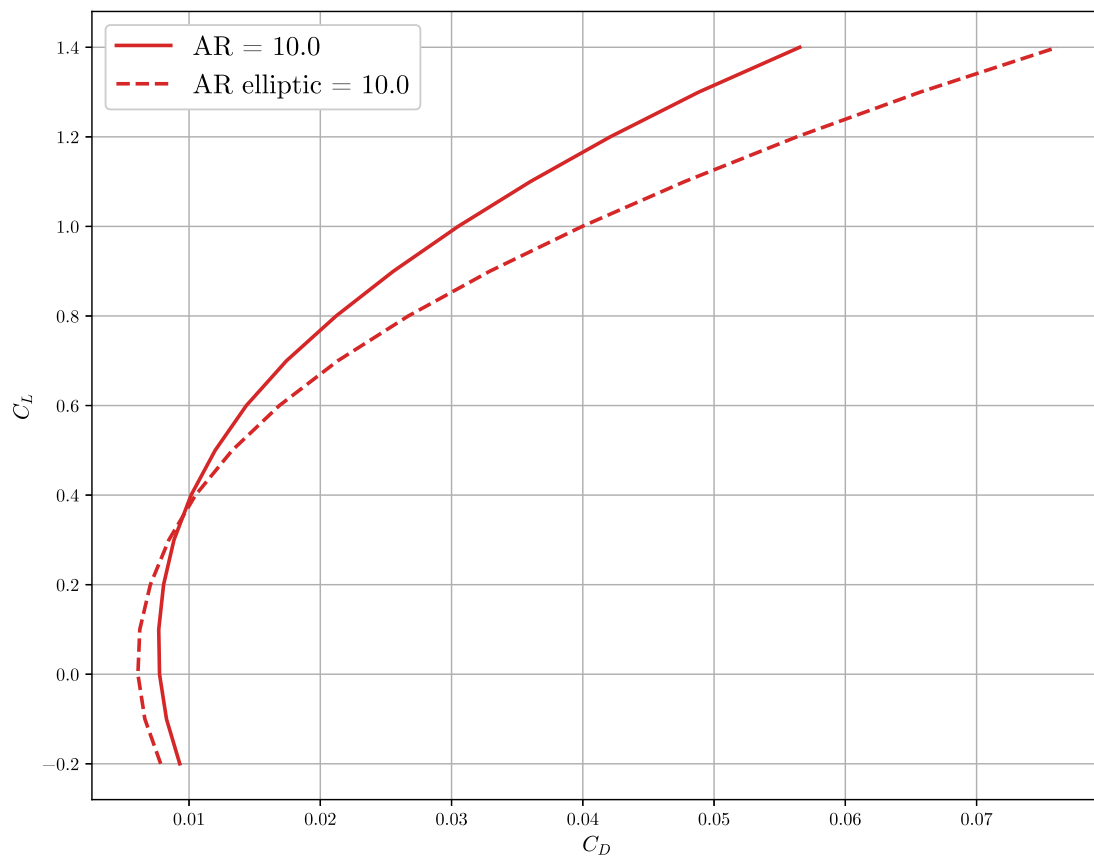


Figure 4.35: Drag polar for $AR = 10$.

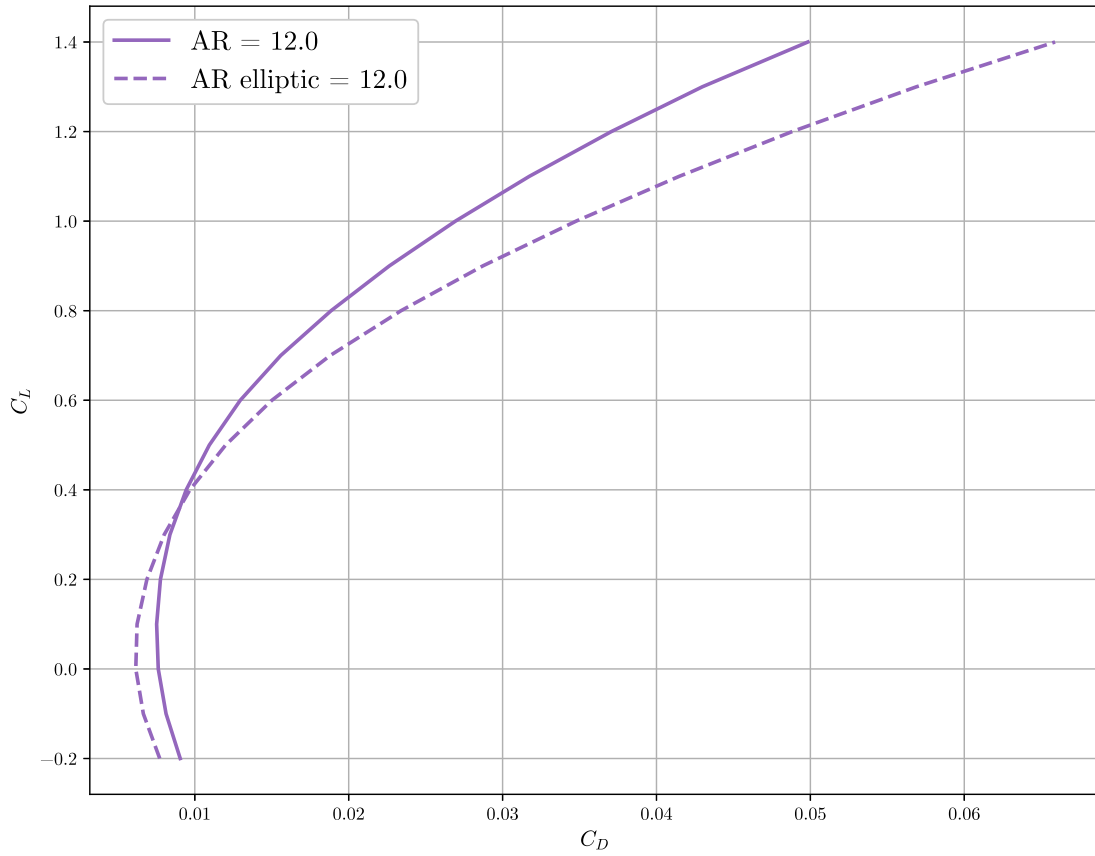


Figure 4.36: Drag polar for $\mathcal{R} = 12$.

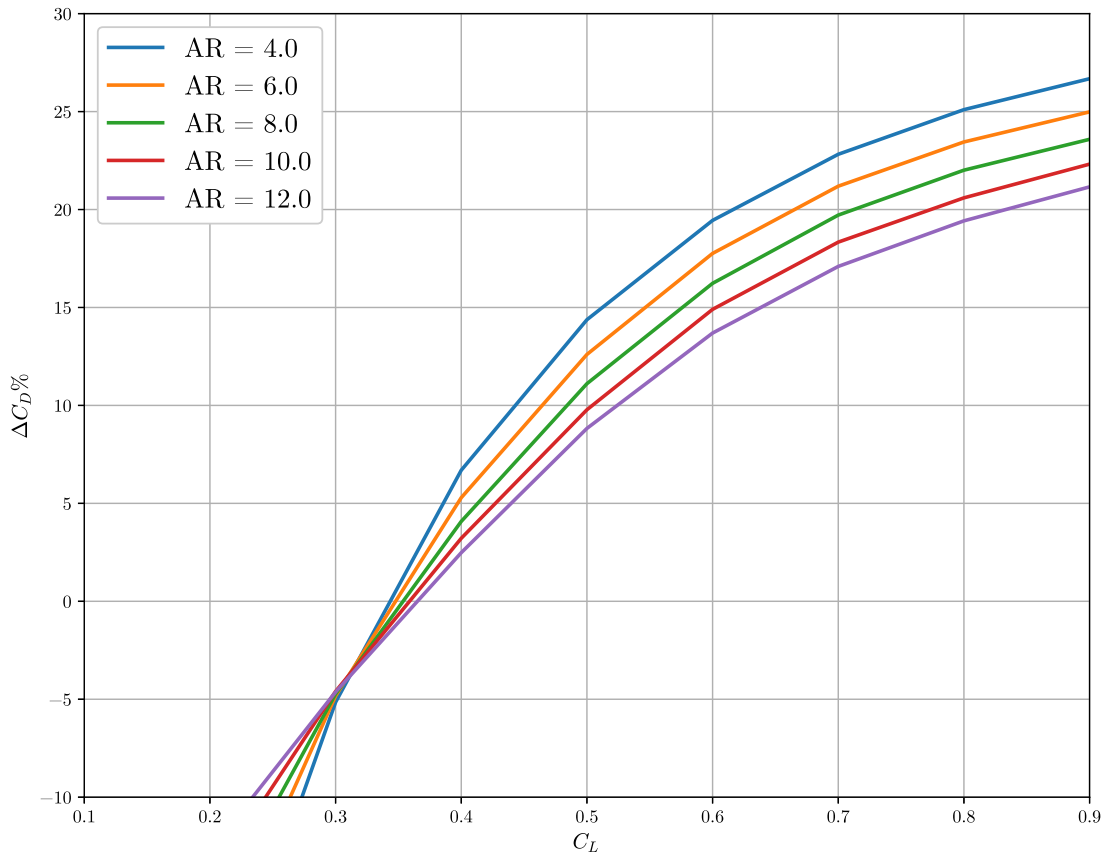


Figure 4.37: Relative drag variation for different $\mathcal{R}$

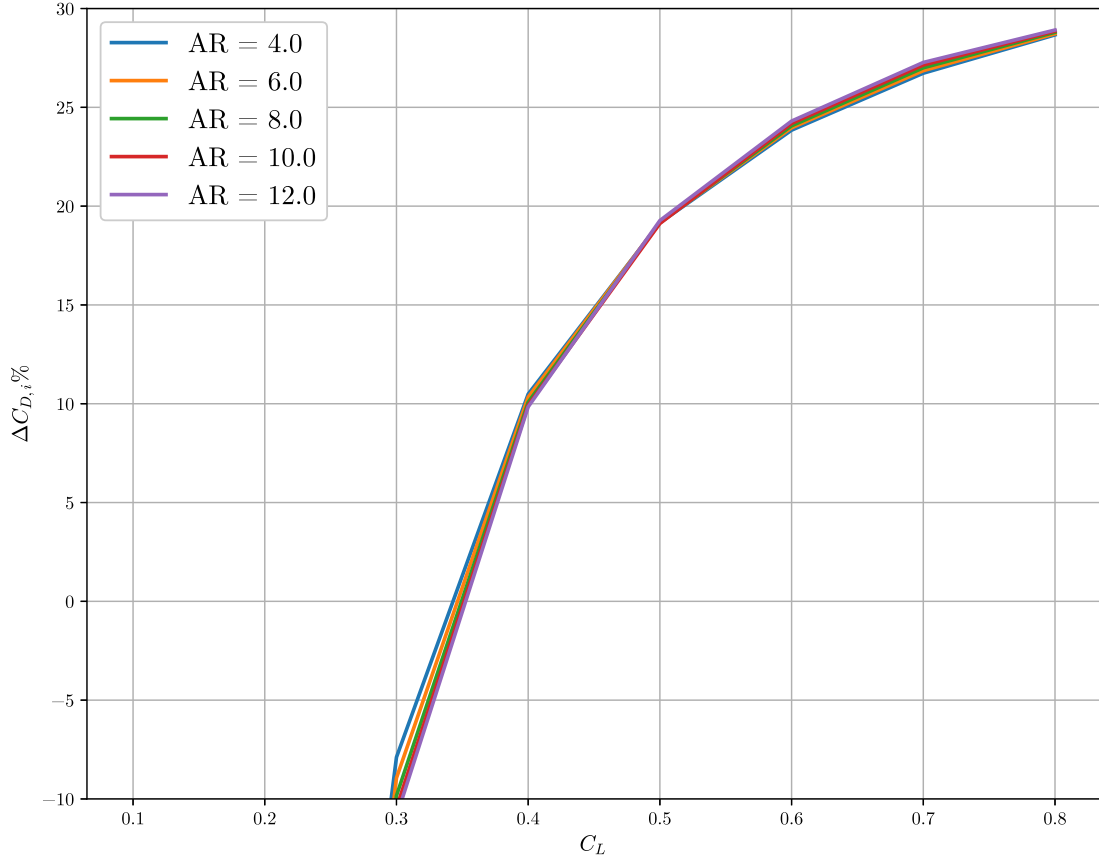


Figure 4.38: Relative induced drag variation for different $\mathcal{AR}$

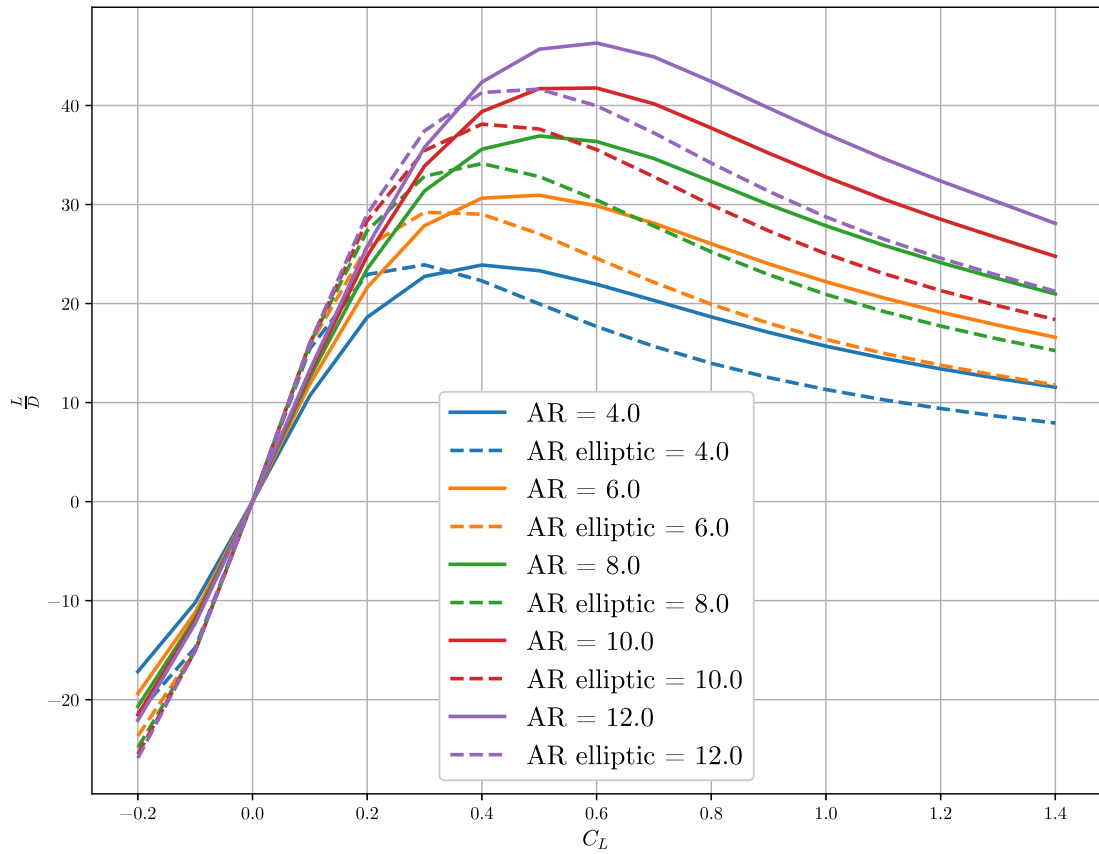


Figure 4.39: L/D for different $\mathcal{AR}$

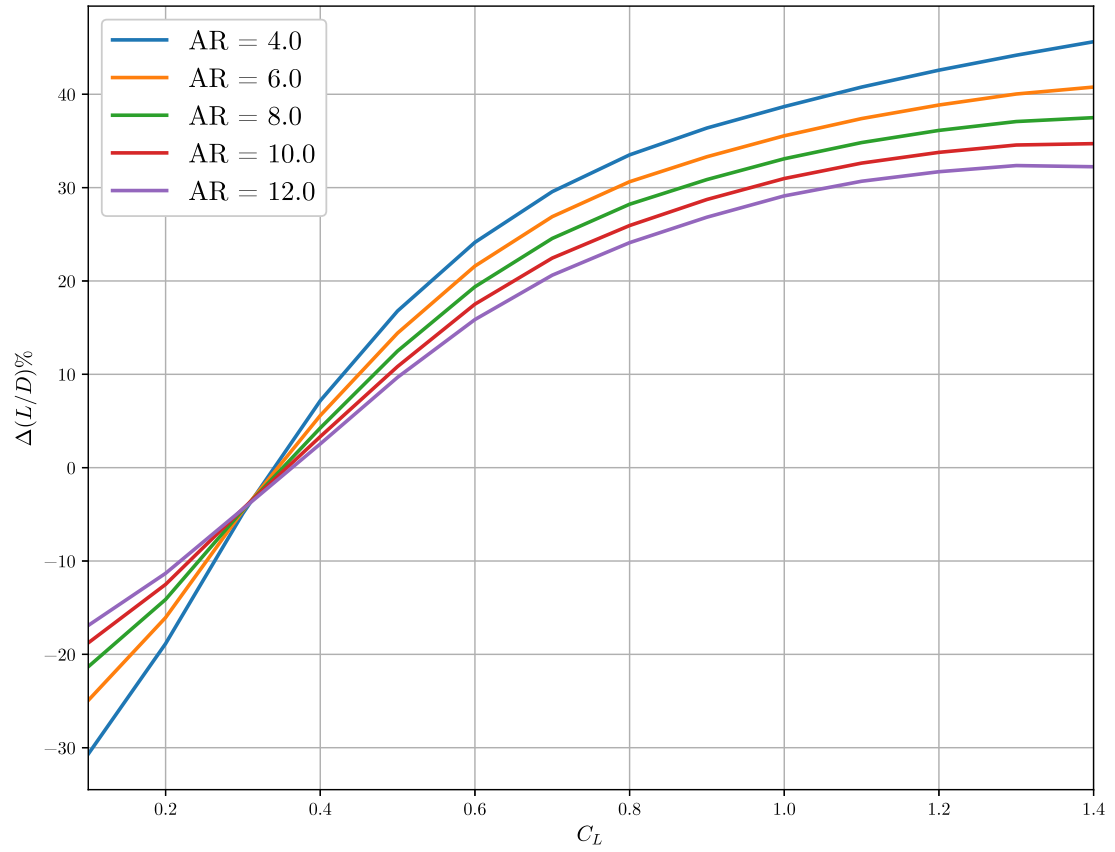


Figure 4.40: Relative L/D variation for different $\mathcal{R}$

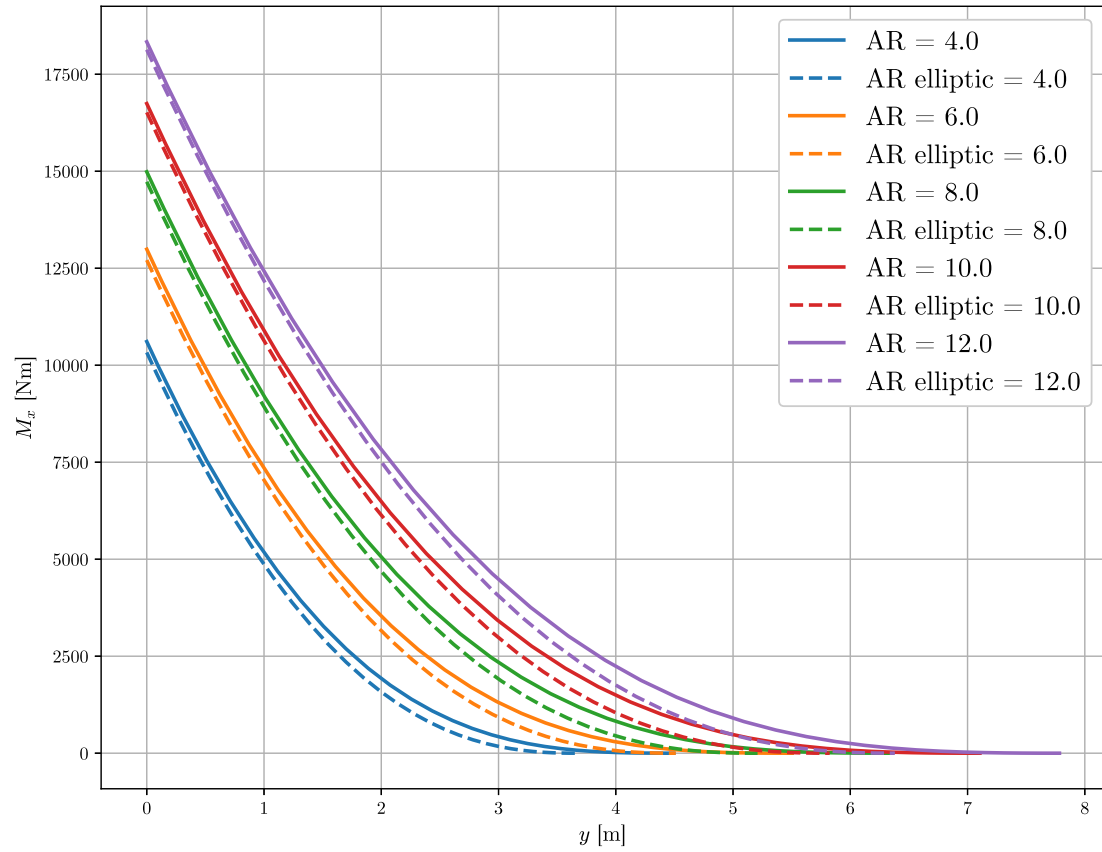


Figure 4.41: Bending moments at design point for different $\mathcal{R}$

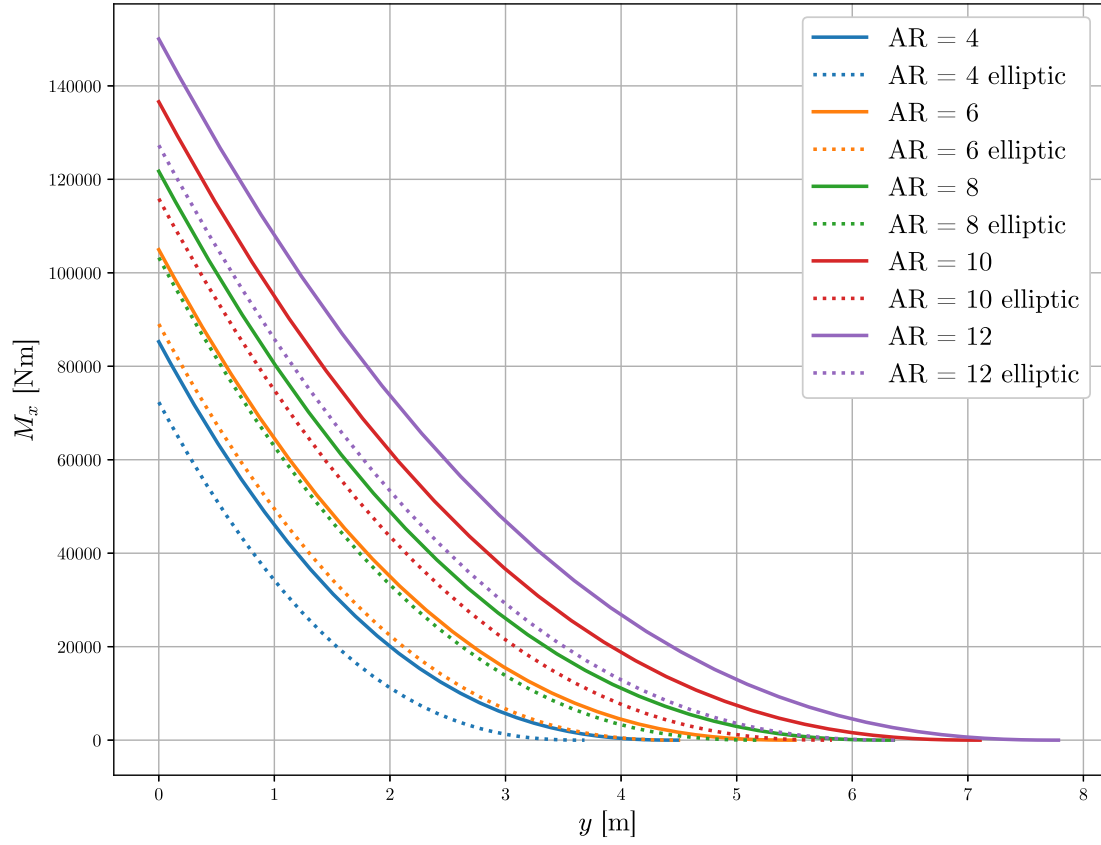


Figure 4.42: Bending moments at V_a for different AR

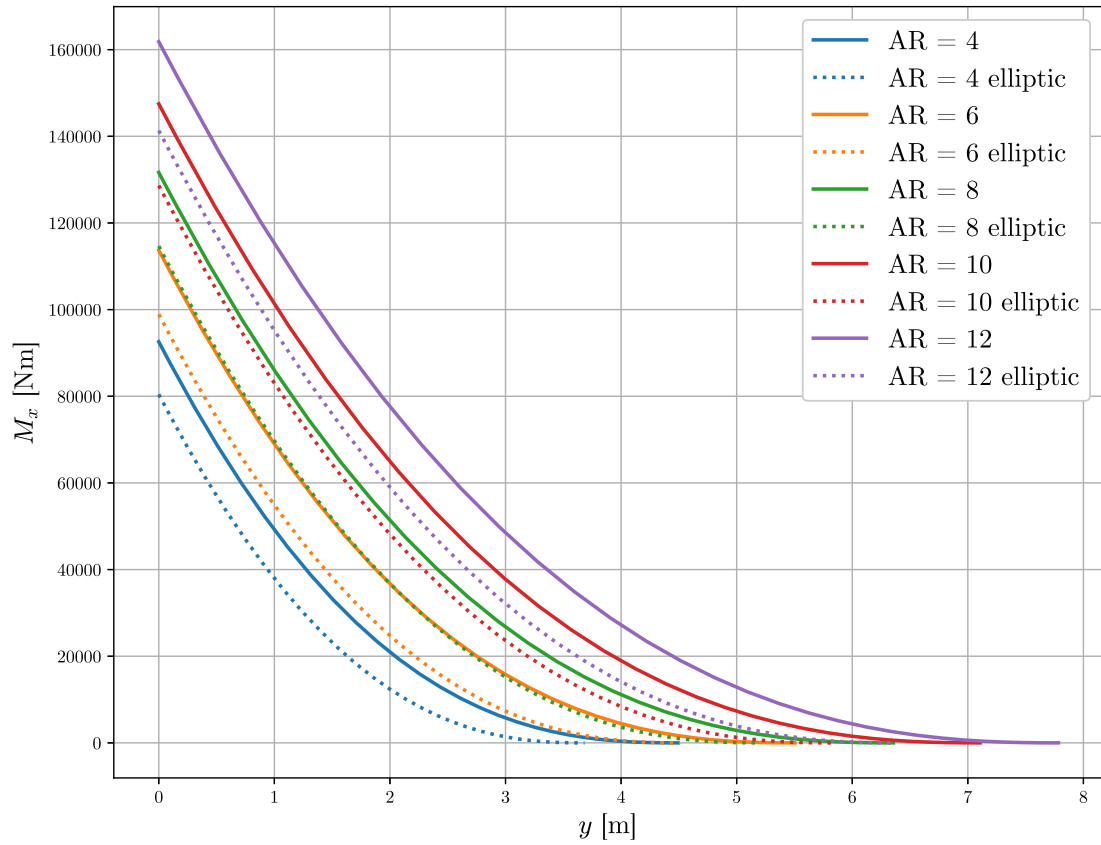


Figure 4.43: Bending moments at V_a for different AR

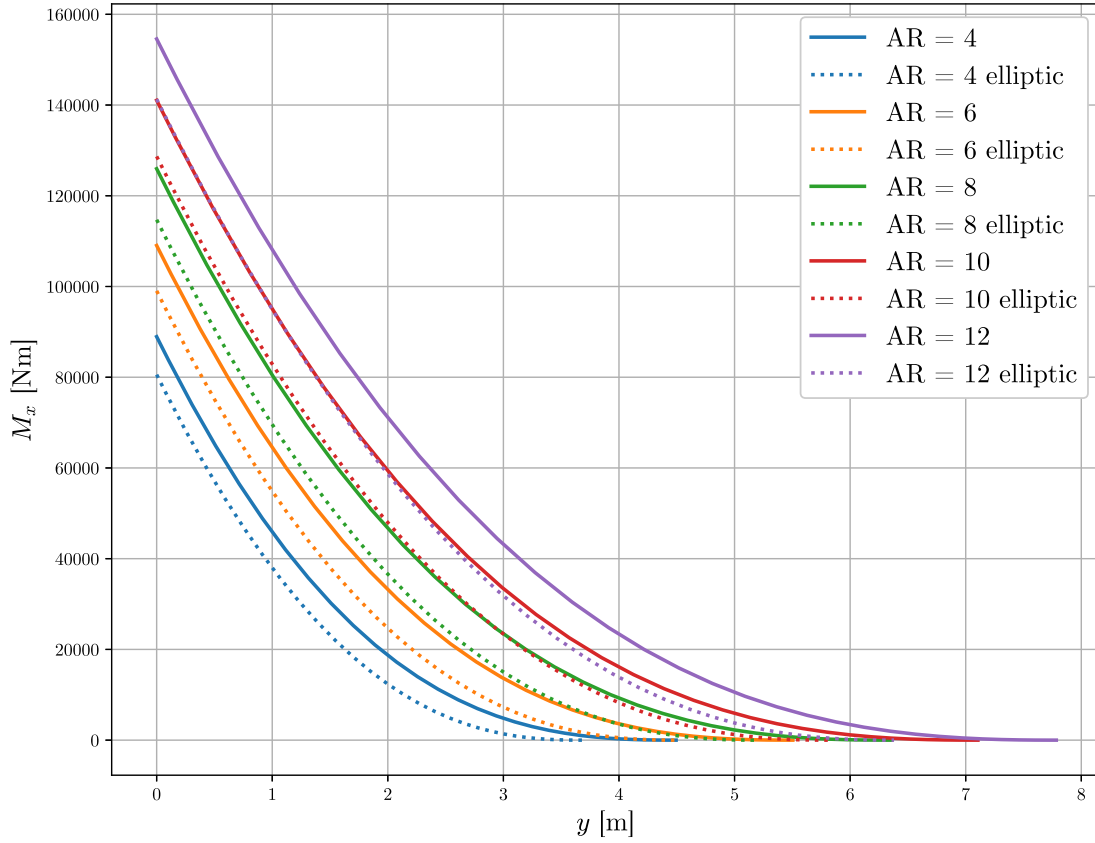


Figure 4.44: Bending moments at V_d for different $\mathcal{R}$

4.4 Effects of λ

Taper ratio effects have been studied on the same reference geometry of Sec. 4.1, with $C_{L,des} = 0.4$ and $\mathcal{R} = 8$. The analyzed λ values are 0.2, 0.4 and 0.6. The resulting aerodynamic polars are shown in Fig. 4.45-4.47, from which relative drag diagrams are obtained (Fig. 4.48 and Fig. 4.49).

The effect of wing taper is to "pivot" the drag polar around the design point: by reducing λ the bell polar is rotated closer to the elliptical one. This effect is also visible in the relative induced drag diagram. L/D instead is not affected significantly by the wing taper (Fig. 4.50).

The pivoting point is at the design lift coefficient if viscous drag is not accounted, while it is shifted of about +0.05 units if viscosity is added in. At small lift coefficients an effect similar to high $C_{L,des}$ is present. If λ is high, induced drag is heavily increased because more lift is present at the tips in this off-design condition. Here strong local negative lift is generated at $C_L \sim 0$, which contribute to the increment of induced drag. Taper aids in unloading this region, and therefore reducing off-design effects.

Furthermore, the tapering of the wing approximates the shape of the bell-lift distribution in the planform, thus reducing the requirement to twist the wing. It's clearly visible from the Oswald efficiency factor diagram (Fig. 4.52), where the most tapered wing has a more flattened increment of e , closer to a constant value,

hence maintaining more the bell-lift distribution shape. The influence of the chord distribution on the spanload is later explored in Sec. 4.5.

Taper ratio has also major effect on the bending moment of the wing. While at the design point, elliptical and bell bending moments are similar (Fig. 4.53), at the structural design cases (Fig. 4.54-4.56) the bending moment distribution has significant differences depending on the taper ratio. As shown in Tab. 4.10, for small taper ratios the root bending moment has a lower increment.

λ	V_a	V_c	V_d
0.2	14.27	12.95	7.33
0.4	15.05	13.49	7.74
0.6	15.83	13.87	8.01

Table 4.10: % M_x increment for different λ

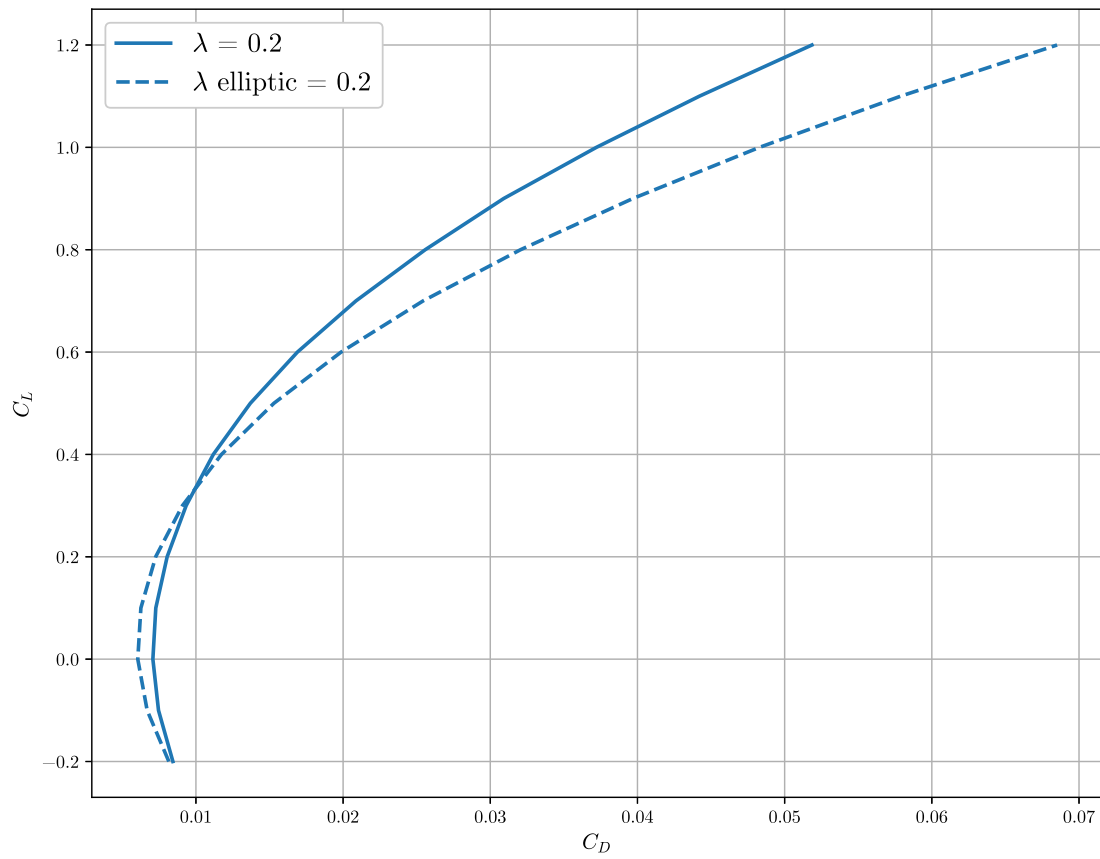
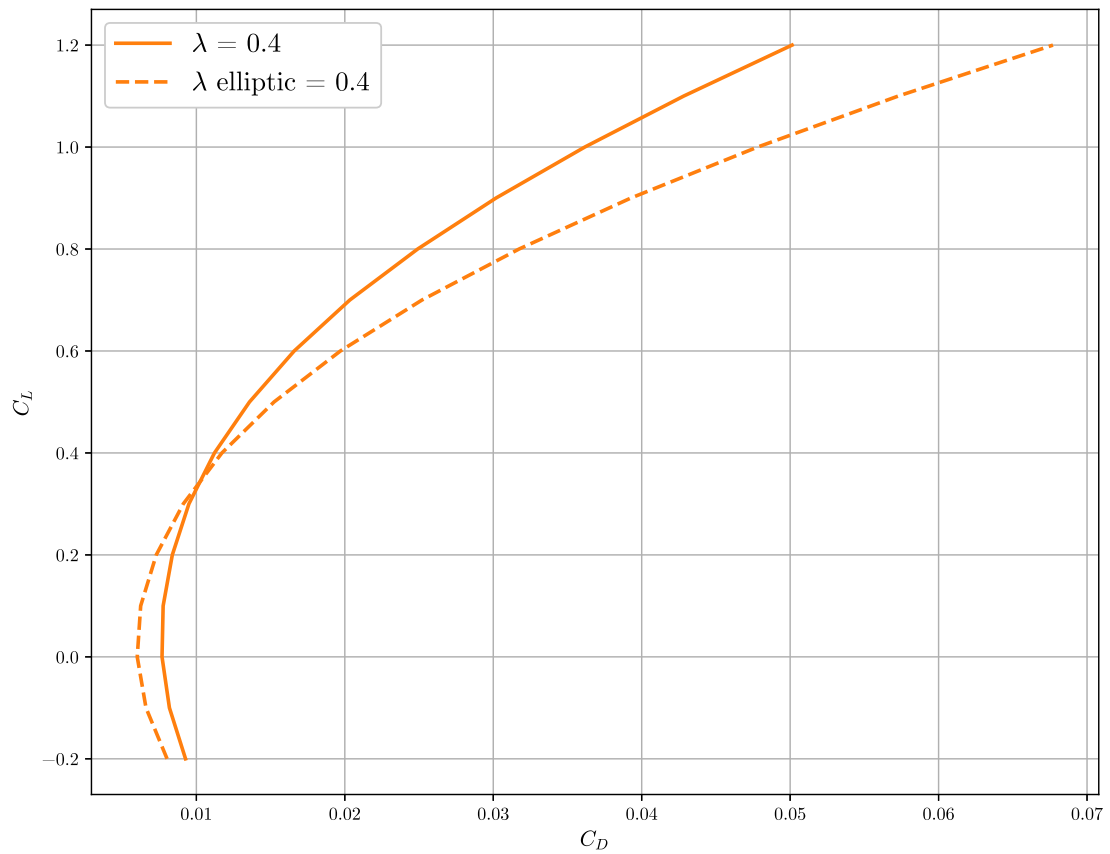
This phenomenon is caused by the lift distribution. A tapered wing has less local lift near the wingtips and more near the root, as shown in Fig. 4.57. With less lift near the wingtips, where the bending arm is the greatest, less bending moment is present. Also, as stated above, the planform shape approximates more the bell-distribution, therefore retaining some of its aerostructural properties outside of the design lift coefficient.

When representative wing masses are calculated and compared, this effect is augmented by a structural effect caused by tapering. Indeed if the root chord is larger, the local thickness is also larger which means a higher sectional moment of inertia for the same panel thickness. Therefore for the same load, less resisting material is required. The results in Tab. 4.11 show how much influence has the taper ratio on the wing mass.

Hence when designing a wing with Prandtl's bell spanload, small taper ratios should be considered for containing the effect of structural weight. However this has to be applied with caution, as it would strongly affect the Reynolds distribution across the wingspan, and therefore requiring a careful airfoil selection.

λ	0.2	0.4	0.6
Elliptical	74.04 kg	83.49 kg	92.06 kg
Bell	114.19 kg	131.66 kg	147.22 kg
% Variation	54.21	57.69	59.92

Table 4.11: Representative wing masses for different λ

**Figure 4.45:** Drag polar for $\lambda = 0.2$.**Figure 4.46:** Drag polar for $\lambda = 0.4$.

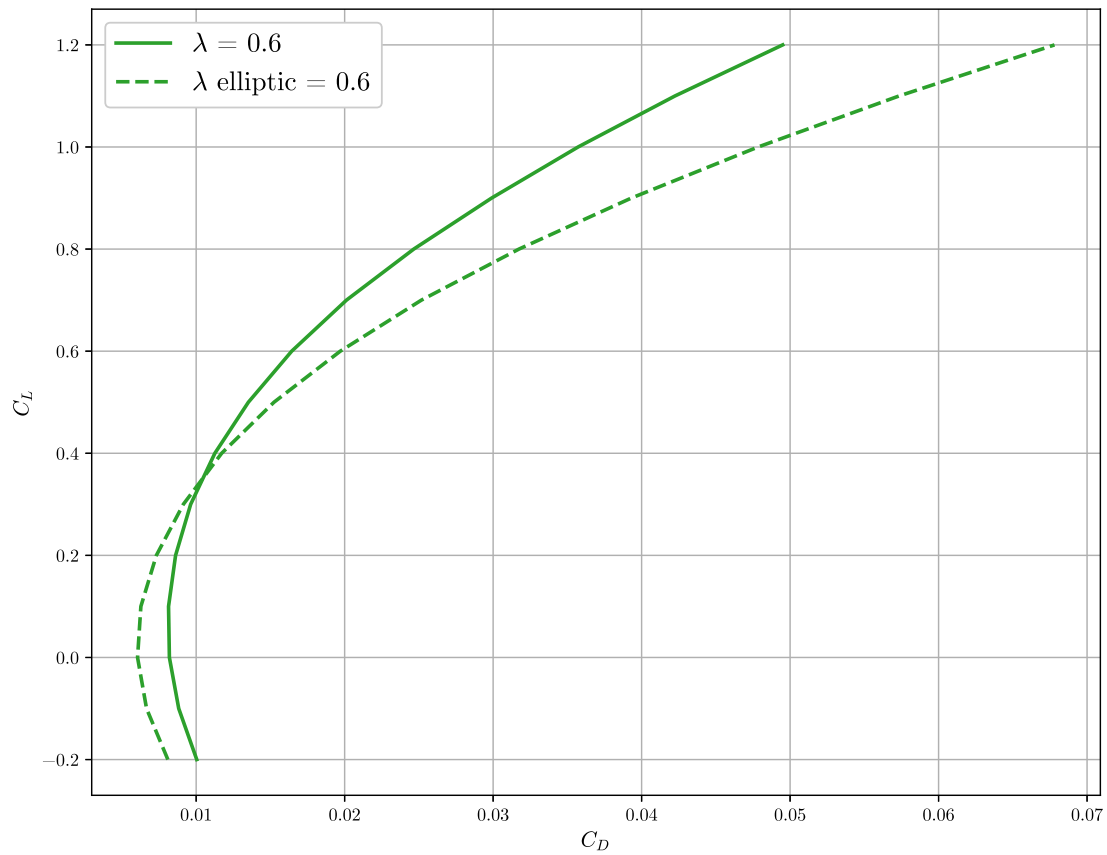


Figure 4.47: Drag polar for $\lambda = 0.6$.

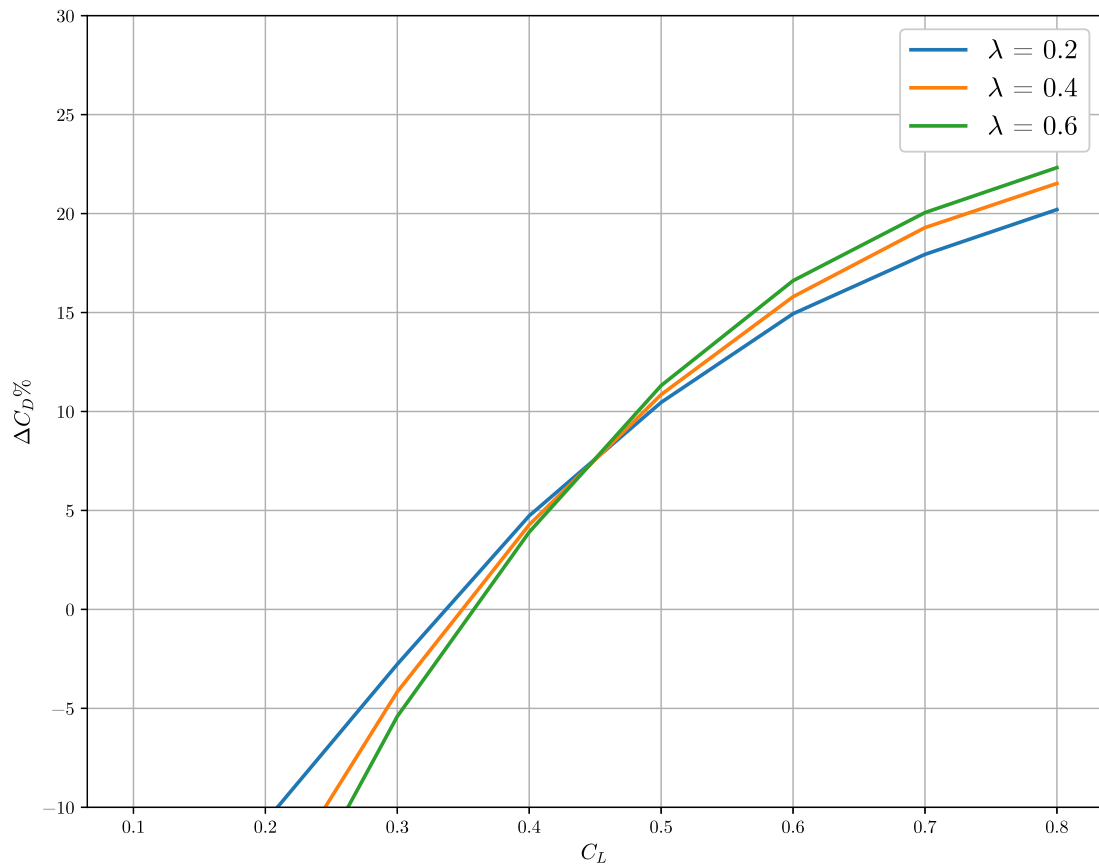


Figure 4.48: Relative drag variation for different λ

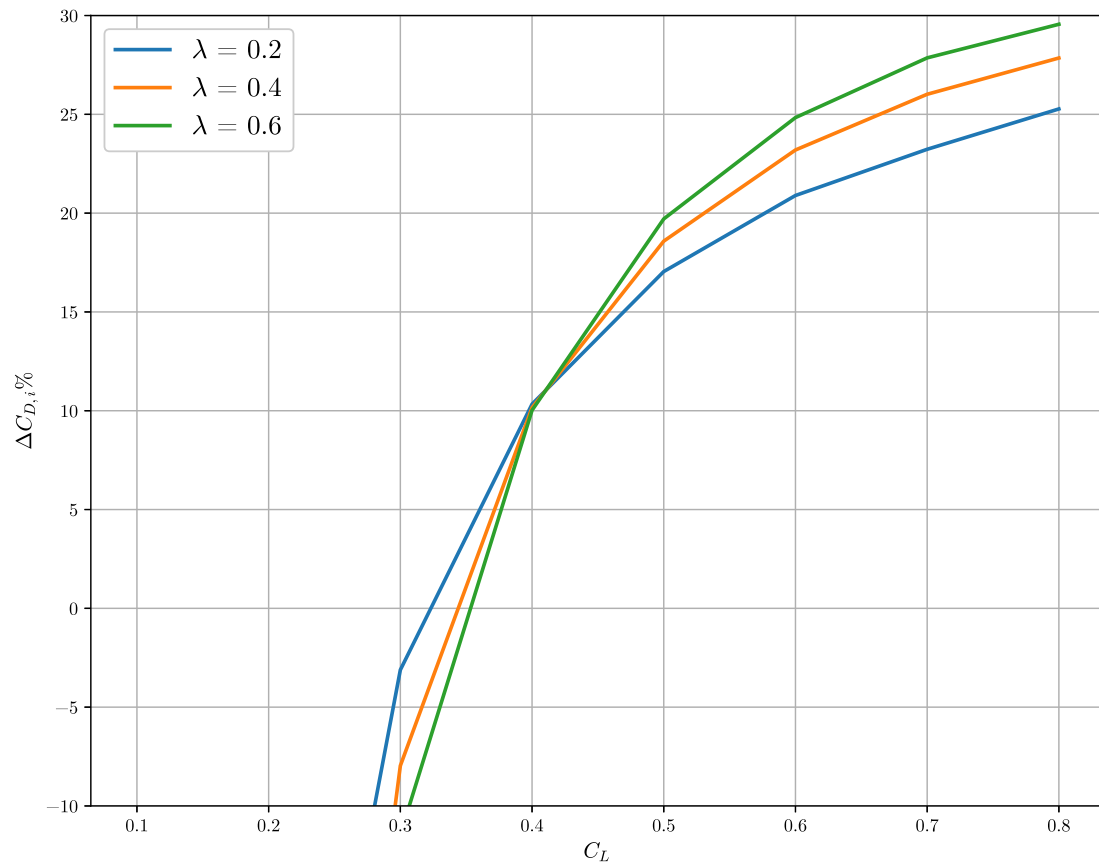


Figure 4.49: Relative induced drag variation for different λ

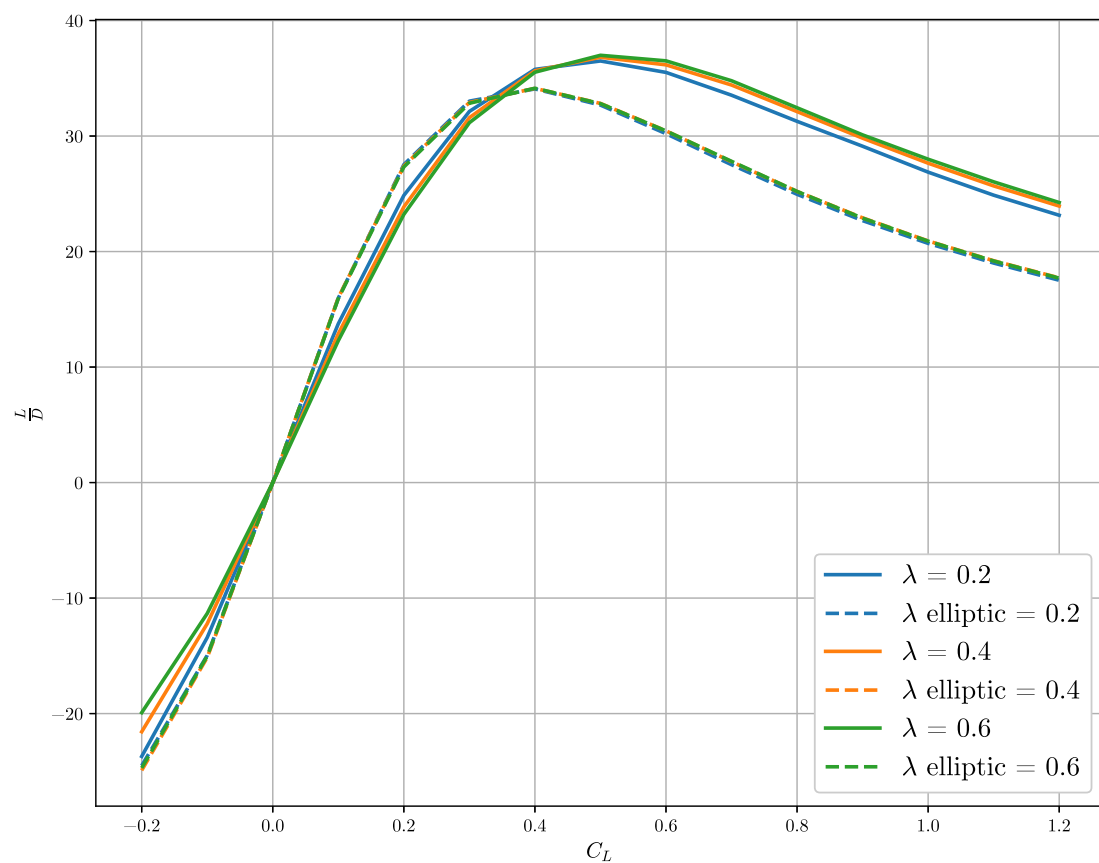


Figure 4.50: L/D different λ

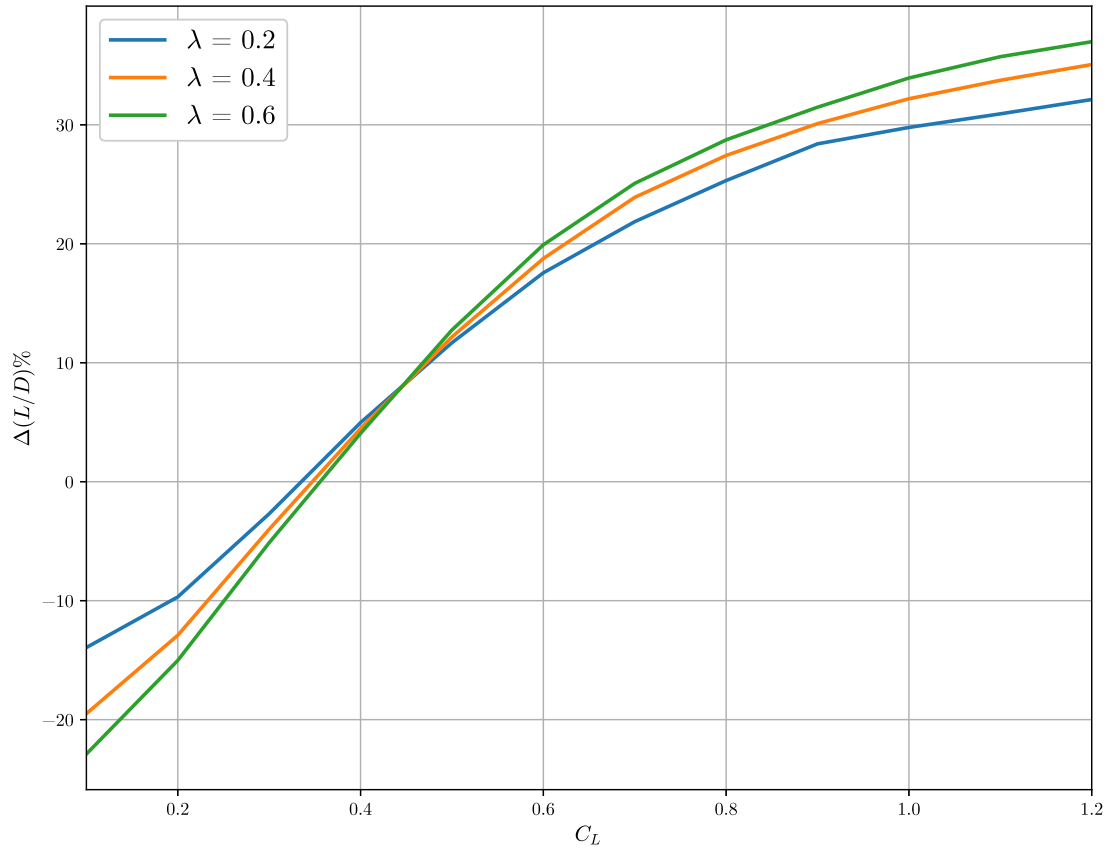


Figure 4.51: Relative L/D variation for different λ

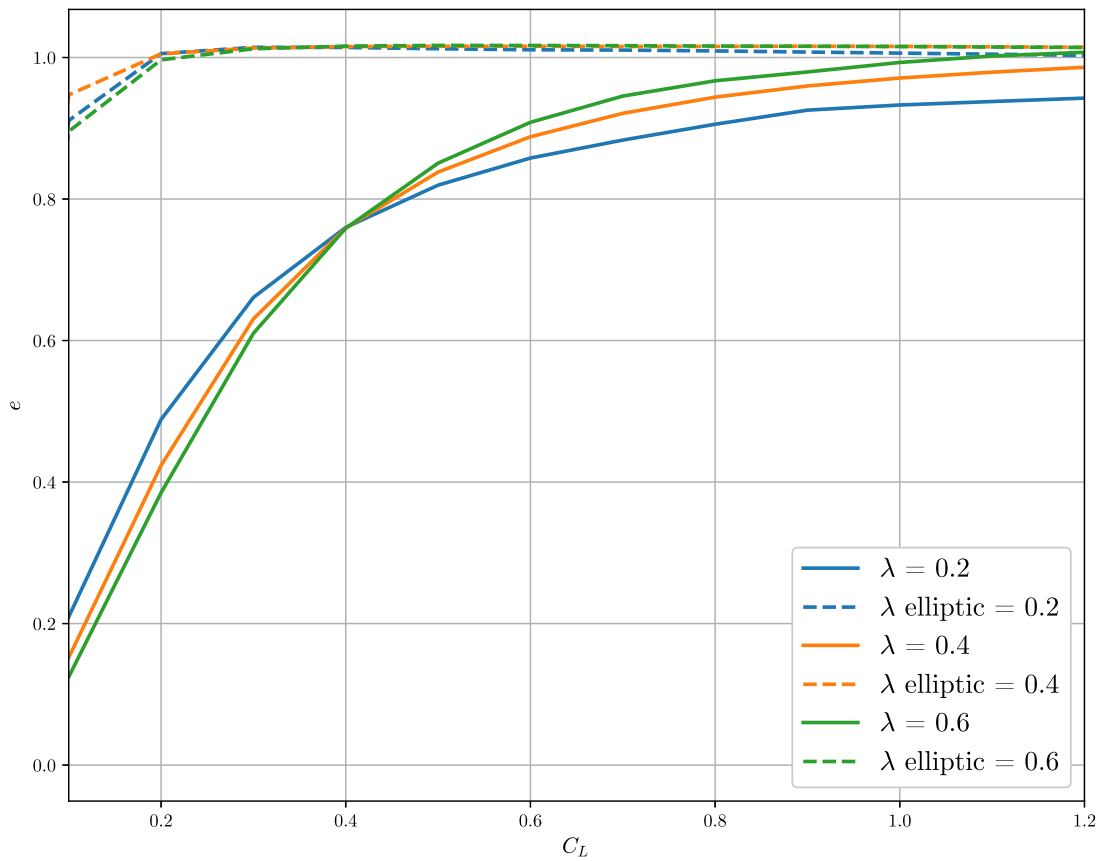


Figure 4.52: Oswald efficiency factor for different λ

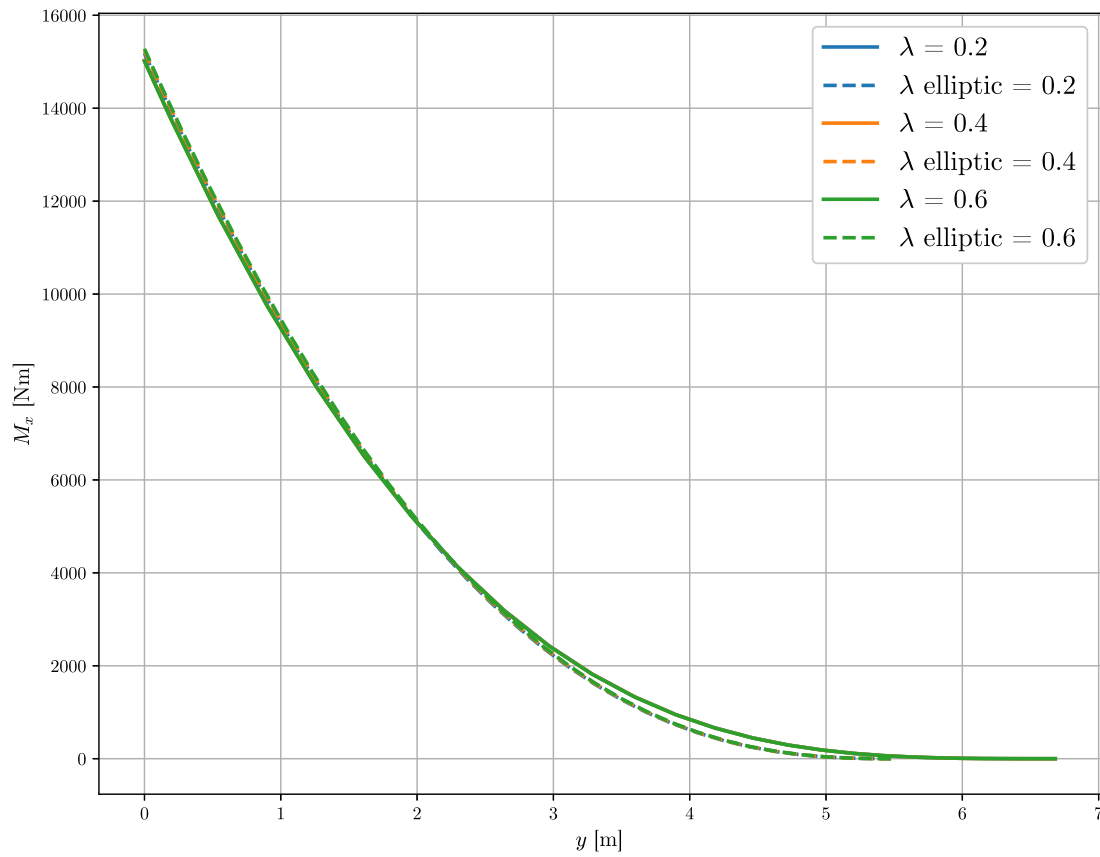


Figure 4.53: Bending moments at design point for different λ

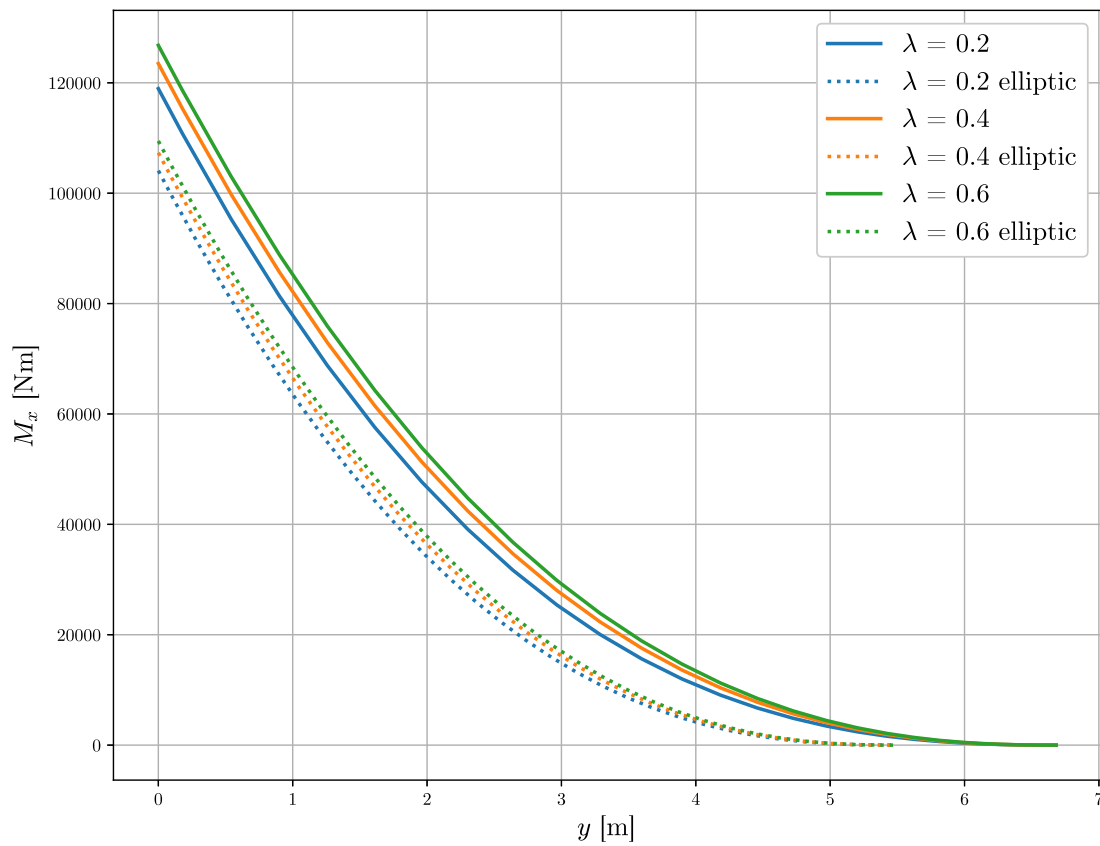


Figure 4.54: Bending moments at V_a for different λ

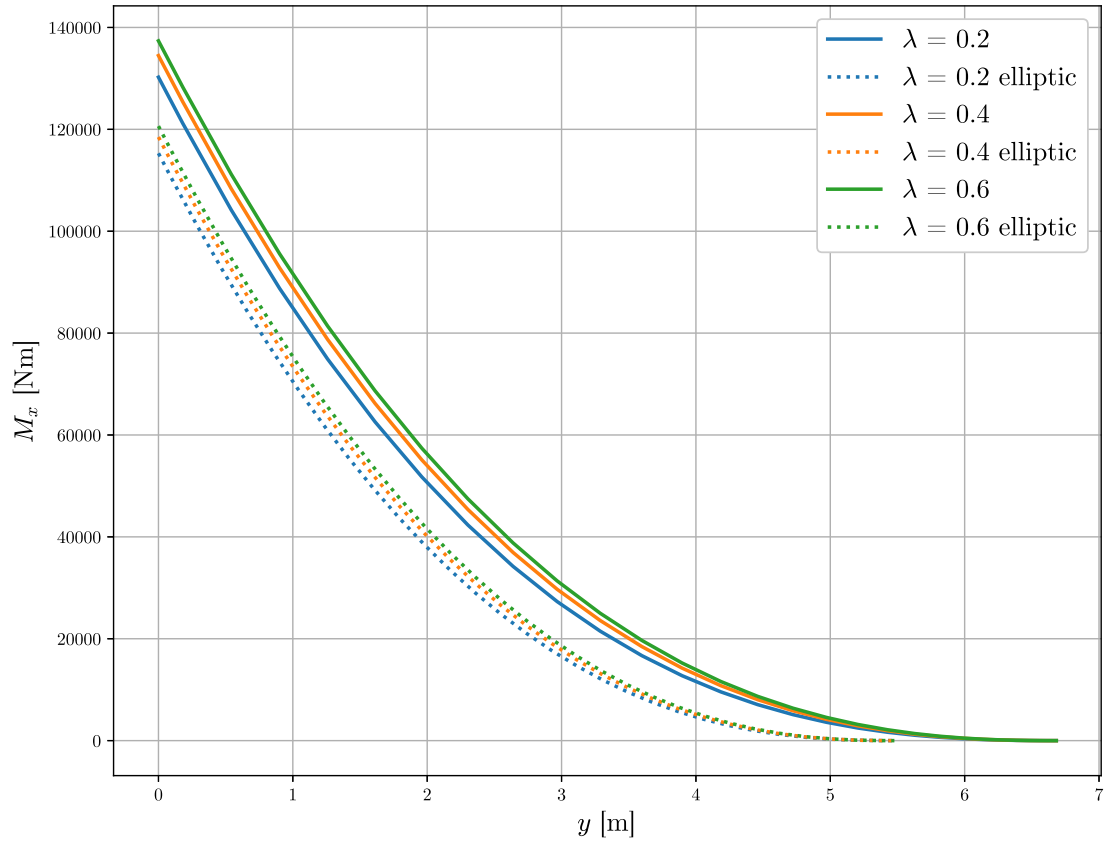


Figure 4.55: Bending moments at V_c for different λ

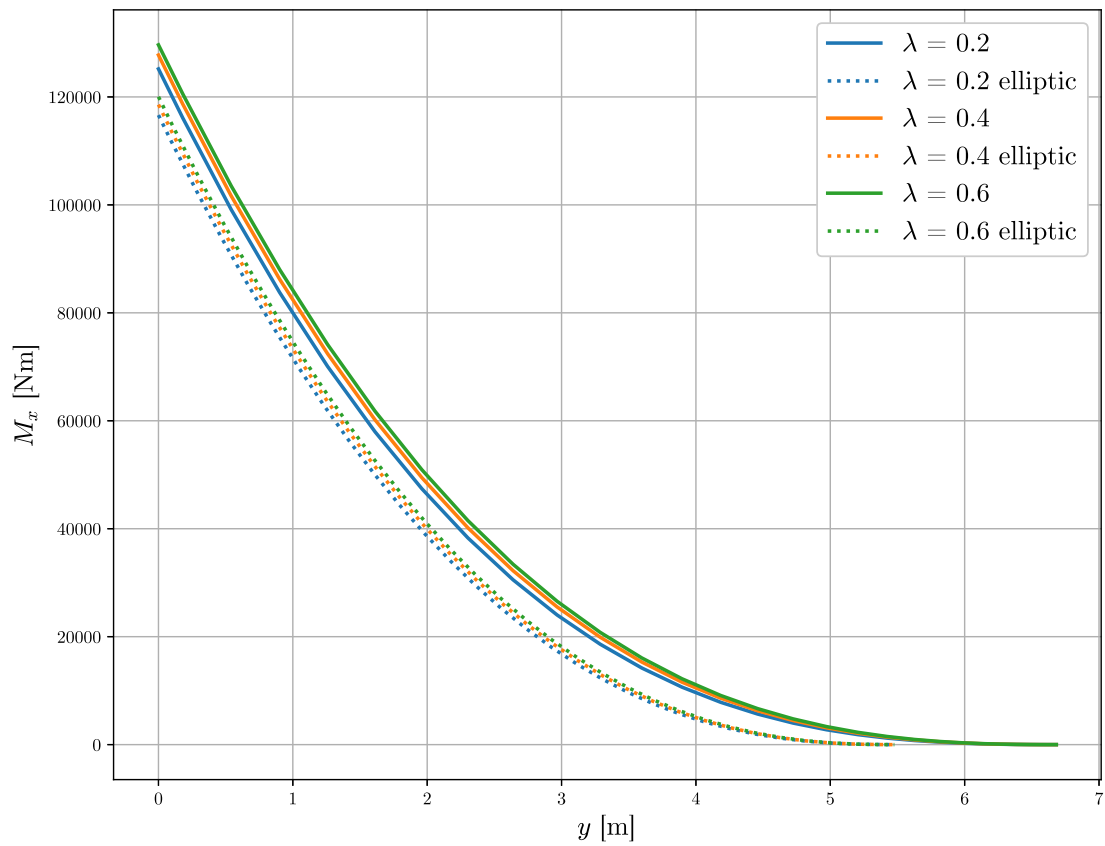


Figure 4.56: Bending moments at V_d for different λ

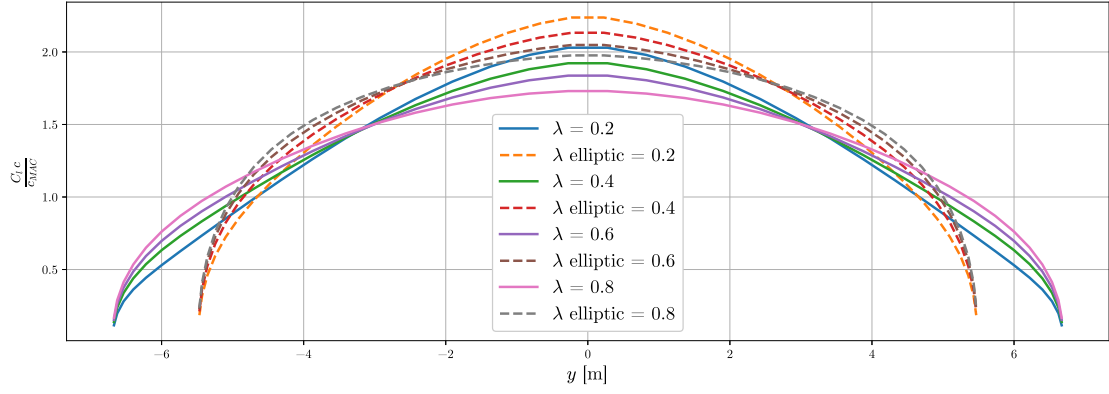


Figure 4.57: C_L distribution at V_a for different λ

4.5 Chord Distribution

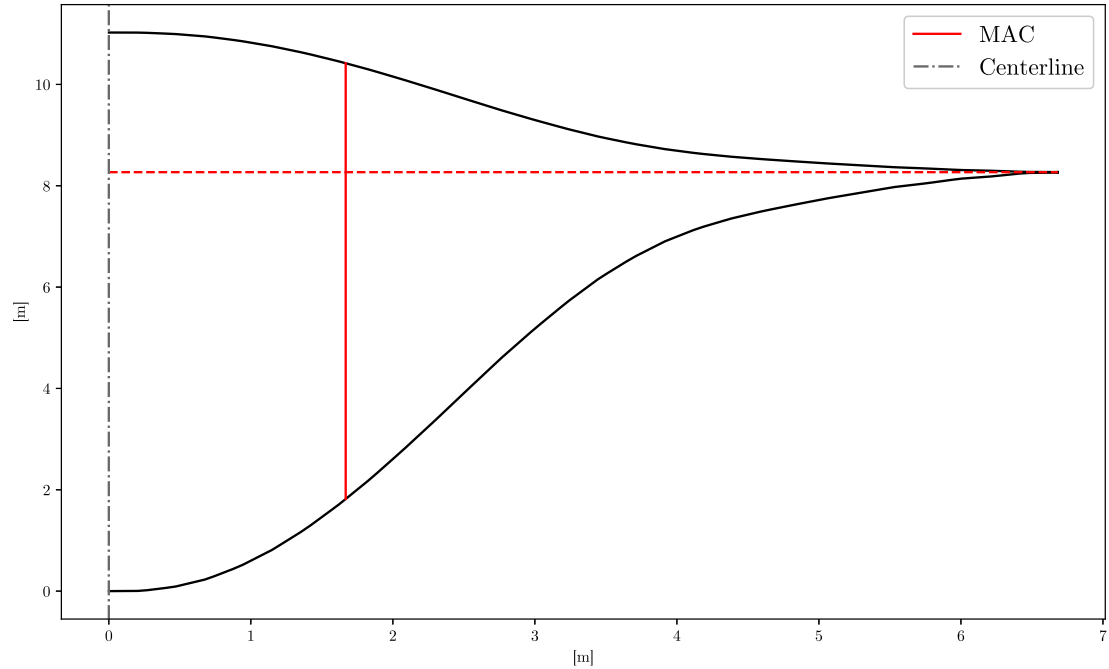


Figure 4.58: Straight untwisted wing with bell chord distribution (axes not at scale).

The original application of the elliptical spanload was by using an elliptical chord distribution, no geometric twist and constant spanwise camber. The advantage of this approach was to obtain an elliptical lift distribution at any wing lift coefficient. Disadvantages were the stalling behavior of the wing, which would stall completely, and the Reynolds distribution, especially at the wingtip where chord would be zero.

This idea was investigated for the bell lift distribution, however due to the tapering of the lift distribution at the wingtips, the obtained shape proved to be of impractical use. The obtained wing planform for the same design conditions in Sec. 4.1 is shown in Fig. 4.58. The cusped tip is impossible to manufacture and the small chord of the outer portion precludes the placement of an aileron.

Instead a different approach was devised. In practical wing design, the trapezoidal tapered wing is the most common shape because it approximates the planform of an elliptical wing without losing much in span efficiency (usually $e \approx 0.85$). Indeed for straight wings the lift distribution for quick analysis is calculated with Schenck's Method, that uses the planform shape as a correction factor to the elliptical lift distribution.

Therefore a similar approach may be taken for the bell lift distribution. Fig. 4.59 shows a non-dimensional bell distribution with two proposed approximations for the chord distribution. Approximation *a* has two points, while approximation *b* and *c* have three points.

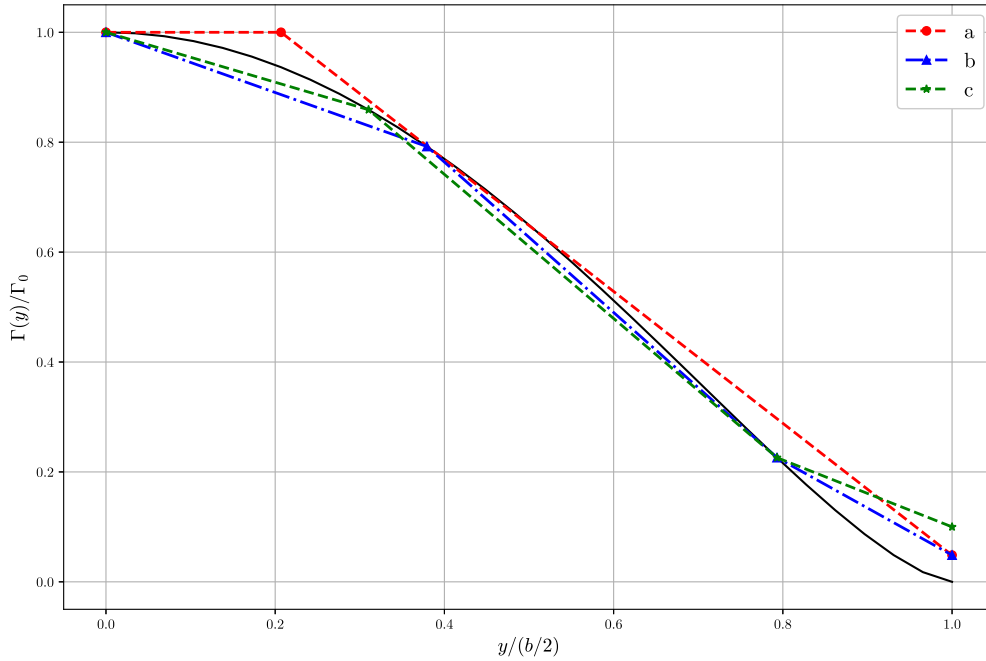


Figure 4.59: Comparison between non-dimensional bell lift distribution and chord distribution.

y/b	c/c_r	y/b	c/c_r	y/b	c/c_r
0	1	0	1	0	1
0.2068	1	0.3793	0.79214	0.3103	0.859
1	0.05	0.7931	0.22596	0.7931	0.223
		1	0.05	1	0.1
(a)		(b)		(c)	

Table 4.12: Non-dimensional chord distributions

Three wings have thus been designed with these proposed chord distributions on top of the design procedure used in Sec. 4.1. The obtained planforms are shown in Fig. 4.60, Fig. 4.61 and Fig. 4.62. The twist distributions are still not linear and with curved bumps and valleys to compensate for the edges in the wing shape (Fig. 4.63).

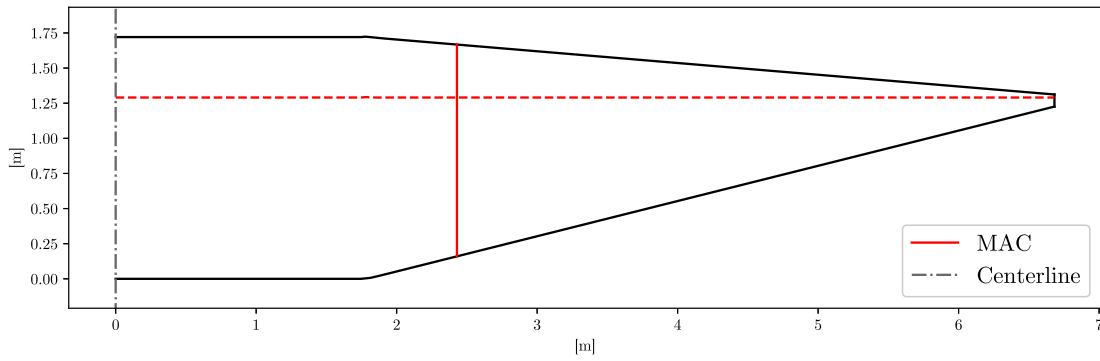


Figure 4.60: Wing with Type A chord distribution

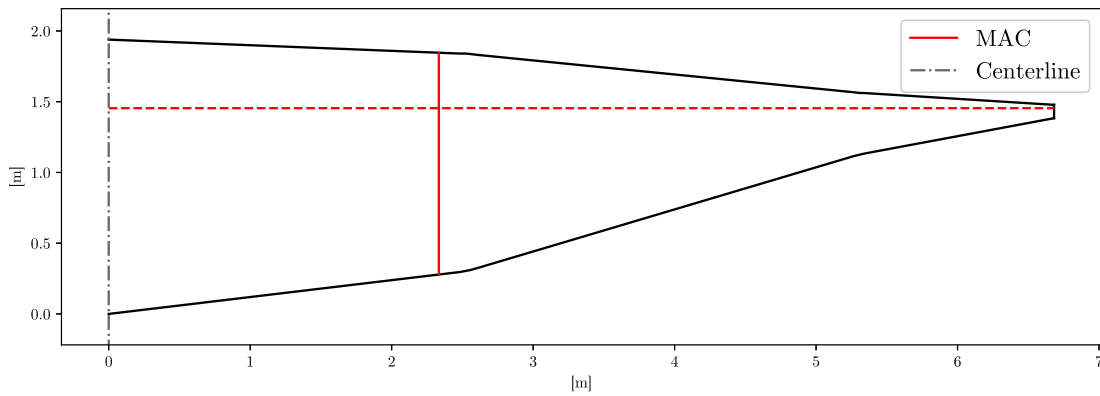


Figure 4.61: Wing with Type B chord distribution

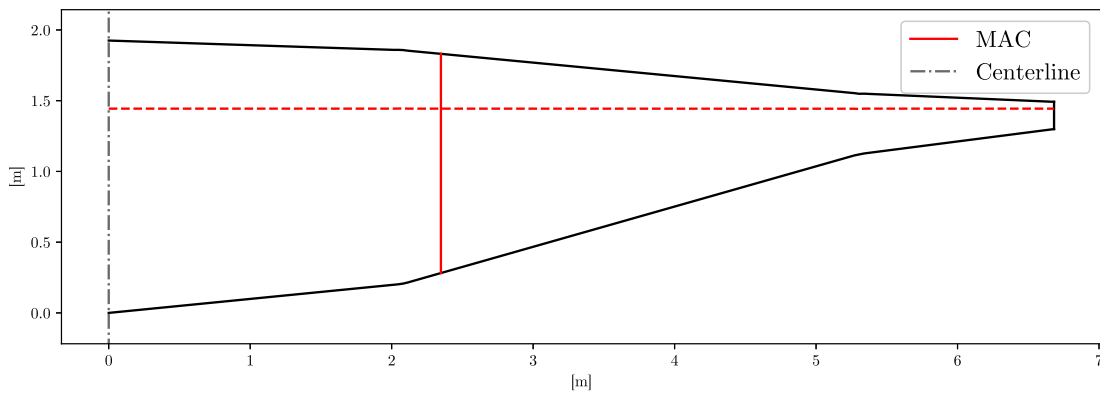


Figure 4.62: Wing with Type C chord distribution

The lift curve and polar diagrams for these new wings and their baseline is shown in Fig. 4.64. The polar and lift curves overlap, except at very high lift coefficients. Table 4.13 provides the aerodynamic properties of the three wings at the design point of $C_L = 0.4$. While the results are similar, Type C gives the best L/D and lowest C_D . An interesting feature of these chord distributions is how induced drag increment at zero lift is small (Tab. 4.14), unlike the designs analyzed in Sec. 4.2. This mitigates the high drag at low lift that was found with a simple tapered wing. The three designs have similar drag reduction at the design point

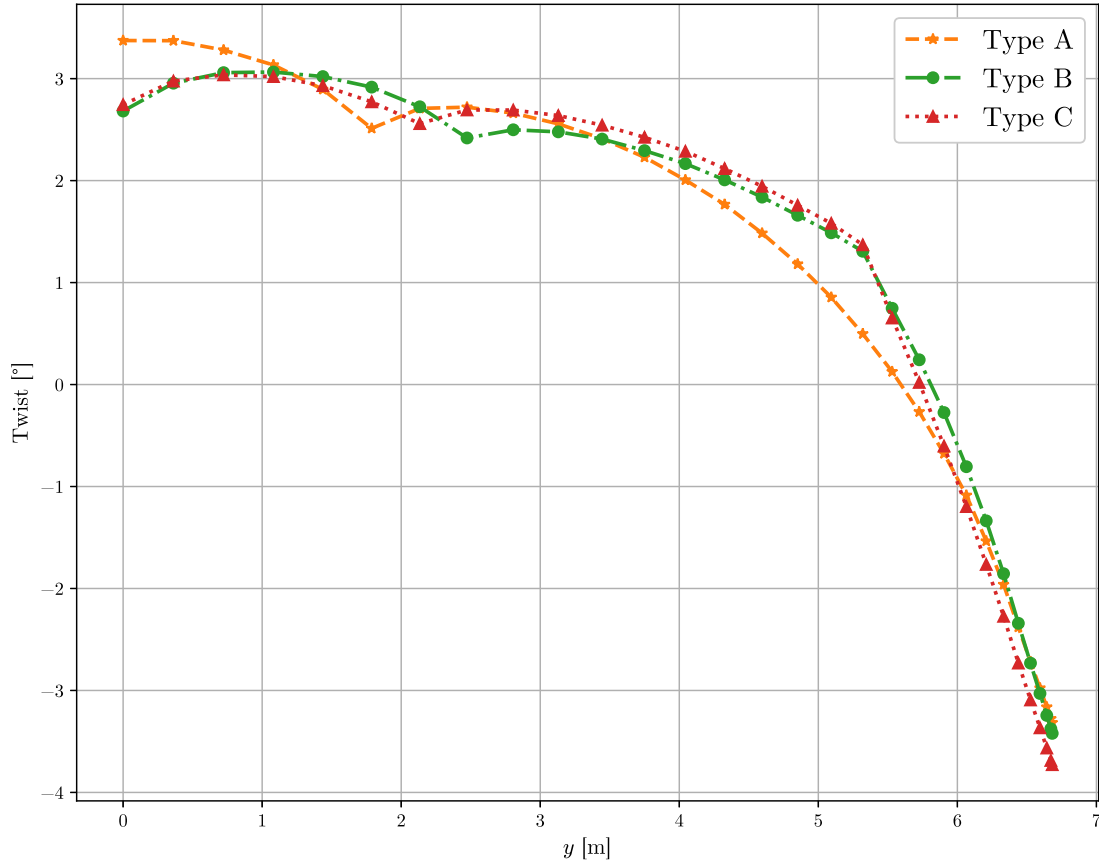


Figure 4.63: Twist distributions for different chord distributions

(Fig. 4.65), however with the A-type design underperforming the expected 10-12% induced drag reduction.

	Original	Type A	Type B	Type C
C_{Di}	0.006275	0.005705	0.005617	0.005608
C_{Dv}	0.005447	0.005522	0.005527	0.005532
C_D	0.011723	0.011227	0.011144	0.011140
L/D	34.12	35.63	35.89	35.91
e	1.0159	0.7497	0.7616	0.7626
$\mathcal{R}$	8	11.9	11.9	11.9

Table 4.13: Aerodynamic properties at design point

The major benefit of approximating the bell distribution with the chord distribution is constraining the bending moment increase outside of the design point. Indeed as shown in Figures 4.70-4.72, even at the maximum lift coefficient (the flight condition at V_a) the bending moment distribution is still close to the elliptical one. Indeed the wing planform helps to retain the bell shape on the lift distribution even outside the design point, as it's shown by the trend of Oswald's efficiency factor e with lift (Fig. 4.68).

	C_{Di}	C_{Dv}	C_D
Original	0.000008	0.005974	0.005983
Type A	0.000356	0.006168	0.006524
Type B	0.000207	0.006152	0.006359
Type C	0.000302	0.006158	0.006459

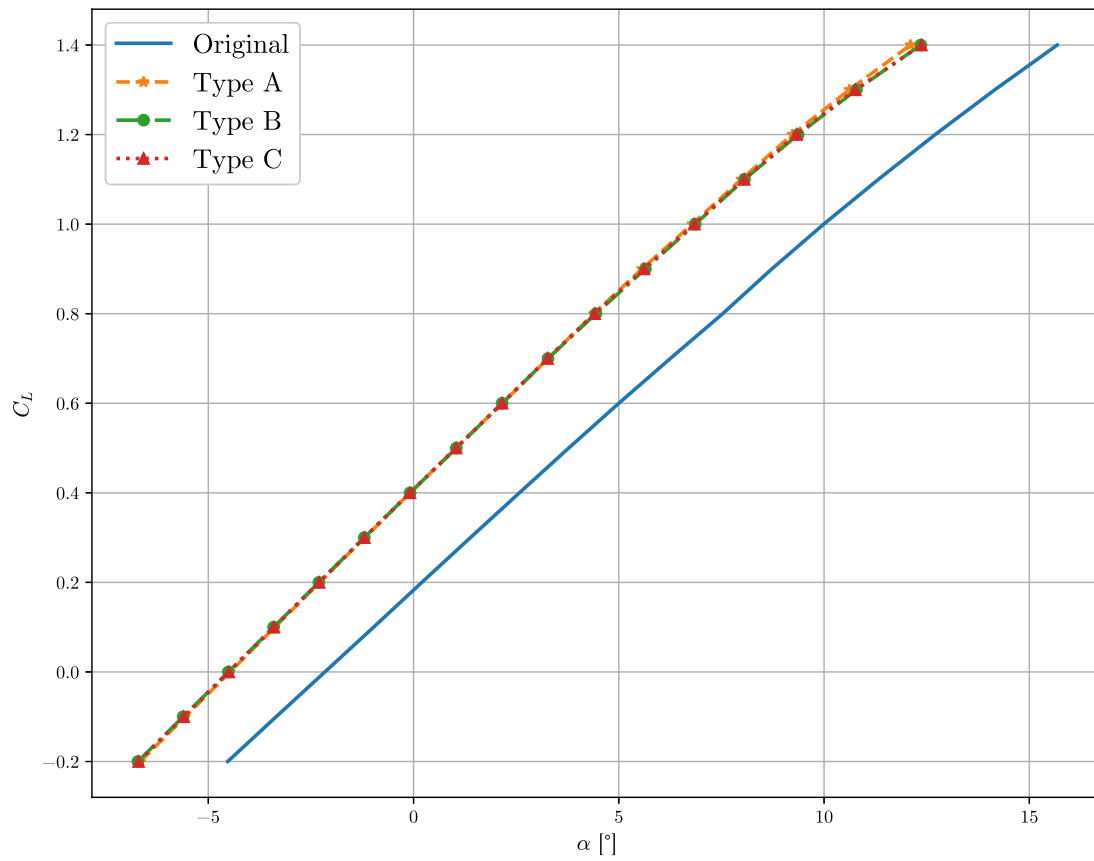
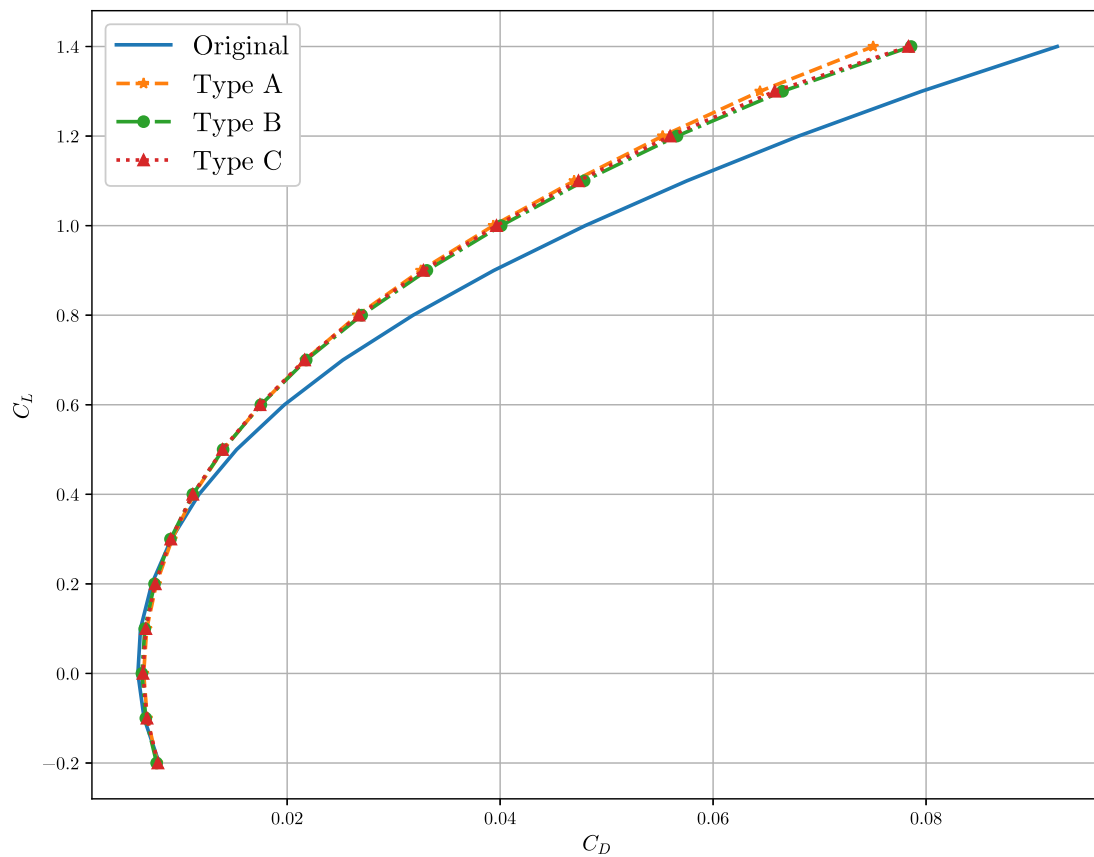
Table 4.14: Drag at zero lift

The calculated representative masses are shown in Tab. 4.15. Unlike the results in the previous sections, the mass increments are smaller, less than 20%. Approximation "B" provides the smallest mass increment, however it is the most tapered wing of the three. This means the airfoils should be properly tailored for the spanwise Reynolds distribution, especially at the tip. A similar issue is encountered on winglets, as these airfoils need to be of high camber at low Reynolds numbers. Stalling characteristics may also be an issue here, however it would require a more careful analysis. Assuming the airfoils would be properly tailored to their local Reynolds, the generated lift distribution would have their highest local C_l near the root and not at the wingtips. Therefore the classical issue with tapered wings, stall near the wingtips, should not be present in this design.

Overall Type C, which features a taper ratio of 0.1, may be a good compromise between weight, aerodynamic performance and manufacturability.

	Mass	% Variation
Original	81.04 kg	
Type A	96.56 kg	19.15
Type B	93.50 kg	15.38
Type C	95.65 kg	18.03

Table 4.15: Representative wing masses for different chord distributions.

(a) $C_L - \alpha$ 

(b) Polar

Figure 4.64: Aerodynamics of tested chord distributions

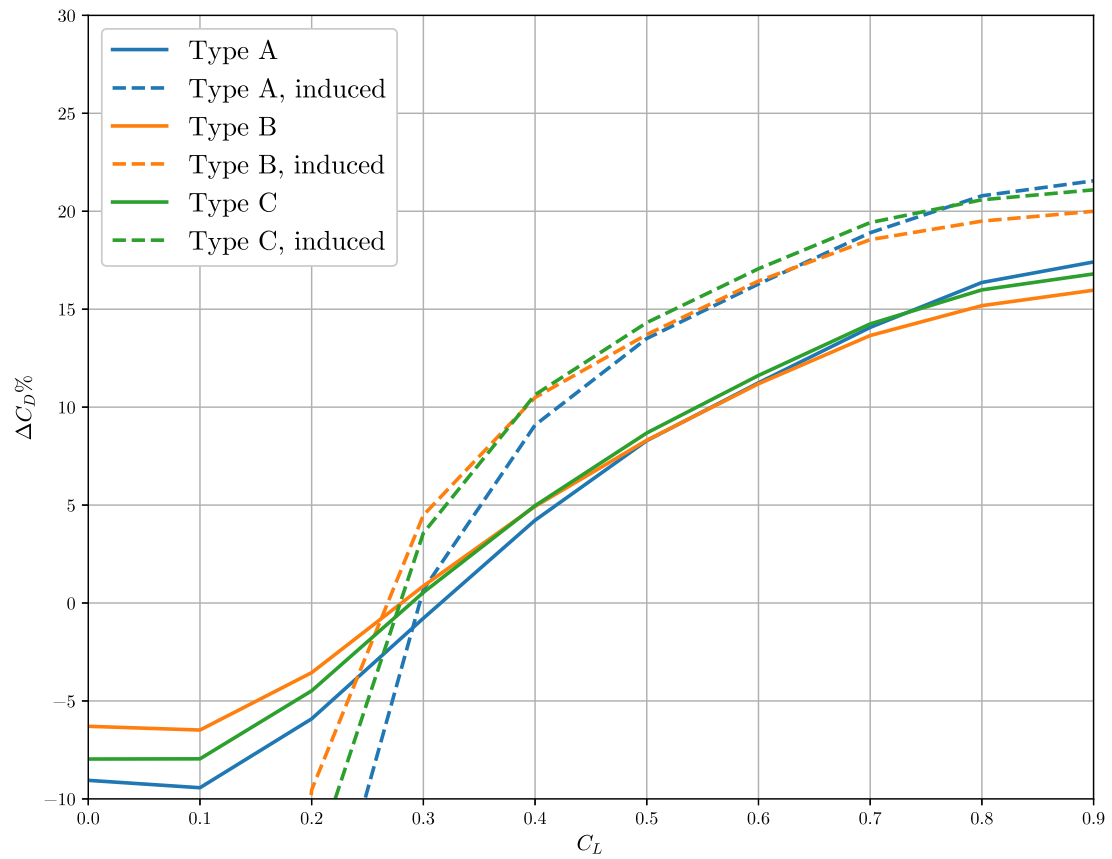


Figure 4.65: Relative drag reduction of tested chord distributions

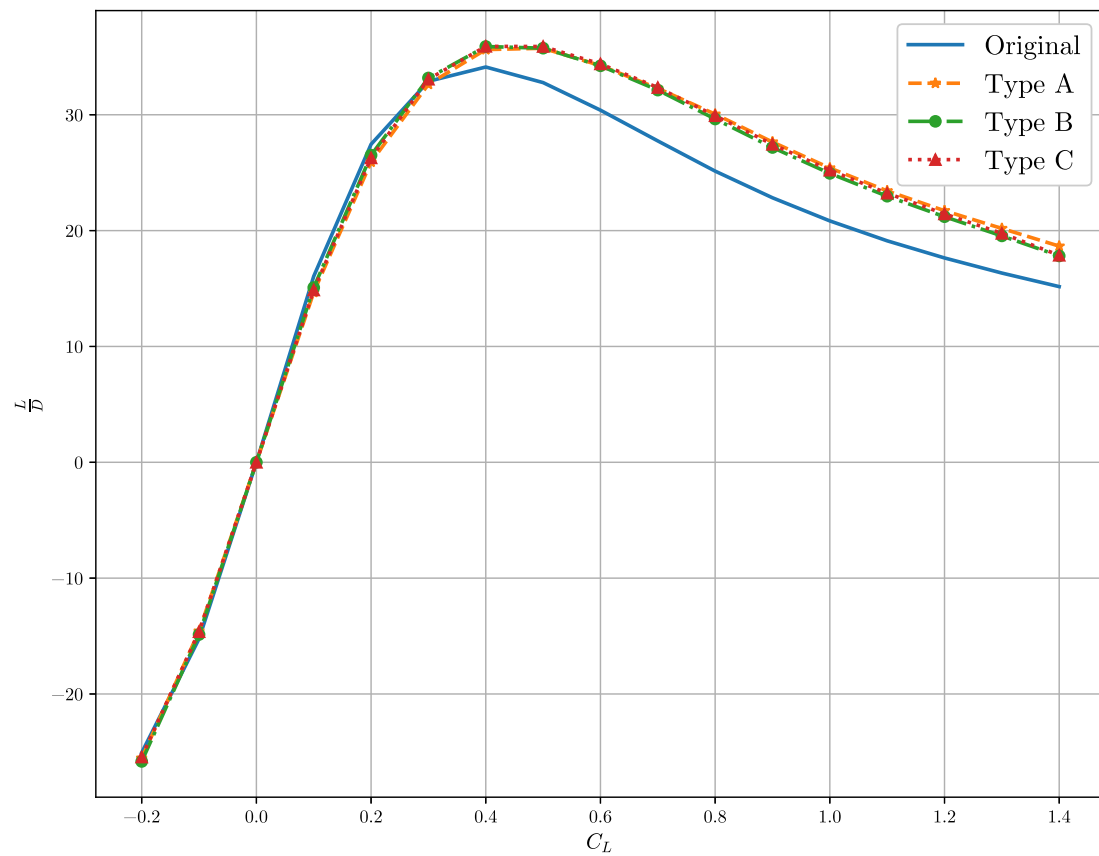


Figure 4.66: L/D of tested chord distributions

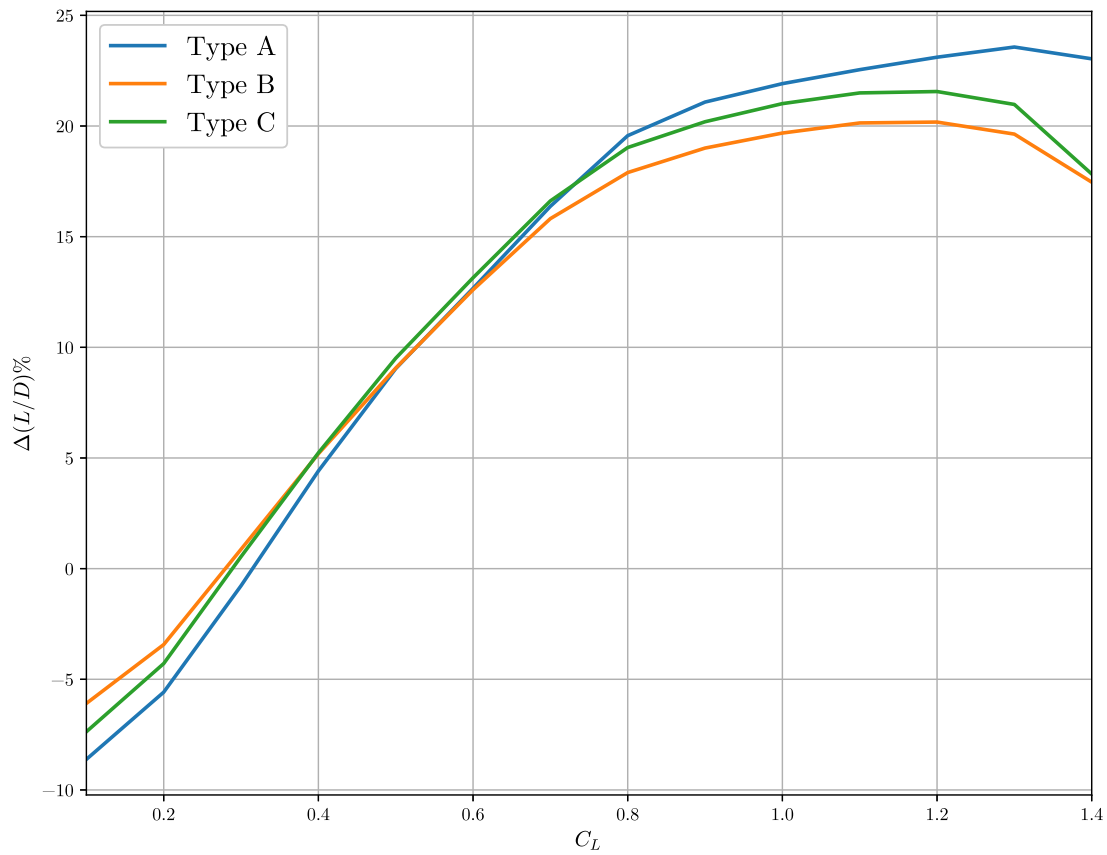


Figure 4.67: Relative variation of L/D of tested chord distributions

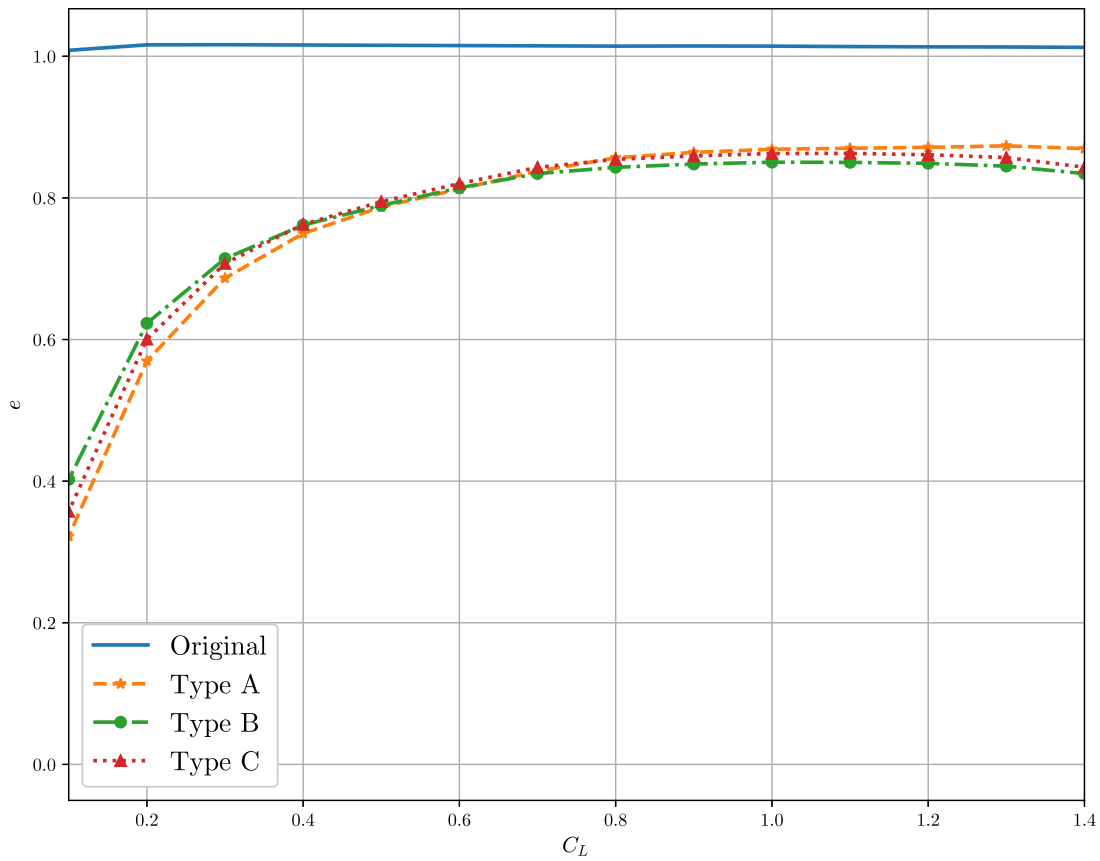


Figure 4.68: Oswald efficiency factor of tested chord distributions

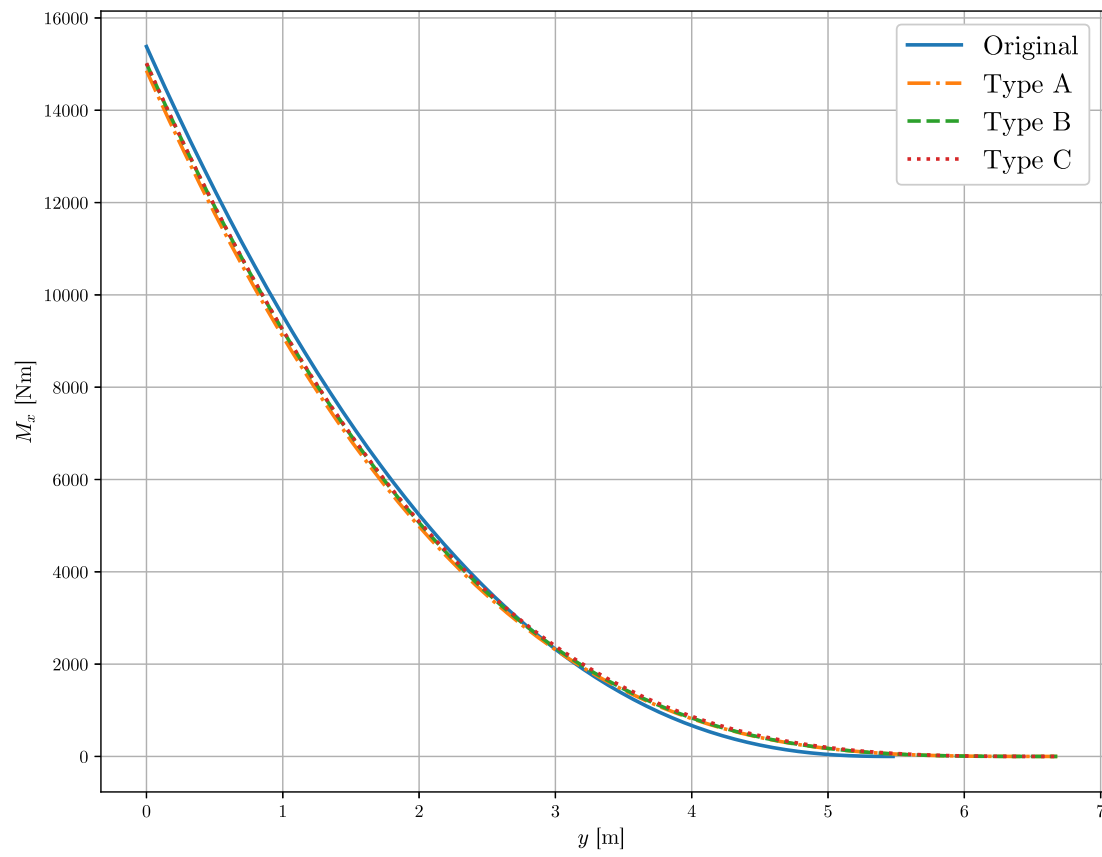


Figure 4.69: Bending moment distribution at design point

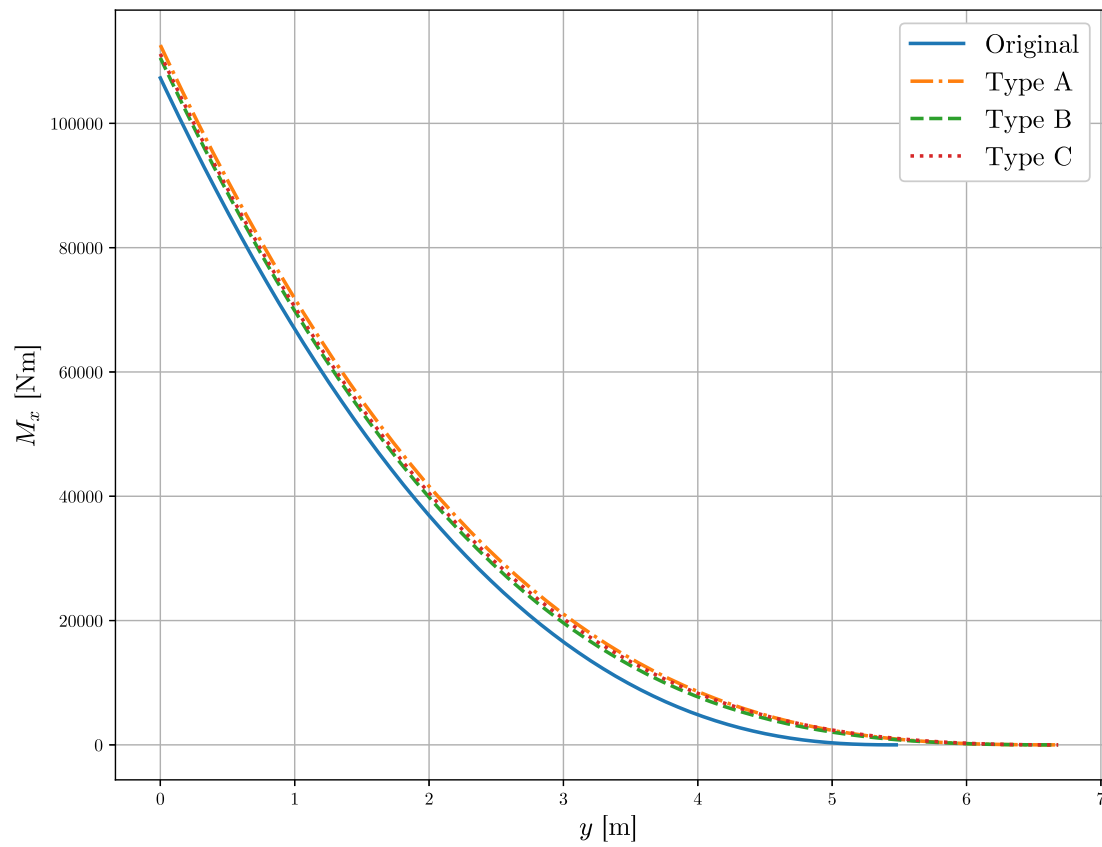


Figure 4.70: Bending moment distribution at V_a

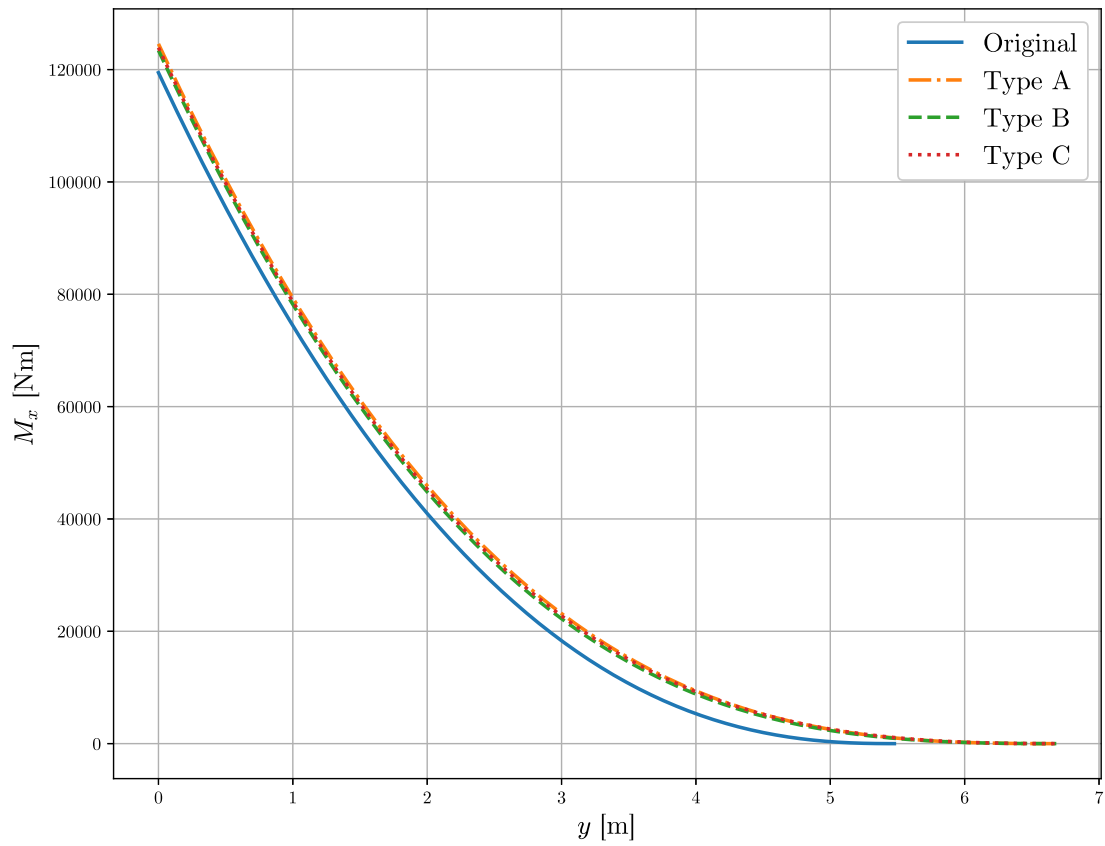


Figure 4.71: Bending moment distribution at V_c

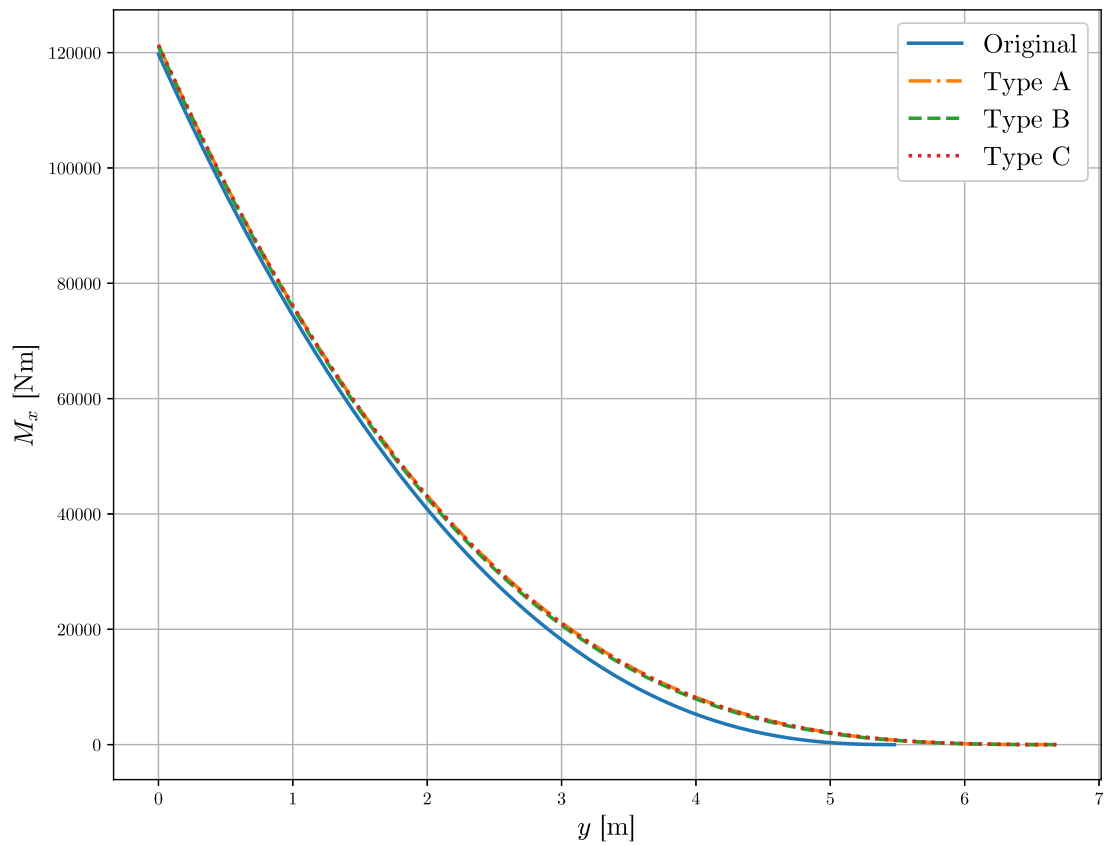


Figure 4.72: Bending moment distribution at V_d

4.6 Aileron Design

One of the interesting aspects of the bell lift distribution is the possibility to exploit the upwash near the wingtips to reduce or nullify adverse yaw. This capability is found if the aerodynamic derivative C_{n,δ_a} is positive. However, the upwash region depends on the lift distribution which is perfectly bell-shaped only at the design lift coefficient. In this section the effect of C_L and aileron span will be analyzed.

With the VLM code, several analysis were performed in which the aileron span was varied to calculate C_{l,δ_a} and C_{n,δ_a} . The span was defined by fixing the relative endpoint $y/(b/2)$ to 0.98 and changing where the first aileron point is set. Two different wings were analyzed: a trapezoidal one with the geometric features of Tab. 4.1, and a "bell-like" one with the Type C chord distribution of Sec. 4.5. Both wings have a design lift coefficient of 0.4, and the ratio between the chord of the aileron and the wing c_f/c is constant at 0.25.

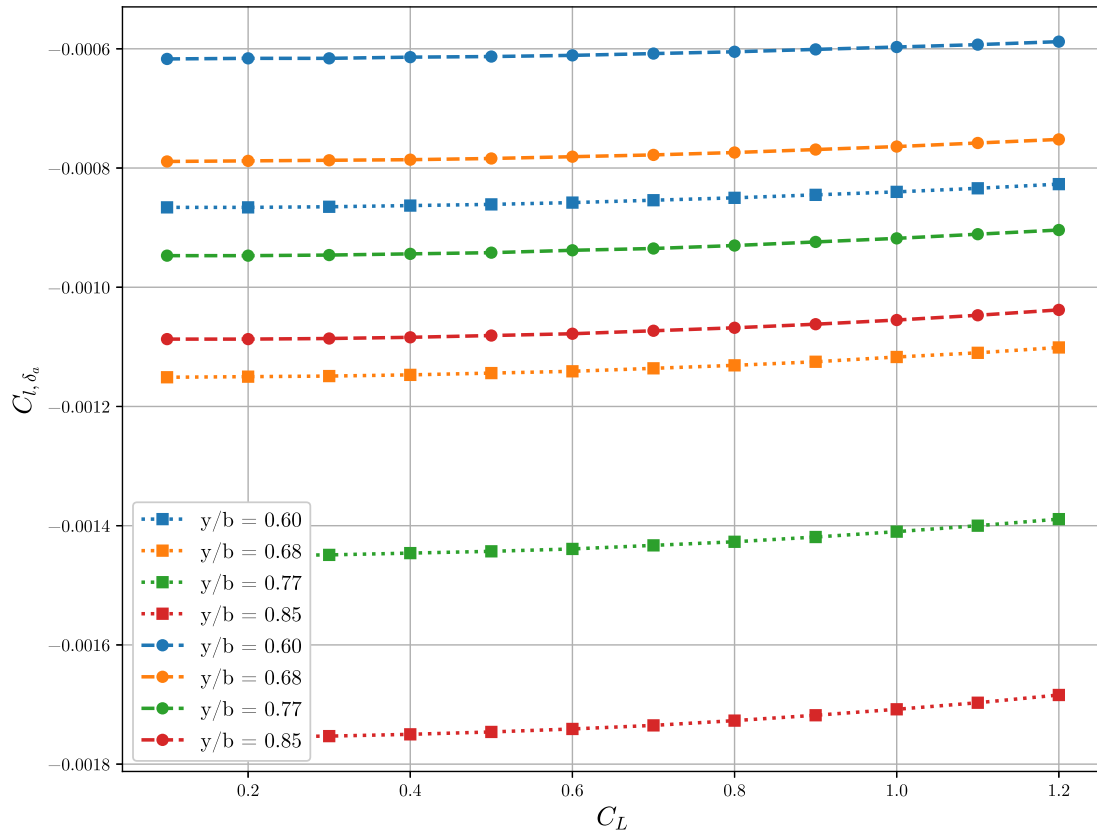


Figure 4.73: C_{l,δ_a} for different aileron spans. Squared markers correspond to trapezoidal wing, circle markers to Type-C wing.

In the resulting graphs, results for the trapezoidal wing have dotted lines with $\square$ markers, while results for the wing with the new chord distribution have a dashed line with $\circ$ markers. Fig. 4.73 shows that the rolling control derivative is almost independent from the lift coefficient, instead is mainly defined by the span of the aileron. It is the most effective for ailerons of small span, where the deflection is

mainly concentrated at the wingtips, benefiting of a larger rolling arm. For the same aileron span, the new chord distribution yields a smaller control derivative. The Type C chord distribution has less lift near the tips, thus deflecting a control surface here generates less rolling authority.

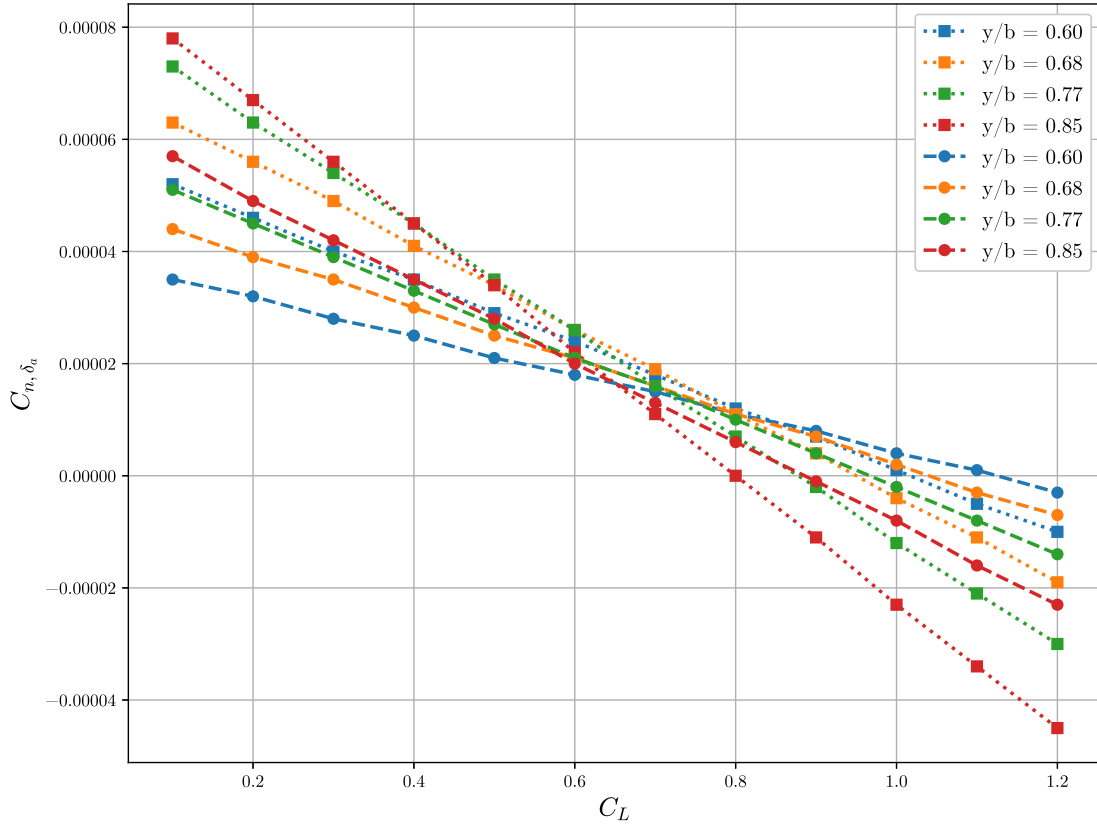


Figure 4.74: C_{n,δ_a} for different aileron spans. Squared markers correspond to trapezoidal wing, circle markers to Type-C wing.

Instead Fig. 4.74 shows that the coupled effect of the aileron changes sign with the lift coefficient. The aileron span instead affects how wide is the range of lift coefficients with proverse yaw. The wider the span, the steeper is the curve and therefore the less constant is the proverse yaw property. Indeed the point where $C_{n,\delta_a} = 0$ shifts to lower C_L the larger is the aileron span. The shape of the chord distribution also affects this property: the "Type C" wing has shallower curves than the respective trapezoidal wing. This is to be expected as the chord distribution was designed to retain the characteristics of the bell lift distribution outside of the design point.

The two derivatives have conflicting requirements. If the ailerons have a smaller span, the range of lift coefficients with positive C_{n,δ_a} will be smaller, even if the rolling control derivative will be greater.

Chapter 5

Practical application of bell spanloads

The bell distribution is found from an optimization problem that assumes an elliptical wing as a baseline. In real world applications, however, wings are seldomly designed with an elliptical lift distribution in mind. Usually designers adopt a trapezoidal wing with a taper ratio of 0.5 and a washout angle of -1 to -2 degrees, in order to prevent a wingtip stall, typical of tapered wings. Therefore the lift distribution of these wings are not perfectly elliptical but near-elliptical.

To analyze an application on a non-elliptic wing, wind tunnel data from NACA TN 1270 [32] is used. It provides experimental data of several straight tapered wings featuring NACA 44xx airfoils at different aspect ratios and taper ratios. The geometric properties of these wings are presented in Tab. 3.1. They were tested in a pressurized wind tunnel with $\rho = 2.834 \text{ kg/m}^3$ and $V_\infty = 38.7242 \text{ m/s}$. This data was used to validate the error of the VLM code in Ch. 3.

Then the bell spanload is used to design a new wing for the Pilatus PC-24 and performance is compared to the original baseline aircraft. The collected characteristics of the baseline aircraft are presented in Appendix B.

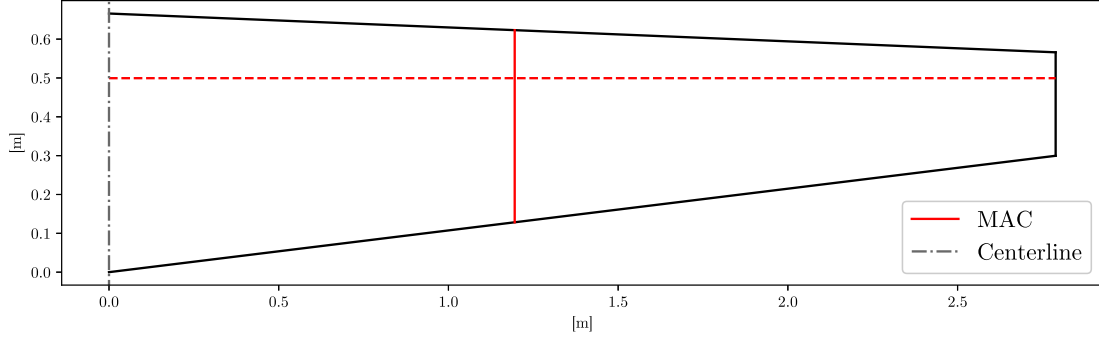
5.1 NACA Wings

The redesign procedure is the one outlined in Ch. 4. The wing area S_w is kept constant, while the wingspan b is increased by 22%. The new geometry is then twisted with the numerical algorithm for the given design lift coefficient and velocity to obtain the bell lift distribution.

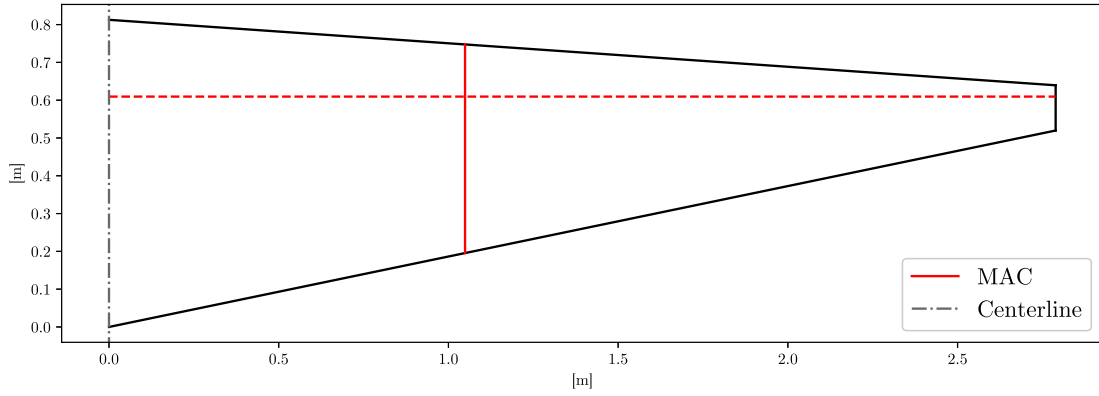
Following the design guidelines discussed in Ch. 4, three new designs are proposed for each NACA wing:

- Design 1 maintains the taper ratio of the original wing;
- Design 2 maintains the root chord;
- Design 3 applies the "Type C" chord distribution of Sec. 4.5, with a quadratic spline smoothing.

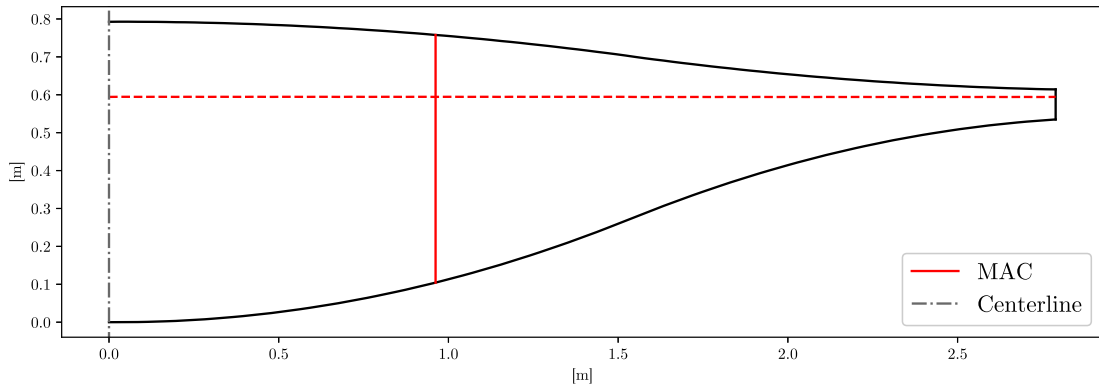
For all the three wings a design lift coefficient of 0.3 was selected. The obtained planforms applied to the first NACA wing are shown in Fig. 5.1, with their respective twist distributions in Fig. 5.2. Notice how the third design has a portion where the twist is almost constant because of the quadratic smoothing of the chord distribution.



(a) Design 1 - λ constant



(b) Design 2 - C_r constant



(c) Design 3 - Type C chord distribution

Figure 5.1: NACA Wing 1 new planforms

Figure 5.3 shows the results of the aerodynamic analysis for the first NACA wing configurations. The lift curve has a slight increased steepness due to the higher aspect ratio. The three designs have a similar drag count near zero lift, while above $C_{L,des}$, the first two designs have lower induced drag. With Fig. 5.6 and Fig. 5.7 the drag characteristics of the three designs are clearer. Unlike the

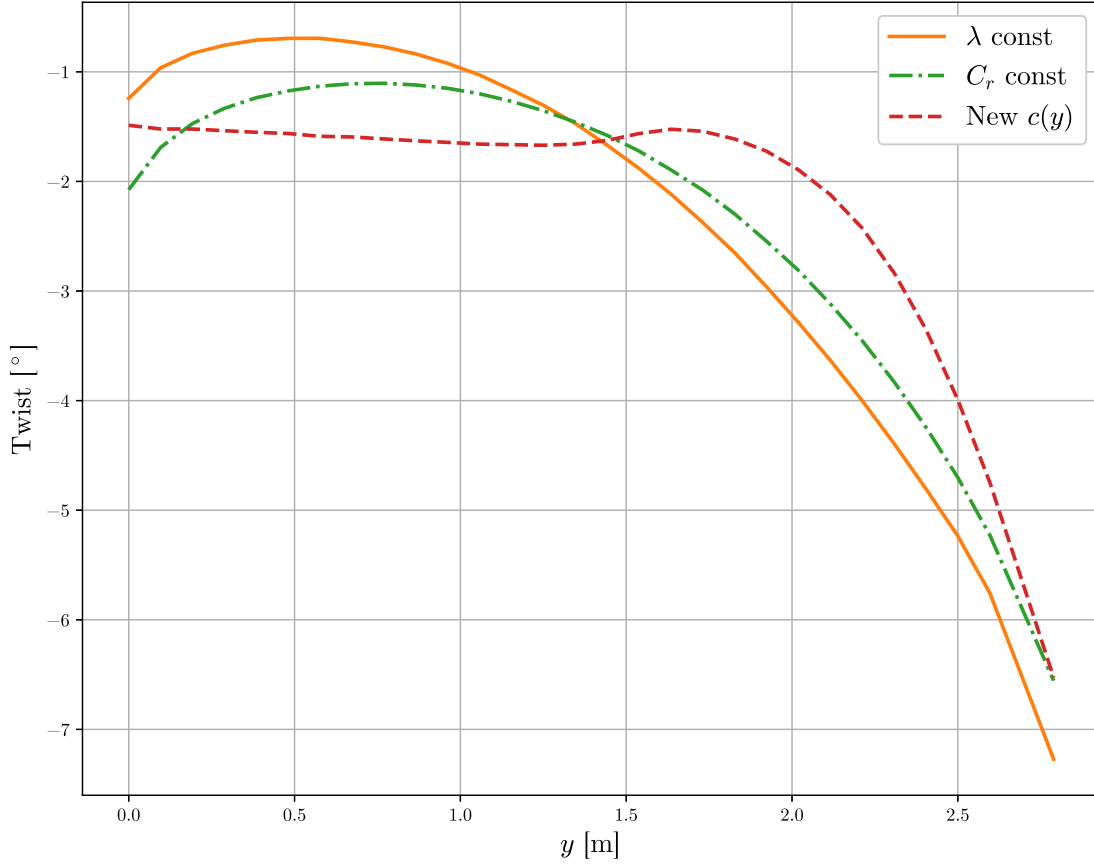


Figure 5.2: NACA Wing 1 Twist distributions

results in the previous chapter, the induced drag reduction at the design point is at 24% for all three, while if viscous drag is included it drops between 8% and 10%. Interestingly the third design lowers significantly the induced drag near $C_L = 0$. This is coherent with the trend of Oswald's efficiency factor (Fig. 5.5), as under $C_L = 0.2$, the third design has the highest value of e . At high lift instead, the third design maintains $e < 1$, retaining some of the bell shape of the lift distribution. While this does hinder the reduction of drag at high lift, and consequently L/D , it helps constraining the structural mass increment as it is shown later. L/D is generally higher for all the three designs (Fig. 5.4), with peak values between 34-36.

	$C_{L,des}$	V_a	V_c	V_d
λ	14.51	21.72	20.23	18.82
C_r	14.64	12.13	14.53	14.56
New $c(y)$	15.02	2.07	8.56	10.63

Table 5.1: Percentile of root bending moment increment for NACA Wing 1

The obtained bending moments are presented in Figures 5.9-5.12. Unlike the results with elliptical wings as baseline, the bending moment at the design point does not match the baseline (Fig. 5.9), but instead is about 15% greater (Tab. 5.1). Indeed the wash-out of the original wing provides a triangular-like lift distribution with most of the lift near the root (Fig. 5.13). So if it is compared to an elliptical wing with the same span, the practical wing would have a lower bending moment.

While at $C_{L,des}$ the third design has the highest root bending moment increase, at the three structural design points it has the lowest increment. Indeed, while the original wing's lift distribution becomes more elliptical-like (as indicated by Oswald's factor approaching 1), the third design retains the bell-like shape even at high C_L , containing the bending moment increase, much like was discussed in Ch. 4. The consequence is the mass of the third design has the lowest increment of all, just 7.78 % as shown in Table 5.2. It should be noted that the representative wing masses are very small because these wings are wind tunnel models.

	Mass	Variation %
Original	6.23 kg	-
$\lambda = \text{const}$	9.97 kg	60.01
$C_r = \text{const}$	8.00 kg	28.44
New $c(y)$	6.71 kg	7.78

Table 5.2: Representative wing mass for NACA Wing 1

NACA Wing 2 differs from the first one for the increased aspect ratio ($\mathcal{R} = 10$) and the thicker root chord ($t/c = 0.2$). Most of the considerations outlined so far apply to this second wing. Fig. 5.14 provides the main aerodynamic analysis results. Induced drag reduction at design point is lower than the previous wing, at 22%, while the total drag reduction is at 5%. The drag characteristics are similar to the ones of NACA Wing 1.

	$C_{L,des}$	V_a	V_c	V_d
λ	12.14	21.25	19.87	17.67
C_r	12.40	11.78	13.10	13.15
New $c(y)$	12.56	3.61	6.94	8.81

Table 5.3: Percentile of root bending moment increment for NACA Wing 2

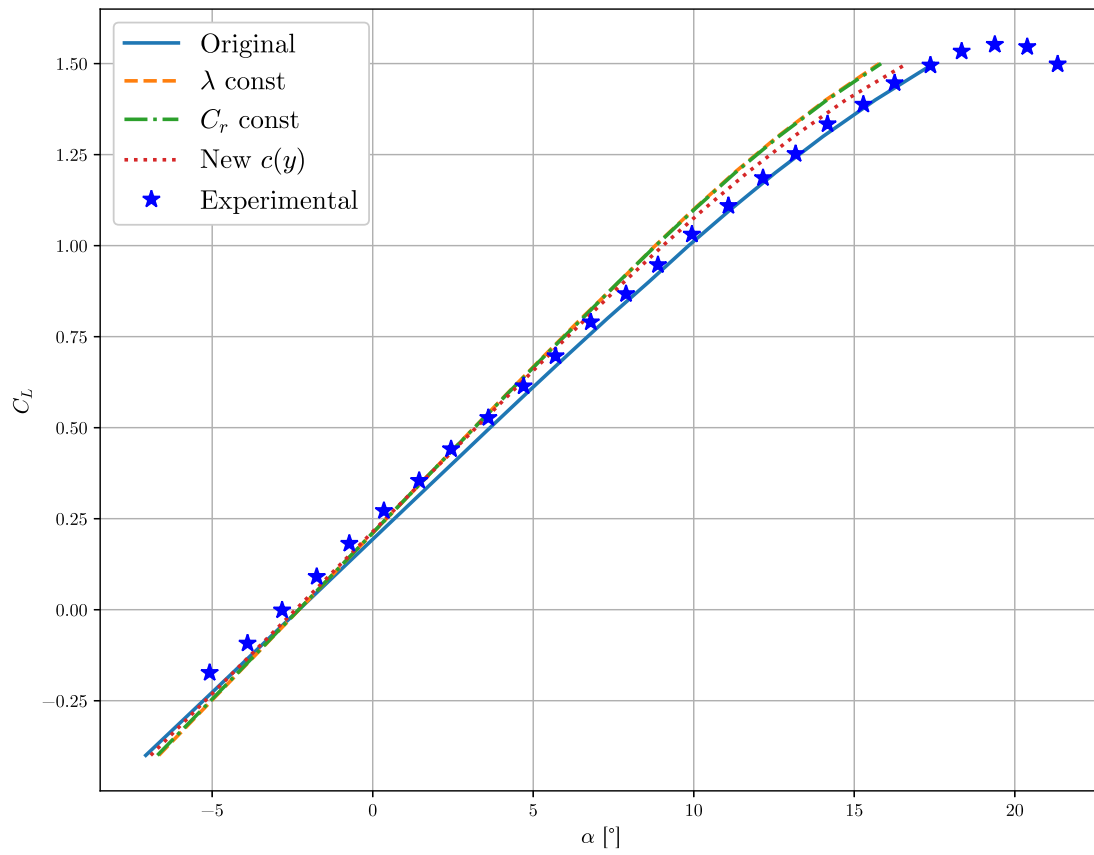
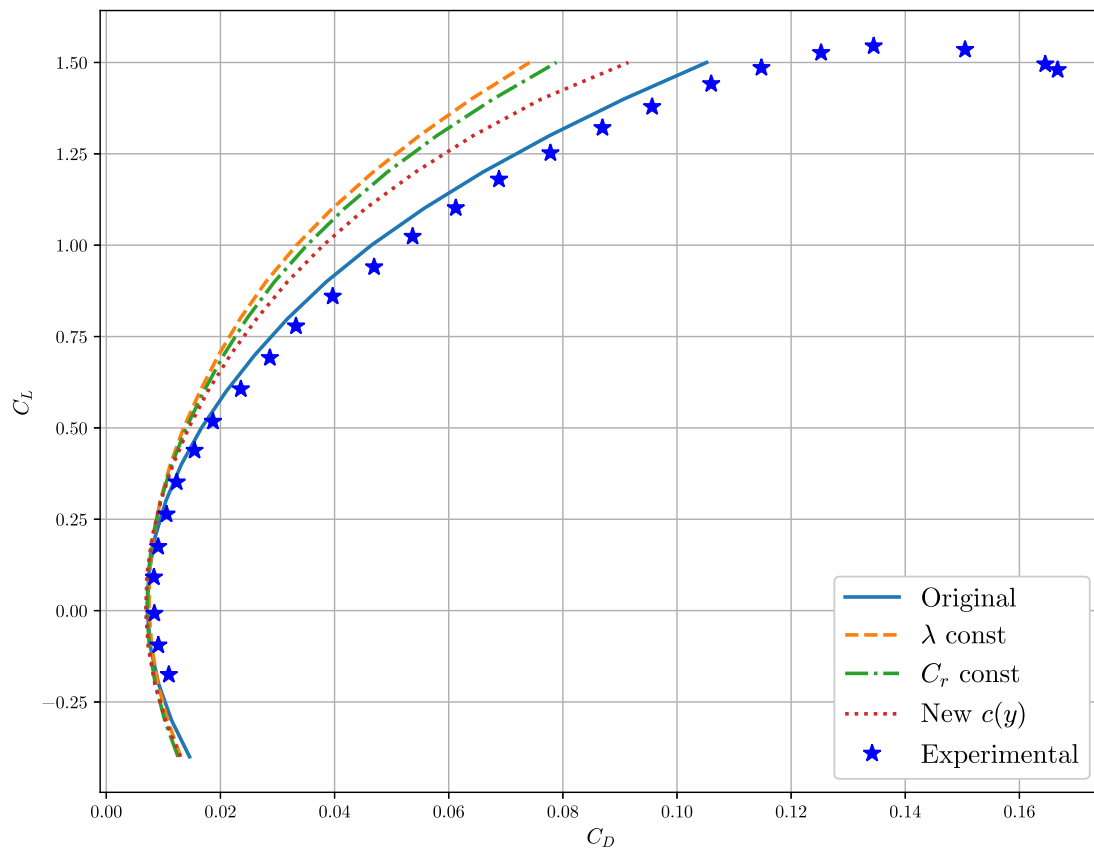
Regarding the bending moments, it's possible to draw conclusions similar to the previous case: the bending moment distribution of the three designs is higher both at the design point (Fig. 5.20) and the limit loads (Fig. 5.21-5.23). The root bending moments are presented in Table 5.3, the trends are similar to the Wing 1 case.

Finally the calculated representative masses are presented in Tab. 5.4. For this wing, the resulting mass increment is smaller than the previous case. Indeed even if

	Mass	Variation %
Original	5.08 kg	-
$\lambda = \text{const}$	8.17 kg	60.81
$C_r = \text{const}$	6.54 kg	28.80
New $c(y)$	5.49 kg	8.03

Table 5.4: Representative wing masses for NACA Wing 2

the original aspect ratio is larger, the root chord is considerably more thick yielding an higher sectional moment of inertia and reducing the overall panel thickness.

(a) $CL - \alpha$ 

(b) Polar

Figure 5.3: NACA Wing 1 aerodynamic data

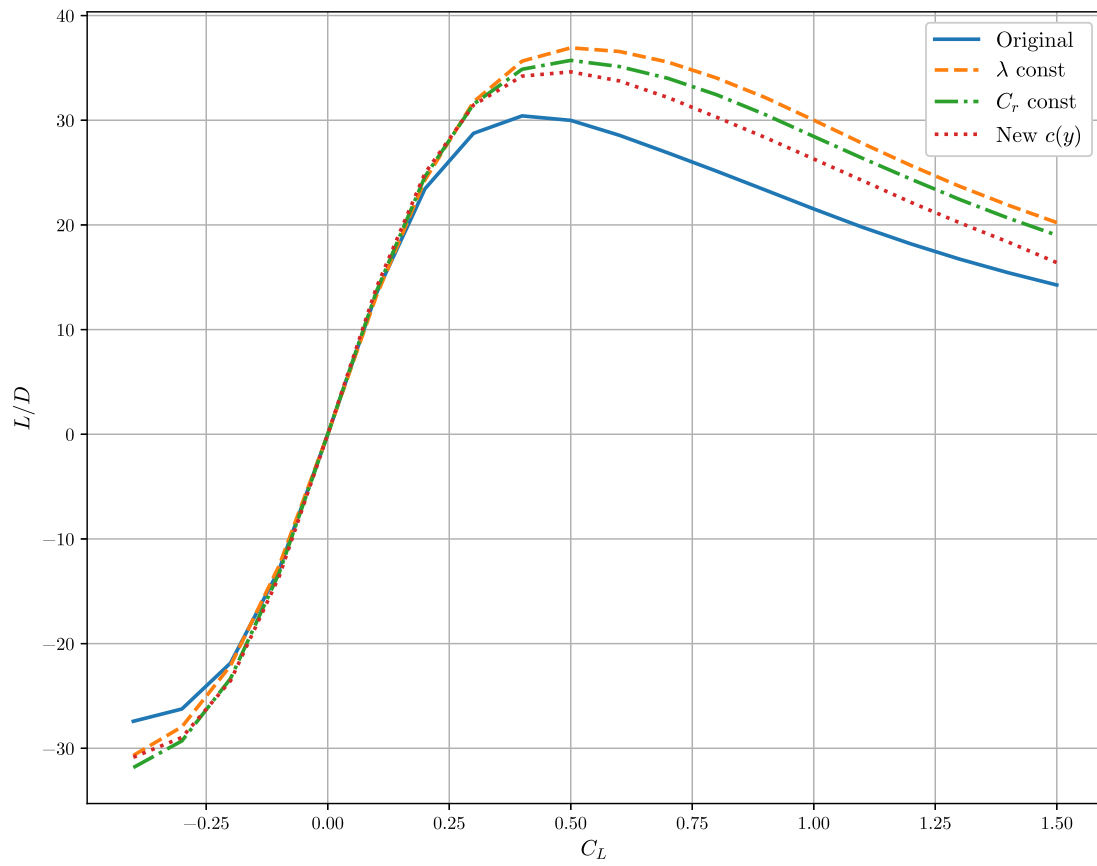
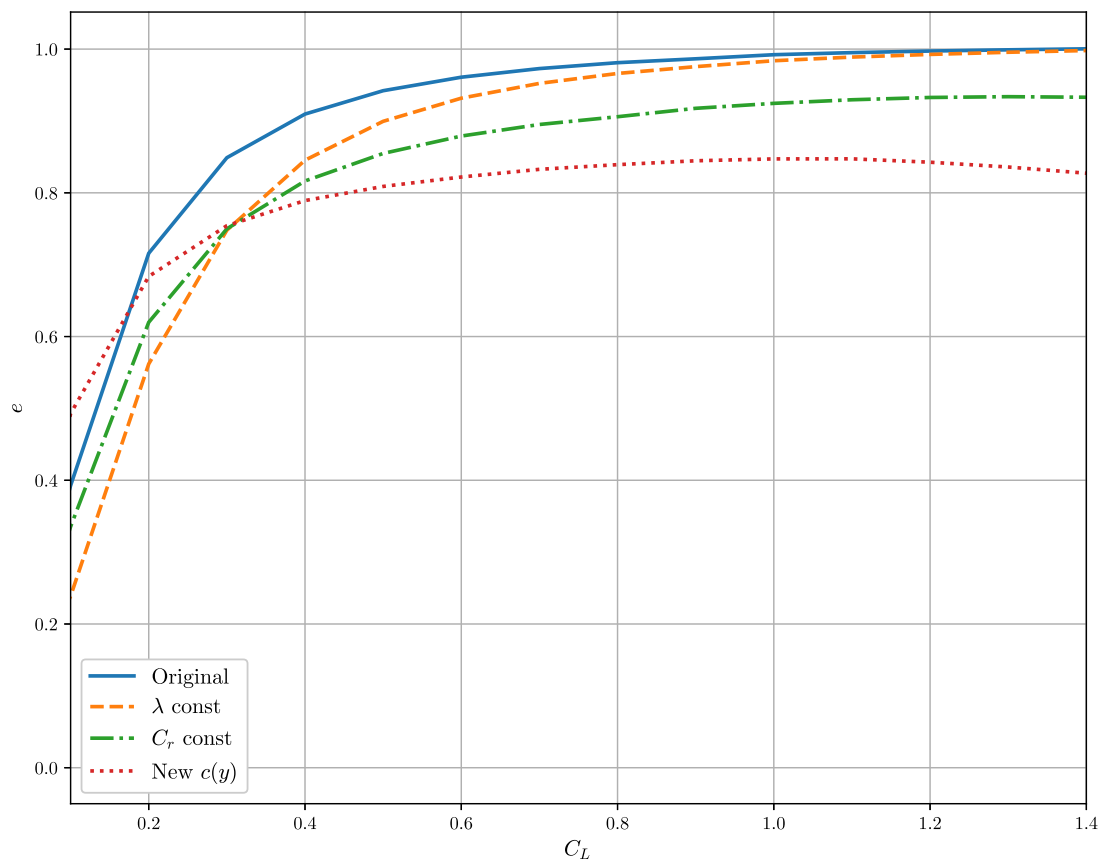
Figure 5.4: NACA Wing 1 L/D 

Figure 5.5: NACA Wing 1 Oswald Efficiency Factor

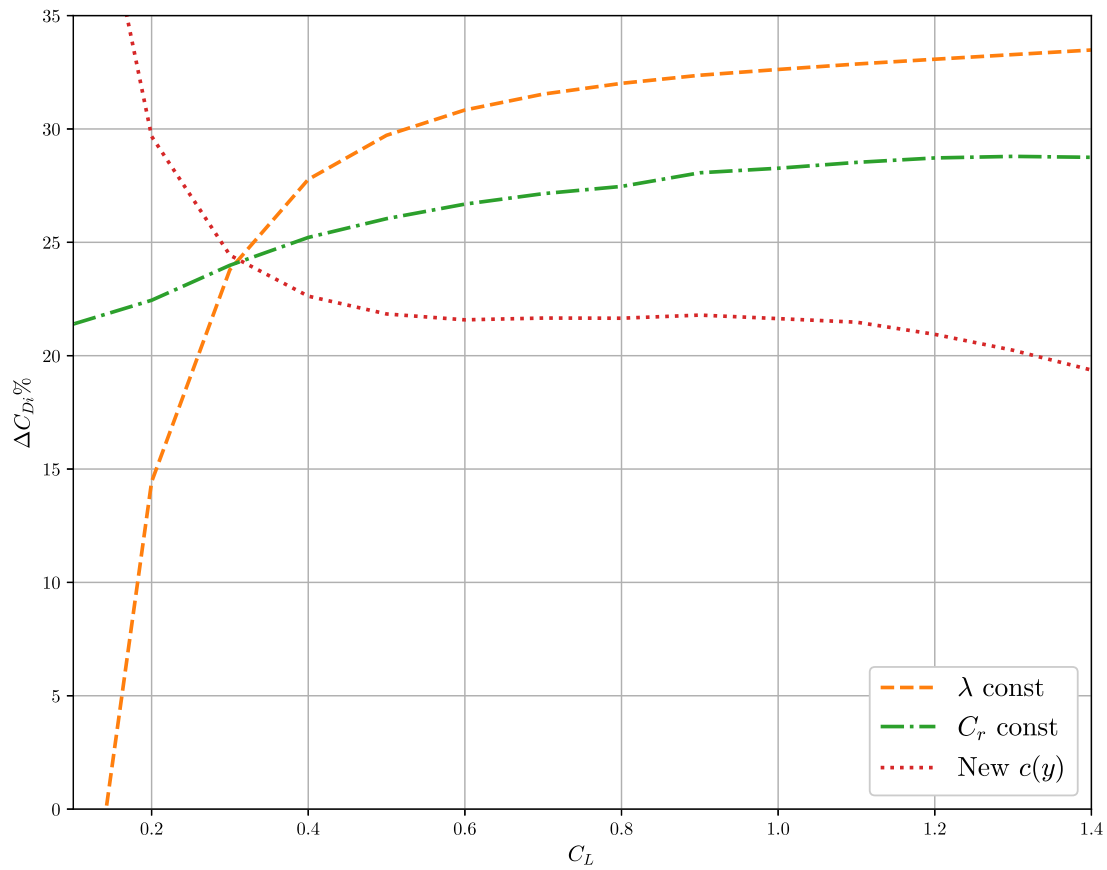


Figure 5.6: Relative C_{Di} variation for NACA Wing 1

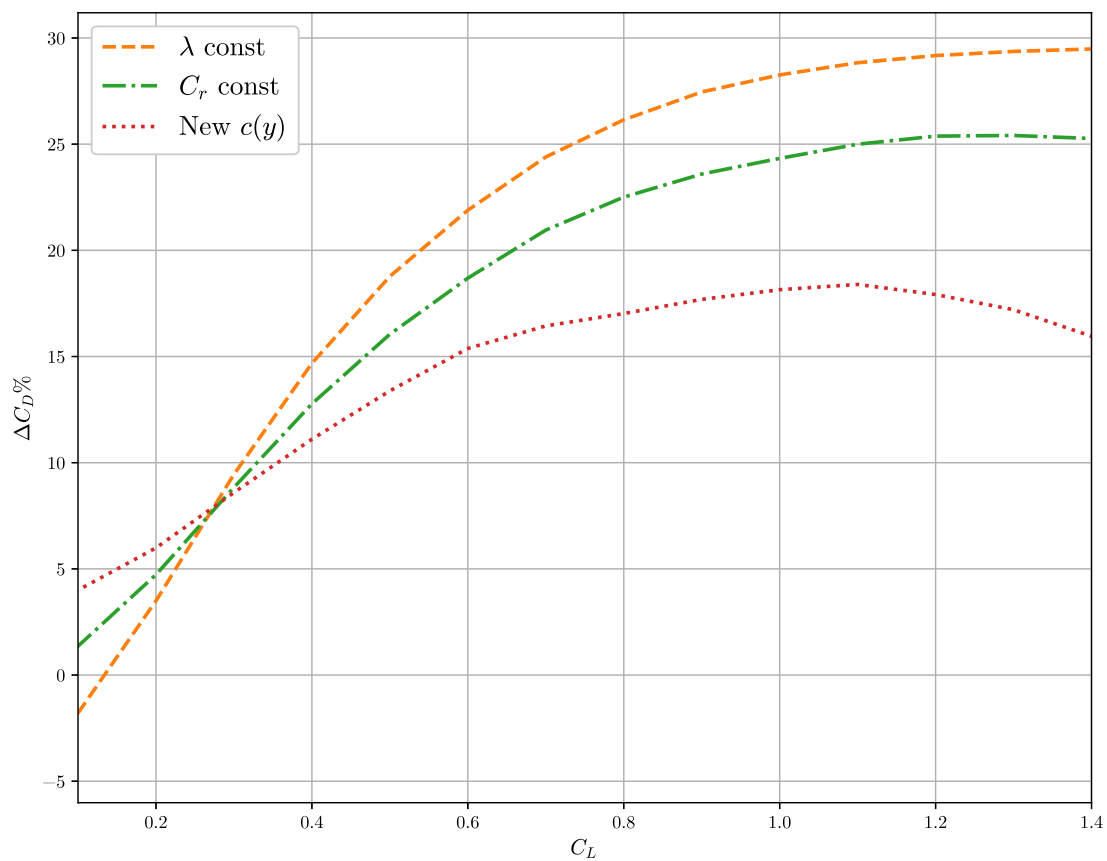


Figure 5.7: Relative C_D variation for NACA Wing 1

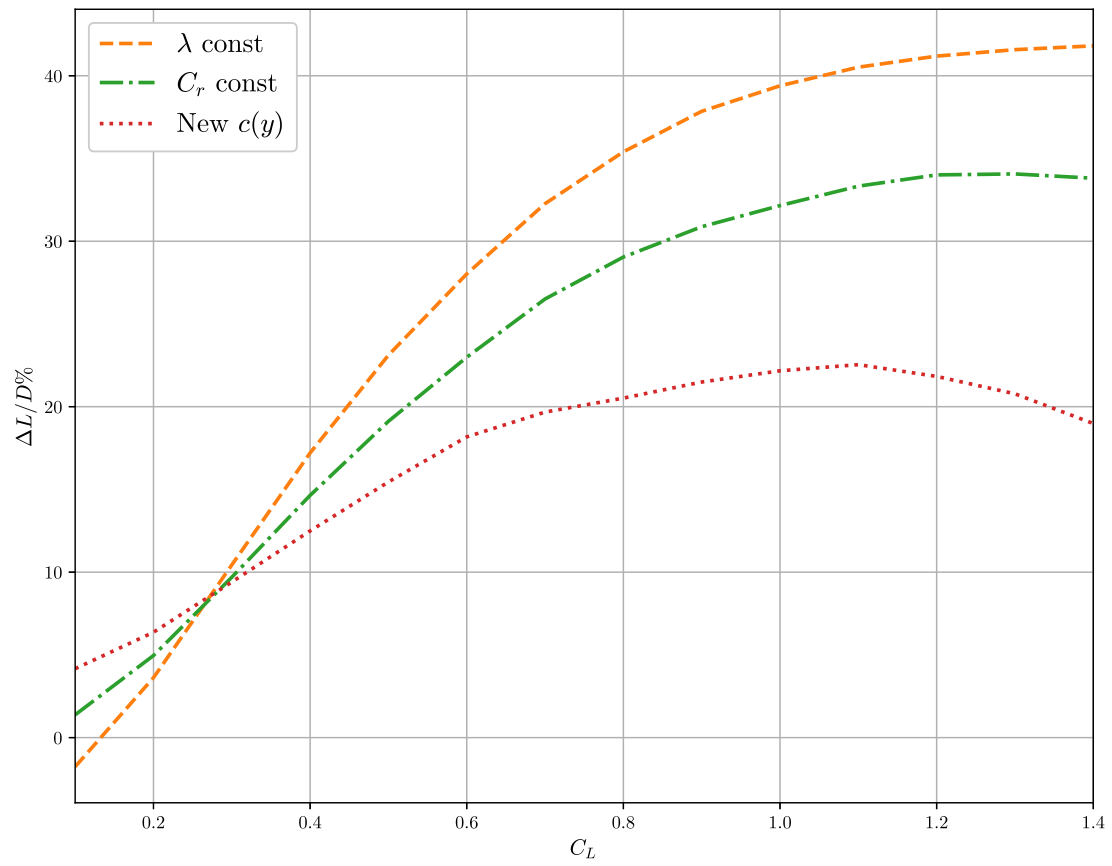


Figure 5.8: Relative L/D variation for NACA Wing 1

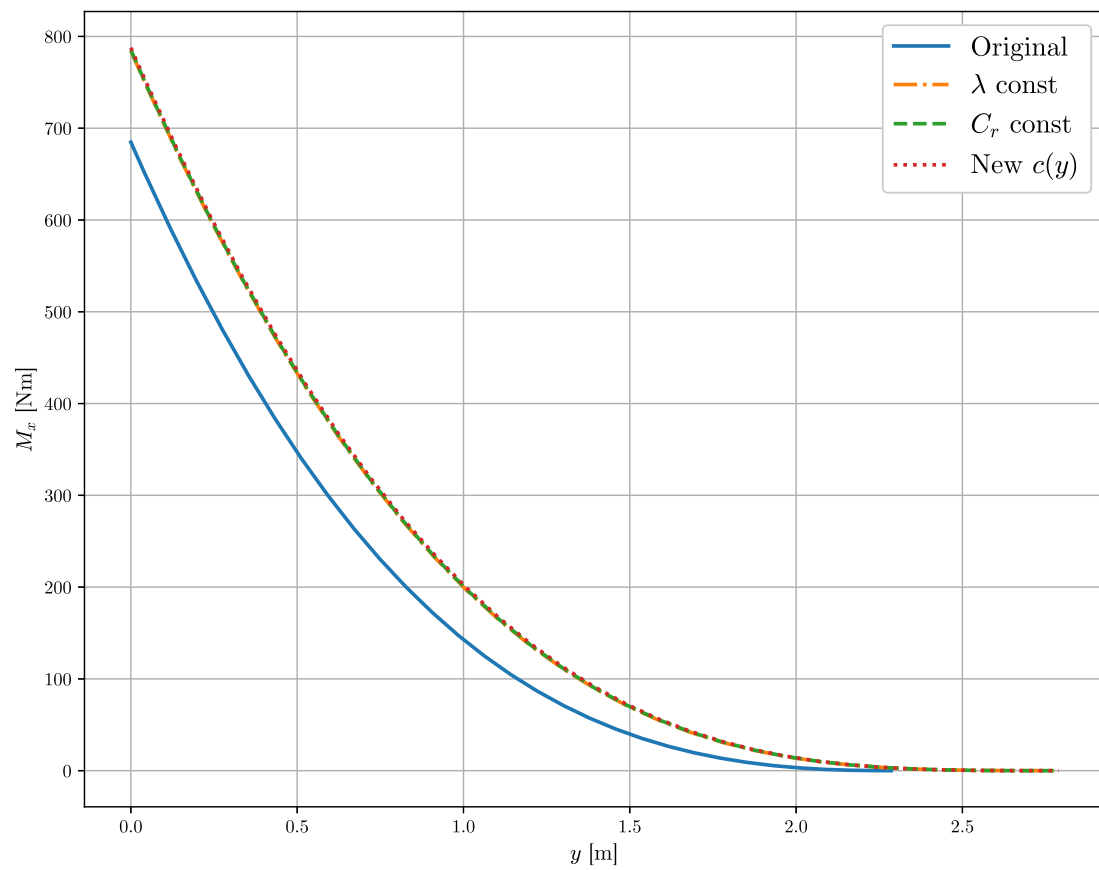


Figure 5.9: M_x at design point for NACA Wing 1

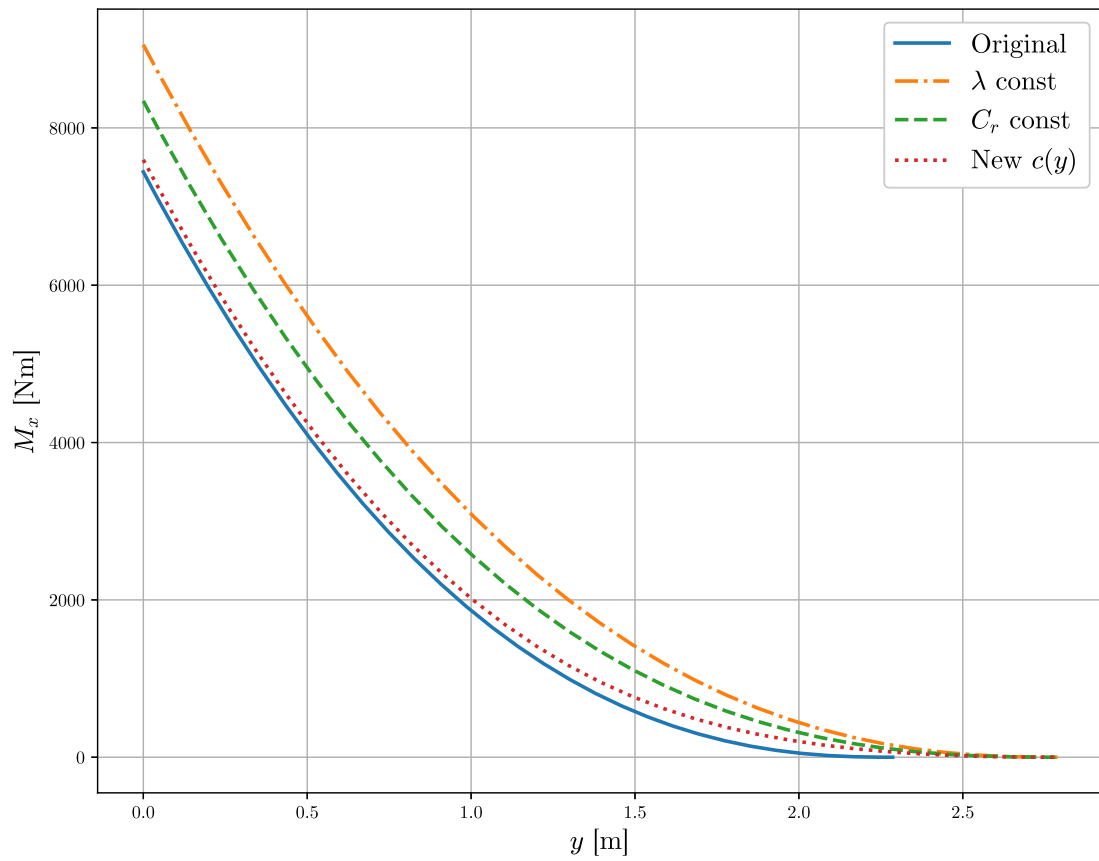


Figure 5.10: M_x at V_a for NACA Wing 1

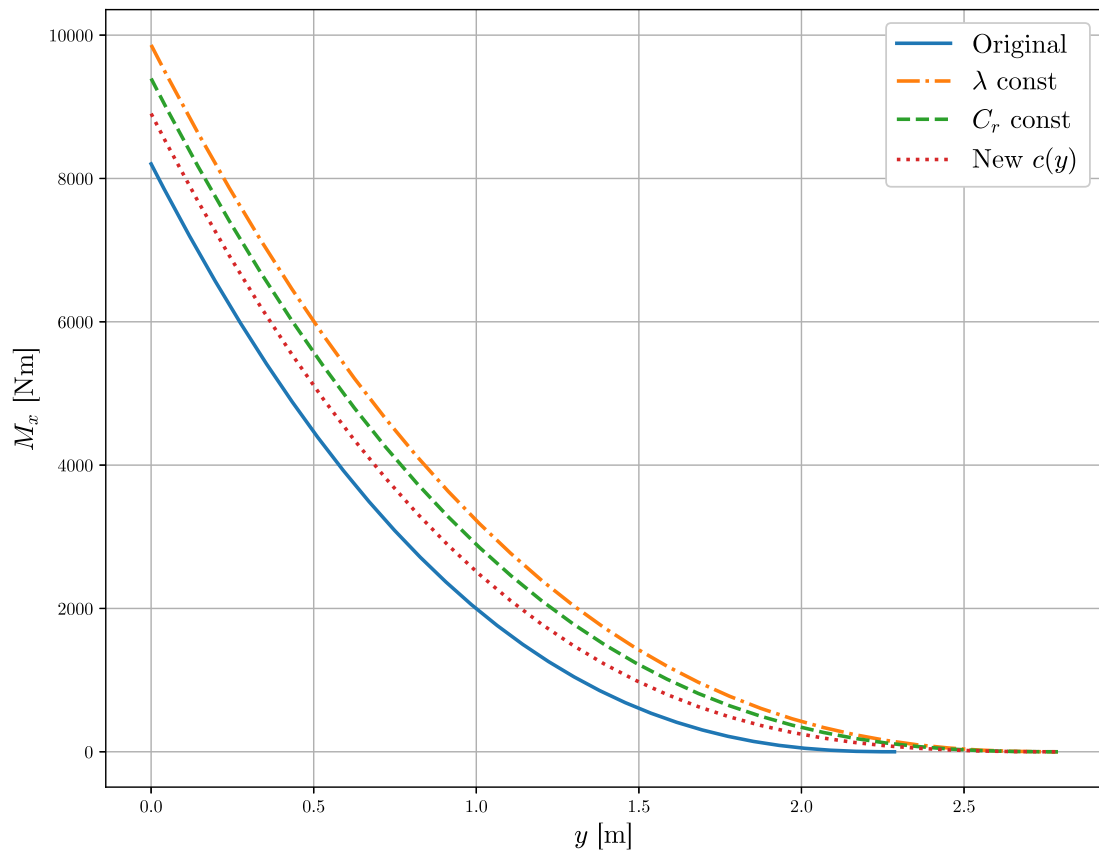


Figure 5.11: M_x at V_c for NACA Wing 1

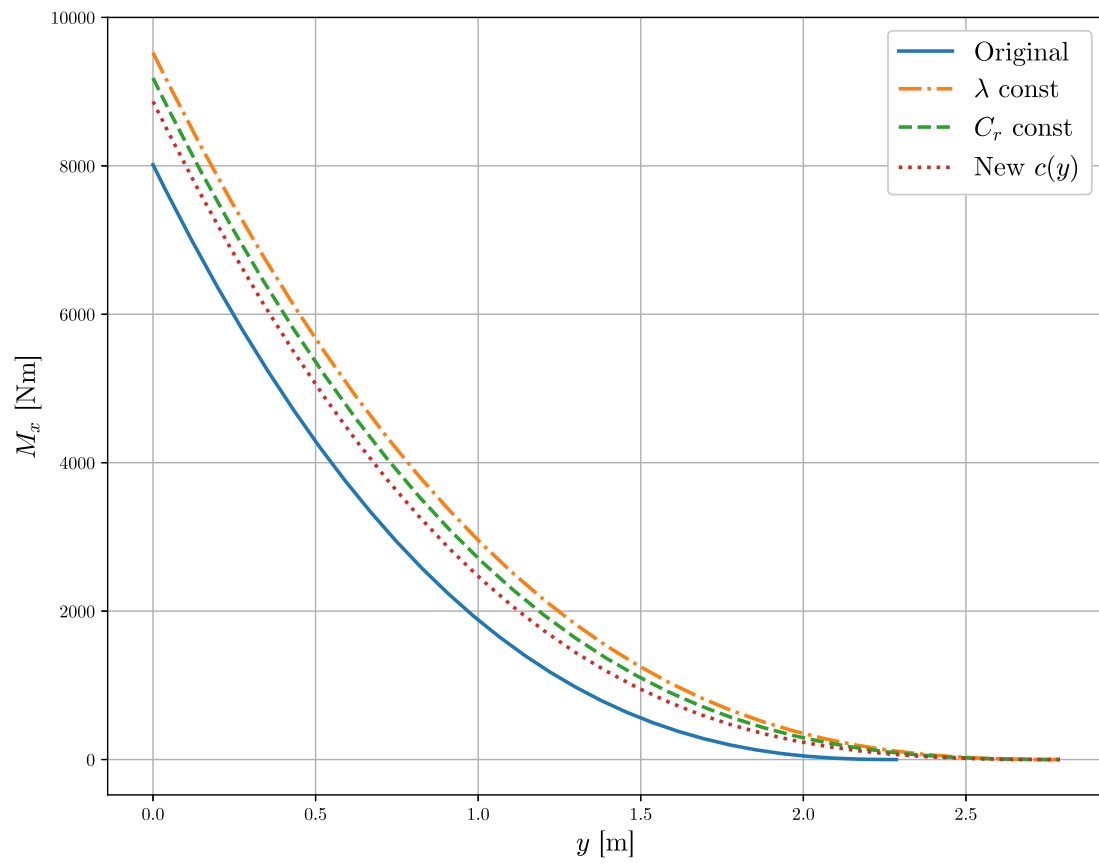


Figure 5.12: M_x at V_d for NACA Wing 1

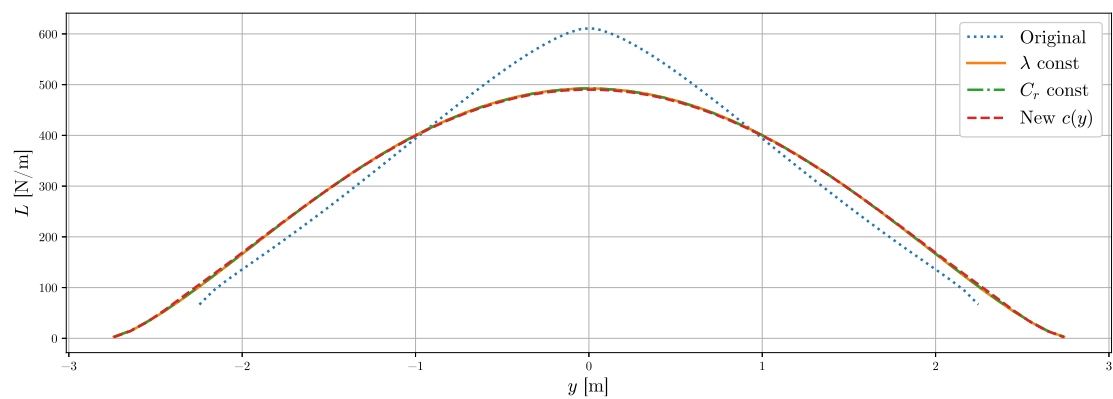
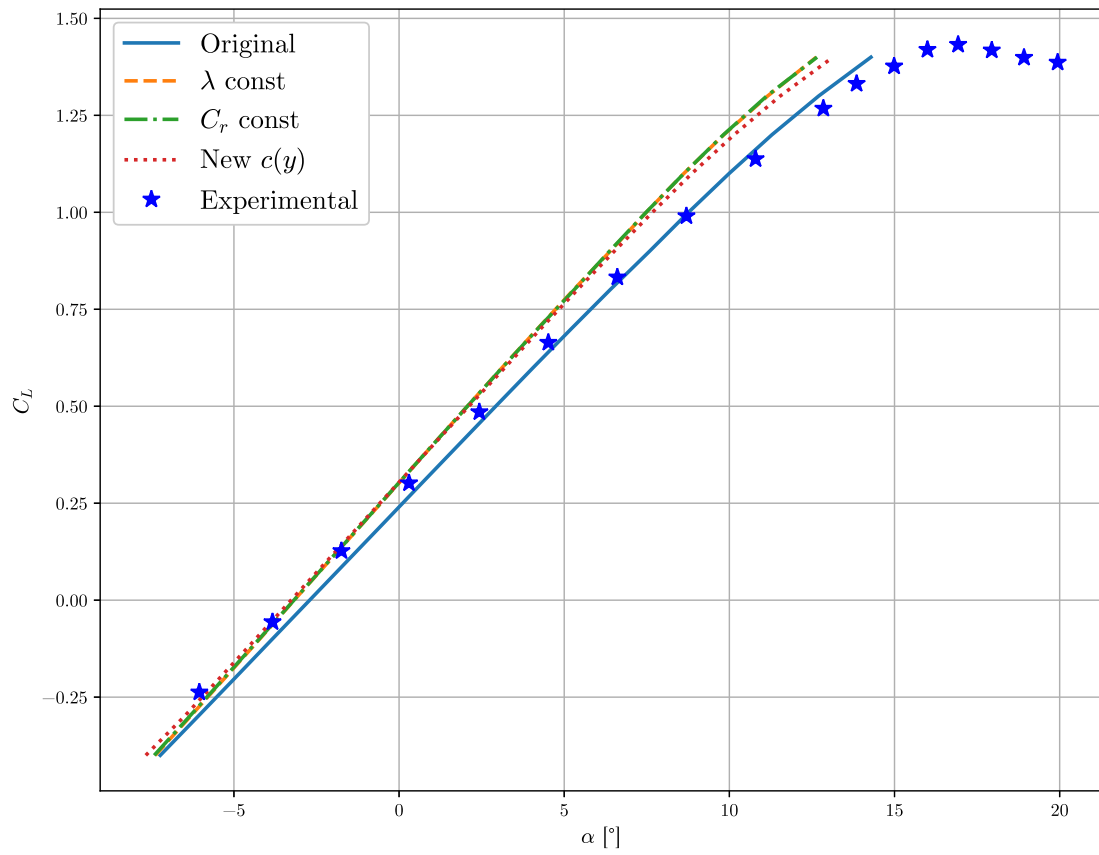
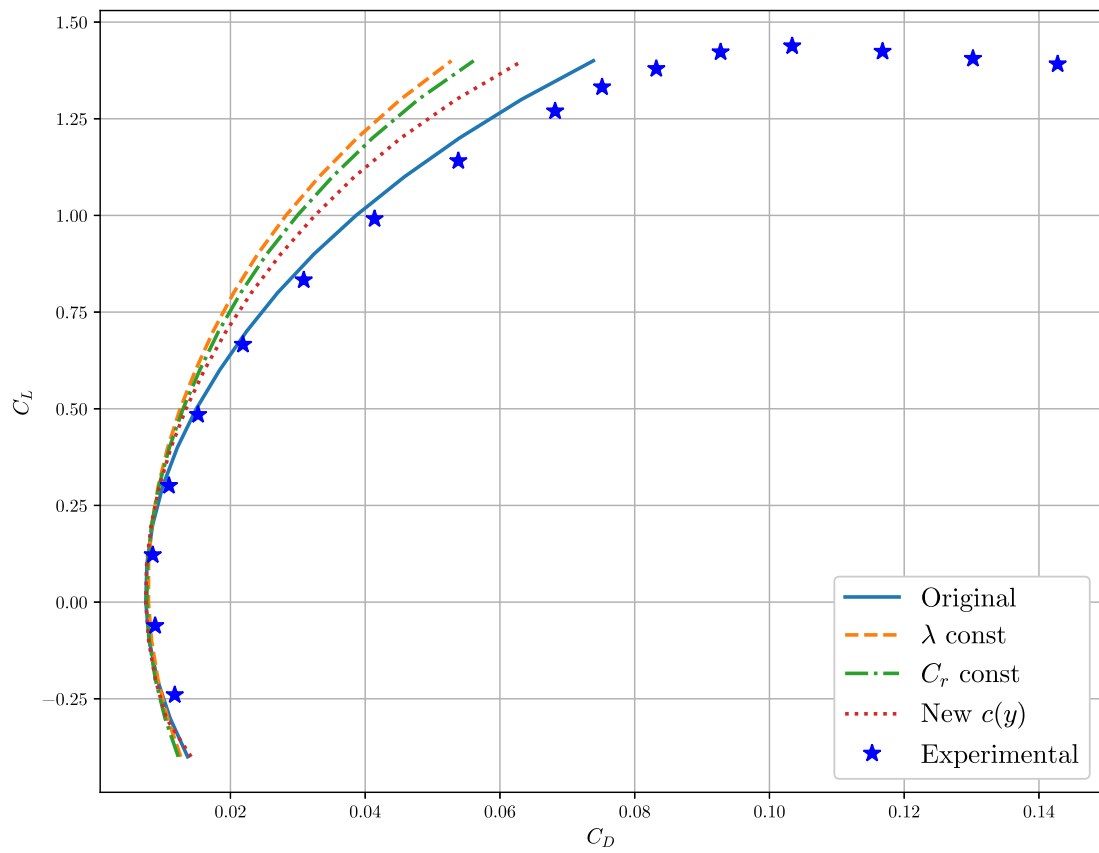


Figure 5.13: Lift distribution at $C_L = 0.3$

(a) $CL - \alpha$ 

(b) Polar

Figure 5.14: NACA Wing 2 aerodynamic data

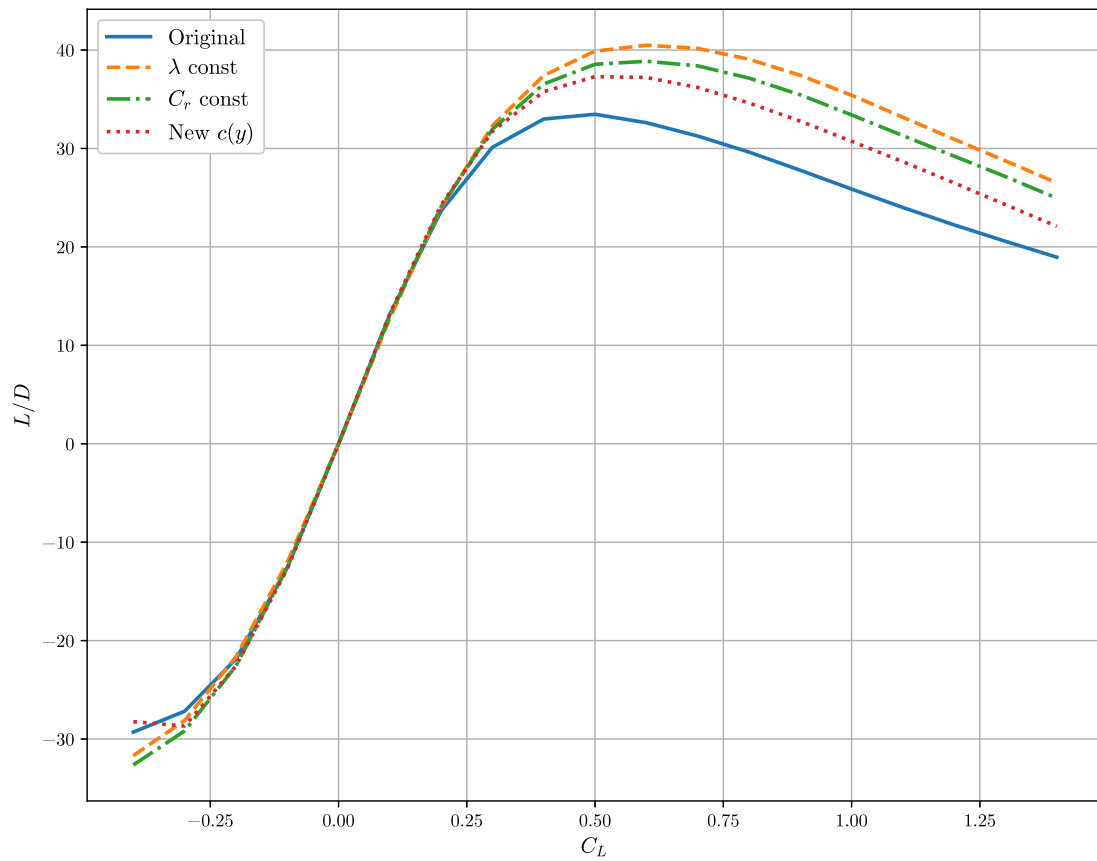
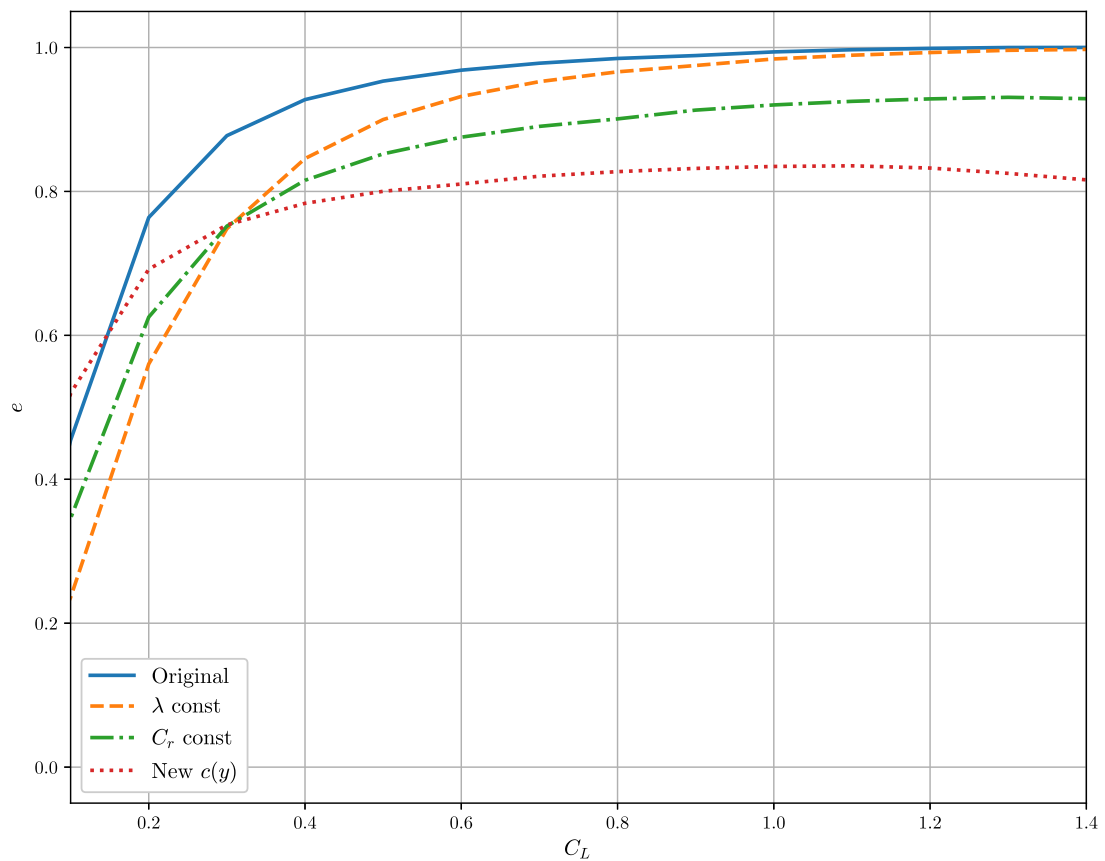
Figure 5.15: NACA Wing 2 L/D 

Figure 5.16: NACA Wing 2 Oswald Efficiency Factor

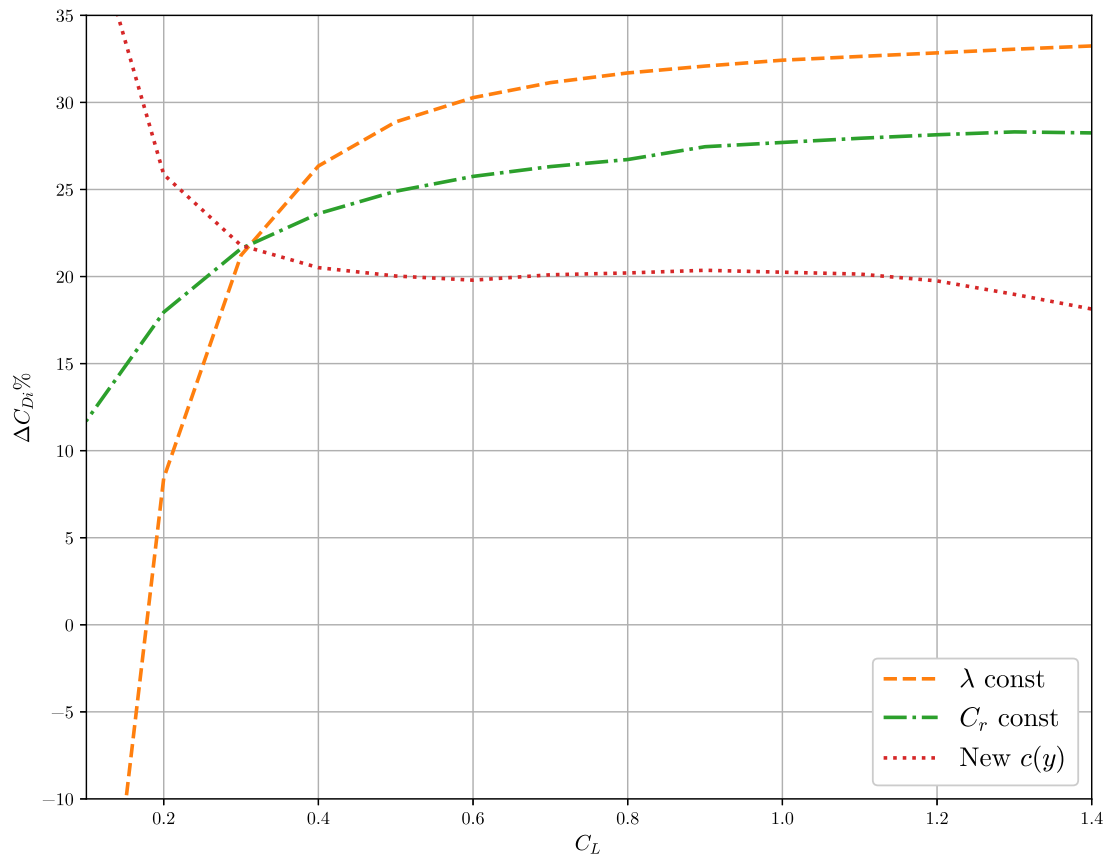


Figure 5.17: Relative C_{Di} variation for NACA Wing 2

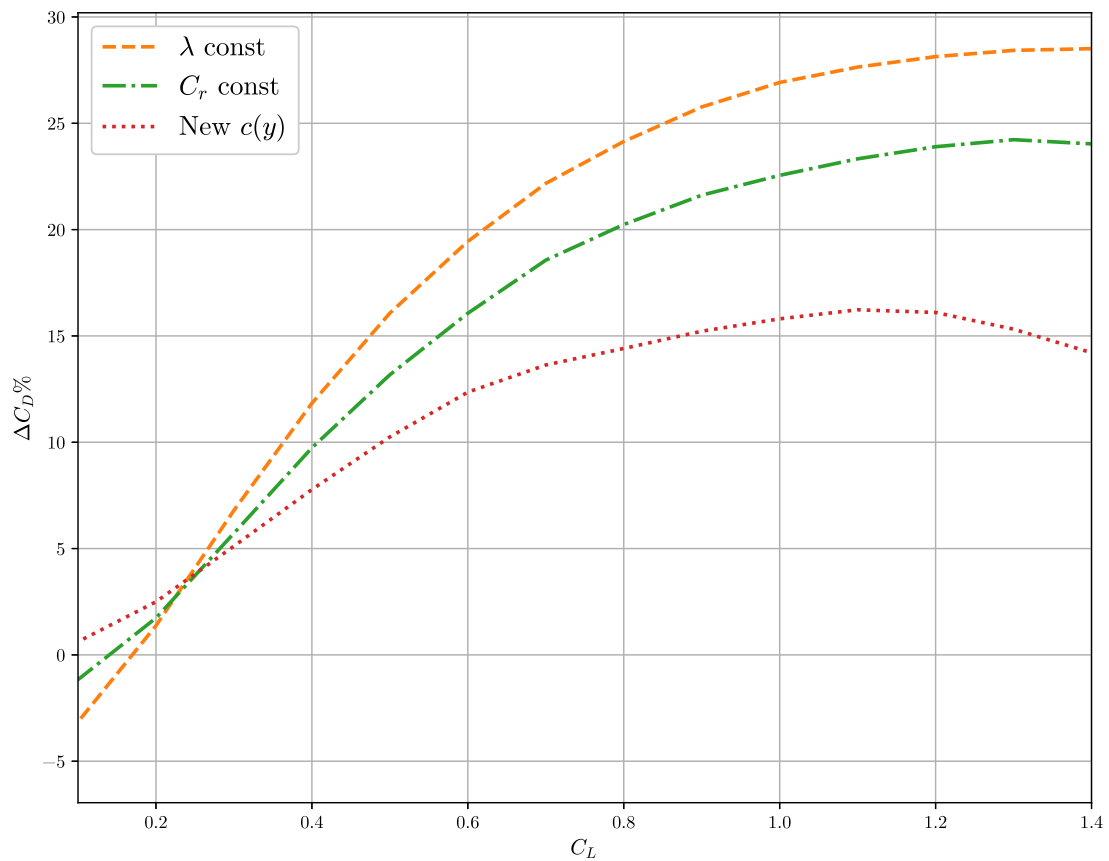


Figure 5.18: Relative C_D variation for NACA Wing 2

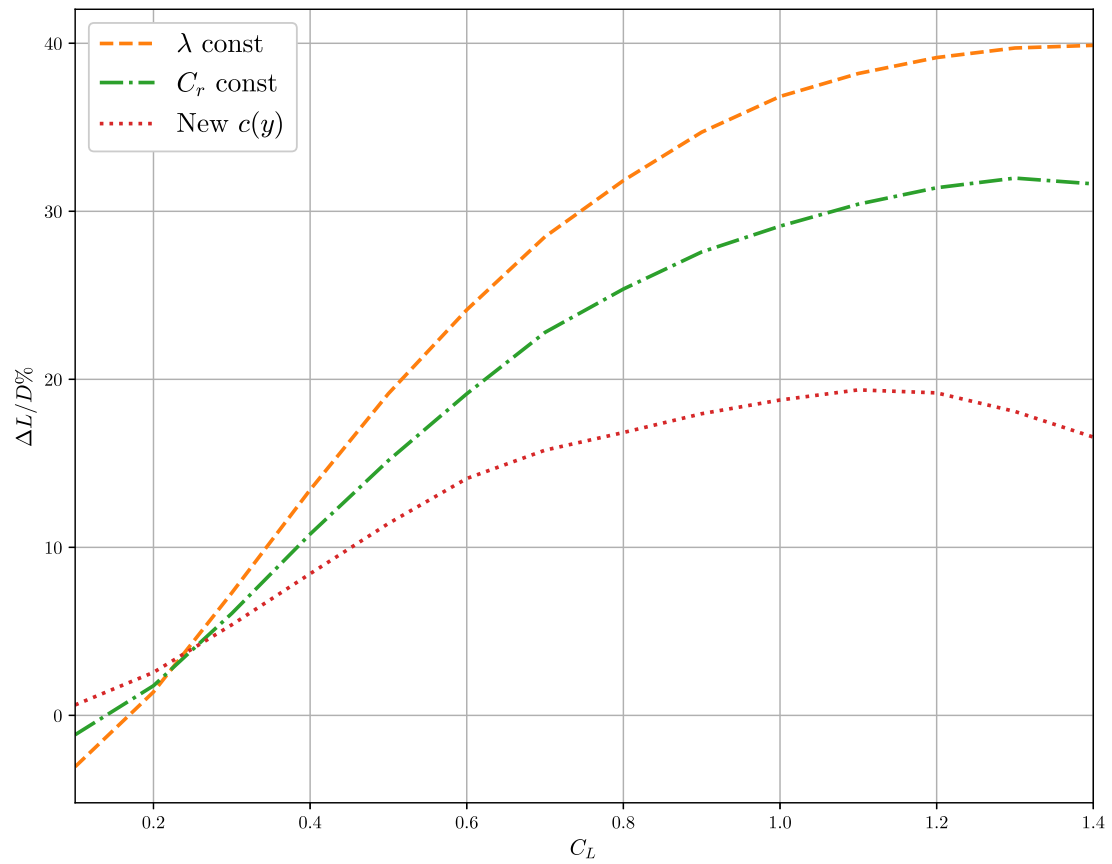


Figure 5.19: Relative L/D variation for NACA Wing 2

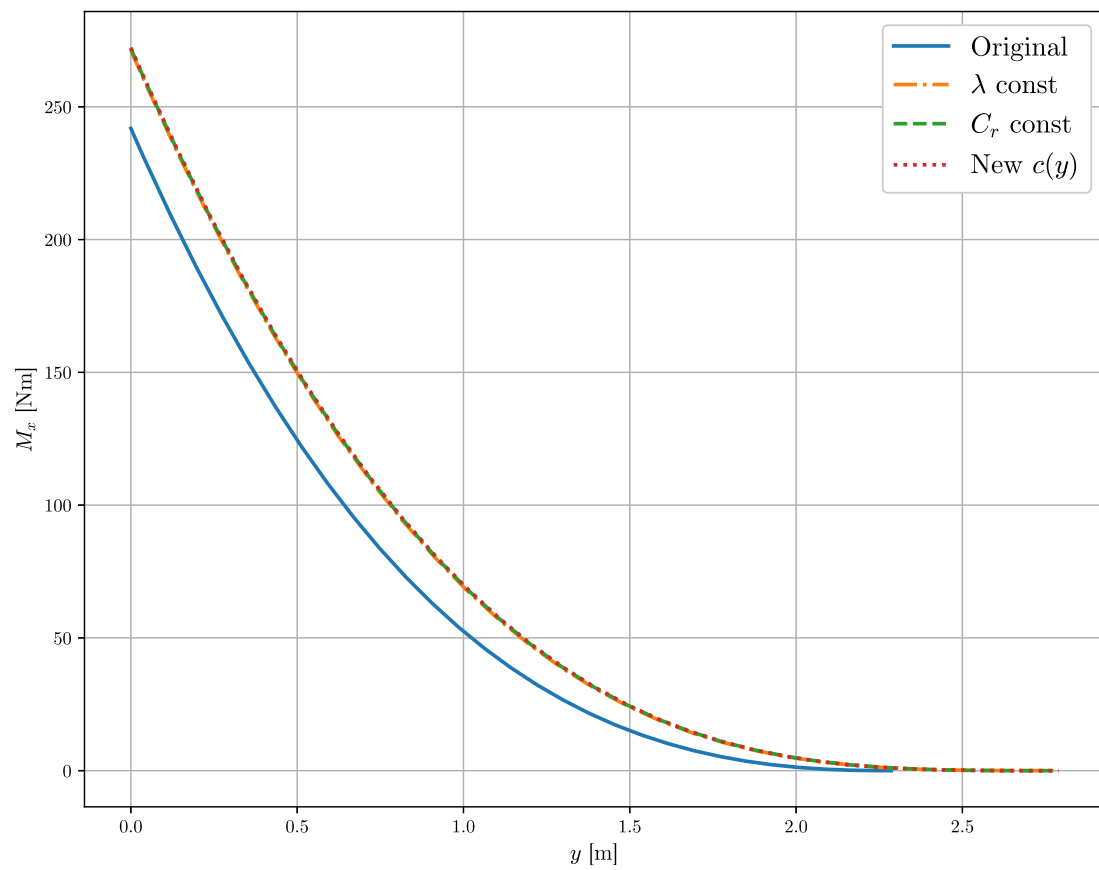


Figure 5.20: M_x at design point for NACA Wing 2

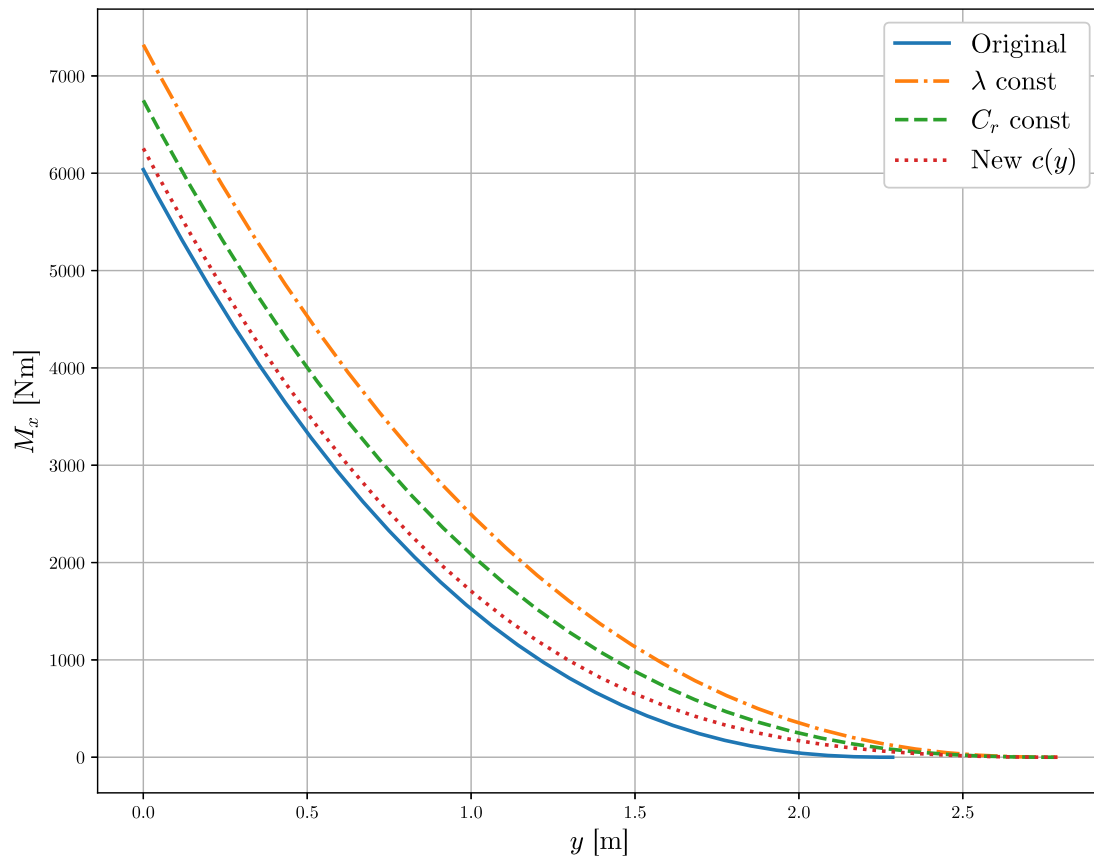


Figure 5.21: M_x at V_a for NACA Wing 2

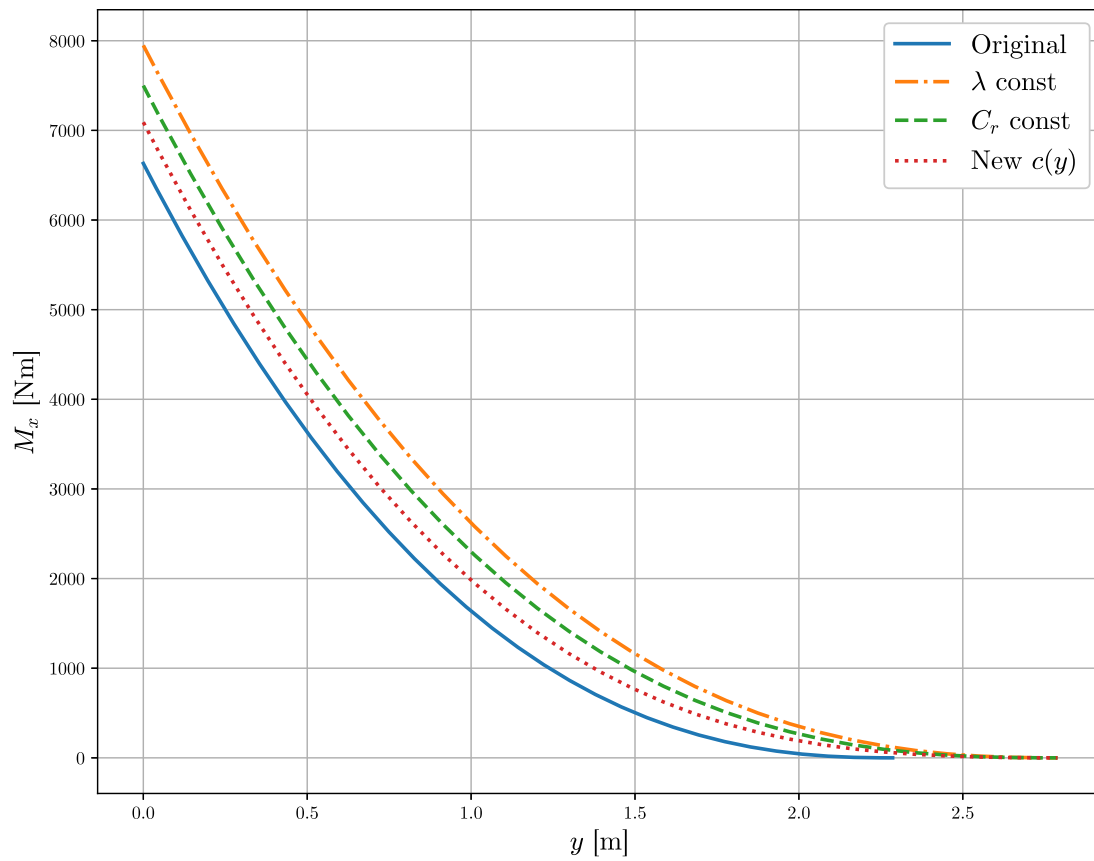


Figure 5.22: M_x at V_c for NACA Wing 2

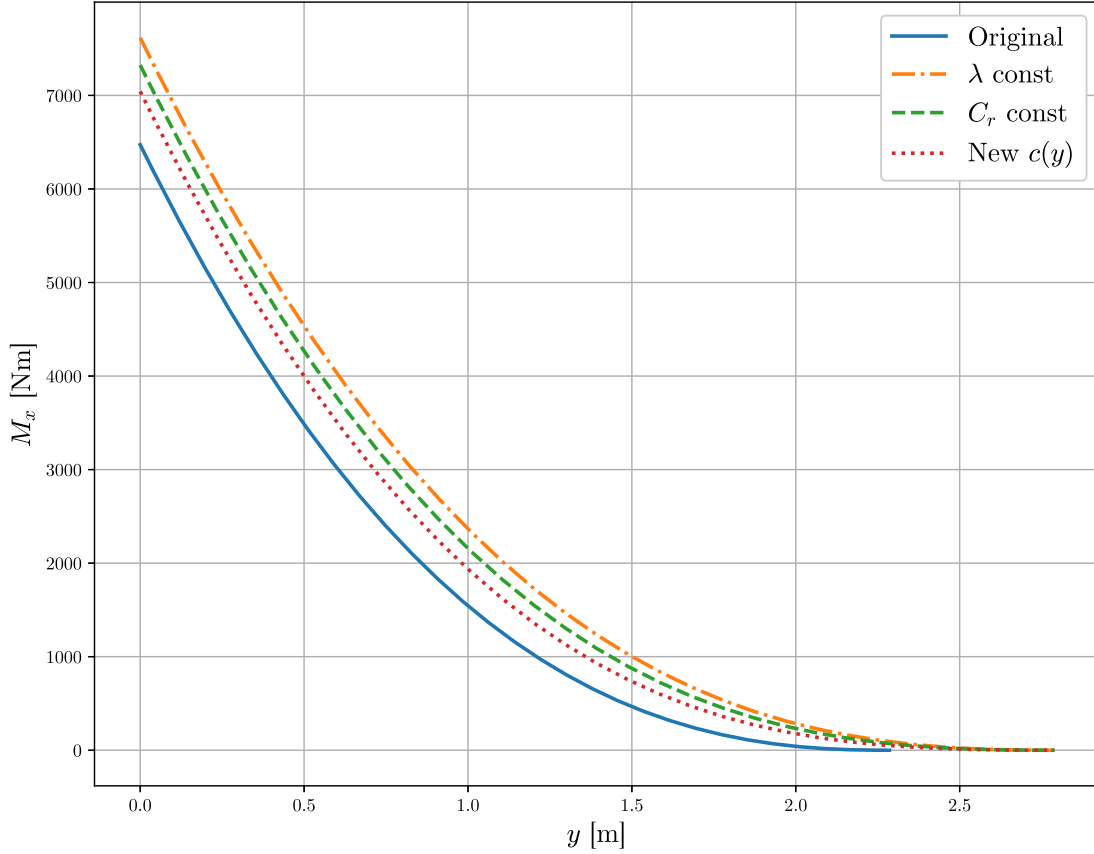


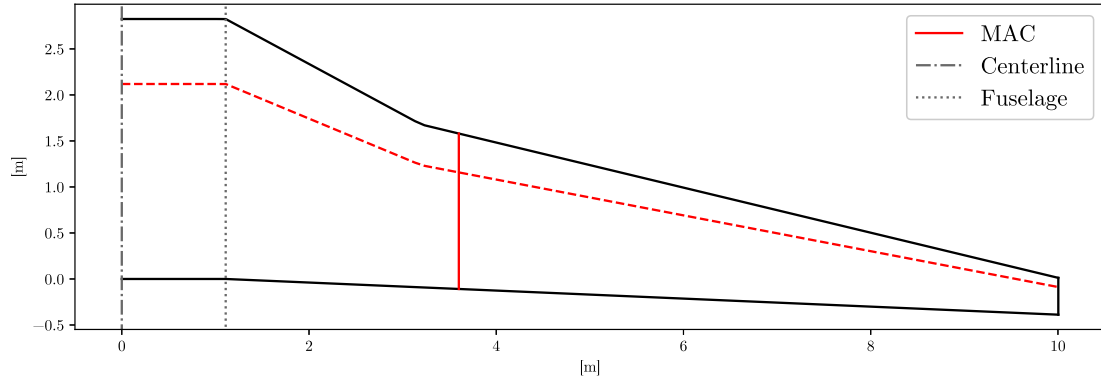
Figure 5.23: M_x at V_d for NACA Wing 2

5.2 PC-24 Wing

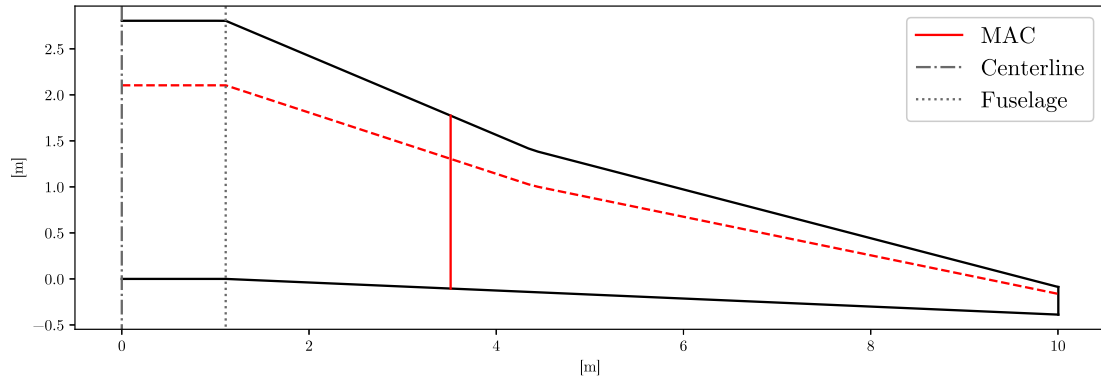
Three different wing designs are proposed to improve the existing configuration, all three maintaining the same S_{wing} and applying the redesign procedure developed in Ch. 4. The first configuration attempts to maintain similar proportions and leading edge sweep (Fig. 5.24a). The second one used a Particle-Swarm Optimization to find the ideal combination of kink position and tip chord that minimized wing weight (Fig. 5.24b). Finally the third configuration presents the "Type C" chord distribution of Sec. 4.5 (Fig. 5.24c), with a quadratic spline interpolation.

For all of them $C_{L,des}$ was set to 0.4, as a trade-off between high speed cruise and economy cruise. In this study however, due to limitations of the aerodynamic code, only the economy cruise condition will be studied ($Ma = 0.65$ at FL 450). The calculated twist distributions are shown in Fig. 5.25. For these wings the same airfoils of the original PC-24 are used.

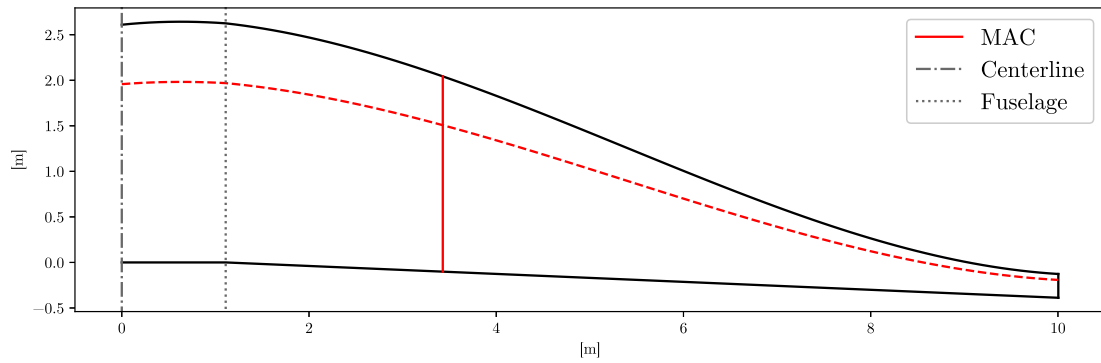
The aerodynamic polar was calculated in the economy cruise flight condition (Fig. 5.26) and was limited to $C_L = [0.0 - 0.9]$ for compressibility limitations. All three designs have less drag over most of the polar, however only the third one has less drag at zero lift. This provides extra L/D at higher cruise speeds, as shown in Fig. 5.27. The relative drag reduction trends also show this behavior (Fig. 5.29 and Fig. 5.30), where at $C_L = 0.2$ the third design has a peak of 15% in total



(a) Design 1 - Similar geometry



(b) Design 2 - PSO Optimized



(c) Design 3 - Type C chord distribution

Figure 5.24: New PC-24 wings

drag reduction. Everything else behaves according to what has been discussed so far. Above the design lift coefficient, the increased aspect ratio and the growing e further reduce induced drag.

Unlike the previous analyses, here the structural design velocities V_a , V_c and V_d are provided in the type certificate document of the PC-24[34], while the limit load factor was calculated with CS-23 regulations[35]. While the design load is calculated in the economy cruise conditions, the structural loads are calculated at ISA sea level. The obtained results are shown in Figures 5.32-5.35. Bending moments are similar for all three designs in all load cases except for V_a , where

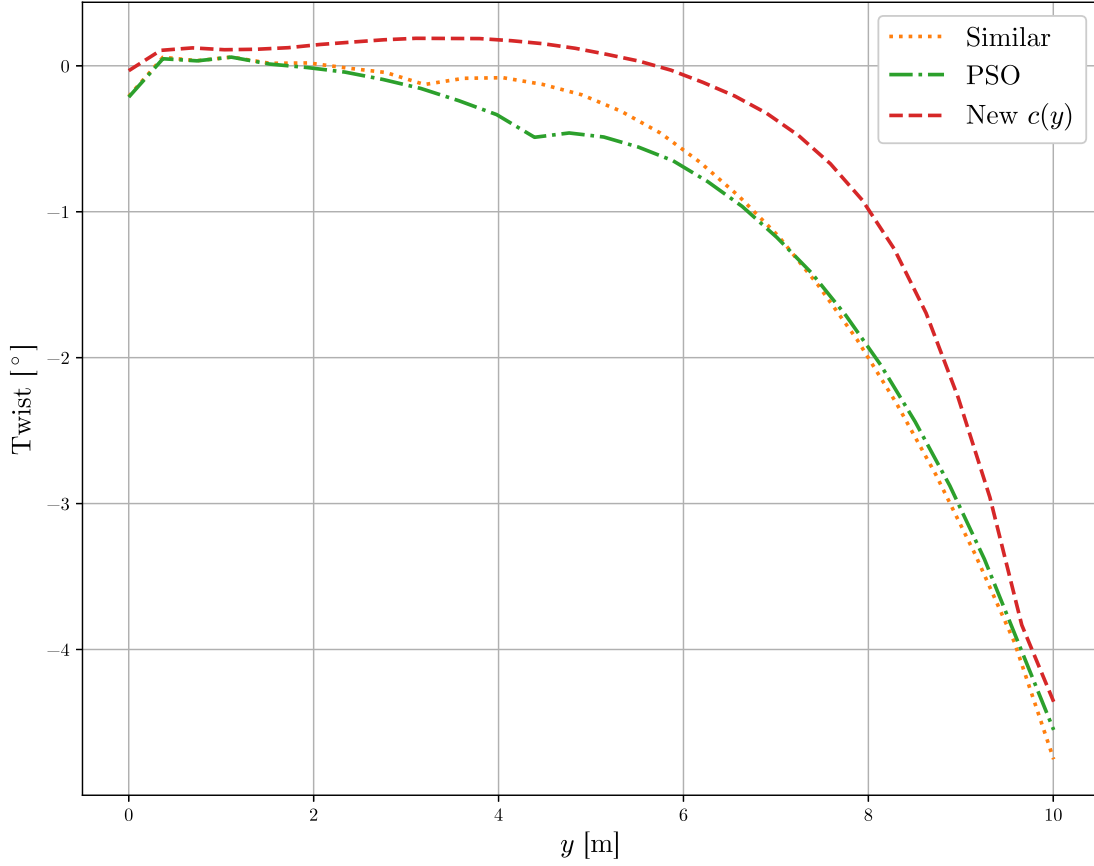


Figure 5.25: PC-24 Twist distributions

the third design is slightly better than the others. If the lift distributions at V_a is inspected (Fig. 5.36), it is clear that the third design has less loaded tips compared to the other two.

Now unlike the previous cases, it was found that the most critical flight condition was at V_a , and thus it is selected for computing masses. The calculated representative masses are presented in Tab. 5.5. While in the previous section the Type C chord distribution yielded a very small mass increment, for this wing the increment is still the smallest yet not insignificant (18.17 %). The cause of this is the thickness of the airfoils, as the PC-24 has thinner supercritical airfoils with t/c ranging from 0.14 at the root to 0.10 at the tip. This reduces the sectional moment of inertia and thus requires thicker (and heavier) panels.

	Mass	% Variation
Original	239.63 kg	-
Similar	359.66 kg	50.09
PSO	353.27 kg	47.42
New $c(y)$	283.17 kg	18.17

Table 5.5: Representative masses of new PC-24 wings

In Ch. 1 it was discussed how weight, propulsion and aerodynamics contribute to the fuel consumption and greenhouse emissions of an aircraft. Let's now evaluate how these three new designs may save fuel in cruise condition.

In order to incorporate the structural mass increase, the wing mass of Tab. B.2 was scaled with the representative mass increments from Tab. 5.5. For this calculation a payload of 6 passengers and one crew is assumed with fuel tanks half full. Then by considering the economy cruise flight conditions ($Ma = 0.65$ at FL450), the aerodynamic data of Tab. 5.6 was calculated by including the drag of the original tail, fuselage and nacelles in trimmed flight.

	C_L	C_D	L/D
Original	0.5295	0.030136	17.57
Similar	0.5722	0.029791	19.21
PSO	0.5655	0.029591	19.11
New $c(y)$	0.5438	0.028637	18.99

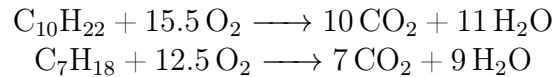
Table 5.6: Aerodynamic data in cruise

With Eq. 1.1 it is possible to estimate the burned fuel. Assuming a specific fuel consumption $SFC = 0.456$ lb/h lbf, the burned fuel mass is calculated for three different flight ranges in Tab. 5.7. With the new chord distribution and the bell spanload it is possible to obtain a 4.8% fuel saving, while the other two designs, even if with higher L/D , produce marginal improvements (0.64% and 0.83%).

Range	500 nm		1000 nm		1500 nm	
Original	208.44 kg		424.26 kg		647.72 kg	
Similar	207.42 kg	-0.49%	421.55 kg	-0.64%	642.61 kg	-0.79%
PSO	207.00 kg	-0.69%	420.72 kg	-0.83%	641.40 kg	-0.98%
New $c(y)$	198.61 kg	-4.71%	403.73 kg	-4.84%	615.56 kg	-4.97%

Table 5.7: Calculated burned fuel for new PC-24 wings.

Let's now evaluate the avoided CO_2 emissions. Jet fuels are kerosene-based compounds, whose main components are n-Decane $C_{10}H_{22}$ (89%) and Toluene C_7H_{18} (11%). Their balanced combustion reactions, ignoring any combustion inefficiency, are:



It is possible to calculate how much CO_2 is released by converting the molar ratios into mass ratios:

$$\frac{m_{CO_2}}{m_{fuel}} = 0.89 \frac{10M_{CO_2}}{M_{C_{10}H_{22}}} + 0.11 \frac{7M_{CO_2}}{M_{C_7H_{18}}} = 3.089 \quad (5.1)$$

therefore 1 kg of burned fuel would release about 3 kg of CO₂. If we assume this aircraft flies 1000nm 6 times a week, then it is possible to estimate how much greenhouse emission is avoided each year (Tab. 5.8). With a fuel saving of 4.8%, about 20 tons of CO₂ were not released into the atmosphere each year.

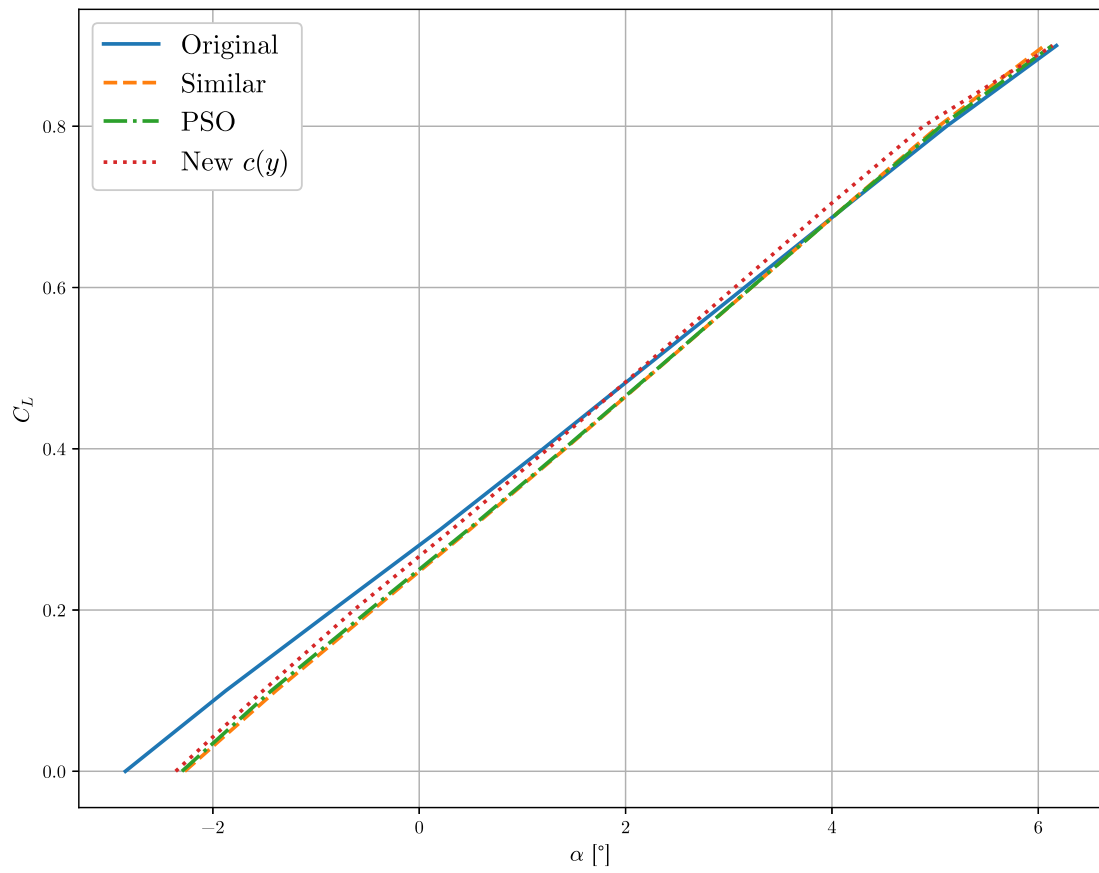
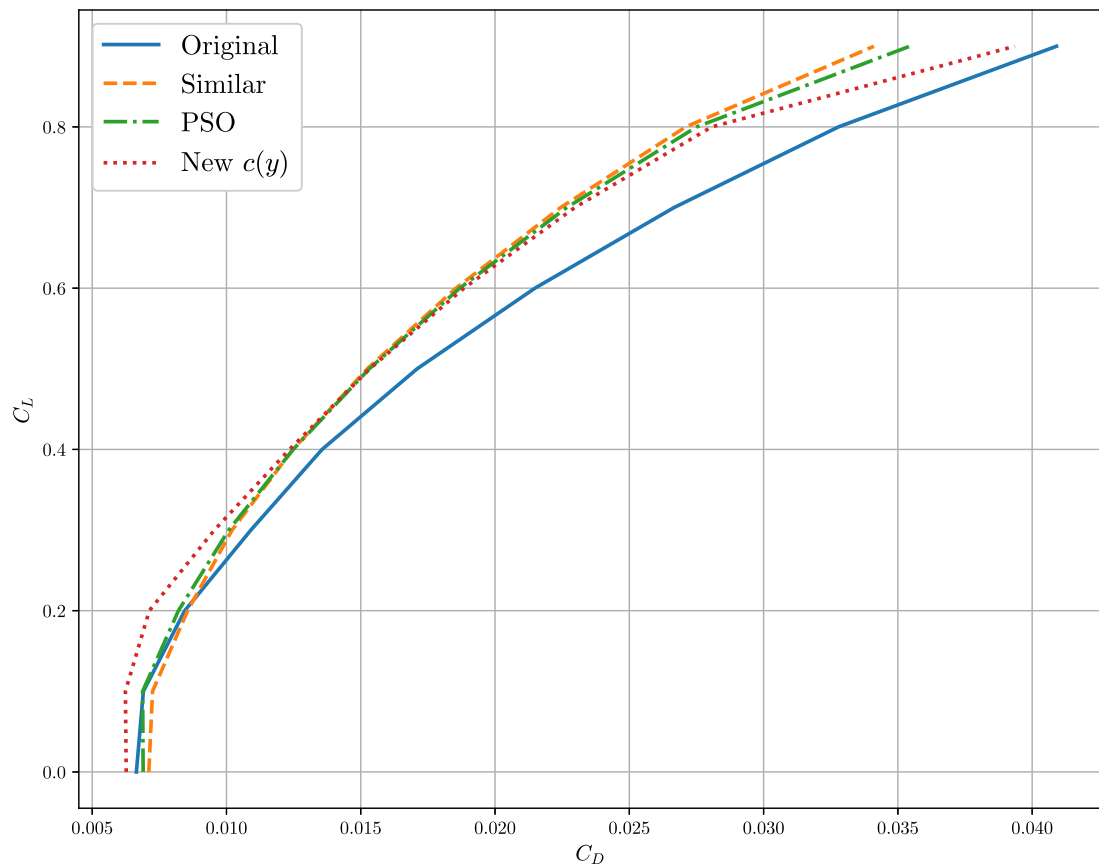
	One Trip		Per Week		Per Year	
Original	1311 kg		7863 kg		424615 kg	
Similar	1302 kg	-8 kg	7813 kg	-50 kg	421902 kg	-2712 kg
PSO	1300 kg	-11 kg	7798 kg	-64 kg	421072 kg	-3542 kg
New $c(y)$	1247 kg	-63 kg	7483 kg	-380 kg	404069 kg	-20546 kg

Table 5.8: Estimated saved greenhouse emissions with new wing design, assuming 6 flights of 1000nm per week.

Another benefit of less burned fuel is the reduction of operating costs. According to IATA, on average fuel accounts for 23.7 % of the total operating expenses of an airline[36]. Let's therefore calculate how much US\$ are saved by these new wing design. Currently (March 2020) the average Jet-A price is 0.42877 US\$/kg[37]. Assuming again 6 flights of 1000nm per week, it is possible to save 2852 US\$ per year of fuel costs with the third wing design (Tab. 5.9).

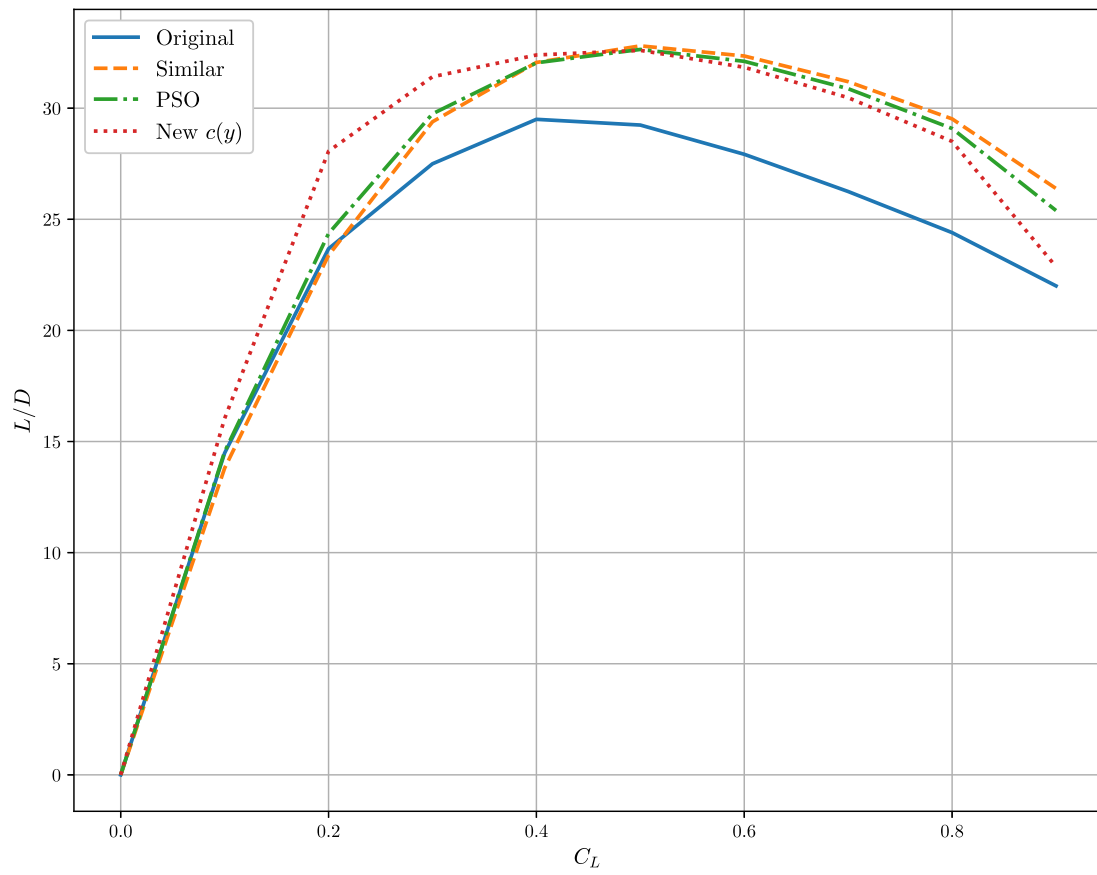
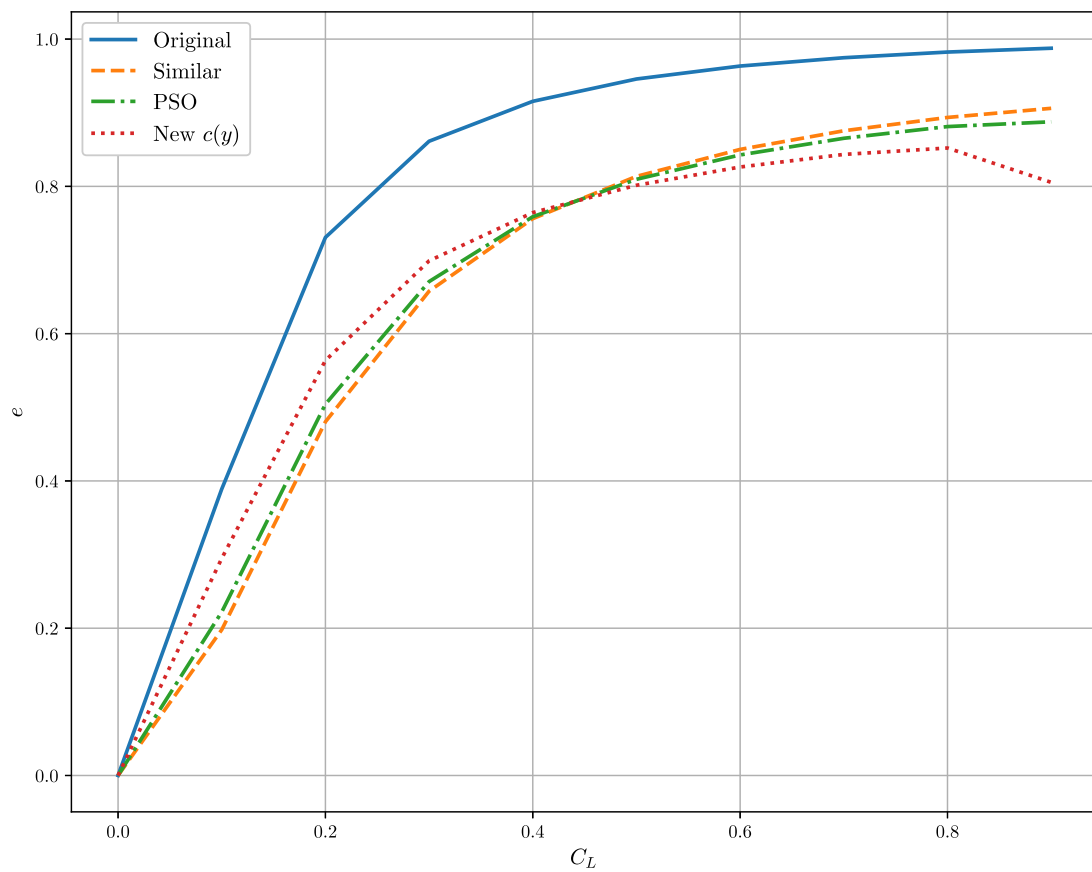
	One Trip		Per Week		Per Year	
Original	181.91 \$		1091.46 \$		58938.83 \$	
Similar	180.75 \$	-1.16 \$	1084.49 \$	-6.97 \$	58562.35 \$	-376.48 \$
PSO	180.39 \$	-1.52 \$	1082.35 \$	-9.11 \$	58447.12 \$	-491.71 \$
New $c(y)$	173.11 \$	-8.80 \$	1038.65 \$	-52.81 \$	56086.96 \$	-2851.87 \$

Table 5.9: Estimated cost reduction with new wing design, assuming 6 flights of 1000nm per week.

(a) $CL - \alpha$ 

(b) Polar

Figure 5.26: Aerodynamic data of PC-24 wings

**Figure 5.27:** L/D of PC-24 wings**Figure 5.28:** Oswald efficiency factor of PC-24 wings

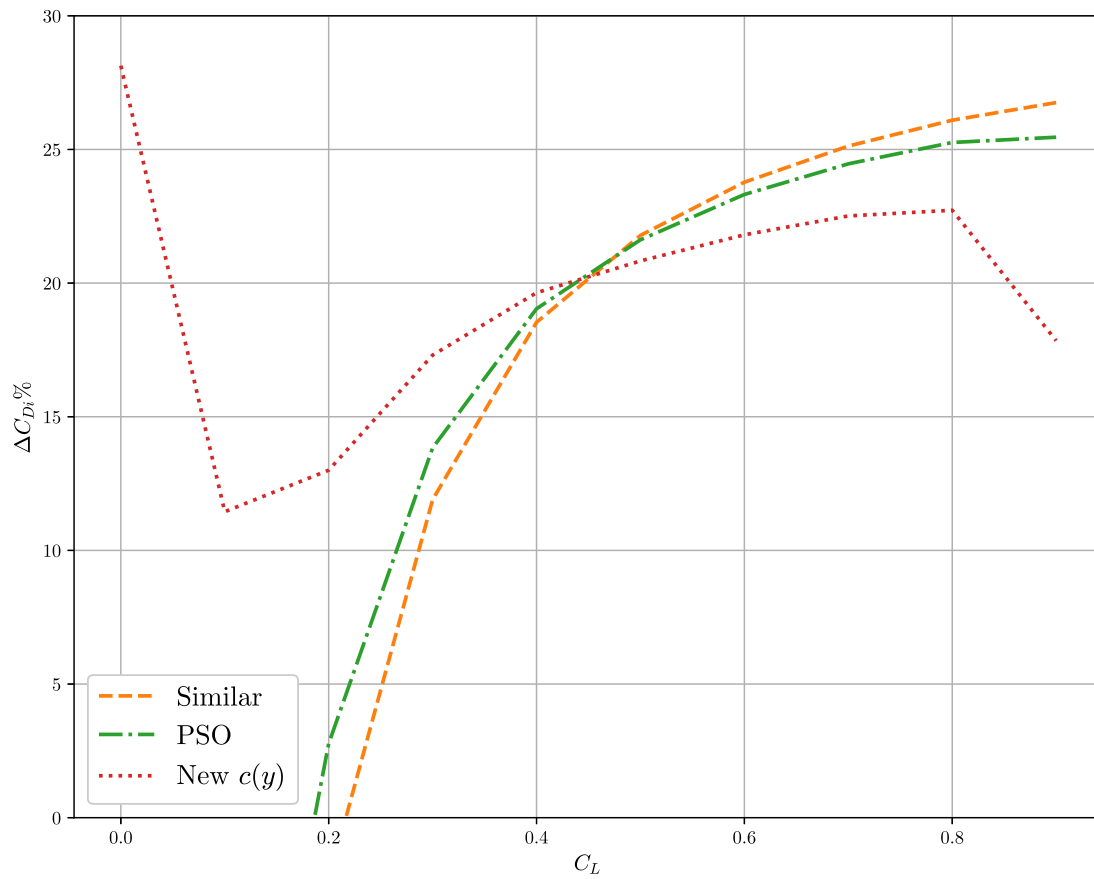


Figure 5.29: Relative C_{Di} variation for new PC-24 wings

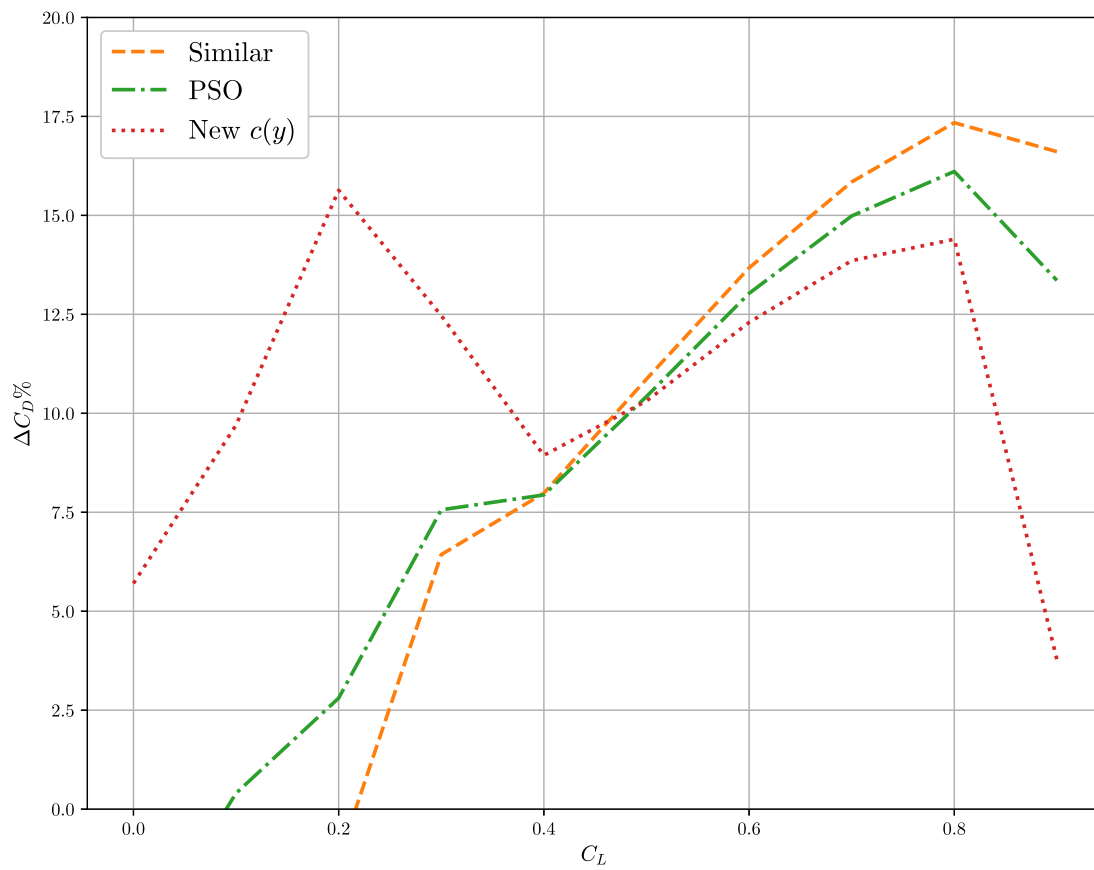


Figure 5.30: Relative C_D variation for new PC-24 wings

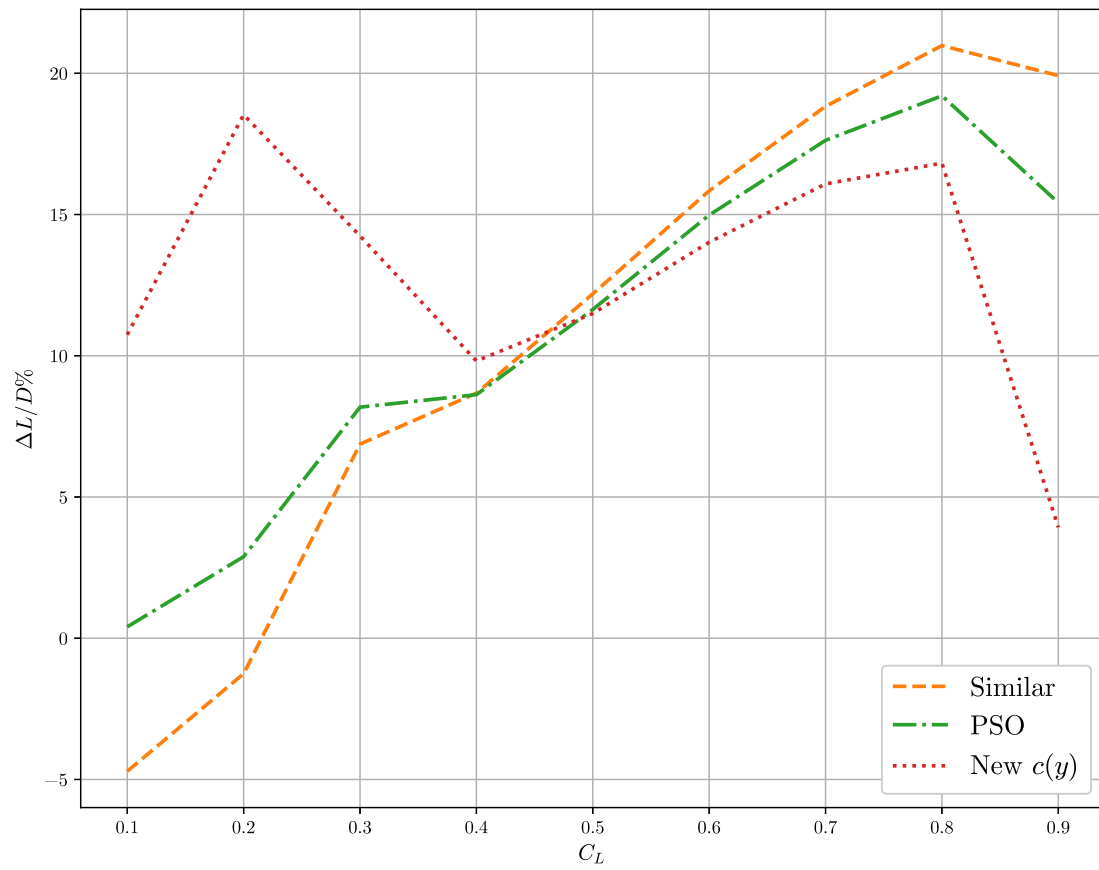


Figure 5.31: Relative L/D variation for new PC-24 wings

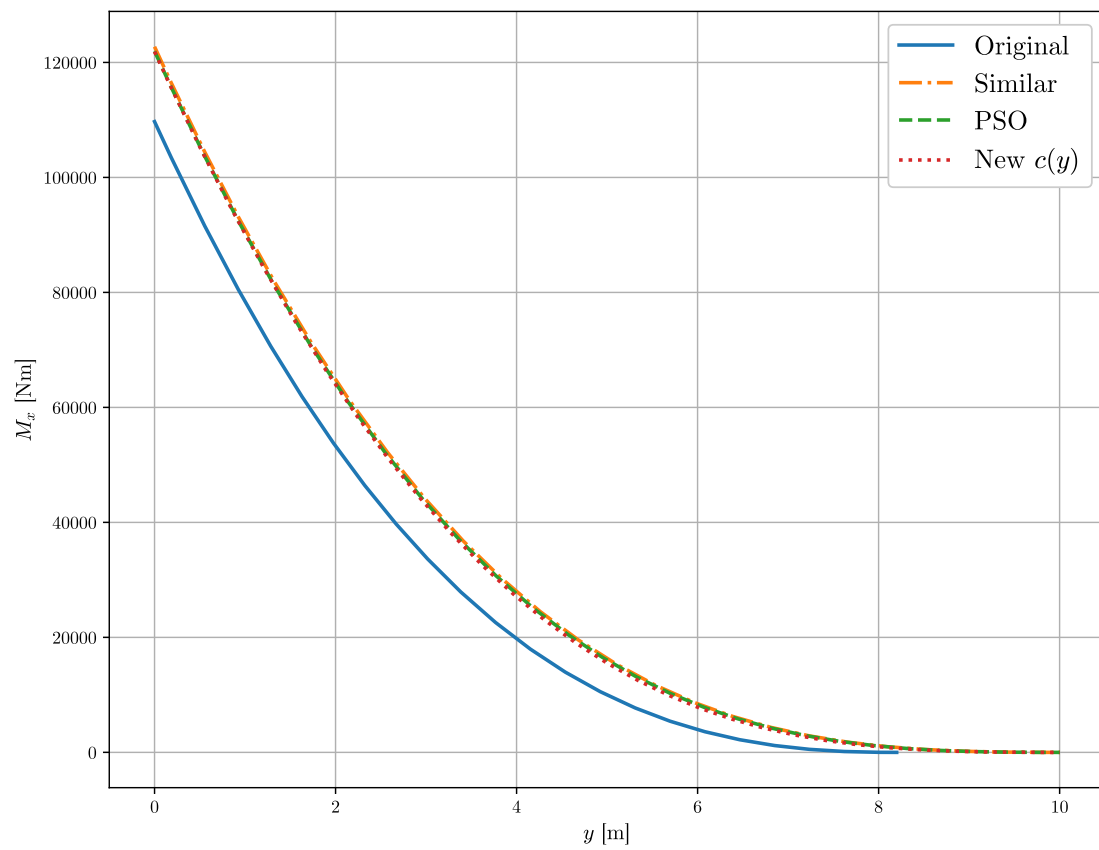


Figure 5.32: M_x at design point for PC-24 wings

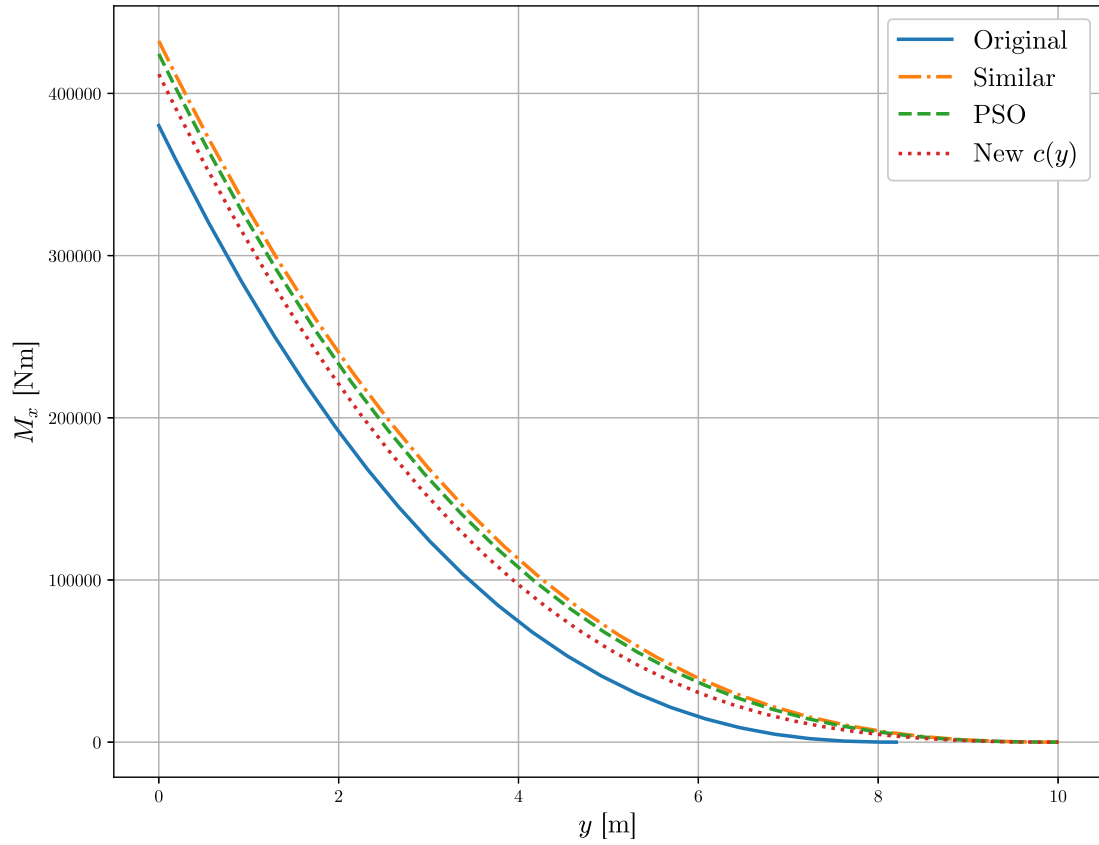


Figure 5.33: M_x at V_a for PC-24 wings

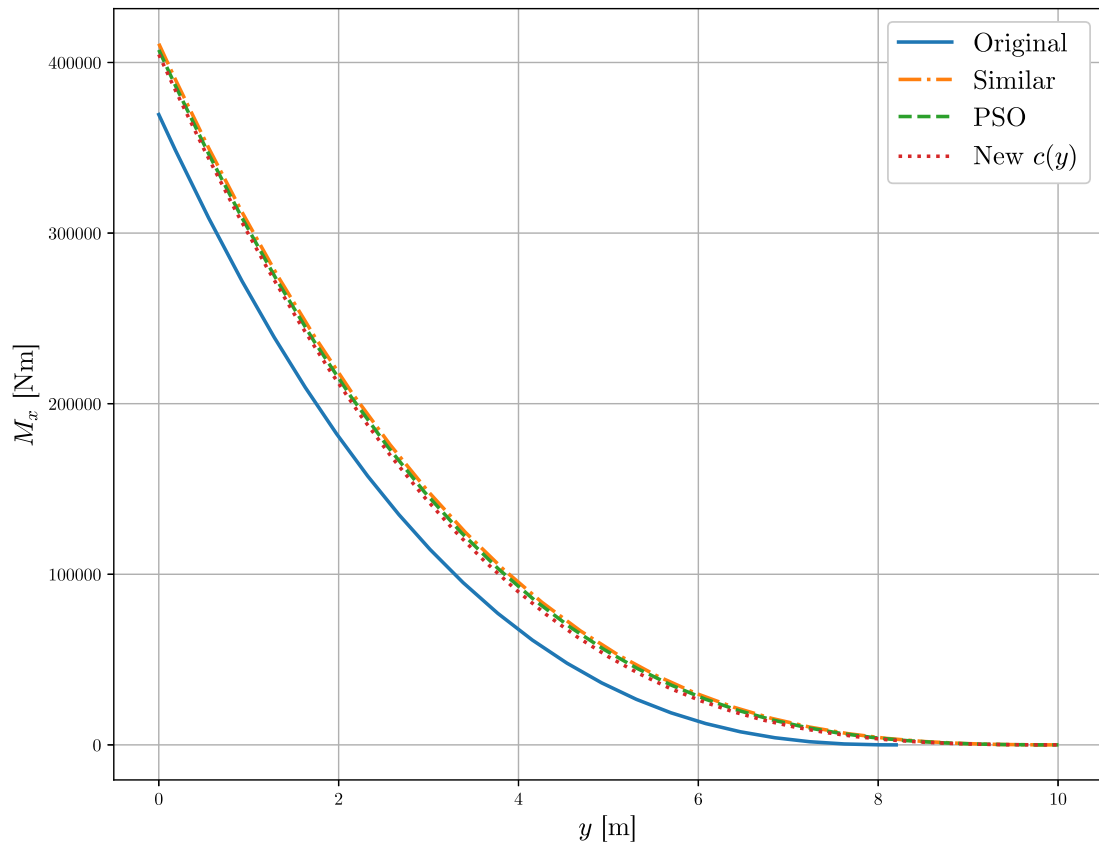


Figure 5.34: M_x at V_c for PC-24 wings

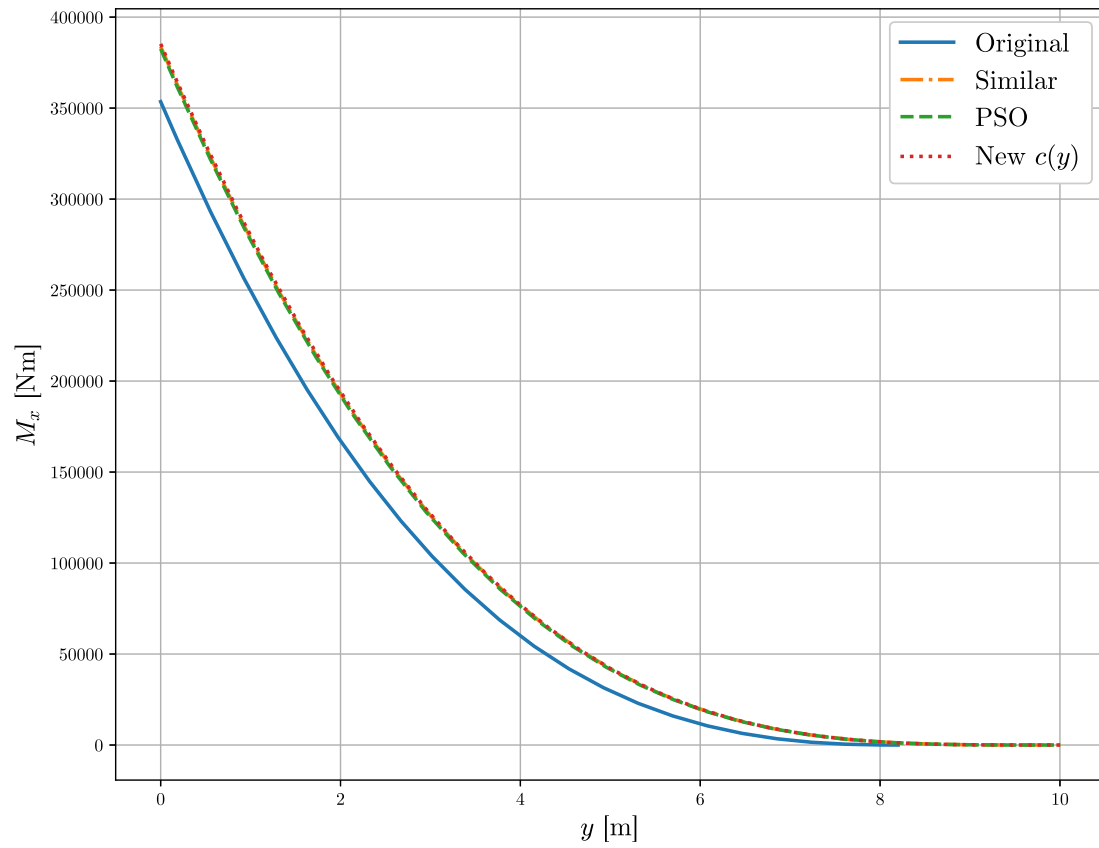


Figure 5.35: M_x at V_d for PC-24 wings

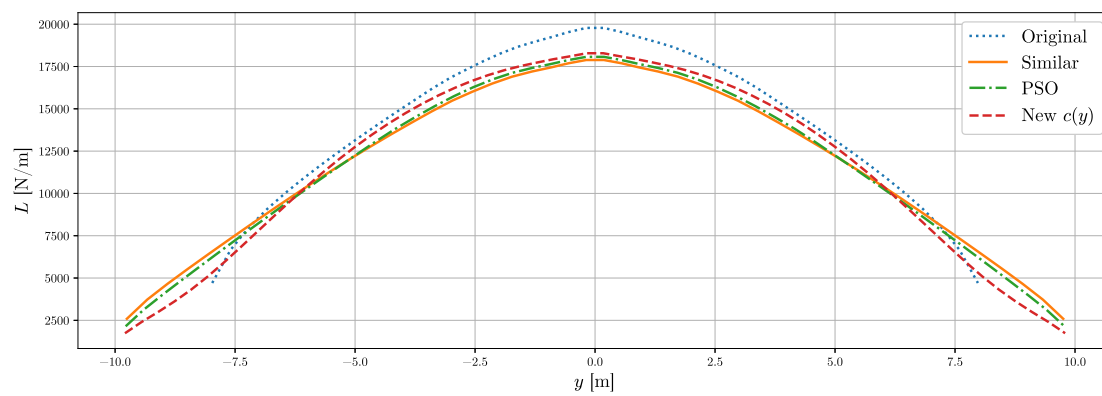


Figure 5.36: Lift distribution at V_a for PC-24 wings

Chapter 6

Vee-Tail Design

Compared to conventional tails, a vee-tail design has potential benefits in reduction of drag and weight because it employs two lifting surfaces instead of three. Despite this, few commercial airplanes had been built with this configuration like the 1947 Beechcraft Bonanza. The Vee-tail arrangement has also been used in military application with long-endurance unmanned aircrafts (UAV) such as the MQ-1 Predator and the MQ-9 Reaper.

The classical method proposed by textbook authors such as Roskam [33] refer to a methodology from a 1945 NACA report [38]. The procedure involves considering the projections of the vee-tail surfaces on the vertical and horizontal plane as a conventional tail. Then the standard tail volume coefficient is used to rapidly estimate the size of these projected surfaces. However tail volume method was developed for conventional tails using statistical data. For unconventional tail layouts, it is less accurate and does not provide insight on the effects of the geometry onto the aircraft stability and control.

Sanchez-Carmona and Cuerno-Rejado [25] proposed a different approach to design a vee-tail for commercial and general aviation aircraft. Commercial airplanes are constrained by airworthiness regulations by FAA or EASA. Guidelines then can be used as a starting point to define the vee-tail geometry and optimize its size while keeping the aircraft within regulations.

6.1 Regulation Requirements

The tail must be designed to keep the aircraft stable and capable of trimming the airplane in all the flight envelope. This includes trimming and controlling the aircraft after an engine failure and when lateral gusts are present. EASA regulations provide standard requirements for these scenarios. For this study the requirements are extracted from EASA CS-23 regulations as the airplane studied carries such certification. Nonetheless, these requirements are similar to the CS-25/FAR-25 requirements from Sanchez-Carmona's study[25].

Stability requirements are prescribed in articles 23-171 and 23-181, for which static stability must be assessed in all flight phases, for both longitudinal and directional planes. Alongside these, all dynamic modes must be stable and damped. Specifically from article 23-181, Dutch roll and Phugoid modes must be damped

	Velocity	Altitude	Payload	Configuration
Cruise (Stability and Trim)	$M = 0.65$	45000 ft	1 Crew + 6 Pax with 50% fuel	No flaps Cruise Thrust
Landing with 90° Crosswind	$1.25 V_{stall}$	50 ft	MTOW	Full Flaps (33°) Idle Engines
Take-Off with One Engine Inoperative	$1.2 V_{stall}$	50 ft	MLW	TO Flaps (8°) Max Thrust

Table 6.1: Test flight conditions

with a 1/10 amplitude in 7 cycles. Static stability is then imposed in the design process, while dynamic stability is assessed later because it requires a new mass estimation of the whole airplane.

Pilots should be able to maintain control of the airplane under critical flight conditions, such as gusts or asymmetric thrust. Regulations prescribe standard cases for which the airplane must be able to comply. In article 23-233, the airplane must be able to maintain controllability in take-off and landing with a 90° crosswind of no less 20% of stalling speed. Usually this is critical in landing, therefore it will be assessed in the landing configuration. Unlike CS-25, CS-23 does not prescribe a critical lateral control requirement due to engine failure in the take off condition, which usually is the most likely situation, but in-flight (article 23.147). However trim regulations (article 23-161), prescribe the airplane should be trimmed without excessive force in the take-off condition with one engine inoperative. These two will be overlapped to synthesize a constraint regarding directional controllability with a failed engine. The same trim regulations prescribe the airplane should be laterally trimmed with less than 5° of bank in all flight conditions, therefore including gusts and engine failures.

Table 6.1 summarizes the flight conditions in which the stability and controllability of the new tail and wing configuration will be assessed. The cruise condition was selected according to available data for the airplane to be used in this study (the Pilatus PC-24).

6.2 Rigid model equations

Analysis of the new tail requires the use of semi-empirical methods alongside the VLM framework. This is to account the lift and pitching moment effects of flaps in landing and take off configuration. The wing and fuselage aerodynamic derivatives are not evaluated with semi-empirical methods due to the non-elliptic lift distribution. A 3D Panel code, VSPAero from NASA's OpenVSP software package[39], is used for estimating the linear stability derivatives of the wing and fuselage. The adopted wing configuration is design 3 from Sec. 5.2. The wingbody 3D panel model used is shown in Fig. 6.1.

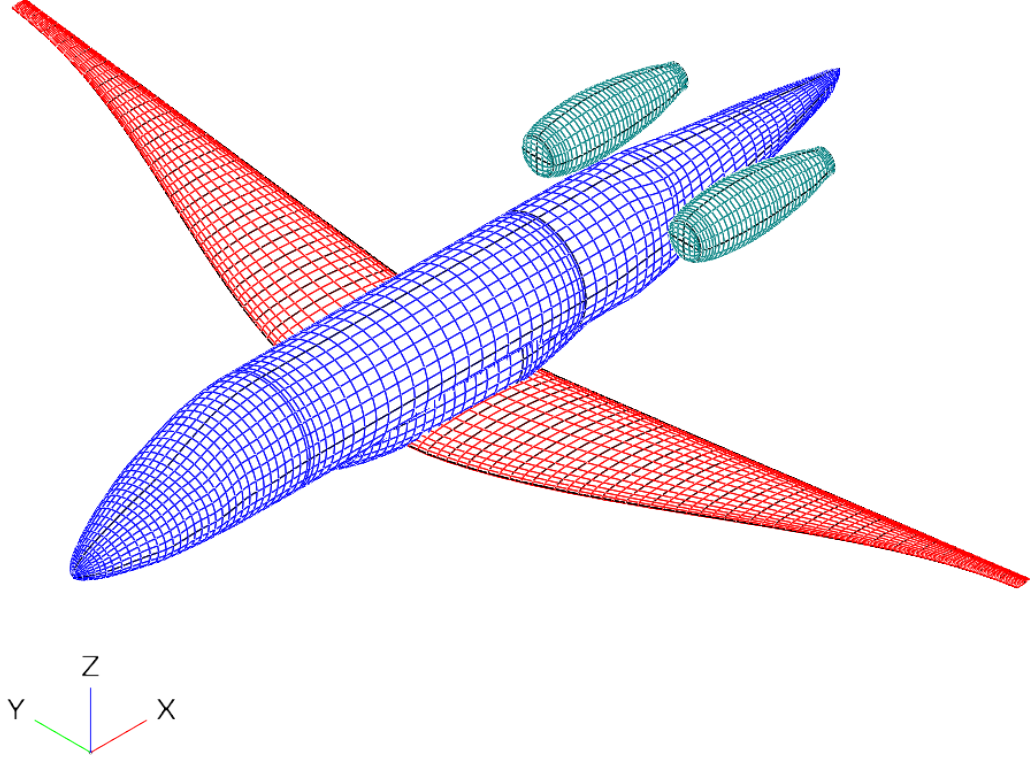


Figure 6.1: PC-24 wingbody 3D Panel model

The aerodynamic derivatives of the tail are estimated by computing a baseline case and perturbing it. The VLM model is composed of tail and wing to estimate the effect of downwash. Finite differences are used to estimate the aerodynamic derivatives of the tail.

To evaluate the tail characteristics, longitudinal and lateral momentum equilibrium equations have to be solved. For a rear-tailed aircraft, the pitch equation is as follows:

$$qS_w c_{MAC} C_{m,wb}(\alpha, \delta_f) - \eta_q qS_w l_{TH} C_{L,t}(\alpha, \delta_e) + Tz_e = 0 \quad (6.1)$$

where $q = \frac{1}{2}\rho V^2$. $C_{m,wb}$ is the pitching moment coefficient of the whole airplane without the tail, here defined as the *wingbody* configuration. The tail lift coefficient $C_{L,t}$ is non-dimensional with respect to the aircraft wing S_w and the tail dynamic pressure $\eta_q q$. The VLM code does indeed estimate the downwash angle effect but does not account for the turbulent flow behind the wing. Hence this effect is modeled with the coefficient $\eta_q = \frac{q_t}{q}$, using standard USAF Datcom semi-empirical

methods. The thrust pitching effect is also included in this analysis with z_e being the distance from the aircraft center of gravity of the thrust-line. Note that δ_e is assumed positive if it generates a nose-down pitching moment.

The wingbody pitching moment coefficient is defined in Eq. 6.2. It is a sum of the pitching moment coefficient of the wingbody at the aerodynamic center, the contribution of the deflection of wing flaps and the position of the center of gravity x_{CG} from the wing's MAC aerodynamic center x_{ac} . In this study DATCOM methods[40] have been used for estimating the flaps pitching effect.

$$C_{m,wb}(\alpha, \delta_f) = C_{m0,wb} + \Delta C_{m,flaps}(\delta_f) + C_L \frac{x_{cg} - x_{ac}}{c_{MAC}} \quad (6.2)$$

C_L is the lift of the entire airplane, which follows from vertical forces equilibrium. It has to be satisfied with the pitching moment equilibrium to obtain longitudinal trim. Thus an additional equation must be solved alongside Eq 6.1:

$$nW = qS_w [C_{L,wb}(\alpha) + \Delta C_{L,flaps}(\delta_f) + \eta_q C_{L,t}(\alpha, \delta_e)] \quad (6.3)$$

where W is the weight of the airplane and n is the load factor in the flight condition at study. As a first approximation, the additional lift from flaps is assumed to be a function of flaps deflection only, that is $\Delta C_{L,f}$.

The expression in square brackets of Eq. 6.3 is the total lift coefficient of the airplane, with its dependencies on the airplane angle of attack and elevator deflection indicated. By combining Eq. 6.2 and 6.3 and solving simultaneously, it is possible to obtain the trim values of angle of attack and elevator deflection. To obtain this, characteristic equations for the lift and pitching moment coefficients have to be provided. In this study we will assume a linear aerodynamic model with aerodynamic derivatives, such as for the tail lift coefficient:

$$C_{L,t}(\alpha, \delta_e) = C_{L,t0} + C_{L,t\alpha}\alpha + C_{L,t\delta}\delta_e \quad (6.4)$$

for which finite differences or semi-empirical methods can be used to calculate the derivatives. This set of equations is finally solved with Newton-Raphson.

For the lateral equilibrium, a similar approach is adopted. Forces and moments applied on the aircraft are shown in Fig. 6.2, from which the lateral equations are developed. The lateral momentum equilibrium is as follows:

$$qS_w b_w C_{n,wb}(\beta) - qS_w l_{VT} C_{Y,t}(\beta, \delta_r) + T y_e = 0 \quad (6.5)$$

where y_e is the distance of the remaining engine from the symmetry plane, in order to include effects of thrust asymmetry. $C_{n,wb}$ is the wingbody yawing moment which is mostly caused by the fuselage in a sideslip. The tail must be able to generate a lateral force to balance the airplane and prevent yawing while keeping the bank angle less than 5° , according to the regulations. By combining the vertical and lateral forces equilibrium and assuming a small angles, it is possible to obtain such value:

$$\varphi = \frac{-(Y_t(\beta, \delta_e) + Y_{wb}(\beta))}{2W} \pm \frac{1}{2W} \sqrt{(Y_t(\beta, \delta_e) + Y_{wb}(\beta))^2 - 4W(W - L)} \quad (6.6)$$

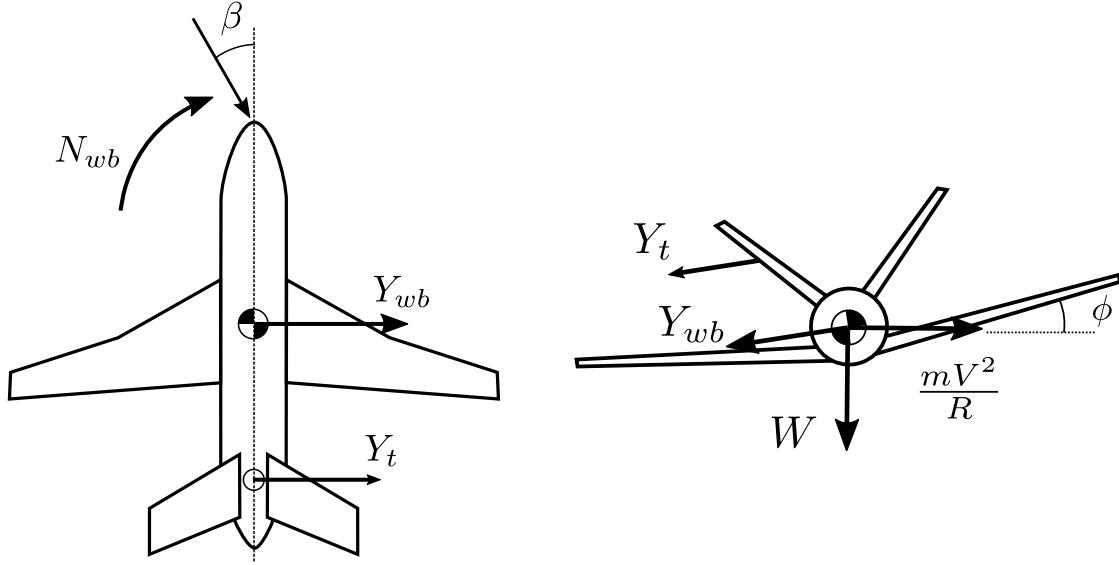


Figure 6.2: Forces and moments arrangement in sideslip

Therefore for a given flight condition and sideslip angle, first Eq. 6.5 is solved to find the control angle then Eq. 6.6 is used to check if the configuration is within the prescribed bank limit.

Finally the stability of the airplane with the new tail must be assessed. Since linear aerodynamic relations are used, the momentum equations can be analytically derived to calculate the $C_{m\alpha}$ and $C_{n\beta}$ of the full airplane (Eq. 6.7 and 6.8). The first represents longitudinal stability and has to be negative, while the second represents directional or weathercock stability, and has to be positive. This must be assessed in cruise condition.

$$C_{m\alpha} = C_{m\alpha}^{wb} + \frac{x_{cg} - x_{ac}}{c_{MAC}} (C_{L\alpha}^{wb} + \eta_q C_{L\alpha}^t) - \eta_q \frac{l_{TH}}{c_{MAC}} C_{L\alpha}^t \quad (6.7)$$

$$C_{n\beta} = C_{n\beta}^{wb} - \frac{l_{VT}}{c_{MAC}} C_{Y\beta}^t \quad (6.8)$$

It is useful to explore the effect of design parameters such as the tail dihedral, surface area and aspect ratio over these coefficients. This can be easily done with the aid of a computer by generating a carpet plot, in which the designer can choose a combination of parameters within constraints. Finally, since the design space is large, an automated multi-point optimization method was used to find a suitable design point. This method is a Particle-Swarm Optimization (PSO) scheme, for which drag and weight minimization was provided as objective along all the constraints discussed above.

6.3 Tail Geometry Parametrization

Control surfaces for a Vee-Tail can be arranged in different manners. The classical design involves the use of a single control surface and the introduction of a "meshing mechanism" to combine the stick and rudder pedals movements.

This control system introduces a potential weight penalty and has been applied in commercial airplane designs such as the Beechcraft Bonanza and the Cirrus SF-50. An alternative arrangement is separating the two control deflections into two different control surfaces, as employed in military UAV applications. While it does not require a meshing control system, the smaller surfaces may have less effectiveness and thus require larger surface deflections.

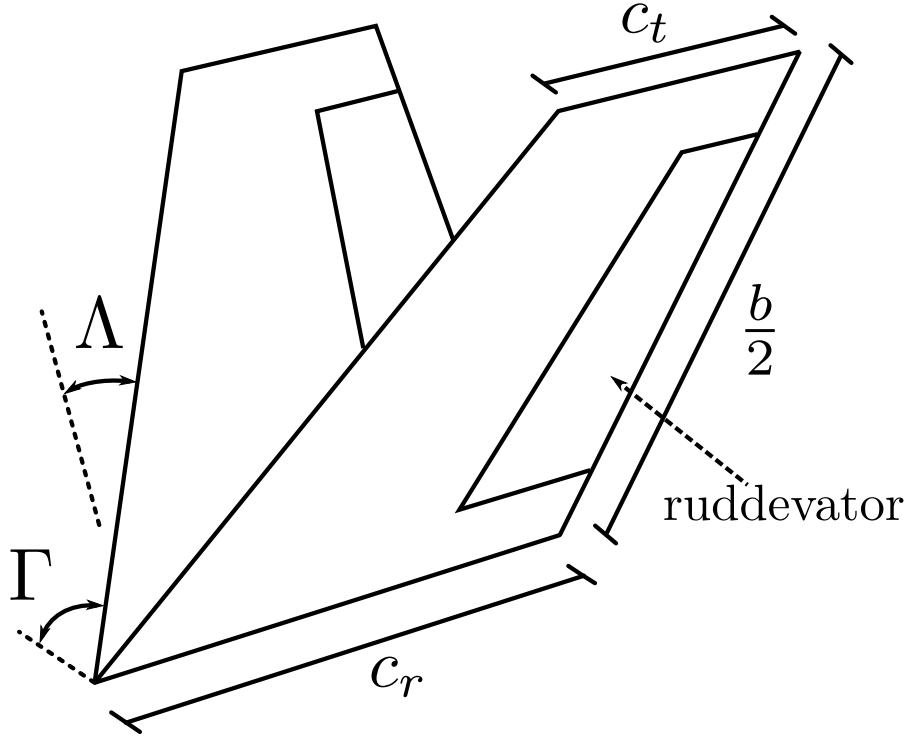


Figure 6.3: Single control surface Vee-Tail design

Both arrangements are proposed in this study. The parameters identifying the classical arrangement are presented in Fig. 6.3. The angle Γ is the dihedral of the tail from the horizontal, as seen from the front of the aircraft, while the angle Λ is the leading edge sweep of the tail, as seen in the plane of the half-tail surface itself. The other parameters in the figure are the length of the extremities and the half span of each surface. In the optimization code however, these are not explicitly set but calculated from three other parameters. Indeed the optimization code sets the aspect ratio $\mathcal{R}_t$, the taper ratio λ_t and the total tail surface S_t , from which the geometric lengths are calculated (Equations 6.9-6.11). Finally the y/b_t position of the outer side of the control surface is used as a design variable for the optimizer, while the inner side is fixed at $y/b_t = 0.1$.

$$b_t = \sqrt{\mathcal{R}_t S_t} \quad (6.9)$$

$$c_r = \frac{S_t}{\frac{b_t}{2} (1 + \lambda_t)} \quad (6.10)$$

$$c_t = \lambda_t c_r \quad (6.11)$$

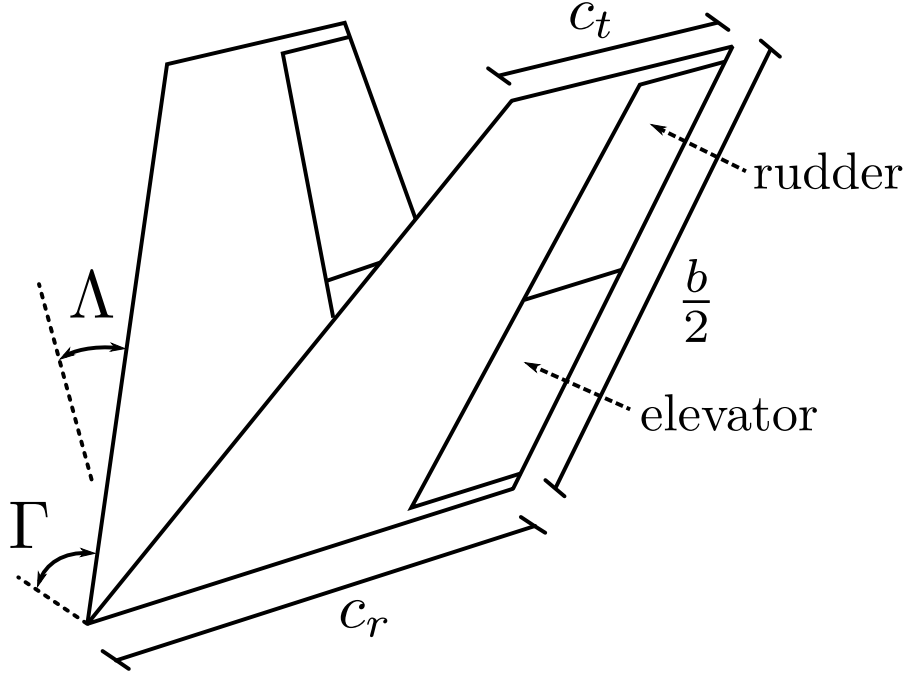


Figure 6.4: Two control surfaces Vee-Tail design

The alternative arrangement is shown in Fig. 6.4. The elevator is placed in the inner side of the span, while the rudder is placed in the outer span. The y/b_t position of the dividing line is introduced as a design variable in the optimization code for this control surface arrangement, while other parameters are the same as the classical arrangement. For both arrangements the control surface hinge position is set at $x/c = 0.63$. Along the constraints put forward by regulations (as discussed in Sec. 6.1), maximum angle of deflection are imposed on the control surfaces. These values are chosen from standard values used in General Aviation (CS-23) airplanes, as suggested by Gundmusson [41], and reported in Tab. 6.2.

Control	Positive Deflection	Negative Deflection
Elevator	+25	-20
Rudder	+25	-25

Table 6.2: Control Surfaces limits

6.4 Particle Swarm Optimization Procedure

Particle Swarm Optimization or PSO for short is an optimization algorithm which does not require the cost function to be differentiable. Therefore it is useful for optimization problems lacking an analytic cost function, such as an aerodynamic optimization. The algorithm is now briefly described with the specific application to the vee-tail optimization. A block-diagram of the process is shown in Fig. 6.5.

More information about PSO can be found in literature[42]. For this study the Python PSO library *pyswarm* was used due to its versatility and application with external tools, such an aerodynamic code[43].

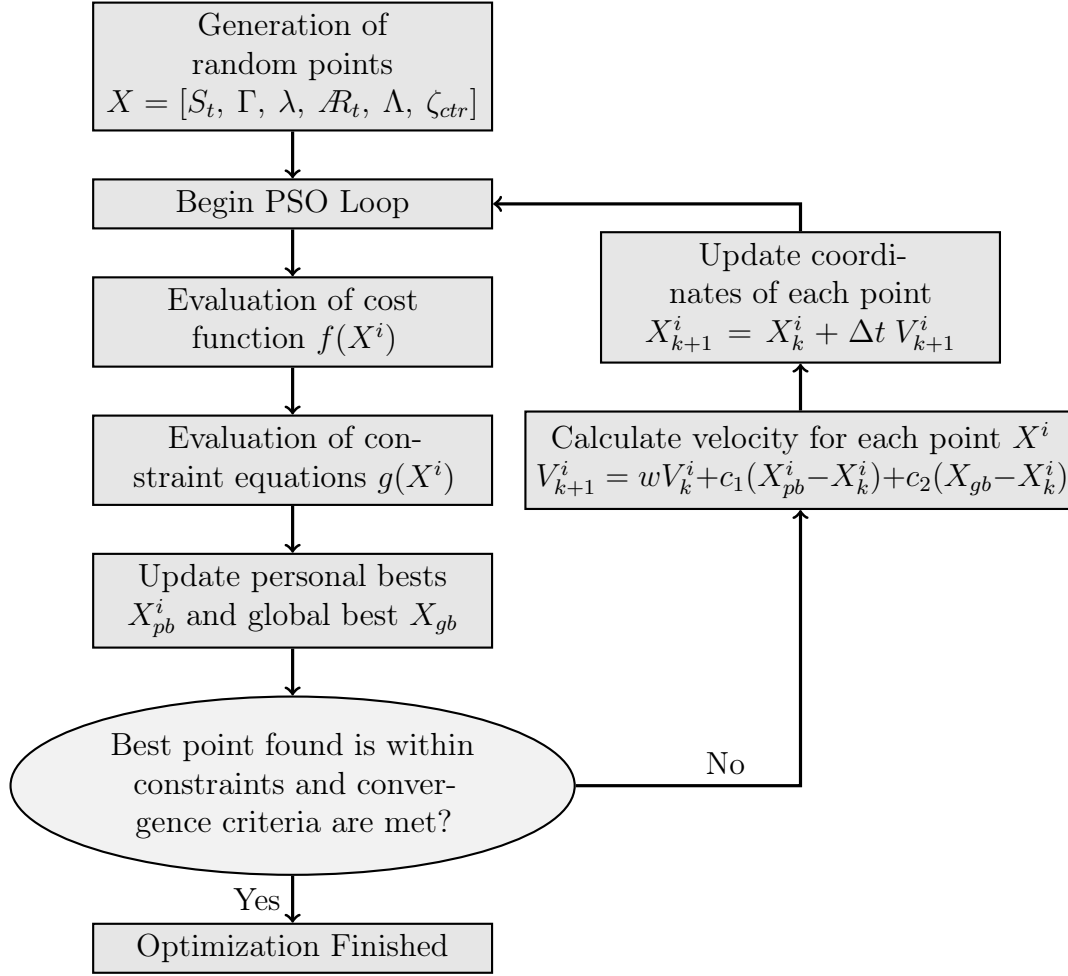


Figure 6.5: Particle Swarm Optimization Algorithm

Given a search-space of n variables, a number N_p of random particles X^i are generated, each with a vector of coordinates of length n . Each particle is evaluated with the cost function $f(X)$ and the constraint equations $g(X)$. In order for a particle to be an acceptable solution of the problem, it must return a positive value from the constraint equations $g(X) > 0$. After the particles had been tested for two iterations, the second one with a random movement of their coordinates, each particle is moved in the search-space according to a calculated velocity V^i . This new velocity vector is calculated as:

$$V_{k+1}^i = wV_k^i + c_1(X_{pb}^i - X_k^i) + c_2(X_{gb} - X_k^i) \quad (6.12)$$

where V_k^i is the velocity vector at the current iteration, X_k^i the current position, X_{pb}^i is the *personal best* of the particle, the coordinates of the best position found so far by the i -th particle, and X_{gb} is the *global best*, the coordinates of the best position of the entire swarm. The three terms are weighted by three coefficients: w is defined as the *inertia factor* of the particle, c_1 as *self confidence* and c_2 as

swarm confidence. These coefficients are generally tuned. Once the new velocity is calculated, each position vector is updated:

$$X_{k+1}^i = X_k^i + \Delta t V_{k+1}^i \quad (6.13)$$

where Δt is a step velocity parameter that decreases with each iteration, as a way to damp oscillations as the swarm approaches the global optimum. The algorithm iterates until one of the convergence criteria are met. Generally these are the minimum variation of the cost function and the minimum average variation of the coordinates. Maximum number of iterations can also be set as a stopping criterium.

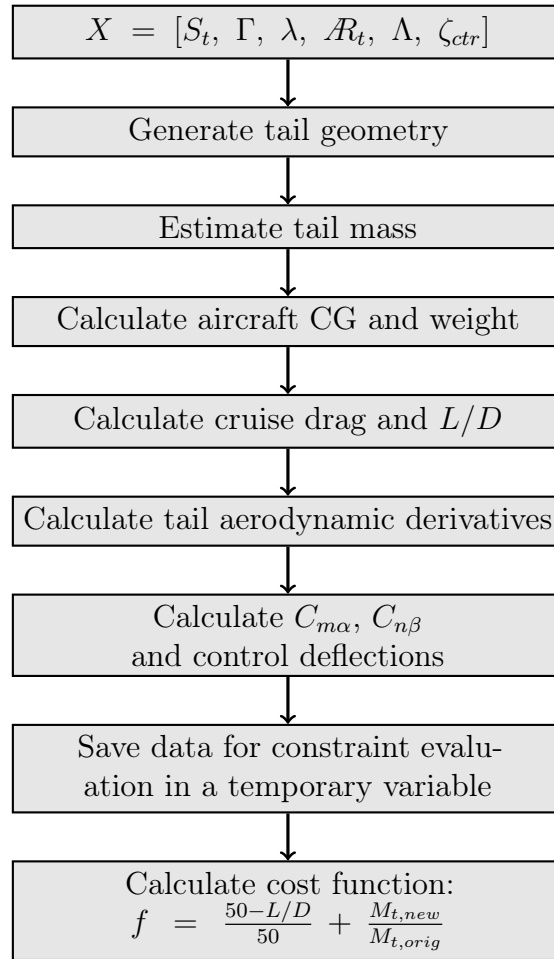


Figure 6.6: Cost Function block diagram

The cost function used for the vee-tail design application is shown in Fig. 6.6. The geometry of the tail is generated from the input variables and used to estimate its structural mass, using the loads and methods of Appendix A. The relative variation of tail mass is then used to scale the tail mass from the baseline aircraft and find the new center of gravity of the airplane. Drag and L/D are calculated in the economy cruise condition of Sec. 5.2. Then with the VLM code, the tail aerodynamic derivatives are estimated and the equations of Sec. 6.2 are used to calculate the stability derivatives and control deflections, which have to be compared

against the regulation requirements. These requirements are defined as constraint equations in the PSO algorithm. Finally the fitness of the design is calculated:

$$f = \frac{50 - L/D}{50} + \frac{M_{t,new}}{M_{t,orig}} \quad (6.14)$$

where L/D is the calculated aerodynamic efficiency in cruise, while $\frac{M_{t,new}}{M_{t,orig}}$ is the ratio between the new tail mass and the mass of the baseline tail. For the specific case of a vee-tail with rudderevators, the calculated tail mass is incremented of 35% to account for the heavier intermeshing control system.

6.5 Optimized design results

Table 6.3 shows the obtained vee-tail geometries from the optimization procedure. Variables are the tail surface S_t , the dihedral angle Γ_t , the tail taper ratio λ_t , the tail aspect ratio $\mathcal{R}_t$, the sweep angle Λ_t , the relative position of control surfaces ζ_{ctr} and finally the x_{root} position of the root leading edge of the tail. The second last one has different purposes depending on the arrangement of control surfaces: for a rudderevator design, it represents the relative $2y/b$ point where the control surface ends (beginning at $2y/b = 0.1$), while for separate controls it represents the relative position of the gap between elevator and rudder (see Fig. 6.4, the flapped portion begins at $2y/b = 0.1$ and ends at $2y/b = 0.9$). Besides the regulatory constraints discussed in Sec. 6.1, a static margin SM of 5% was imposed, a conventional figure for commercial aircraft.

	Design with separate surfaces	Design with rudderevators
S_t	11.3 m^2	10.19 m^2
Γ_t	39.81°	37.6°
λ_t	0.356	0.33
$\mathcal{R}_t$	4.726	4.69
Λ_t	29.58°	32.17°
ζ_{ctr}	0.44	0.67
x_{root}	13.664 m	14.364 m

Table 6.3: Optimized Vee-tail designs

These two designs are applied on the full aircraft and compared to the baseline and the new wing design of Sec. 5.2. The aerodynamic properties were evaluated in the "economy cruise" condition ($Ma = 0.65$, FL 450) with the aircraft trimmed. These results are shown in Tab. 6.4. Masses of structural components such as wing and tail group was scaled with the representative mass variation, as was previously done in Sec. 5.2. Even with an increased empty mass, therefore higher cruise C_L , the new configurations with vee-tails produce better L/D ratios.

	C_L	C_D	L/D	Empty Mass Change
Original	0.5295	0.030131	17.57	
New Wing	0.5439	0.028660	18.99	+18.17% Wing
Tail 1	0.5477	0.027010	20.28	+18.17% Wing, +17.17% Tail
Tail 2	0.5490	0.026828	20.42	+18.17% Wing, +23.14% Tail

Table 6.4: Aerodynamic data in cruise with empty weight variations.

In the same flight conditions a fixed- C_L analysis was carried out, whose main results are shown in Fig. 6.7. The introduction of the new Vee-tail over the conventional one prompted a reduction from 7 to 17 Drag Counts from the design with the new wing only. The greatest difference is in the region of cruise lift coefficients, while at high speeds and zero lift the drag advantage is smaller. Indeed the new tail increases induced drag in this region, as shown in Fig. 6.10. This is caused by root incidence of the tail that is set to -1° , while the original configuration had an horizontal tail with no incidence. Overall, the reduced wetted area of the tail compensates any induced drag disadvantage in both drag (Fig. 6.12 and L/D (Fig. 6.13).

Let's now calculate the burned fuel using the same procedure described in Sec. 5.2 for comparing the new wing designs. The results are shown in Tab. 6.5. Both new vee-tails have a fuel saving of 10%. Interestingly the tail with rudderevators has the best performance results even if it's the heavier design (due to the heavier control system).

	500 nm		1000 nm		1500 nm	
Original	208.44 kg		424.26 kg		647.73 kg	
New Wing	198.63 kg	-4.71%	403.77 kg	-4.83%	615.61 kg	-4.96%
Tail 1	187.43 kg	-10.08%	380.59 kg	-10.29%	579.67 kg	-10.51%
Tail 2	186.63 kg	-10.46%	378.94 kg	-10.68%	577.09 kg	-10.91%

Table 6.5: Calculated burned fuel for new PC-24 configurations.

With the saved fuel mass it's possible to calculate the avoided CO_2 emissions, with the same procedure of Sec. 5.2. Compared to the previous design with the new wing only, the addition of the vee-tail saves even more CO_2 per year, with 45 tons of avoided emissions.

Similar results are found regarding fuel cost savings, as shown in Table 6.7. The new vee-tails yield a saving between 6000-6300 US\$ per year, assuming 6 flights of 1000nm per week.

While static stability, both longitudinal and directional, was imposed as a constraint in the particle swarm optimization, dynamic stability must be assessed afterwards. Therefore two flight conditions have been examined: cruise and approach. Indeed approach is when the aircraft is most exposed to gusts and other disturbances from the environment, hence good dynamic stability is essential. While

	One Trip		Per Week		Per Year	
Original	1311 kg		7863 kg		424618 kg	
New Wing	1247 kg	-63 kg	7483 kg	-380 kg	404103 kg	-20515 kg
Tail 1	1176 kg	-135 kg	7054 kg	-809 kg	380911 kg	-43708 kg
Tail 2	1171 kg	-140 kg	7023 kg	-840 kg	379255 kg	-45363 kg

Table 6.6: Estimated saved greenhouse emissions of new configurations, assuming 6 flights of 1000nm per week.

	One Trip		Per Week		Per Year	
Original	181.91 \$		1091.47 \$		58939.34 \$	
New Wing	173.12 \$	-8.79 \$	1038.73 \$	-52.73 \$	56091.69 \$	-2847.65 \$
Tail 1	163.19 \$	-18.72 \$	979.12 \$	-112.35 \$	52872.46 \$	-6066.87 \$
Tail 2	132.48 \$	-19.43 \$	974.86 \$	-116.60 \$	52642.69 \$	-6296.65 \$

Table 6.7: Estimated cost reduction of new configurations, assuming 6 flights of 1000nm per week.

the cruise condition is the same used to assess fuel burn consumption, the approach flight condition consists of a clean aircraft at $V_{app} = 1.45 V_{st}$, maximum landing weight and an ISA altitude of 5000 ft. These modes are calculated with AVL, after an estimation of aircraft inertia was performed. Such estimation was made combining the methods of Roskam[33] and the inertia estimation capabilities of OpenVSP[39].

The calculated dynamic modes are shown in Fig.6.14 for cruise and Fig. 6.15 for approach. In both flight conditions all modes are stable. The behavior of each configuration remains similar in both flight conditions. Both new vee-tails increase the damping and frequency of the Dutch Roll mode, while decreasing the damping and frequency of the Short Period mode. This is probably an effect of the vee-tail dihedral: the tail arrangement presents two inclined surfaces increasing the vertical projected tail area helping with the Dutch Roll damping, at the cost of a reduced projected horizontal tail area, which contributes to Short Period damping.

The oscillation period of the Phugoid is unaffected by the different configurations, however damping is reduced with both vee-tail designs, with the configuration featuring the rudderevators being the worst. In cruise configuration the Phugoid mode has overall less damping due to the high speed, while it is more damped in approach. The Spiral mode is stable in all configurations and flight conditions, but it has the least damping with the separate control surfaces vee-tail design.

Regulations require that long period oscillations such as Phugoid and Dutch Roll reduce their amplitude by 1/10 after 7 cycles. Table 6.8 and 6.9 show the calculated cycles to damp the oscillations as indicated, in cruise and approach respectively. At first glance the original tail configuration does not meet regulations, as it requires 12.06 cycles to damp the Dutch Roll mode in cruise. However the original airplane features a ventral fin (see Fig. B.1), which has been omitted from this analysis: it

is possible that the purpose is to damp Dutch Roll oscillations besides increasing lateral stability. Both vee-tail designs can meet the Dutch Roll damping requirement in cruise without any ventral fin: this would yield an extra reduction in wetted area and therefore drag. The Phugoid mode damping is not within regulations in cruise for all configurations, while in approach it seems only the vee-tails are affected. Since the difference is at most one cycle, a phugoid-suppressing control system should be enough to satisfy the regulation requirements without requiring a full re-design of the tail.

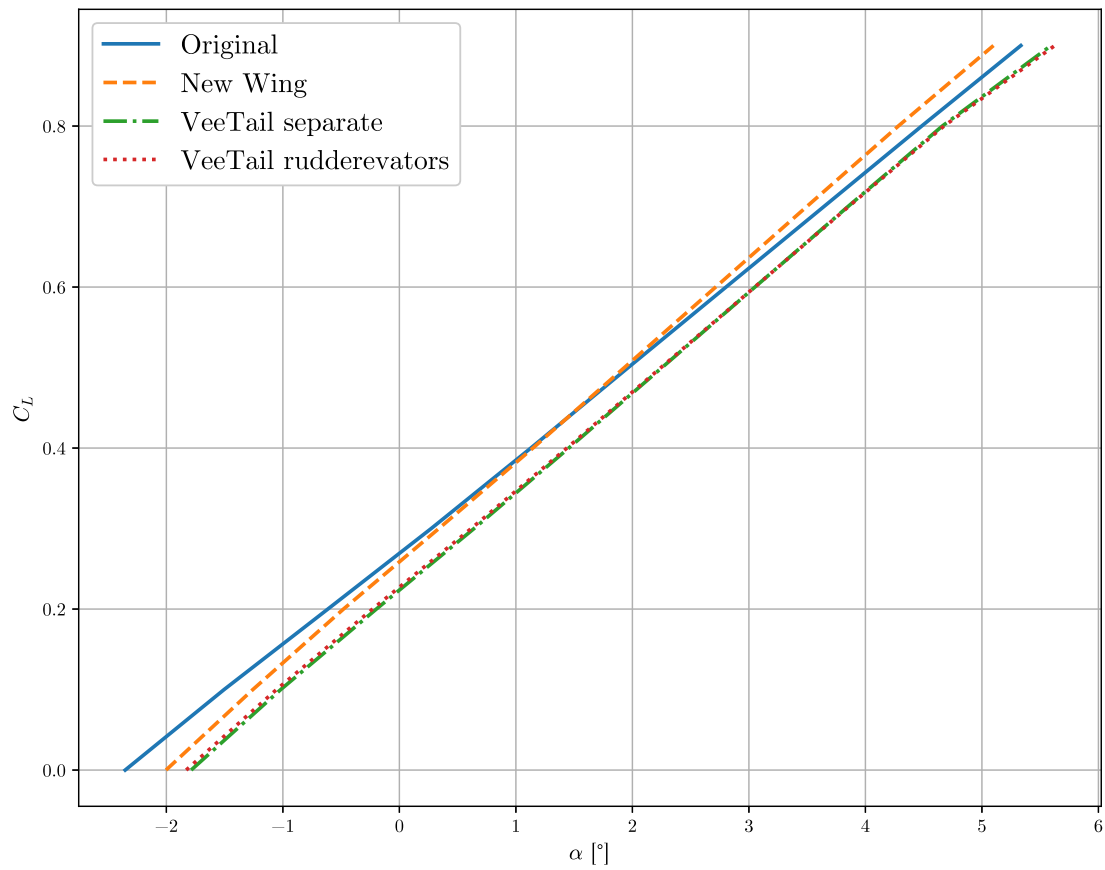
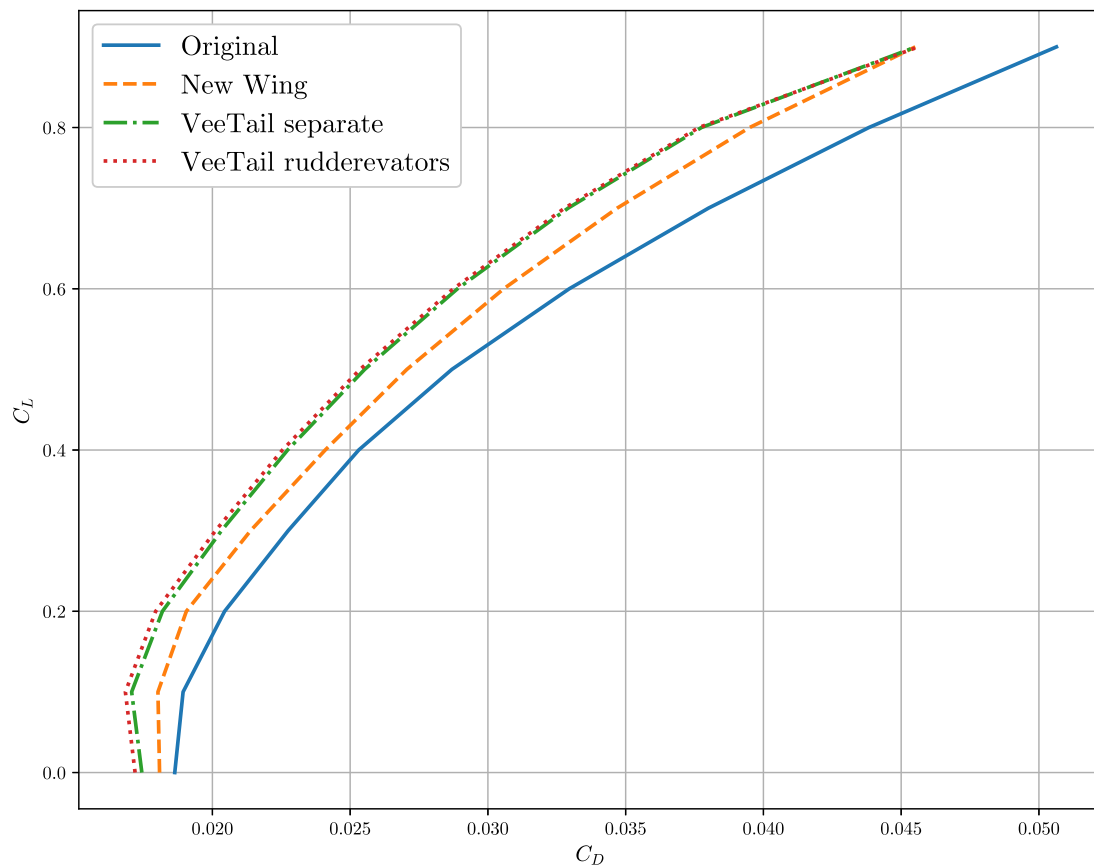
	Original	New Wing	Vee-tail 1	Vee-tail 2
Dutch Roll	12.06	12.65	6.37	6.2
Phugoid	7.12	7.57	8.09	8.32

Table 6.8: Cycles required to reduce by 1/10 the oscillations of dynamic modes in cruise.

	Original	New Wing	Vee-tail 1	Vee-tail 2
Dutch Roll	4.18	4.01	2.91	2.79
Phugoid	4.52	4.92	7.67	7.86

Table 6.9: Cycles required to reduce by 1/10 the oscillations of dynamic modes in approach.

Nonetheless the separate control surface configuration for the vee-tail is selected as a final configuration because of the lowest empty mass increment and higher dynamic mode damping.

(a) $CL - \alpha$ 

(b) Polar

Figure 6.7: Aerodynamic data of new PC-24 configurations

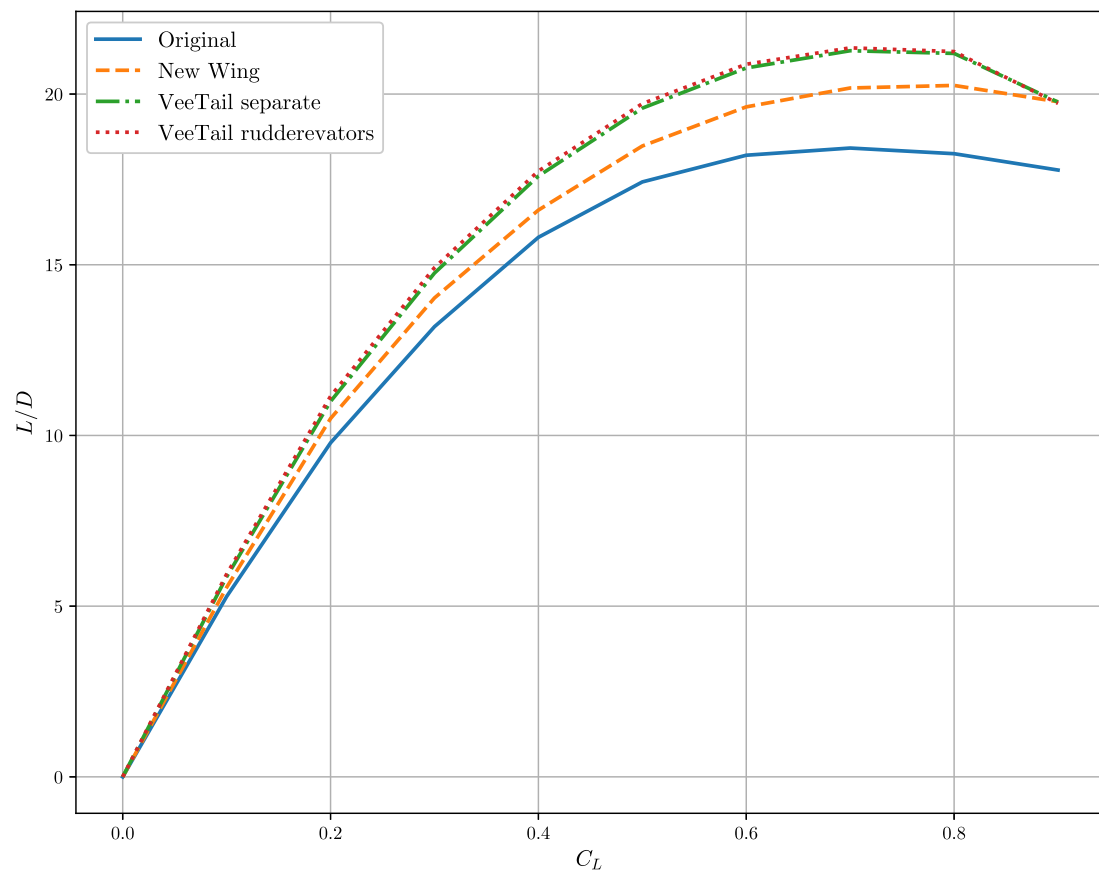


Figure 6.8: L/D of new PC-24 configurations

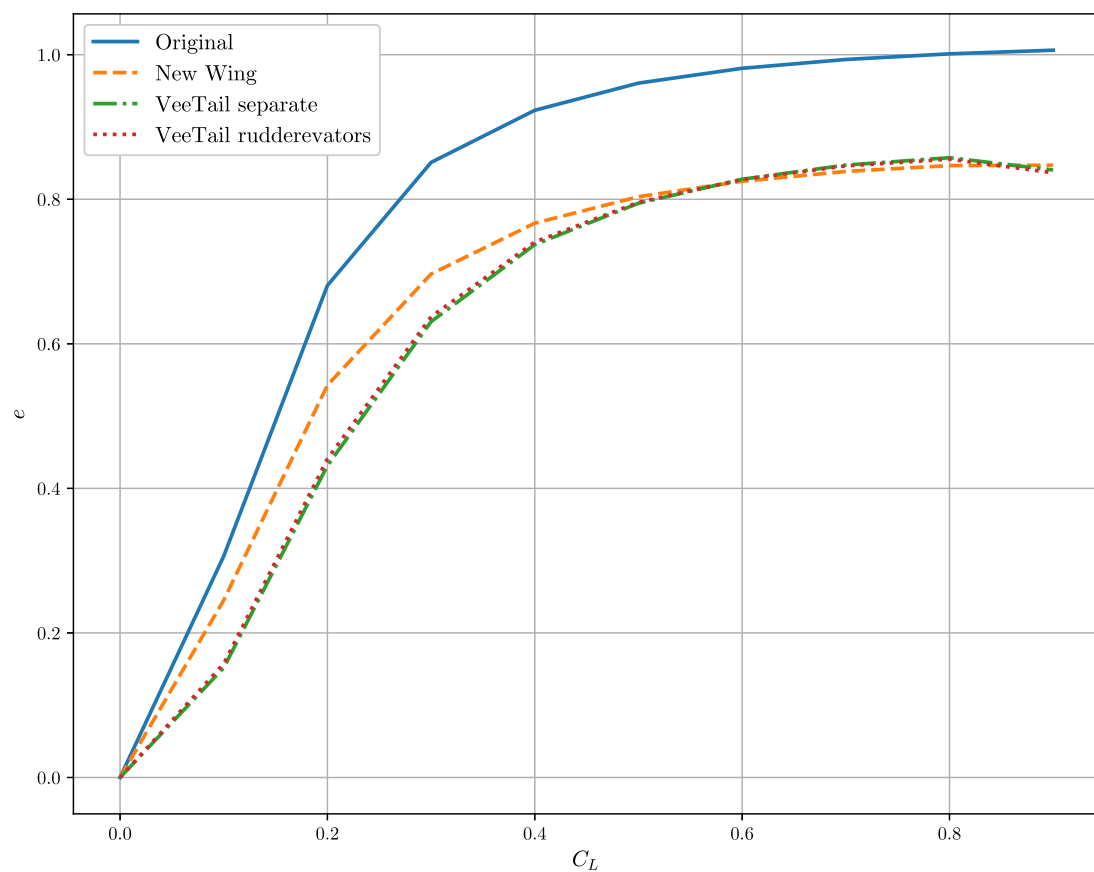


Figure 6.9: Oswald efficiency factor of new PC-24 configurations

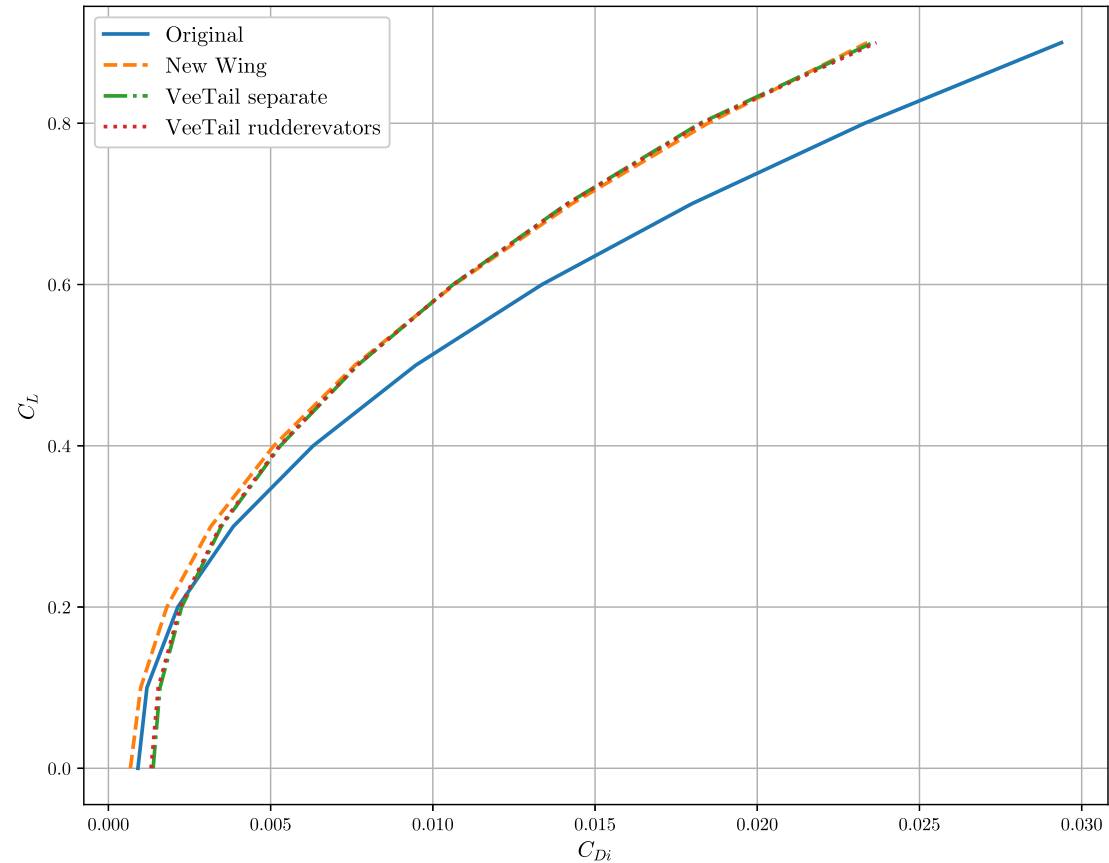


Figure 6.10: Inviscid Polar for new PC-24 configurations

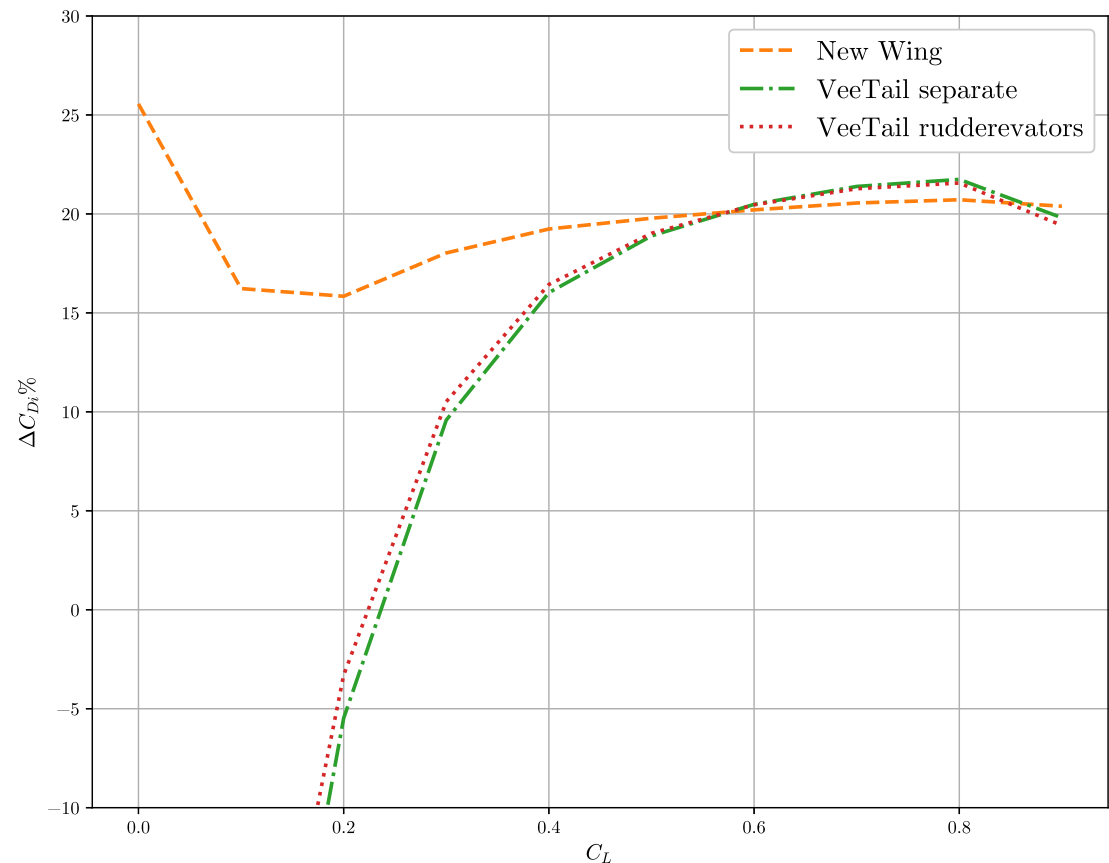


Figure 6.11: Relative C_{Di} variation for new PC-24 configurations

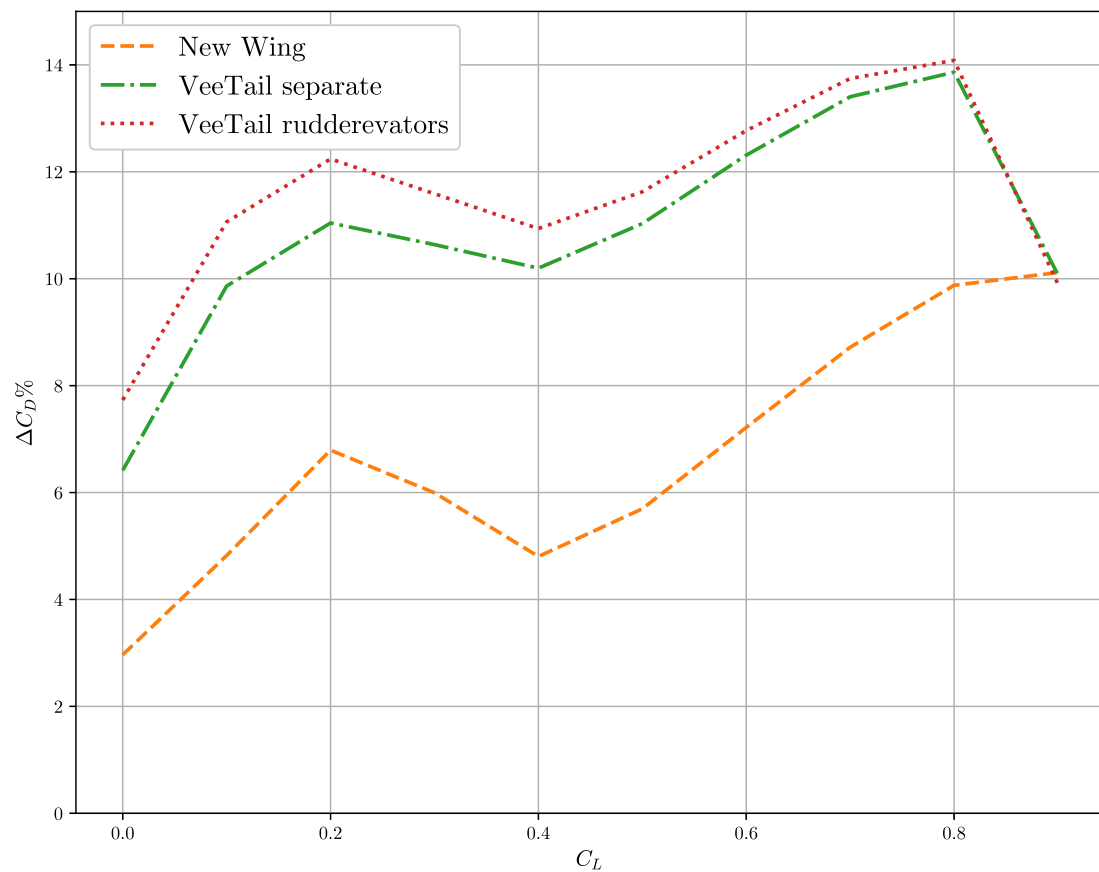


Figure 6.12: Relative C_D variation for new PC-24 configurations

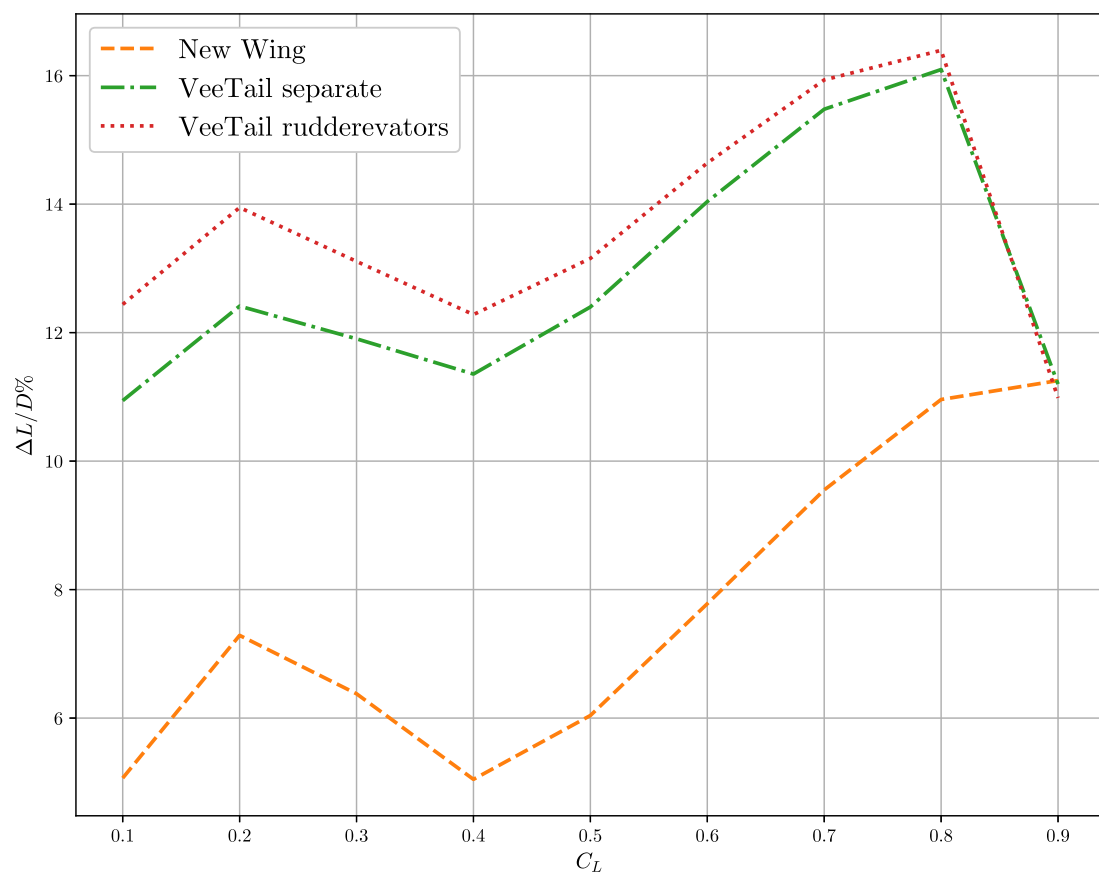
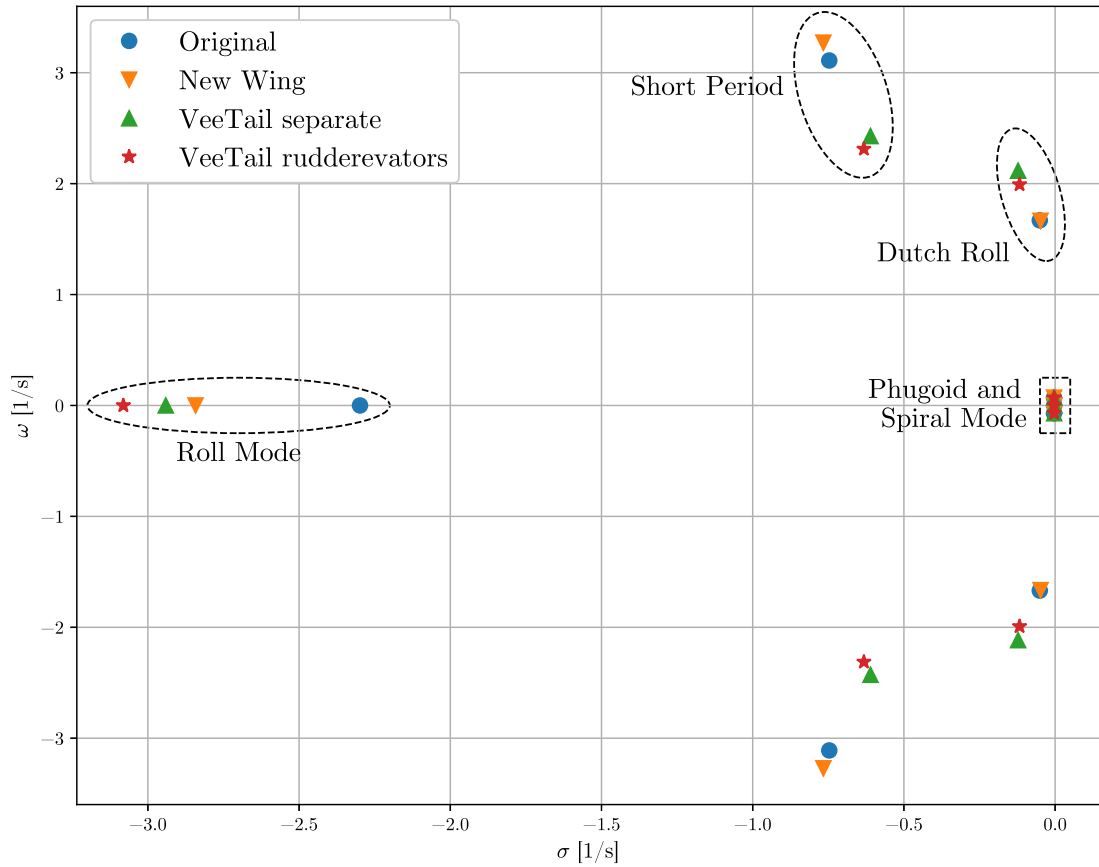
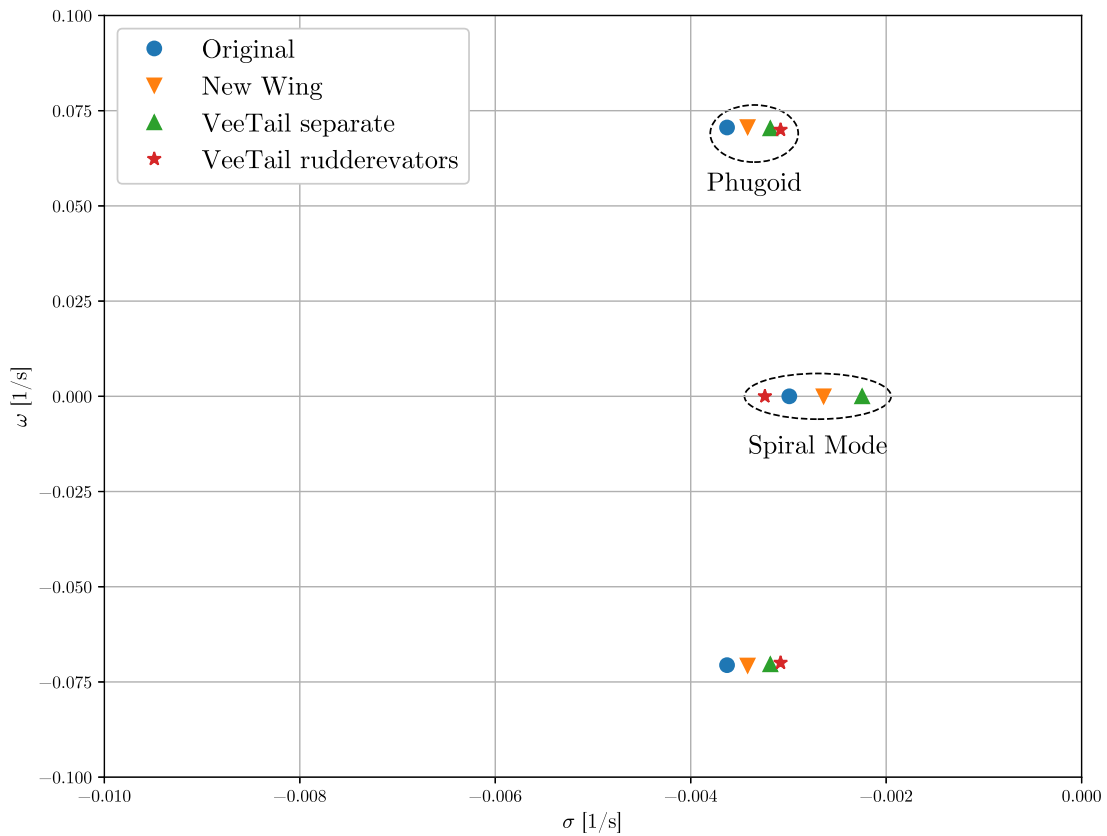


Figure 6.13: Relative L/D variation for new PC-24 configurations

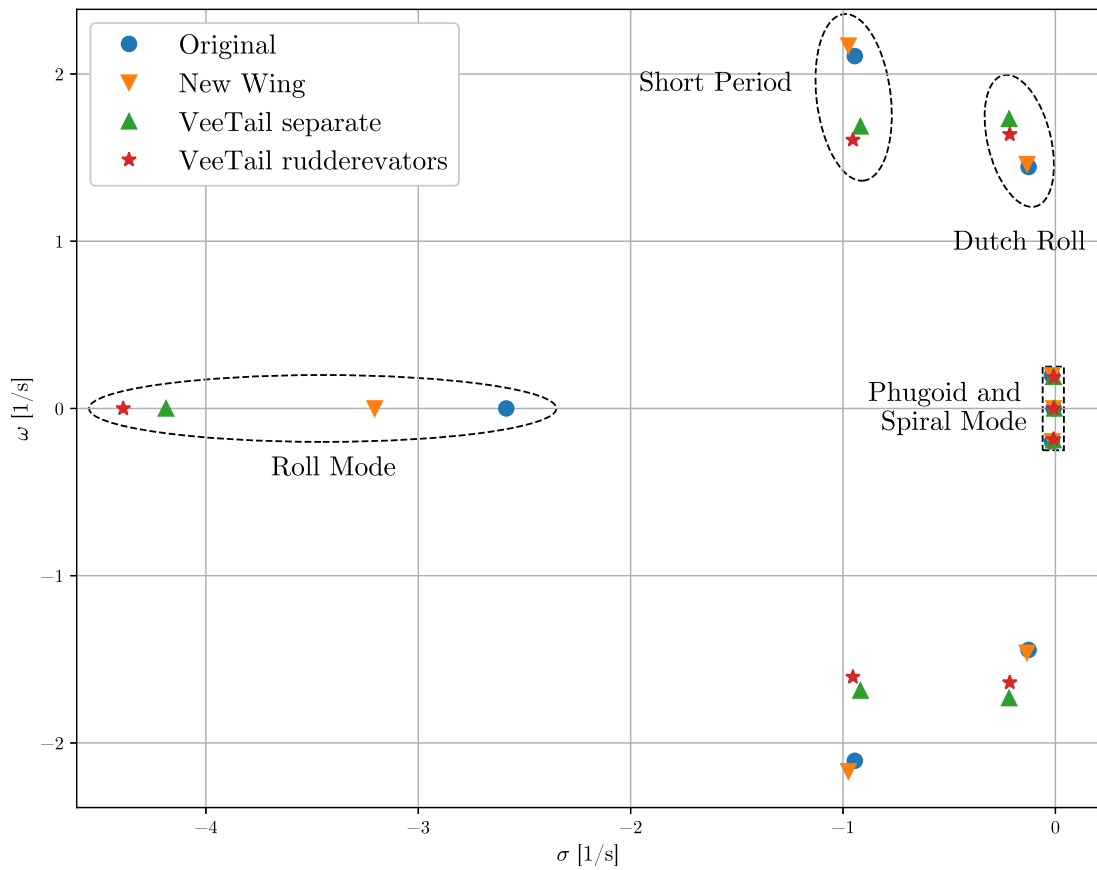


(a) Dynamic modes in cruise

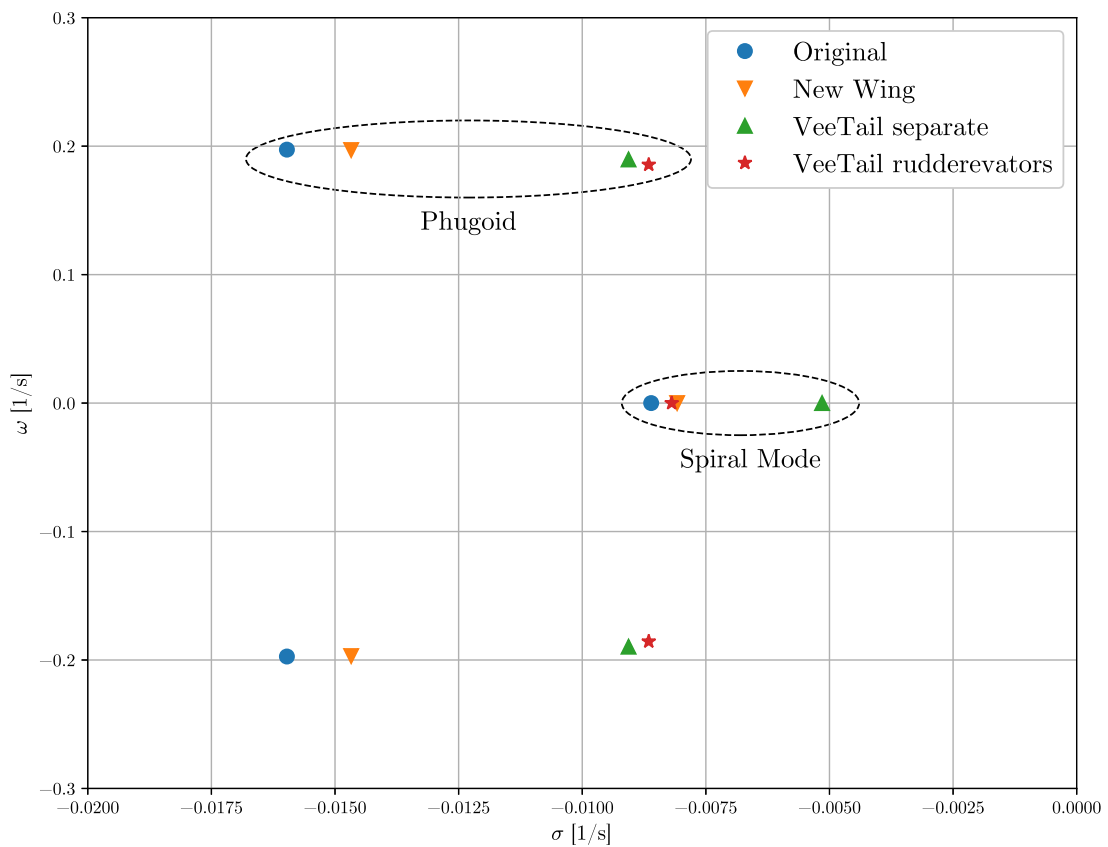


(b) Phugoid and Spiral Mode in cruise

Figure 6.14: Cruise Dynamic Modes for the new configurations



(a) Dynamic modes in approach



(b) Phugoid and Spiral Mode in approach

Figure 6.15: Approach Dynamic Modes for the new configurations

Chapter 7

Conclusions and Future Developments



Figure 7.1: Final PC-24 redesign

Figure 7.1 shows the final configuration obtained from this study. The combination of the bell spanload wing (design 3 from Sec. 5.2) and a vee-tail (design 1 from Sec. 6.5) yielded a 10% reduction in fuel burn, a dramatic result. However due to these modifications, the total empty mass increased by 4.53% therefore reducing the useful payload for the same MTOW. Figure 7.2 presents a three-view projection of the final configuration.

Prandtl's bell lift distribution opens a new perspective on optimum design.

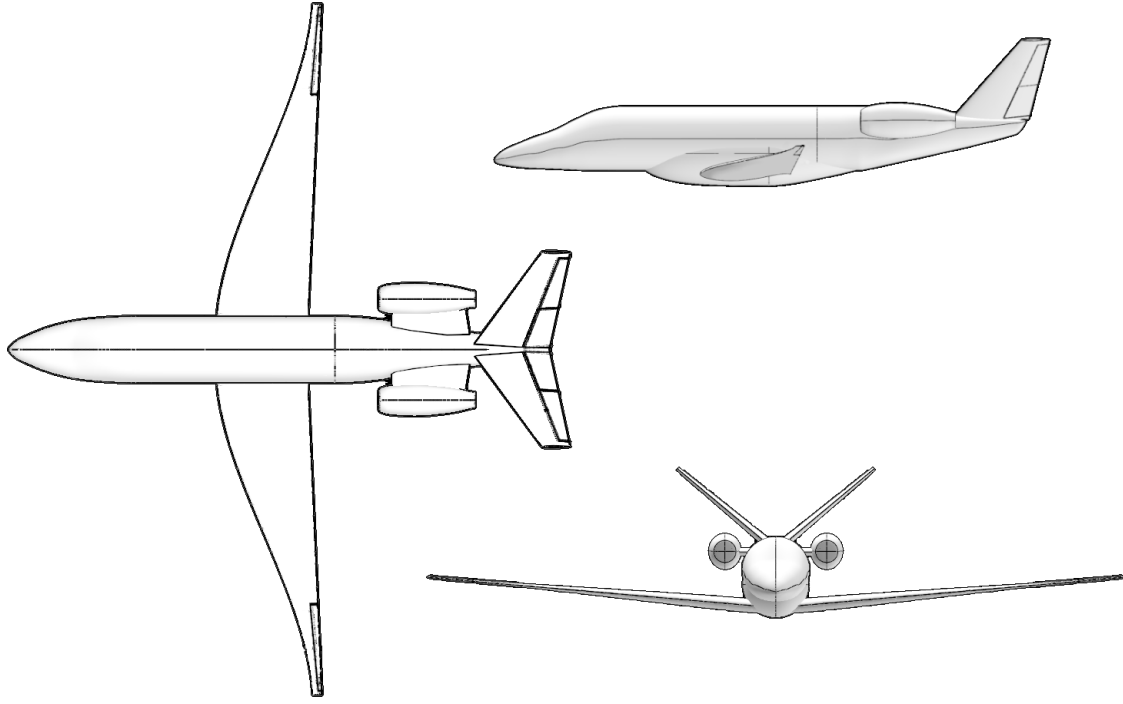


Figure 7.2: Final PC-24 redesign

Indeed is the first attempt at considering a wing design in a multi-disciplinary perspective by including bending moments into the problem. However it is still a limited solution since it is optimized for one design point only, while an airplane can fly in wide range of flight conditions. Current scientific literature lacks analysis of the characteristics of this lift distribution on all the operating flight conditions, from which it's possible to derive some design best practices. Chapter 4 was dedicated to this purpose.

If the design method is applied on a generic trapezoidal wing, the overall characteristics of the wing are a mild improvement from the elliptical wing baseline. Indeed the increment of $\mathcal{R}$ does yield lower induced drag, and the reduction would increase with C_L , however the shape of the lift distribution is fixed by the geometric twist to one specific design point. Therefore the structural disadvantages become evident at the critical loads in the maneuvering envelope, such as the maneuvering point V_a or the design cruise point V_c . Wing structural mass would increase by 50% compared to the elliptical spanload wing.

From the sensitivity analysis some design guidelines were found. The design lift coefficient should lie between 0.35 and 0.5. This is the sweet spot between excessive structural mass growth and high drag at high speed. Induced drag is also more relevant if $C_L > 0.2$: at high speed viscous drag is the predominant contribution, hence it makes little sense to adopt small design lift coefficients. Aspect ratio $\mathcal{R}$ of the original wing plays a similar role: diminishing returns are present with ever increasing $\mathcal{R}$. Indeed if the original wing has an aspect ratio above 10, it is possible that the reduction of induced drag is marginal. In that situation, strategies to reduce viscous drag should be sought instead.

While the wing area and wingspan are specified by the design method, the chord distribution is not explicitly prescribed. Indeed it is a free variable of the

problem. In Ch. 4 the effect of taper ratio of a trapezoidal wing was analyzed: high taper yielded better results on a wider range of lift coefficients, moderating both the structural mass increment and the high induced drag at low lift. When taper is high, the chord distribution is more similar to a bell distribution. This insight was developed further by approximating the shape of the bell distribution with straight lines. The new bell-like chord distributions feature two or three taperings with a finite, albeit small, tip chord. These yielded remarkable results as structural mass increase dropped to 15-19% and zero lift induced drag was eliminated. The chord distribution helped retaining some of the design point advantages in off-design conditions.

These guidelines were applied in Ch. 5 on the re-design of practical wings. Induced drag was reduced even further, more than 15%. Structural mass varied from a 7.78% increase to a 18.17% increase: this disparity is caused by the thickness of the airfoils. The PC-24 wing features thinner airfoils, hence the higher structural mass increase. Nonetheless the best design yielded a 4.8% burned fuel saving in the reference cruise condition. While the result is remarkable, the method used in this study does not account for possible wave drag. It was observed in a preliminary analysis of the airfoils that the critical Mach number was about 0.7, hence a second order compressibility correction like Karman-Tsien was deemed enough. Nonetheless a further development of this study would be analyzing the final configuration with CFD software to better estimate the compressibility effects.

One of the characteristics of the bell lift distribution is wingtip upwash which can be used for correcting adverse yaw in turning flight. This was briefly analyzed in Sec. 4.6, where it is clear that C_{l,δ_a} is not positive in all flight conditions, but is affected by lift, much like the other properties. Using a chord distribution shaped like the bell distribution helps in maintaining this property for a wide span of C_L . This characteristic produced the idea of eliminating the vertical tail or reducing its size. For this purpose a Vee-tail was conceived as a viable replacement of a traditional tail. Chapter 6 elaborated a method for designing a Vee-tail from airworthiness regulations with the aid of a global optimization algorithm, in order to contain structural mass increment and reduce wetted area. The overall result yielded a 10% fuel burn decrease. Dynamic stability was assessed and checked too.

In this study the tools developed were low fidelity. This is indeed a limitation and a point of further development. As mentioned above, in order to further characterize the bell spanload and its properties, CFD software should be used. This would reduce uncertainty regarding compressibility, wave drag, high lift devices and the control surface effectiveness. Then the effects on structural weight require further detailed analysis. In this study the internal structural layout was ignored for simplicity sake, replaced with a simplified skin-panel method, however the position and size of structural members like spars and ribs do affect weight in less obvious ways. It was assumed that ultimate stress in the maneuvering envelope as a structural design condition: in reality gusts may generate higher loadings.

One final detail that was ignored in this study was aeroelastic phenomena: the wing was assumed rigid, while in reality the aerodynamic forces change its shape with dramatic results. Hence the logical next step in this study is modeling the aeroelasticity of wings of bell spanload and analyzing the static and dynamic

aeroelastic properties. Indeed it is possible that the rigidity requirements may be stricter than the strength requirements used in this study to estimate structural mass. It should be noted, however, that aeroelasticity may be used at an advantage with morphing or active control mechanisms, especially for adapting the lift distribution depending on the flight condition. This is at the time of writing an active research field all over the world.

Appendix A

Direct mass estimation method

As described in the introduction, it is required to assess the effect on the aircraft weight caused by the redesign of the wing or the tail. Since classical statistical equations may not be appropriate due to the unconventional designs proposed, it was chosen to estimate masses from structural requirements. The method implemented is the skin-panel method which is suitable for wings and tails, as they are dominated by bending loads. This method is presented in Jay Gundlach's book of UAV aircraft design[44].

The developed method, however, carries some limitations due to its simplified structural model. Masses of stiffeners, frames and rivets are ignored alongside any non-structural component such as the leading edge and trailing edge panels. Hence compared to the statistical equations the calculated mass is much lower (see Section A.5).

Because of this expected underestimation, in the comparative analysis only the relative mass change is analyzed, and these calculated values are referred to as *representative masses*. In Section 5.2 and Section 6.5, the structural mass of each component is calculated by scaling the respective baseline mass with the relative increment calculated with the direct estimation method, in order to evaluate the load effects.

A.1 Determination of wing design loads

In order to obtain a reasonable structural mass estimation, critical loads should be identified. In Ch. 4 mass comparisons are carried out to evaluate the impact upon weight of the wing designs. Because these wings are not related to any specific aircraft design, it is necessary to assume a wing loading W/S and a maximum load factor n_{max} in order to generate a $V - n$ diagram of the wing. Given $C_{L,max}$, the stalling velocity in horizontal steady flight, V_{st} is calculated first:

$$V_{st} = \sqrt{\frac{2 W/S}{\rho_0 S_w C_{L,max}}} \quad (A.1)$$

This velocity is calculated with respect to the standard air density at sea level $\rho_0 = 1.225 \text{ kg/m}^3$, hence it is an *equivalent airspeed*. Note that in this equation W/S was assumed to be N/m^2 , although usually in SI units the wing loading is

specified in kg/m^2 . With V_{st} is possible to calculate the *design maneuvering speed* V_a :

$$V_a = V_{st} \sqrt{n_{max}} \quad (A.2)$$

whose lift coefficient is $C_{L,max}$. In order to determine the *cruise design speed* V_c and the *design dive speed* V_d , an empirical procedure from Roskam is used [33]. In Volume 5 of his Airplane Design book series, Roskam provides an empirical procedure to estimate the $V - n$ diagram. For calculating V_c :

$$V_c \geq k_c(W/S) \sqrt{W/S} \quad (A.3)$$

where the function k_c is defined as such (converted to SI units):

$$k_c(W/S) = \begin{cases} 141.74, & W/S < 97.65 \\ 102.37 - 0.0484 W/S, & 97.65 < W/S < 488.24 \\ 122.84, & W/S > 488.24 \end{cases} \quad (A.4)$$

While V_d is calculated as:

$$V_d \geq 1.25 V_c \quad (A.5)$$

The respective lift coefficients, used to then calculate the wing loads, are finally calculated:

$$C_{L,V_c} = \frac{2 n_{max} W/S}{\rho_0 V_c^2 S_w} \quad (A.6)$$

$$C_{L,V_d} = \frac{2 n_{max} W/S}{\rho_0 V_d^2 S_w} \quad (A.7)$$

After performing the VLM analysis to calculate the bending moment M_x and the shear force V_z distributions, these are multiplied by the factor of safety 1.5 to convert them from maneuvering limit loads to ultimate loads, as prescribed by airworthiness regulations.

A.2 Determination of tail design loads

Tail loads are determined from a combination of maximum control deflection and flight conditions at n_{max} . The adopted approach is to create testing matrices for each type of tail surface, by combining different flight conditions and surface deflections, and then finding the most critical load. For horizontal tails, the three structural limit conditions (V_a , V_c and V_d) were tested with both the maximum deflections of the elevator, positive and negative. For the vertical tail, on top of the same conditions for the horizontal tail a sideslip $\beta = 12^\circ$ was added.

For the Vee-tail a combination of the flight conditions for the vertical and horizontal tail has been selected. Since the most critical condition is at maximum deflection, the three structural limit conditions have been combined with maximum positive deflection of both rudder and elevator (if two surfaces are present) or of

the rudderevator (if just one control surface is present). Finally a sideslip of 12° was added.

Much like the wing, tail bending moment and shear distributions are multiplied by the factor of safety 1.5 to convert them from limit loads to ultimate loads.

A.3 Materials

In all this study the material is assumed to be an Aluminium Alloy ALCLAD 2024-T3, whose properties are presented in Table A.1[45].

Ultimate Axial Stress σ_U	448 MPa
Ultimate Shear Stress τ_U	276 MPa
Density ρ_{mat}	2780 kg/m ³

Table A.1: 2024 Alloy Properties.

A.4 Structural mass estimation algorithm

The wing is first divided into a set of structural segments along the span. For each of them, the method divides the structure into a set of members that react bending moment and a set that reacts shear. These are determined from the geometry of the physical section, as shown in Fig. A.1. It is assumed that the bending-resisting material is allocated in the upper and lower skin panels of the section, while the shear-resisting material is allocated in a web at in the deepest portion of the section. The idealized structure is therefore composed of a set of n straight panels that follow contour of the airfoil, each panel bounded by two nodes and divided into *Top* panels and *Bottom* panels.

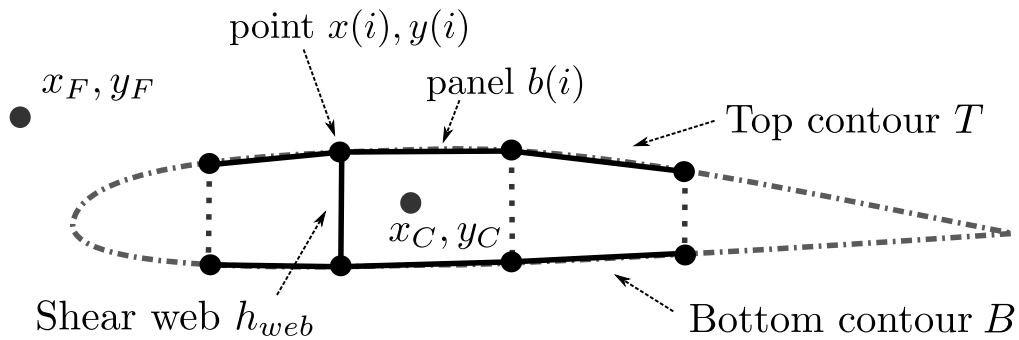


Figure A.1: Cross-section geometry

Let $x(i), y(i)$ be the coordinates of each node in the section plane, then it is possible to calculate the length of each panel and the midpoint with equations A.8-A.9.

$$b(i) = \sqrt{[x(i+1) - x(i)]^2 + [y(i+1) - y(i)]^2} \quad (\text{A.8})$$

$$x_m(i) = \frac{x(i+1) + x(i)}{2}, \quad y_m(i) = \frac{y(i+1) + y(i)}{2} \quad (\text{A.9})$$

Then the centroid coordinates x_c, y_c and moments of inertia of the section are calculated. It's assumed that the skin thickness is equal for the contour panels and that the shear material does not contribute to the centroid or moments of inertia. Equations A.10 and A.11 are used to calculate such coordinates.

$$x_c = \frac{\sum_{i=1}^n b_T(i)x_{m,T}(i) + \sum_{i=1}^n b_B(i)x_{m,B}(i)}{\sum_{i=1}^n b_T(i) + \sum_{i=1}^n b_B(i)} \quad (\text{A.10})$$

$$y_c = \frac{\sum_{i=1}^n b_T(i)y_{m,T}(i) + \sum_{i=1}^n b_B(i)y_{m,B}(i)}{\sum_{i=1}^n b_T(i) + \sum_{i=1}^n b_B(i)} \quad (\text{A.11})$$

The panel midpoint coordinates are then calculated with respect to the centroid coordinates:

$$x'_m(i) = x_m(i) - x_c \quad (\text{A.12})$$

$$y'_m(i) = y_m(i) - y_c \quad (\text{A.13})$$

These coordinates are then used to calculate the moments of inertia of each panel, normalized by the thickness, and summed to obtain the moment of inertia of the section. The angle ε is the orientation of the panel with respect to the horizontal axis, as shown in Fig. A.2.

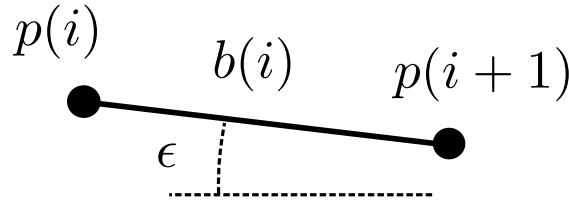


Figure A.2: Geometry of a skin panel

$$I_{xx,t}(i) = \frac{I_{xx}(i)}{t} = \frac{b(i)^3 \sin^2(\varepsilon)}{12} + b(i) y'_m(i)^2 \quad (\text{A.14})$$

$$I_{yy,t}(i) = \frac{I_{yy}(i)}{t} = \frac{b(i)^3 \cos^2(\varepsilon)}{12} + b(i) x'_m(i)^2 \quad (\text{A.15})$$

$$I_{xy,t}(i) = \frac{I_{xy}(i)}{t} = \frac{b(i)^3 \sin(2\varepsilon)}{12} + b(i) x'_m(i) y'_m(i) \quad (\text{A.16})$$

$$I_{xx,t} = \sum_{i=1}^n I_{xx,t}^T(i) + \sum_{i=1}^n I_{xx,t}^B(i) \quad (\text{A.17})$$

$$I_{yy,t} = \sum_{i=1}^n I_{yy,t}^T(i) + \sum_{i=1}^n I_{yy,t}^B(i) \quad (\text{A.18})$$

$$I_{xy,t} = \sum_{i=1}^n I_{xy,t}^T(i) + \sum_{i=1}^n I_{xy,t}^B(i) \quad (\text{A.19})$$

Given the section bending moment M_x , the thickness-normalized axial stresses σ_t are calculated for each panel:

$$\sigma_{z,t}(i) = \frac{M_x(y) [I_{yy,t} y'_m(i) - I_{xy,t} x'_m(i)]}{I_{xx,t} I_{yy,t} - I_{xy,t}^2} \quad (\text{A.20})$$

Finally the highest non-dimensional stress is used to calculate the skin thickness of the contour panels (Eq. A.21), given the material's ultimate stress σ_U .

$$t_{bend} = \frac{\text{Max} [|\sigma_z(i)|]}{\sigma_U} 1.5 \quad (\text{A.21})$$

The shear member is assumed to be a single web located at the maximum thickness of the section. The maximum shear stress occurs at the neutral axis, assumed to be at the centroid. To be conservative, it is assumed that the shear stress q_{sh} is constant and equal to the value at the centroid along all the web:

$$q_{sh} = - \left(\frac{-F_y I_{xy,t} t_{bend}}{(I_{xx,t} I_{yy,t} - I_{xy,t}^2) t_{bend}^2} \right) \sum_{i=1}^n t_{bend} b_T(i) x'_{m,T}(i) \quad (\text{A.22})$$

$$- \left(\frac{F_y I_{yy,t} t_{bend}}{(I_{xx,t} I_{yy,t} - I_{xy,t}^2) t_{bend}^2} \right) \sum_{i=1}^n t_{bend} b_T(i) y'_{m,T}(i) \quad (\text{A.23})$$

The thickness of the shear web is calculated from the ultimate shear stress of the material:

$$t_{shear} = \frac{|q_{sh}|}{\tau_U} 1.5 \quad (\text{A.24})$$

The mass of the structural members is finally calculated. For the skin panels, given the density of the material ρ and the length of the section Δz :

$$M_{bend} = \rho t_{bend} \Delta z \left[\sum_{i=1}^n b_T(i) + \sum_{i=1}^n b_B(i) \right] \quad (\text{A.25})$$

while for the shear web, being h_{web} the height of the spar and L_{web} the length of the section:

$$M_{shear} = \rho t_{shear} h_{web} L_{web} \quad (\text{A.26})$$

The mass of a full wing is therefore:

$$M_w = 2 (M_{bend} + M_{shear}) \quad (\text{A.27})$$

A.5 Comparison to statistical equations

Table A.2 presents a comparison of the calculated wing mass of the elliptical wing in Section 4.1 with several statistical mass estimation equations of current use. It is clear that the method at hand underestimates the wing mass between 55% and 70%, with an average value of 61.11%. As explained before, this is caused by the neglect of the structural detail of the wing, such as ribs and currents. Therefore the calculated value is not used as an absolute estimation, but relative to the baseline in order to assess the variation of structural mass caused by the configuration analyzed. It should be noted, however, that also the mass estimation methods present an high variability, with a standard deviation of 51.34 kg on an average of 225.03 kg.

Direct Estimation Method	87.51 kg	
Cessna[33]	295.10 kg	-70.34%
USAF GA[33]	210.01 kg	-58.33%
Torenbeek GA[33][46]	192.13 kg	-54.45%
Raymer GA[31]	168.79 kg	-48.15%
Howe[47]	259.13 kg	-66.23%
Average of Statistical methods	225.03 kg	-61.11%

Table A.2: Comparison of mass estimation methods with elliptical wing of Sec. 4.1.

Appendix B

Baseline Aircraft Characteristics

B.1 Introduction

The Pilatus PC-24 was selected for this study, a small CS-23 certified general-aviation jet because of the availability of geometric and mass data in the public domain. It has a standard configuration for its class: engines mounted on the rear of the fuselage and a T-tail. Fig. B.1 is a three-view of the airplane in consideration. Data was taken from Pilatus drawings for a scale model of the airplane [48] and the Type-Certificate Datasheet from EASA [34].

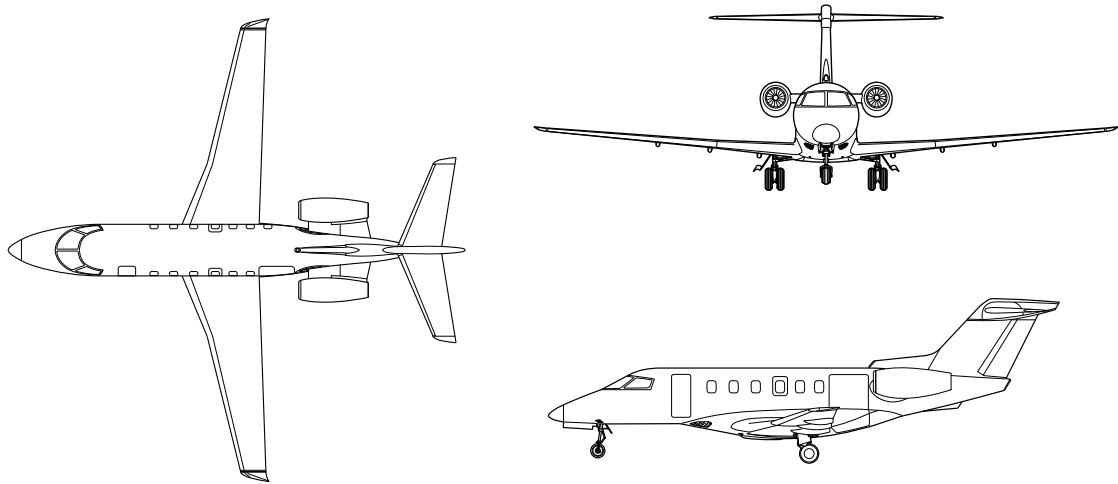


Figure B.1: Pilatus PC-24 three-view

B.2 Geometric Properties

Table B.1 contains the geometric properties of the PC24.

The geometry of the wing features a kink about one quarter of the span, which divides it into two trapezoidal sections. A figure of this arrangement is shown in Fig. B.2.

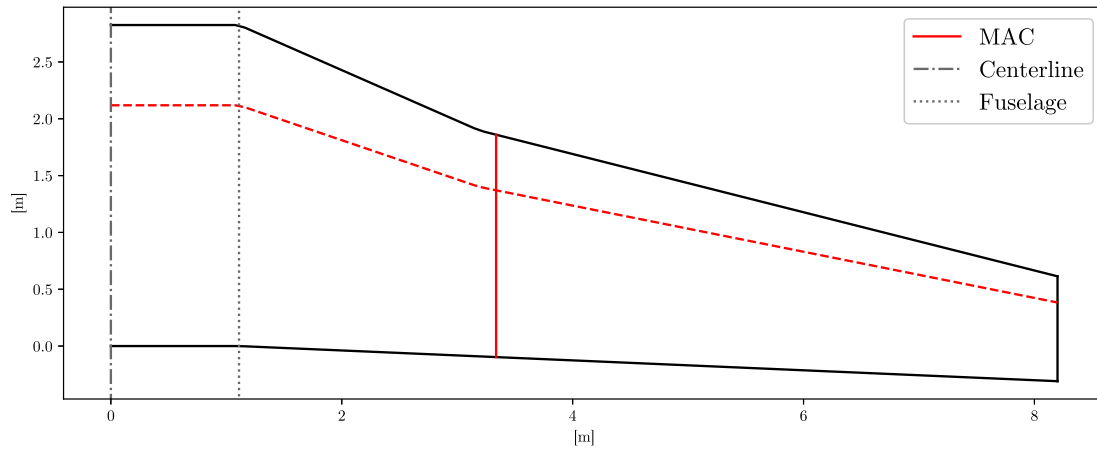
Fuselage	
Length	15 <i>m</i>
Diameter	1.901 <i>m</i>
Finess Ratio	7.89
Engine Nacelle length	2.616 <i>m</i>
Nacelle Diameter	0.95 <i>m</i>
Wing	
Wingspan	16.4 <i>m</i>
Root Chord	2.805 <i>m</i>
Tip Chord	0.923 <i>m</i>
Wing Area	30.1 <i>m</i> ²
Dihedral	5 °
Root Incidence	1.19 °
Tip Washout	-1.063 °
Flaps span	4.5264 <i>m</i>
Flaps relative chord c_f/c	0.2
Aileron span	1.955 <i>m</i>
Aileron relative chord c_f/c	0.266
Horizontal Tail	
Span	6.632 <i>m</i>
Root Chord	1.665 <i>m</i>
Tip Chord	0.755 <i>m</i>
Area	8.02472 <i>m</i> ²
Sweep	23.27 °
Incidence	0 °
Elevator span	3.316 <i>m</i>
Elevator relative chord c_f/c	0.34
Vertical Tail	
Span	2.07 <i>m</i>
Root Chord	3.069 <i>m</i>
Tip Chord	2.153 <i>m</i>
Area	5.4047 <i>m</i> ²
Sweep	43.64 °
Rudder span	1.7331 <i>m</i>
Rudder relative chord c_f/c	0.35

Table B.1: Geometric Properties

B.3 Mass Properties

Structural and Mass properties of the airplane are presented here. Mass and center of gravity positions were estimated with Raymer's weight formulas [31] and adjusted to match the empty mass calculated from available data.

The $V - n$ diagram characteristic velocities are all Equivalent Airspeeds (EAS).

**Figure B.2:** PC-24 Wing

Empty Mass	5233.23 kg
Fuel Tank Capacity	3363 L
Max Fuel Mass	2707 kg
Max Take Off Mass	8300 kg
Max Landing Mass	7665 kg

Table B.2: Mass Properties

Component	Mass	X_{CG}	Z_{CG}
Fuselage	1614.56 kg	7.802 m	2.423 m
Wing	1013.98 kg	8.318 m	1.677 m
Horizontal Tail	188.77 kg	15.703 m	5.427 m
Vertical Tail	119.65 kg	14.484 m	4.274 m
Engines	587.98 kg	12.273 m	3.076 m
Main Landing Gear	303.77 kg	9.719 m	1.642 m
Nose Landing Gear	45.5 kg	1.31 m	1.963 m
Systems	1359 kg	7.514 m	2.175 m

Table B.3: Empty Mass Breakdown

$+n_{max}$	2.84
V_a	95.17 m/s
V_c	149.18 m/s
V_d	185.2 m/s

Table B.4: Structural Load Properties

B.4 Propulsion Properties

These are the characteristics of the installed engines.

Name	Williams FJ44-4A
Number	2
Type	Twin-Spool Turbofan with 1-stage fan
Compressor	3-stage axial LP + 1-stage HP centrifugal
Turbine	1-stage HP + 2-stage LP
Uninstalled Thrust	16.0 kN
Uninstalled Weight	295 kg
Length	1.34 m
Diameter	0.64 m
Specific Fuel Consumption @ ISA Sea Level	0.456 lb/(h lbf)

Table B.5: Propulsion Properties

B.5 Performance Properties

Here are the declared performance from the constructor website. Maximum lift coefficients were estimated from the declared landing weight and speed plus the estimated $\Delta C_{L,flaps}$ from the DATCOM methods.

Balanced Field Length @ MTOW, Sea Level	893 m
Landing distance over 50 ft obstacle @ MLW, Sea Level	724 m
Maximum Rate of Climb (S.L, 200 KCAS)	20.7 m/s
Maximum Cruise Speed (FL280)	440 KTAS
Range with 4 passengers (with 100 nm IFR reserves + 30 min VFR reserves)	2000 nm
Maximum certified altitude	45'000 ft
Stall Speed (landing configuration, MLW)	82 KIAS
$C_{L,max}$, full flaps	2.2873
$C_{L,max}$, clean configuration	1.4295

Table B.6: Performance Properties

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